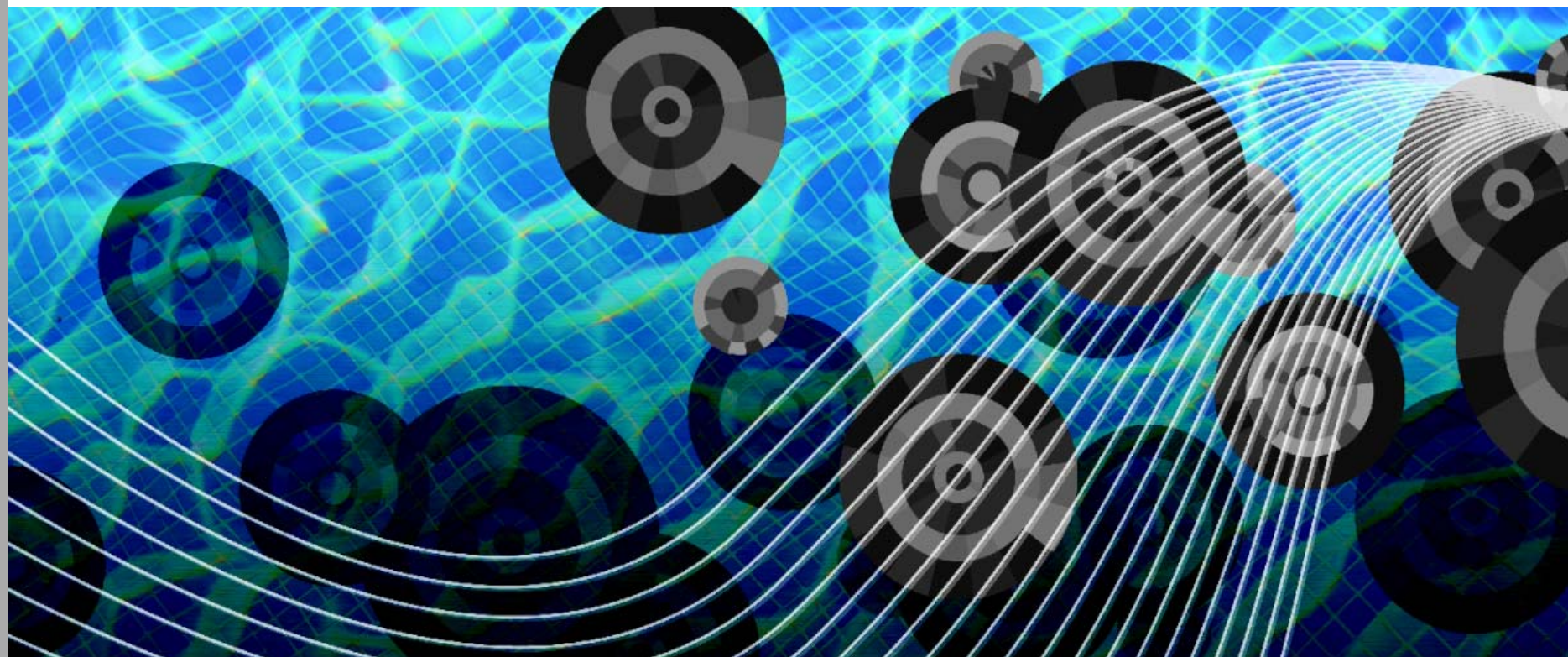
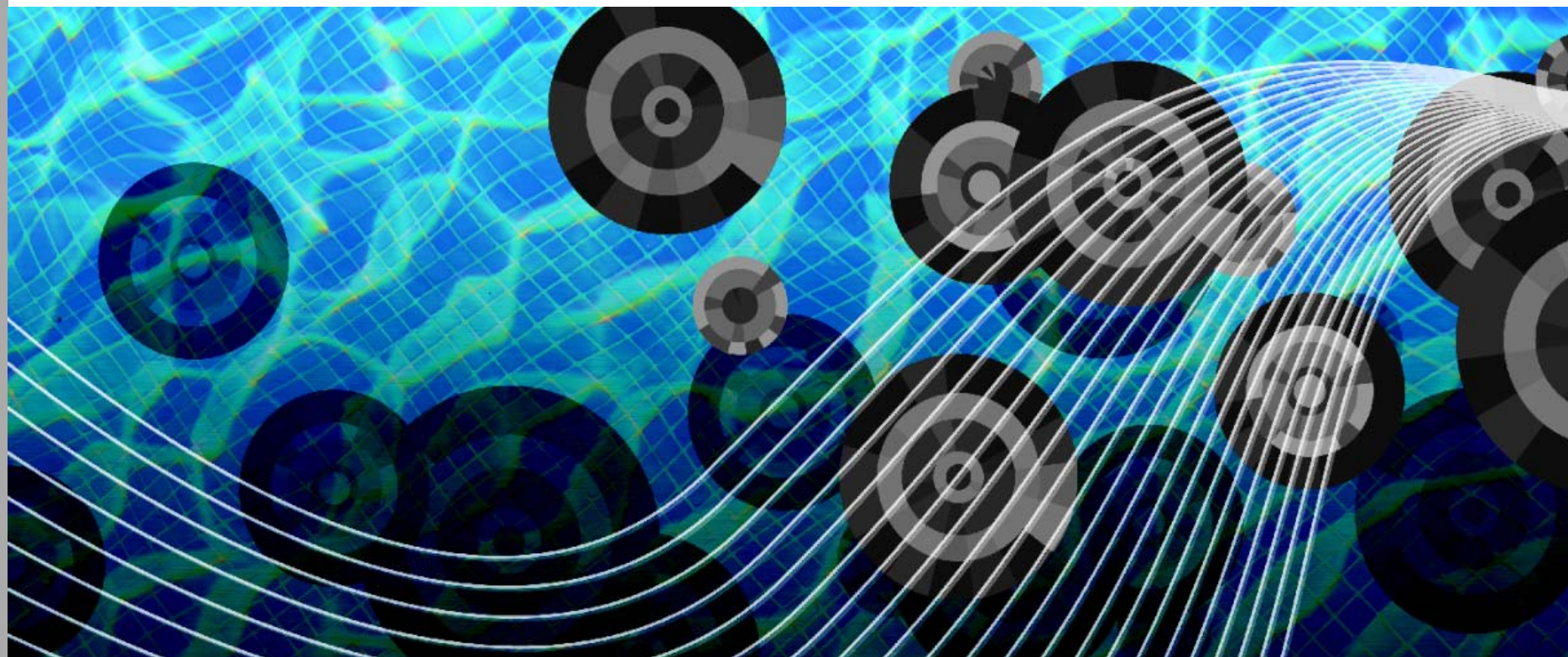
专注高频线路板pcb生产（板材有：F4B,Rogers, TACONIC，ARLON, TP-2等），13823677869 QQ116425136 叶工

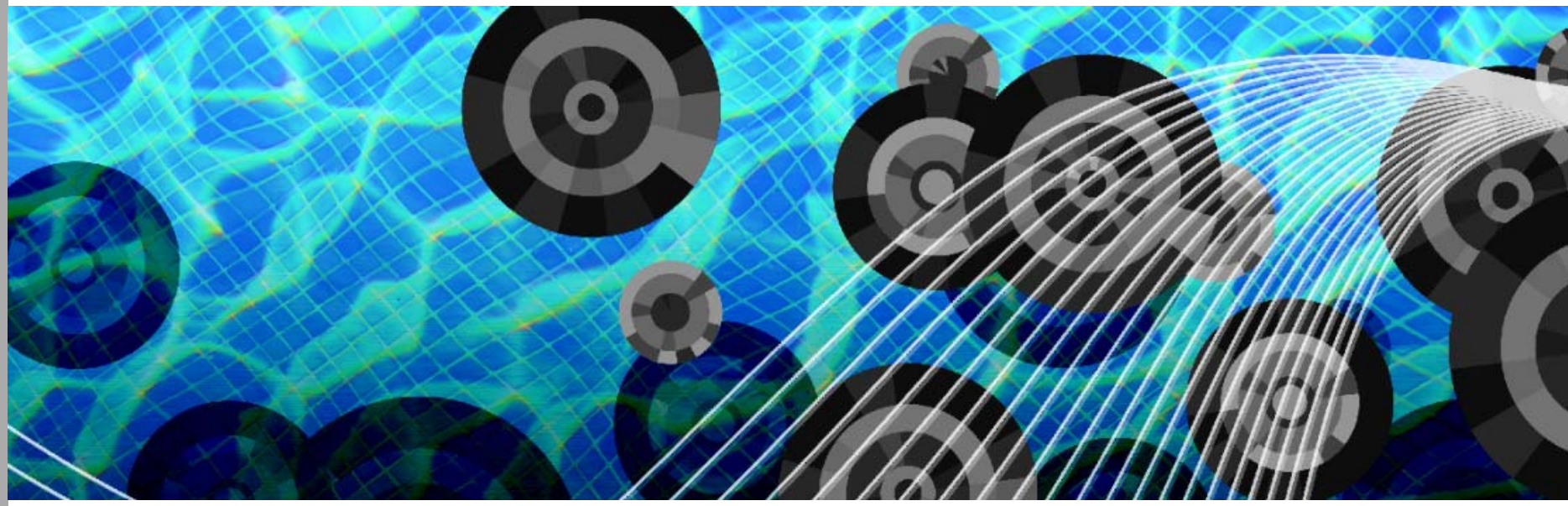
CST – 各类传输线仿真报告



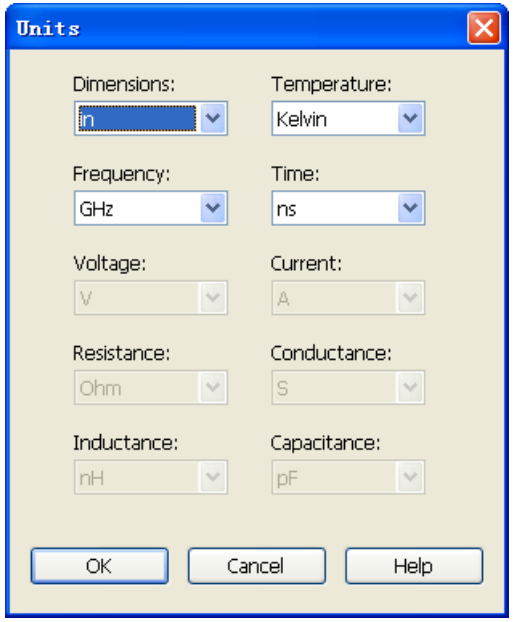
带状线

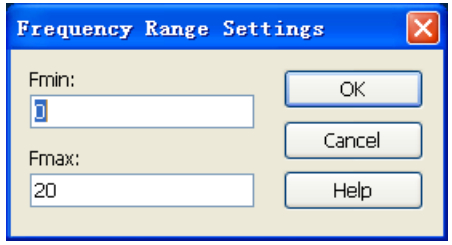


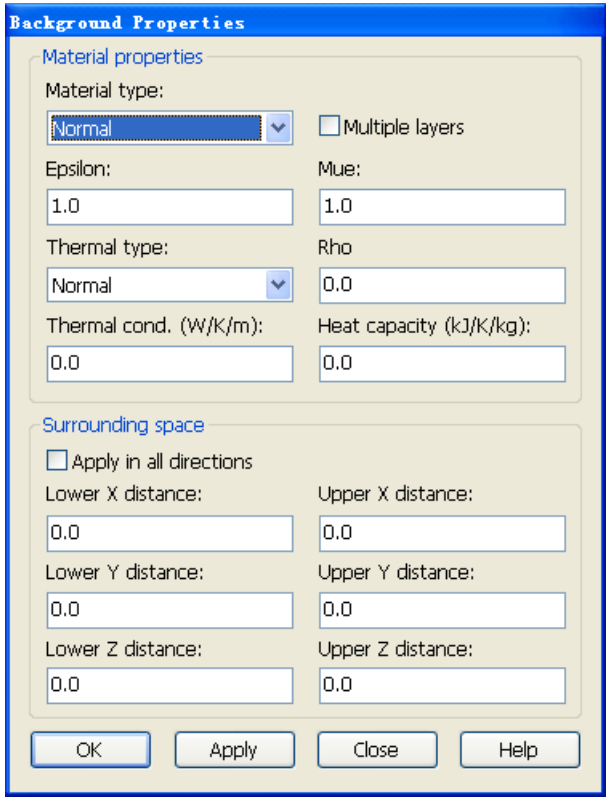
对称带状线



定义单位、背景条件

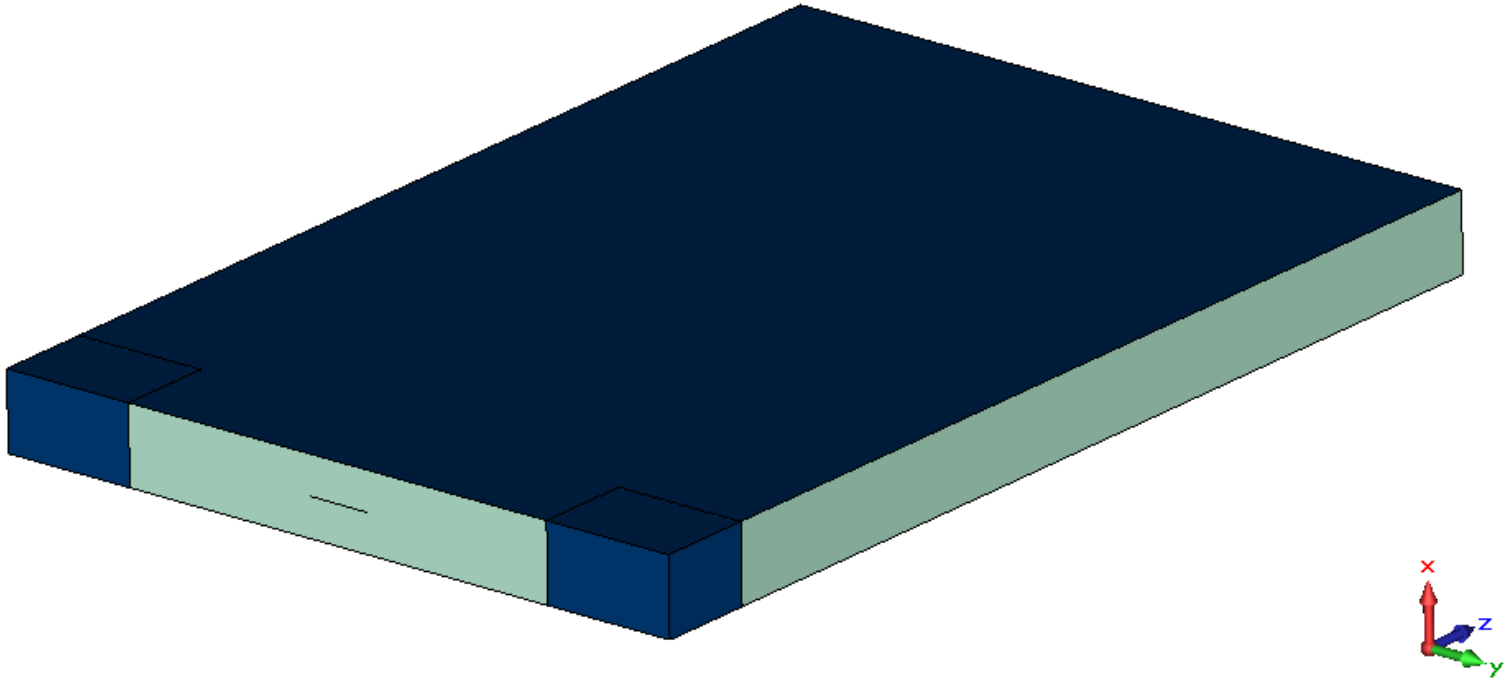




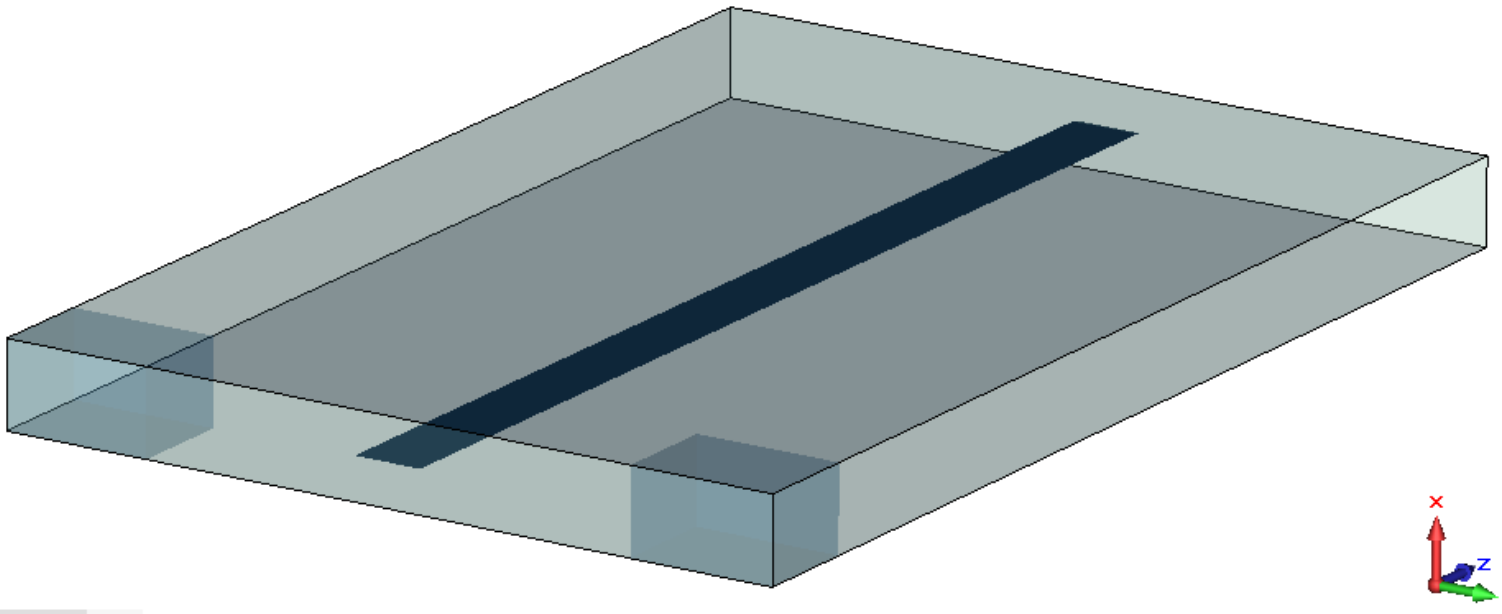


选择系统单位，这里将长度单位定义为inch，频率单位定义为GHz。

带状线模型



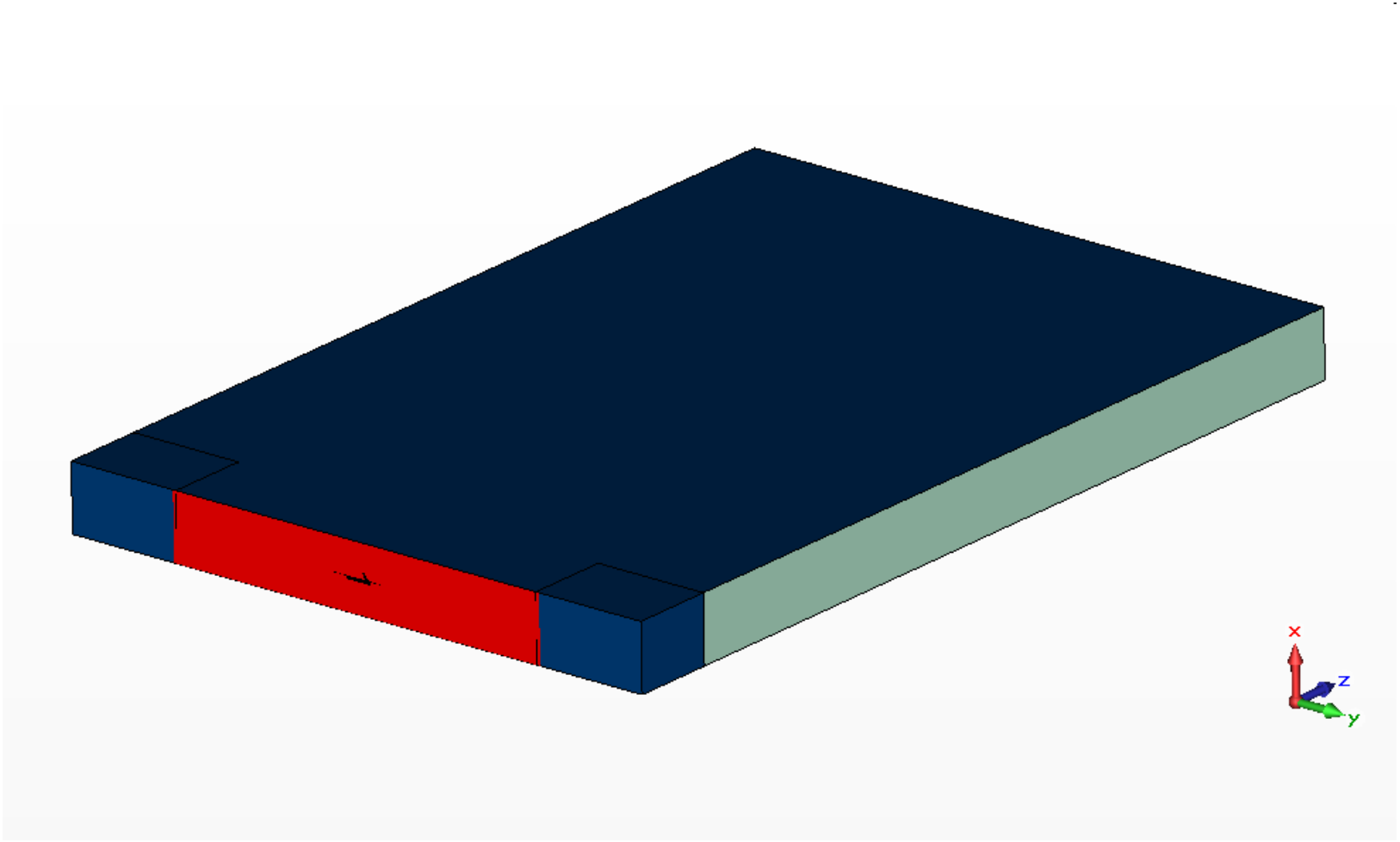
带状线模型



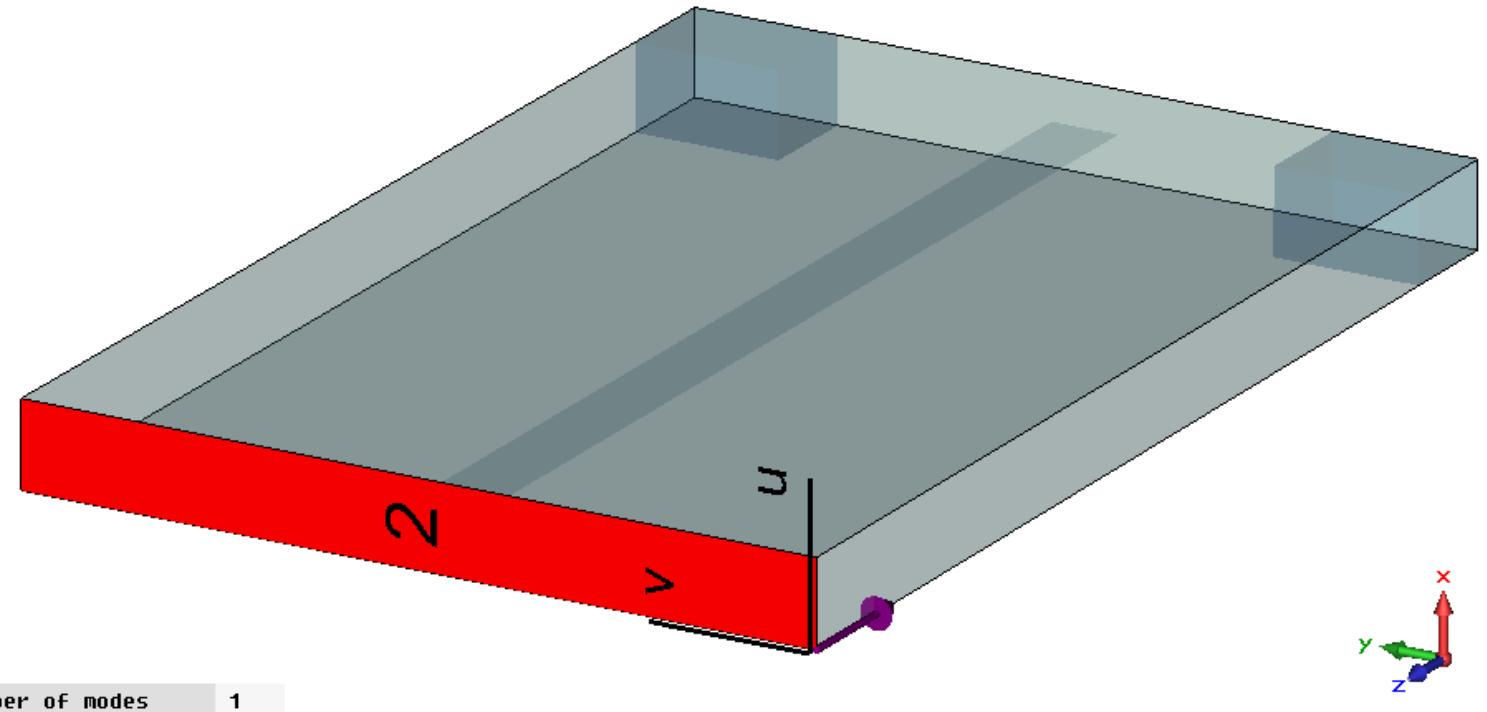
Material	PEC
Type	PEC
Therm.cond.	PTC

介质层相对介电常数2.2。

1端口激励



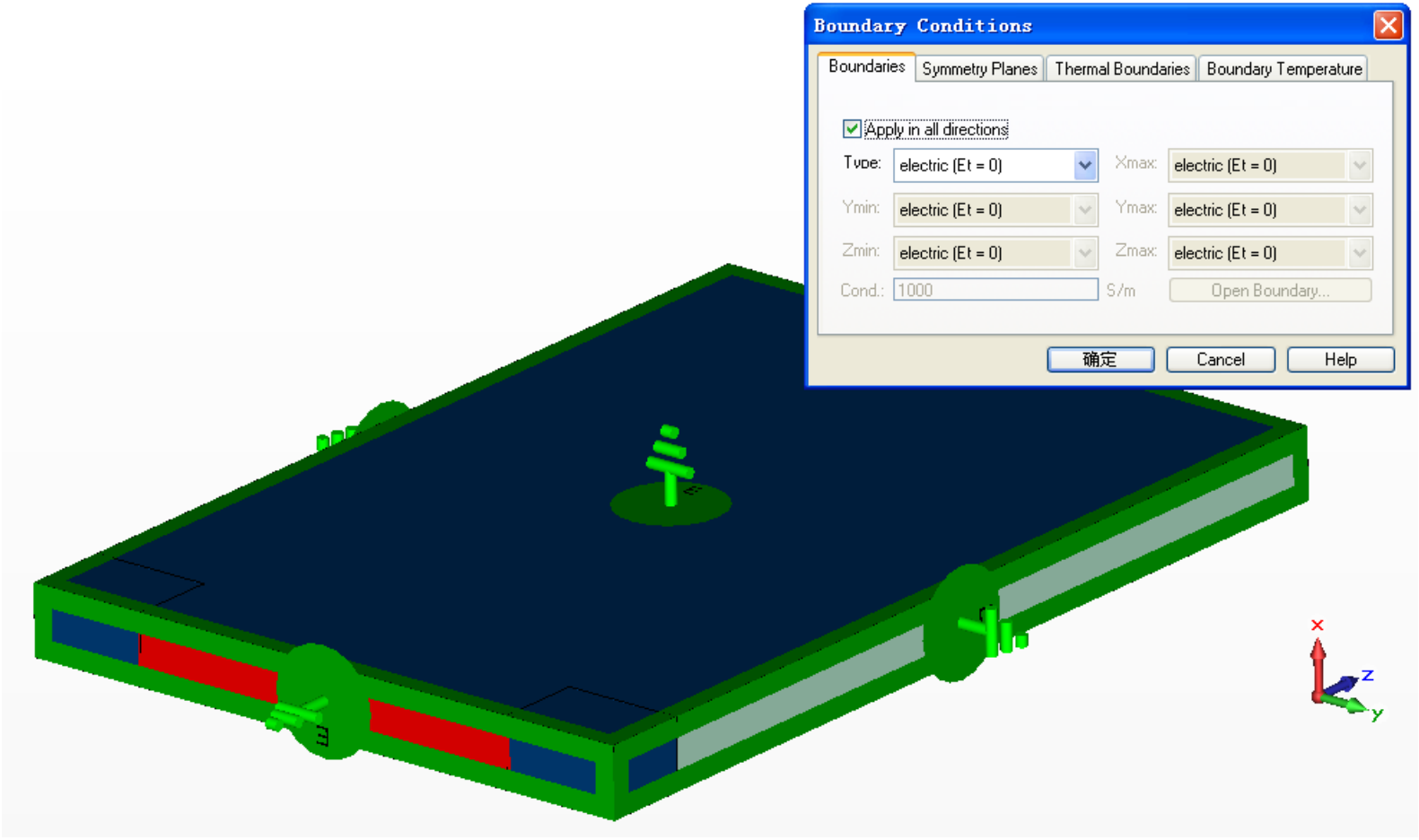
2端口



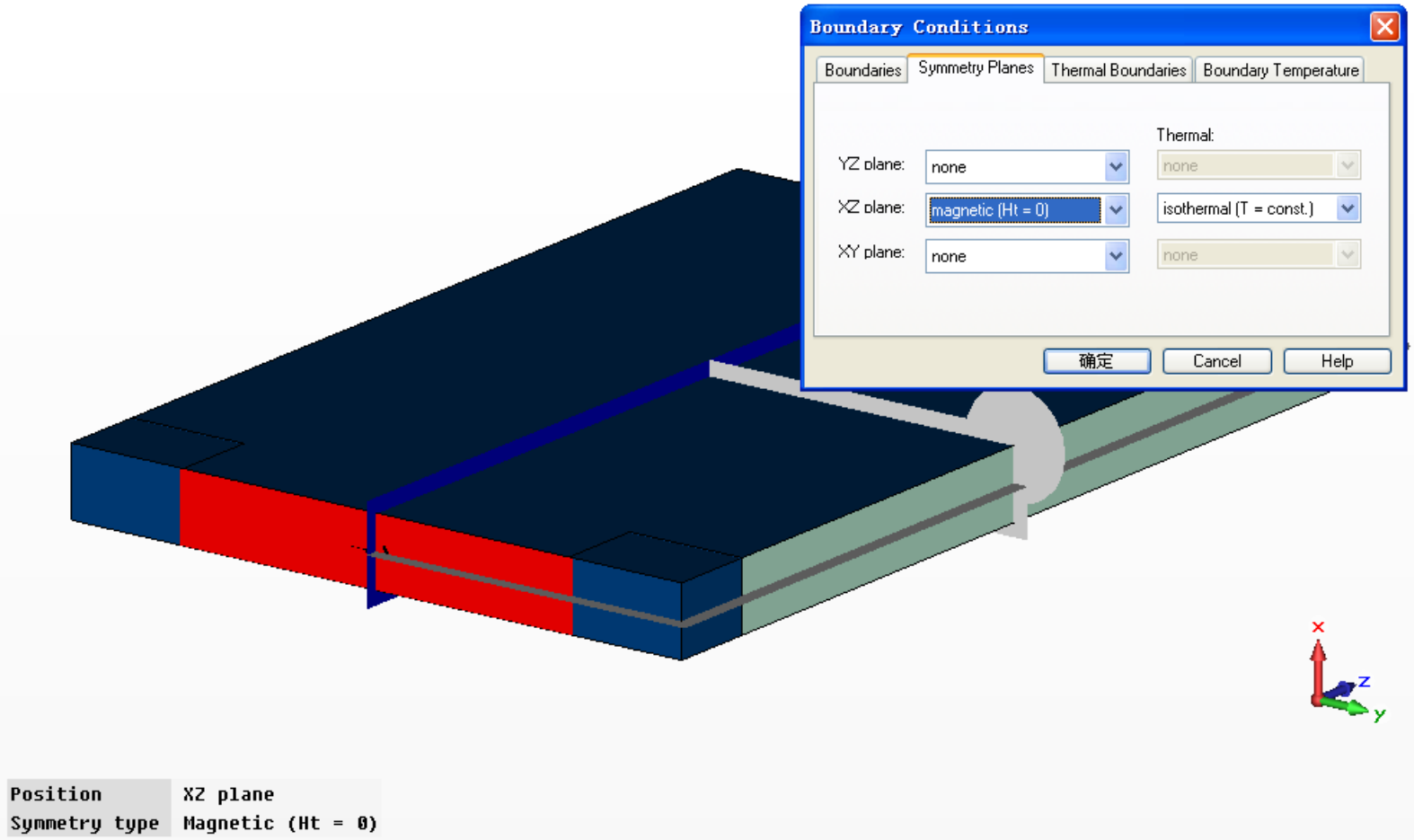
Number of modes	1
Polarization angle	none
Dist. to ref. plane	0

2端口作主要用来作为匹配

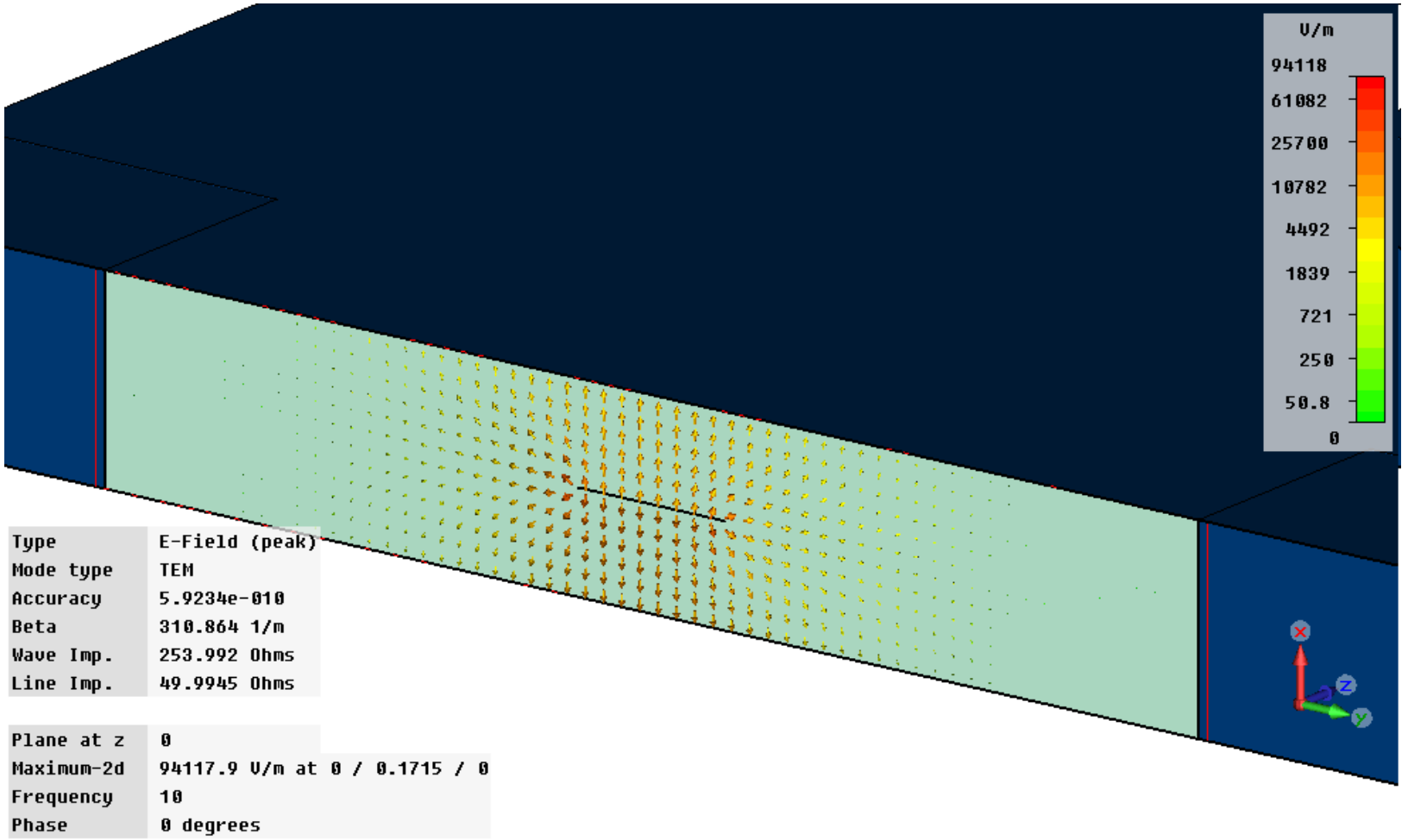
边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Type:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help
x
z
y

使用对称面加快仿真速度

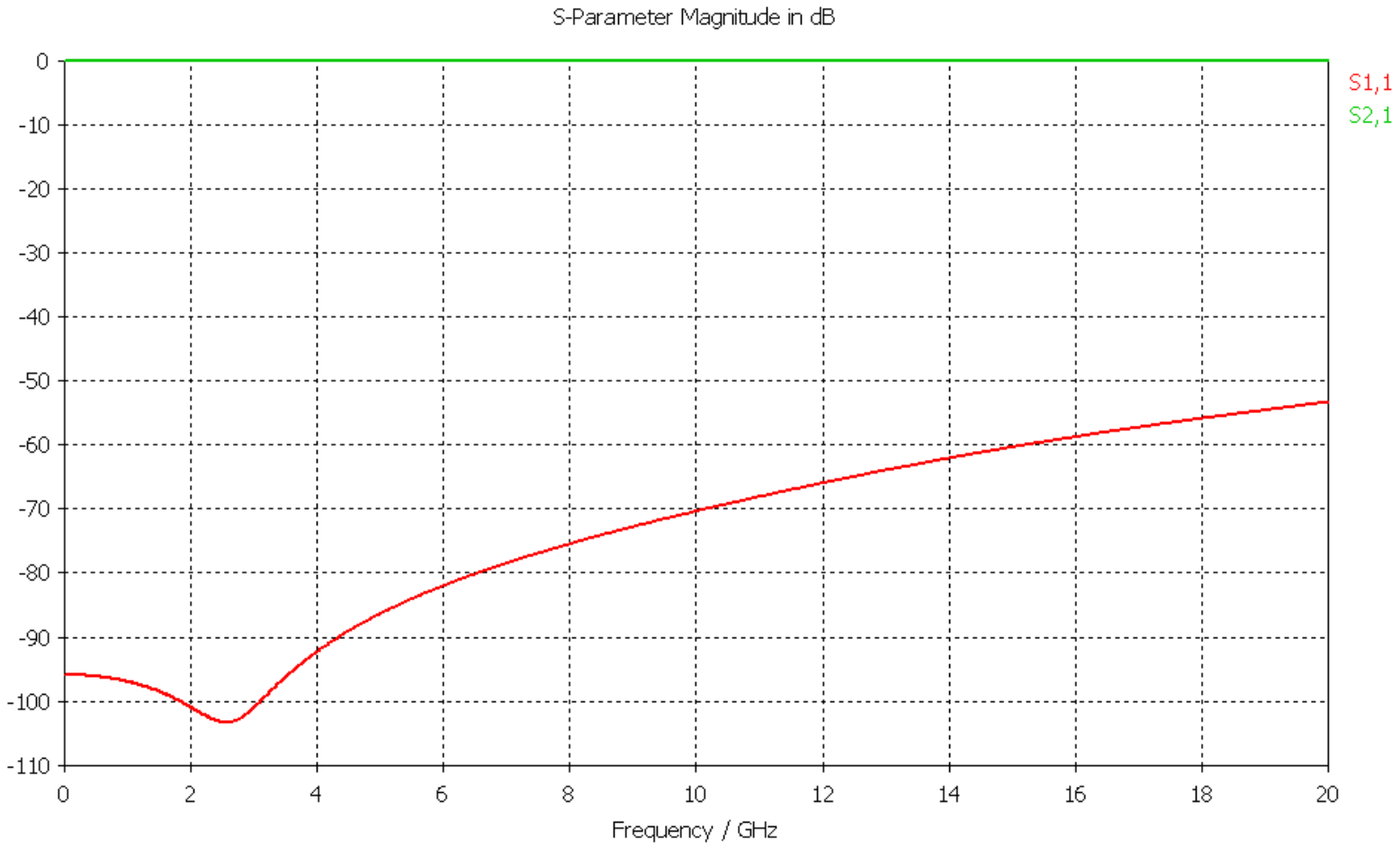


1端口电场端口模式

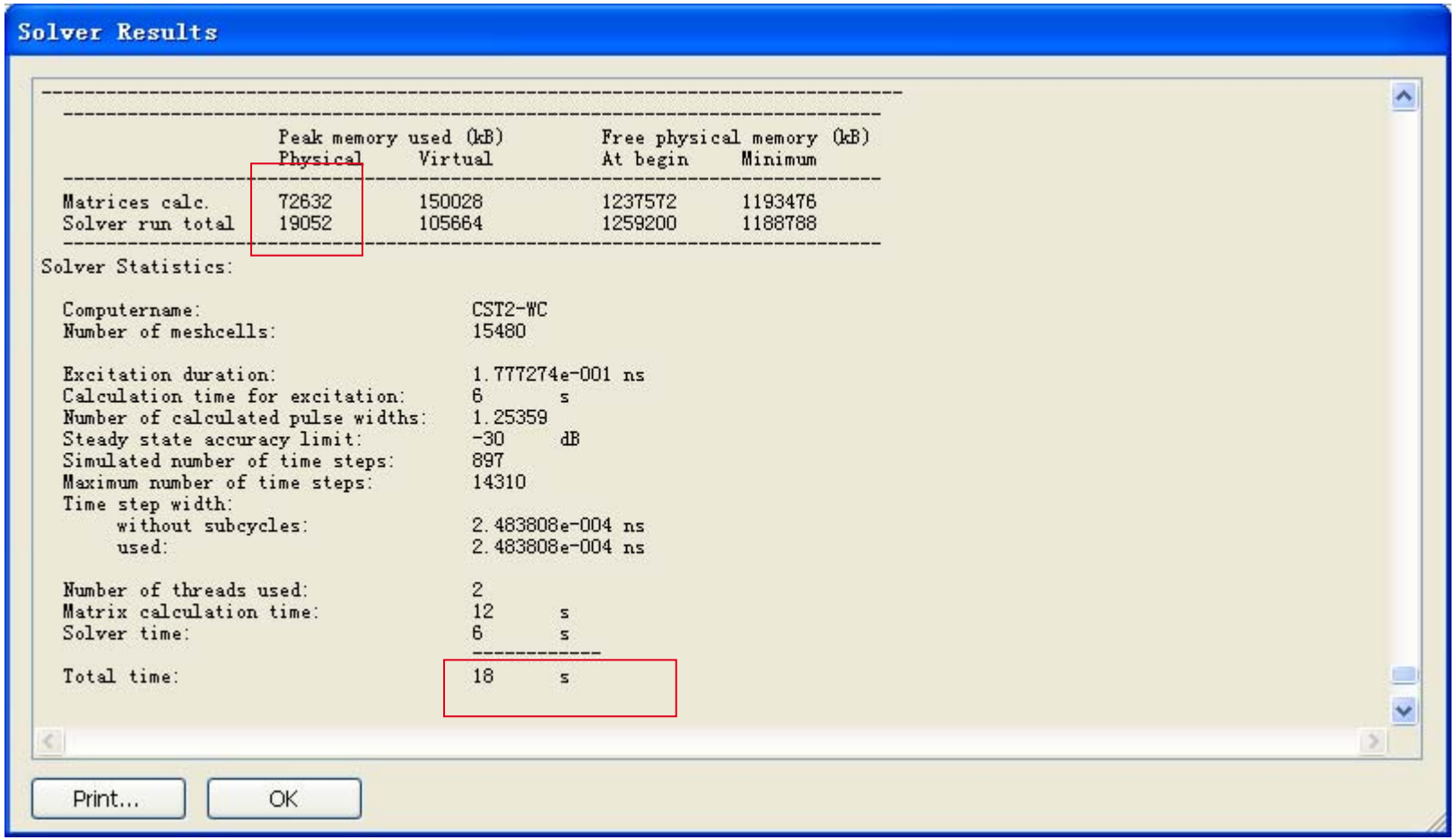


端口阻抗50欧姆。

S参量 dB

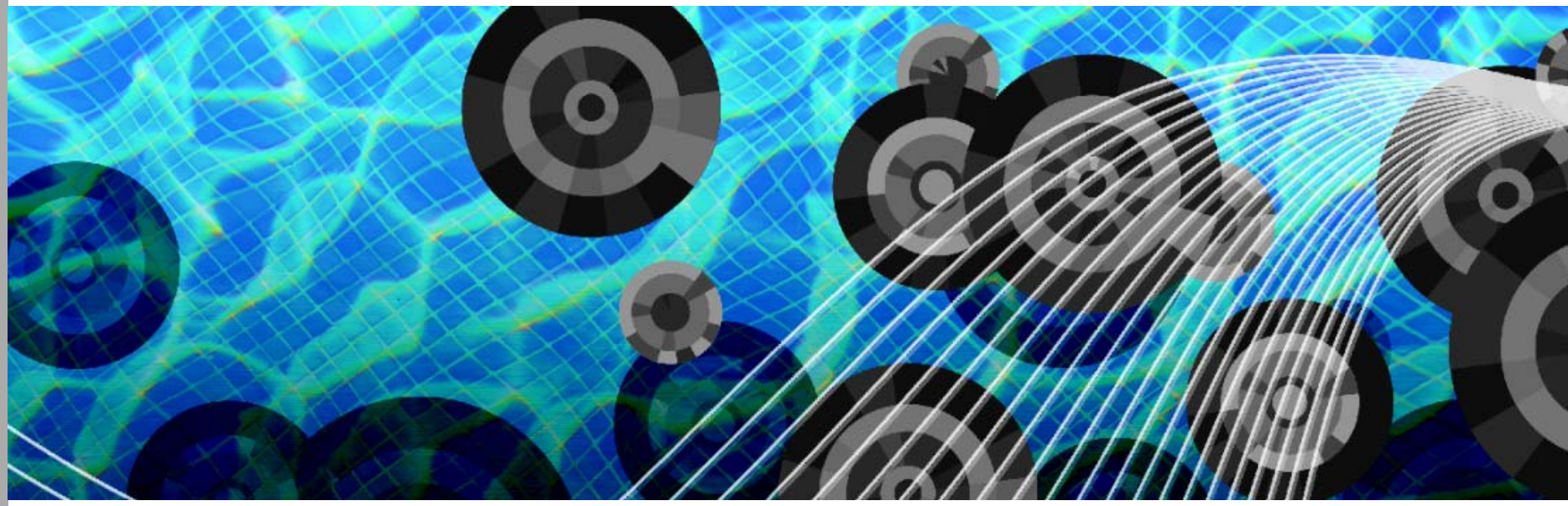

S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-110
0
2
4
6
8
10
12
14
16
18
20
Frequency / GHz
S1,1
S2,1

占用系统资源

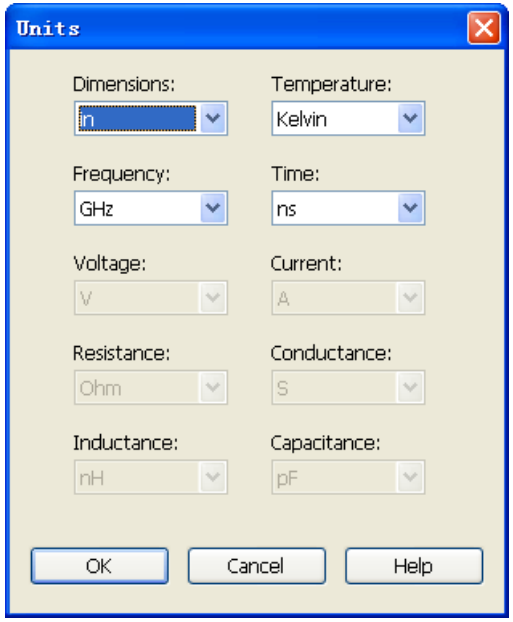


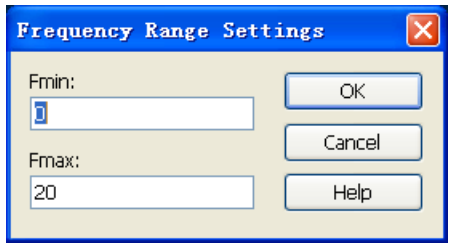
建立矩阵占用内存72M，求解占用内存19M，求解器计算时间18s。

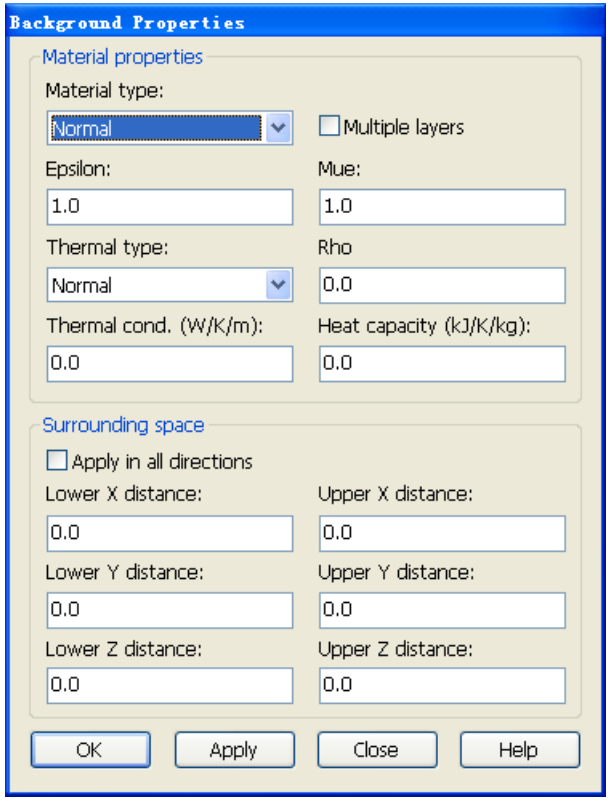
非对称带状线



定义单位、背景条件

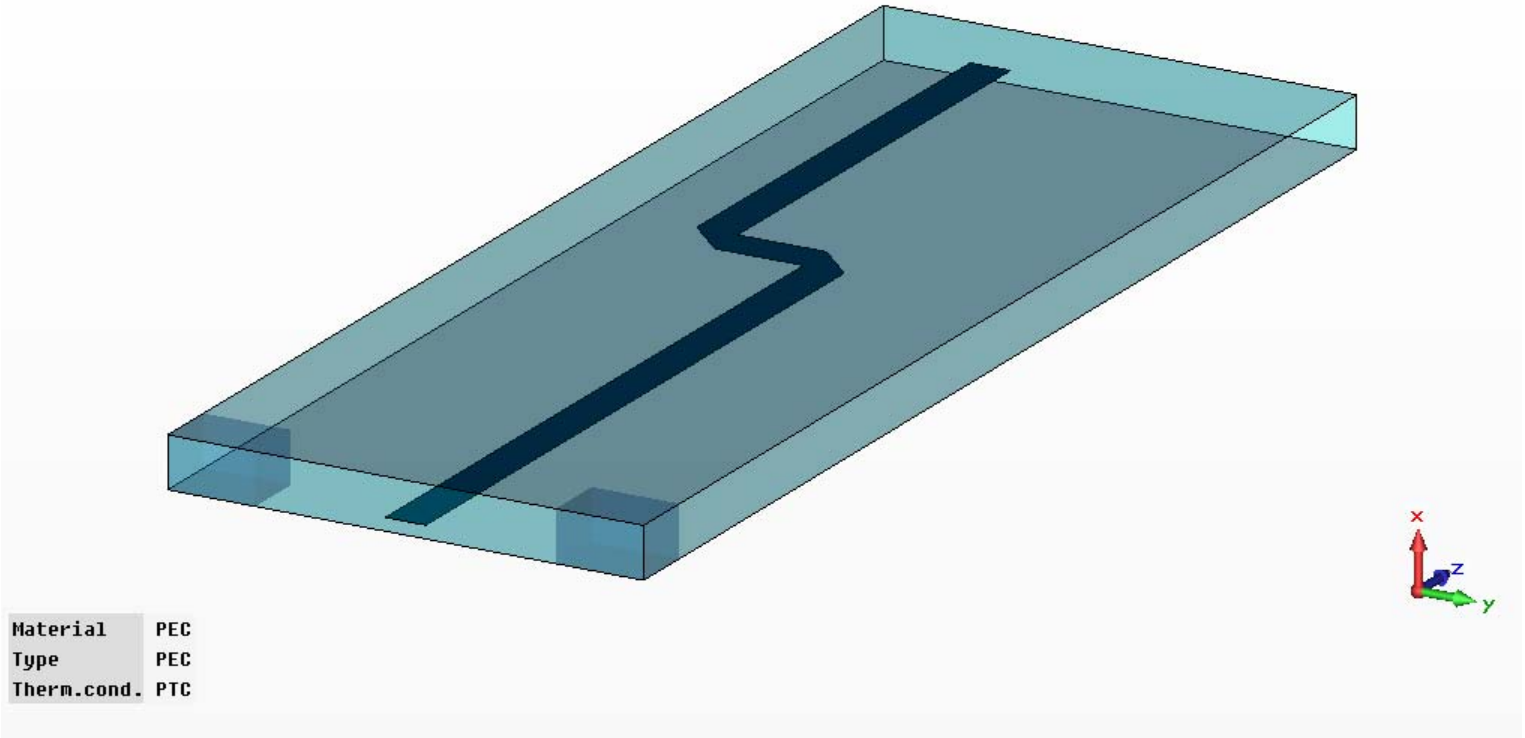






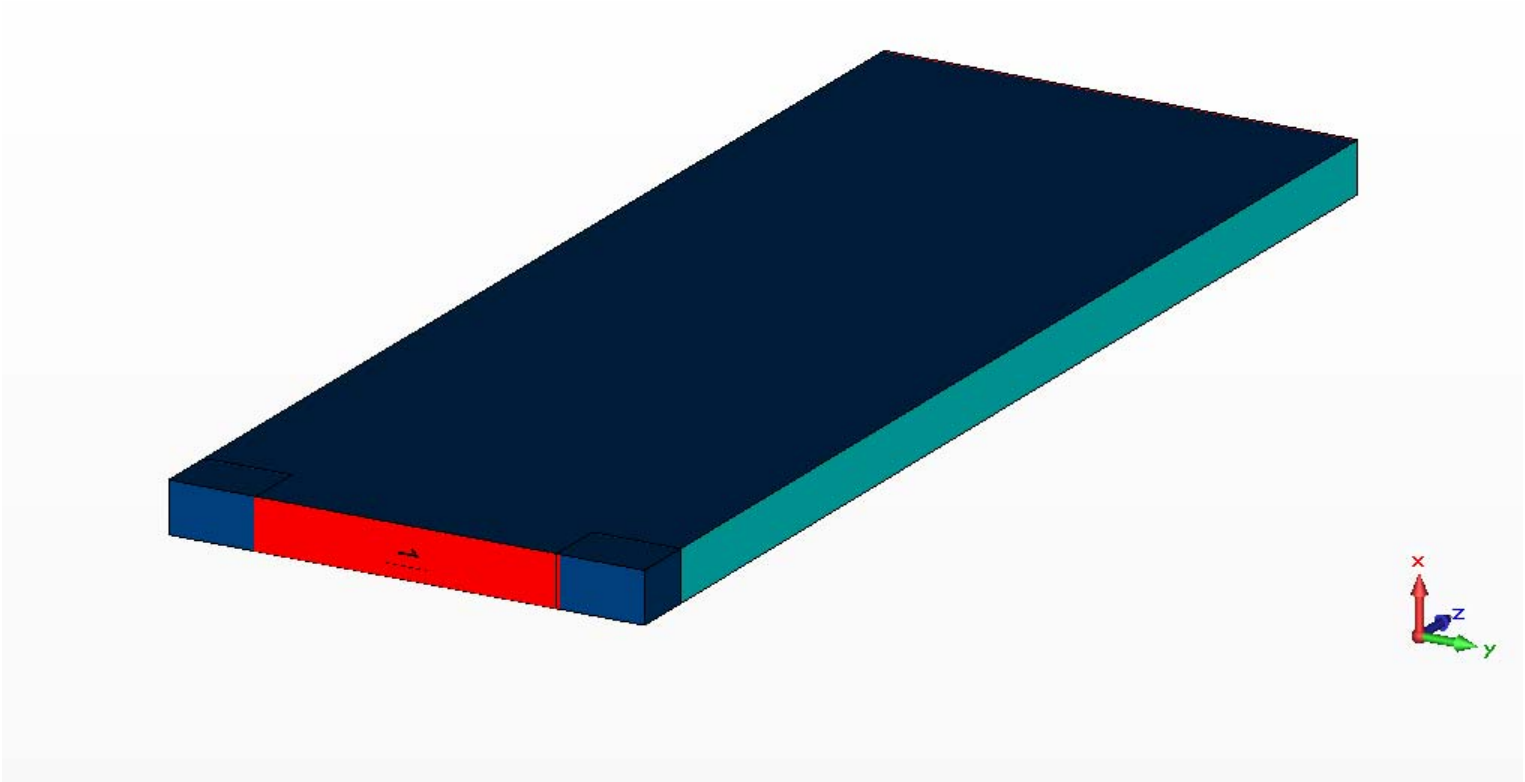
选择系统单位，这里将长度单位定义为inch，频率单位定义为GHz。

模型：非对称带状线

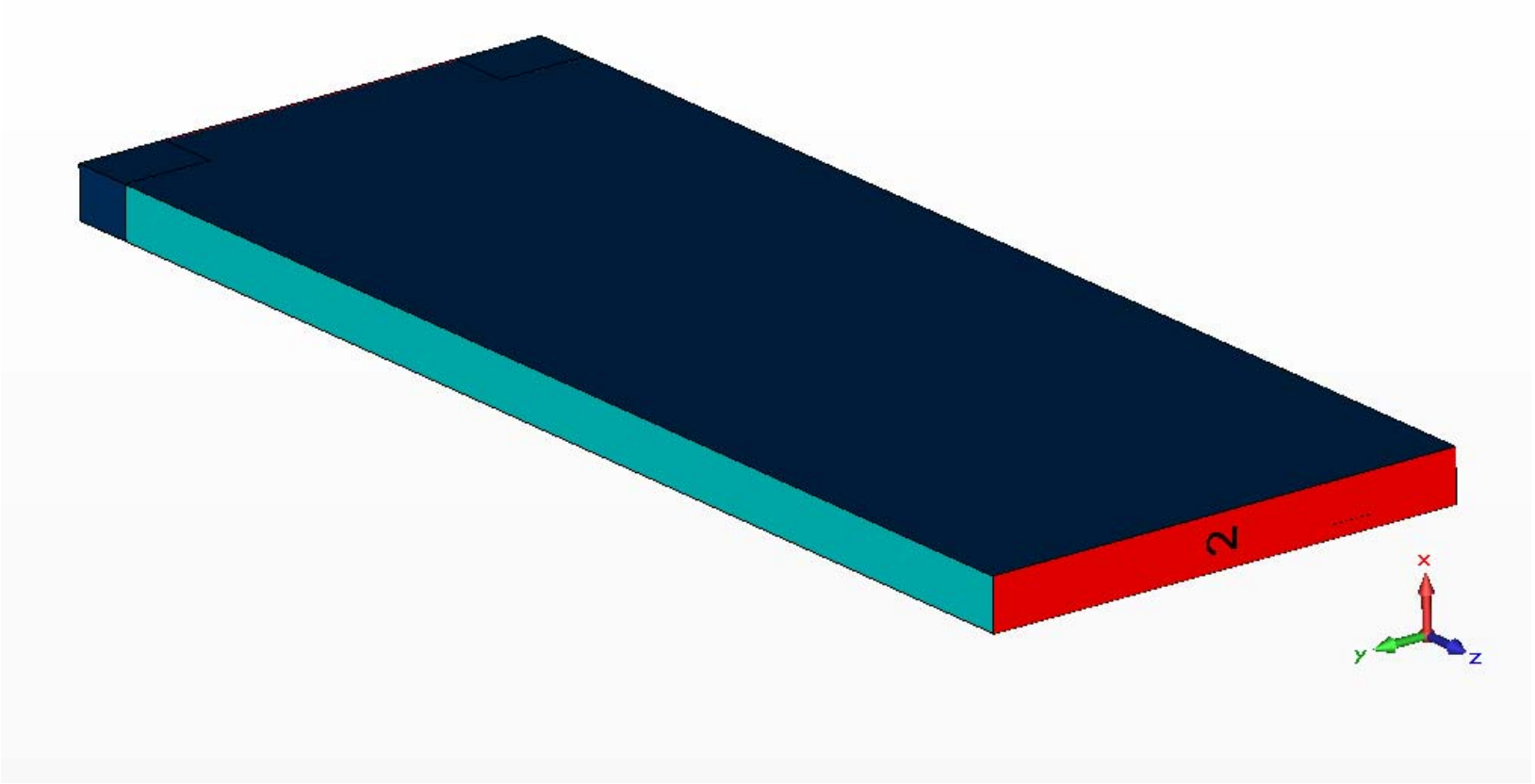


中心为非对称带状线馈线部分，介质板相对介电常数1.5。

1端口激励

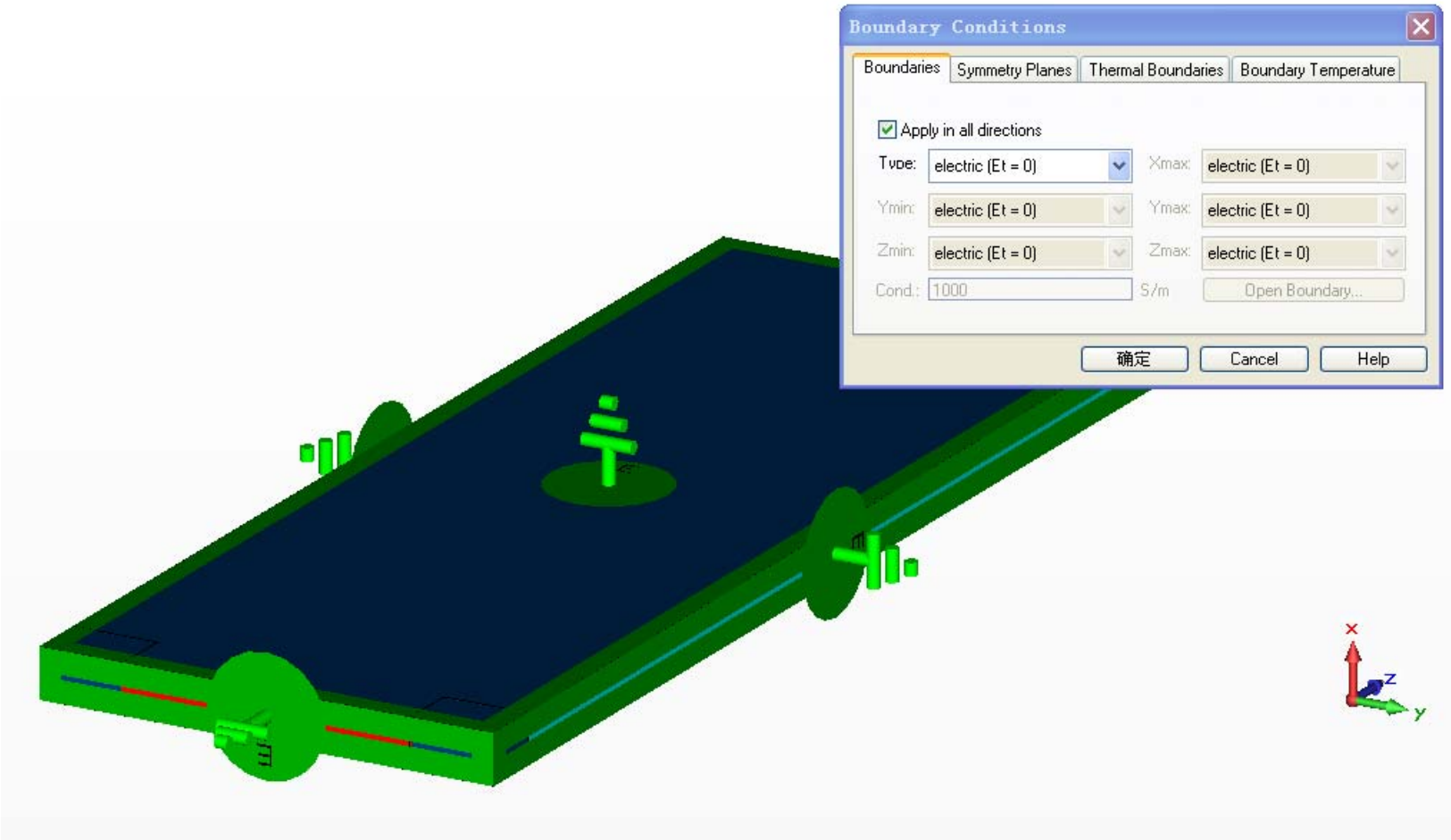


2端口激励

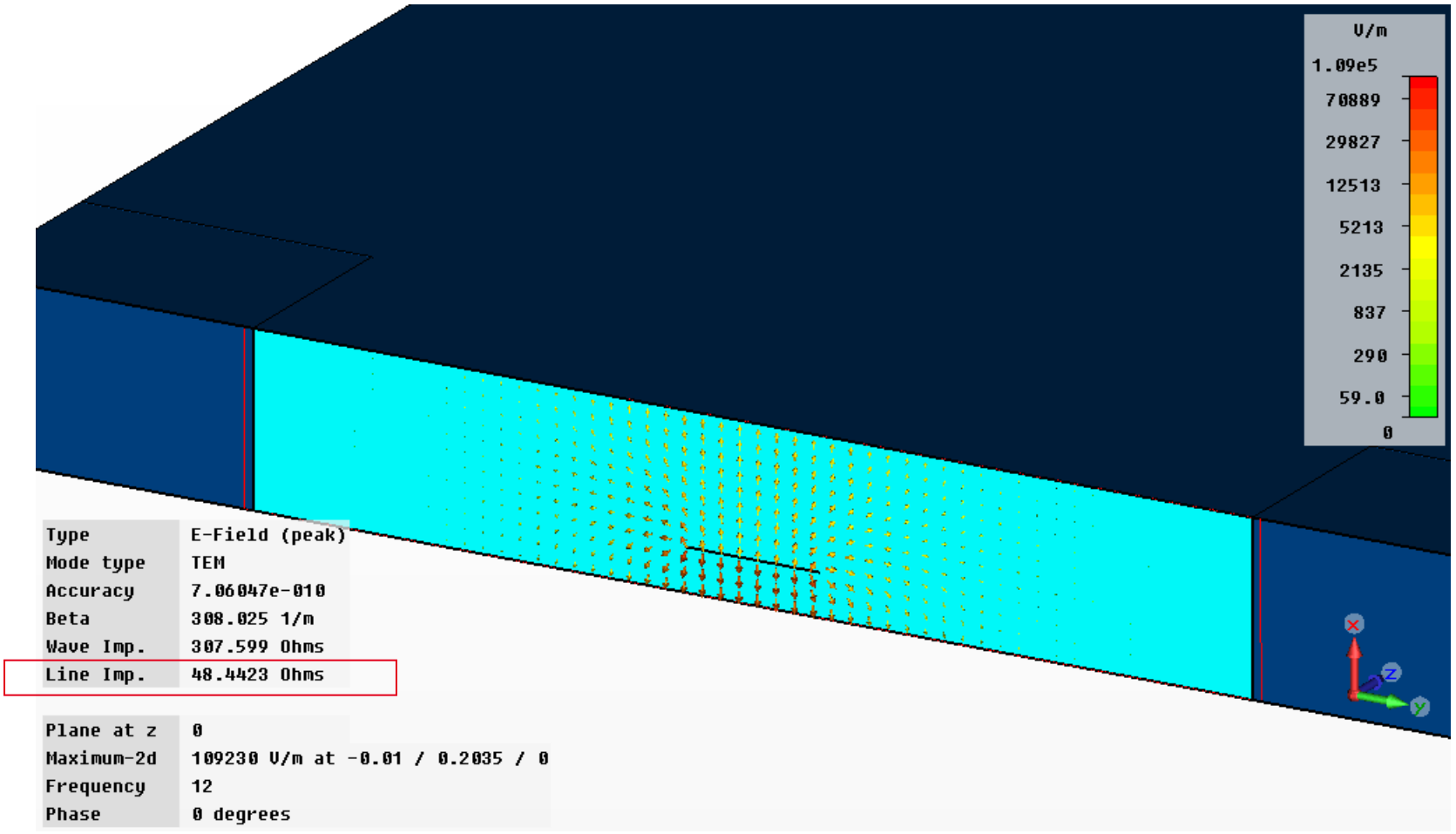


2端口主要用来做匹配。

边界条件

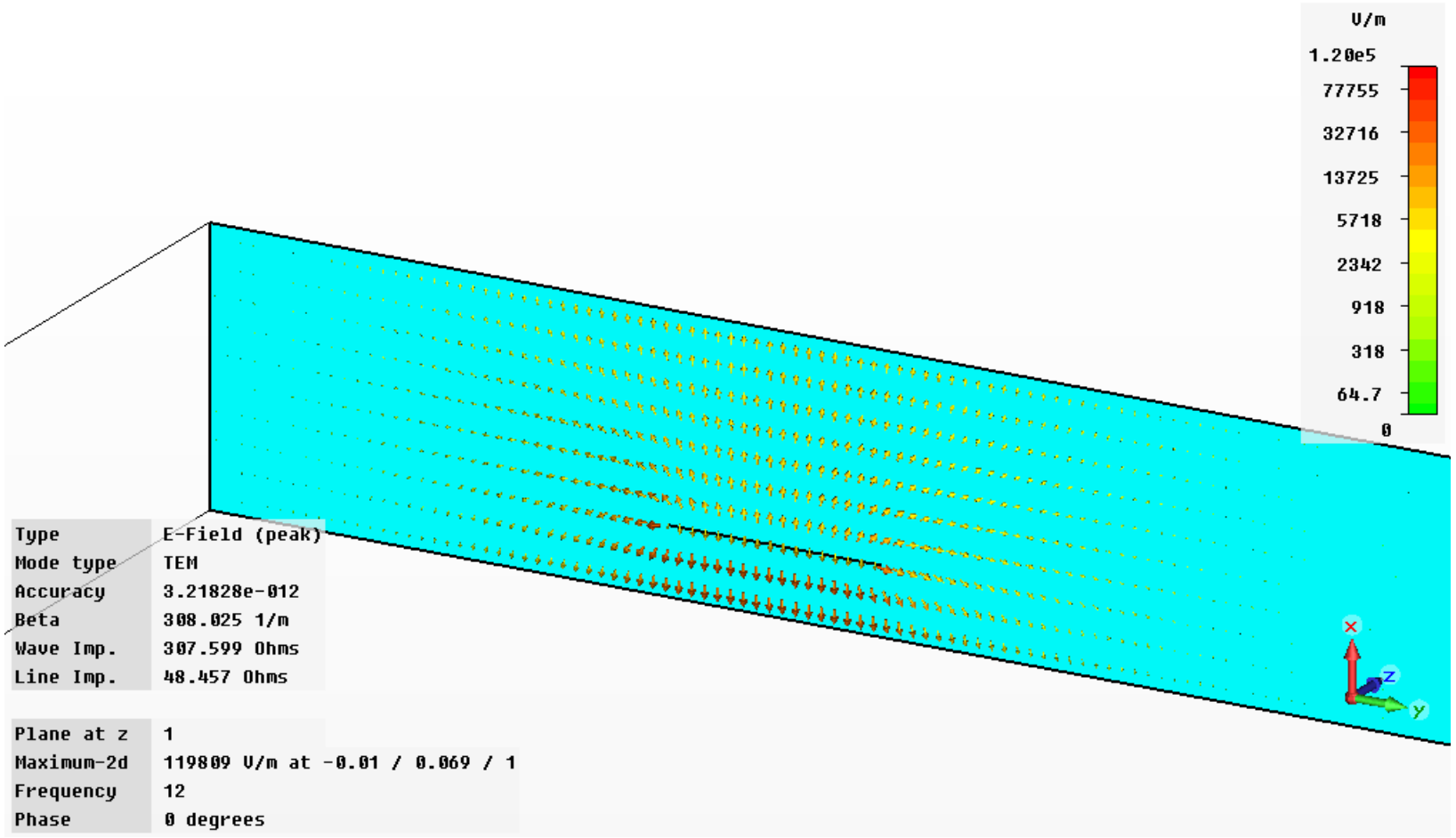

Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Type:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help
x
z
y

1端口电场端口模式



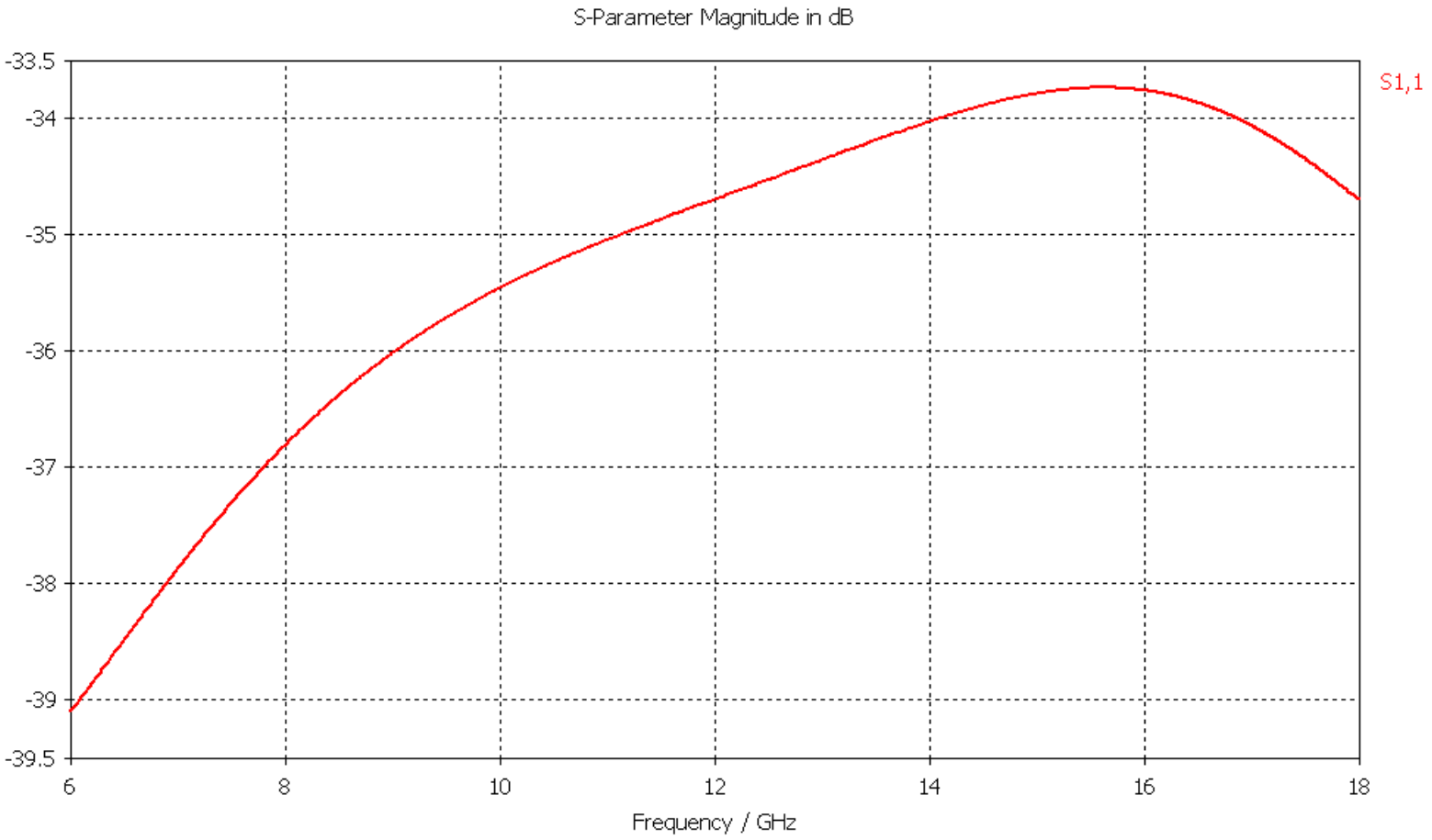
非对称带状线端口阻抗48欧姆。

2端口电场端口模式

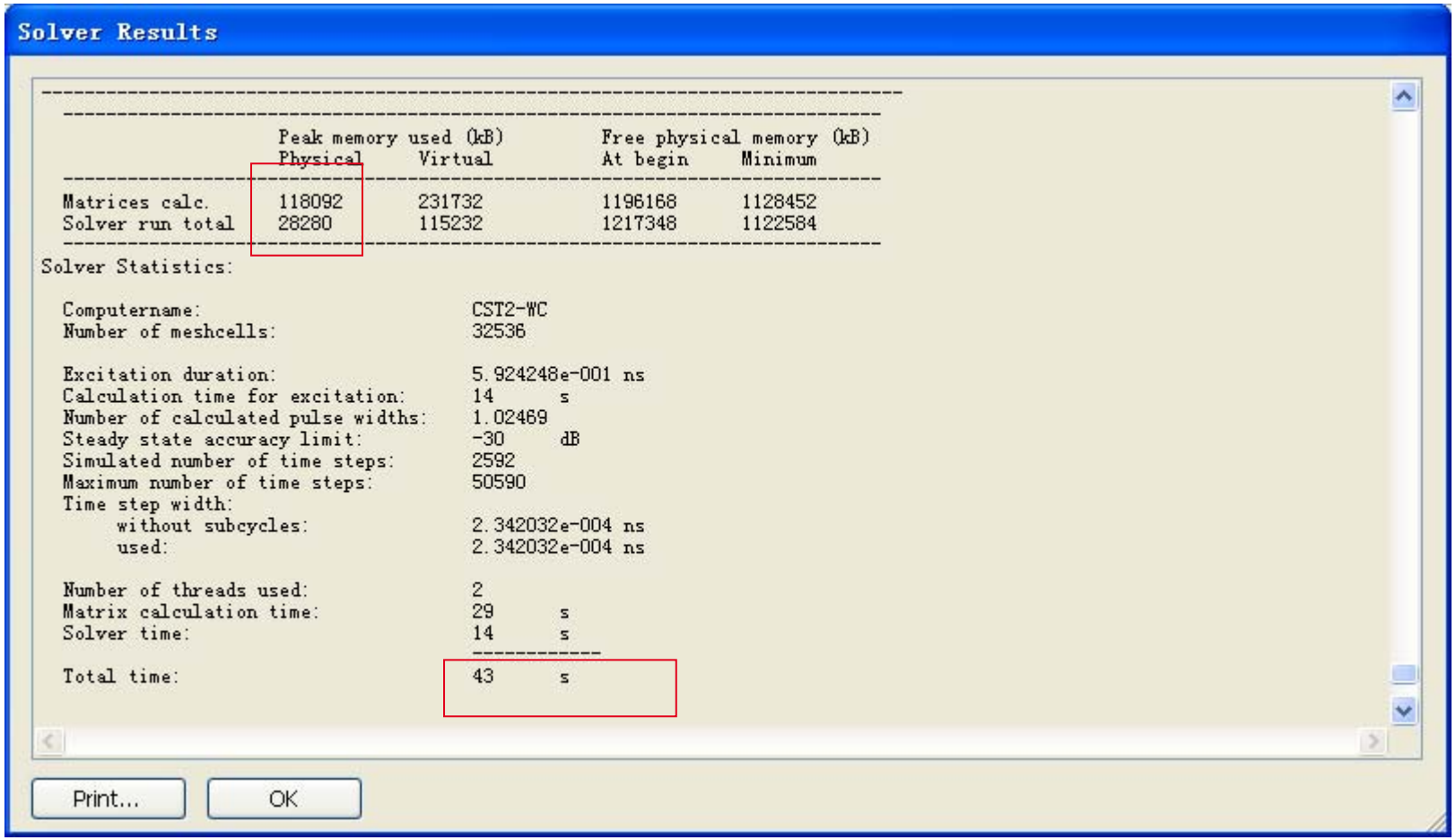


2端口用来做匹配。

1端口S参量（dB）

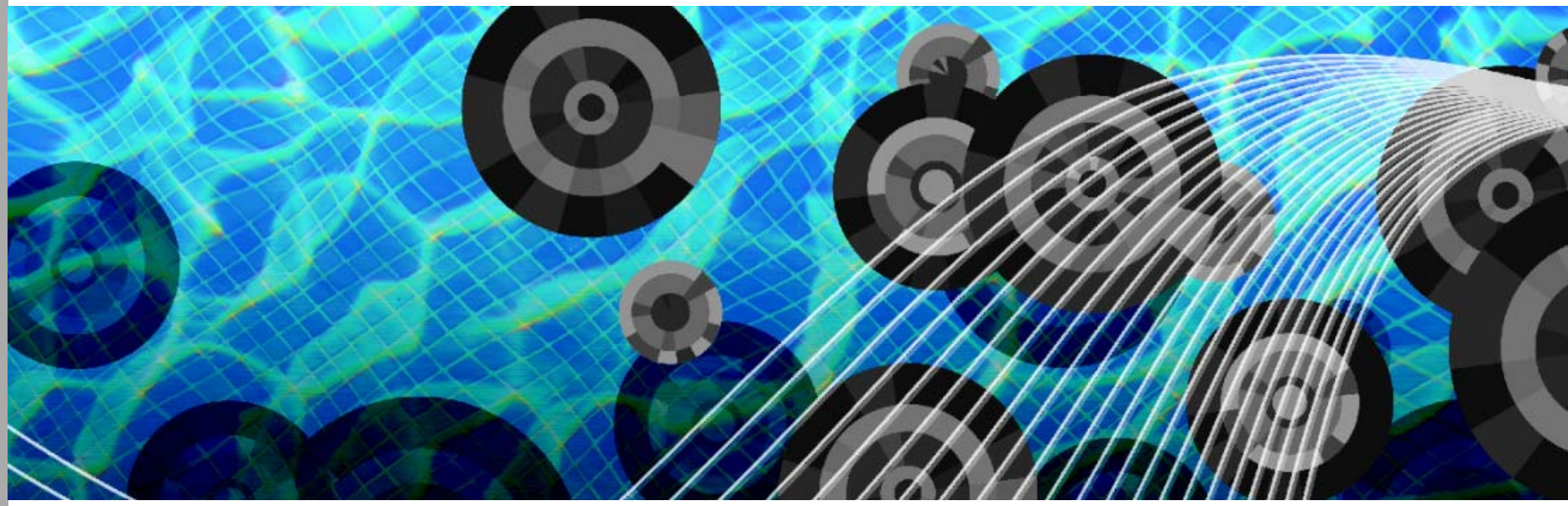

S-Parameter Magnitude in dB
-33.5
-34
-35
-36
-37
-38
-39
-39.5
6
8
10
12
14
16
18
Frequency / GHz
S1,1

占用系统资源

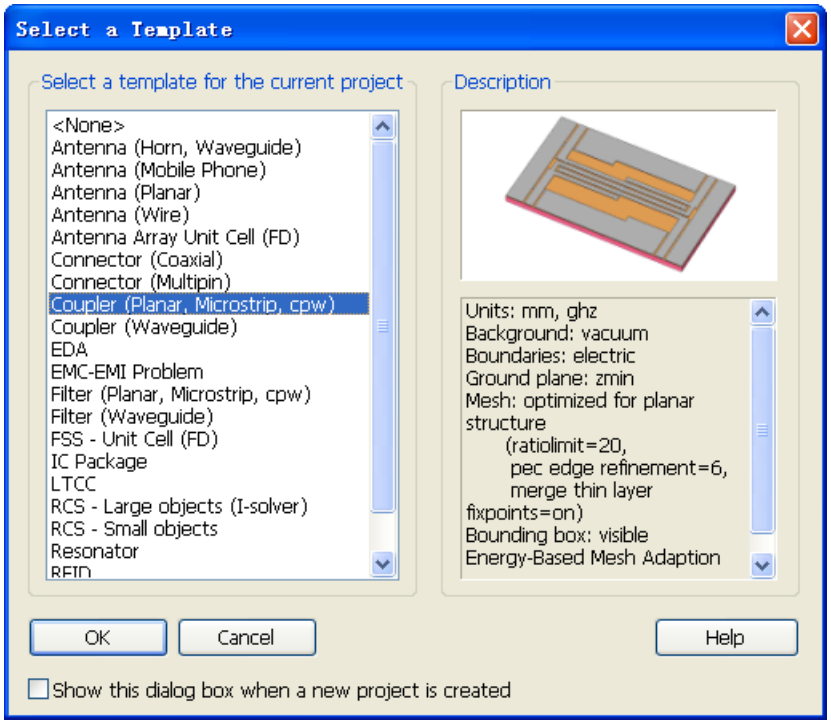


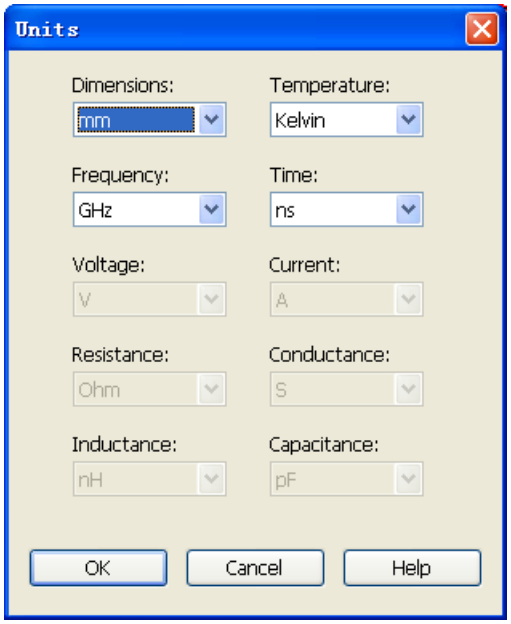
建立矩阵占用内存118M，求解占用内存28M，求解器计算时间43s。

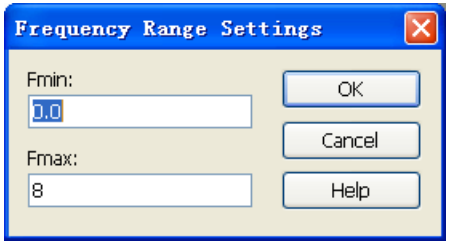
渐变耦合带状线



选择模板、背景条件、频带范围

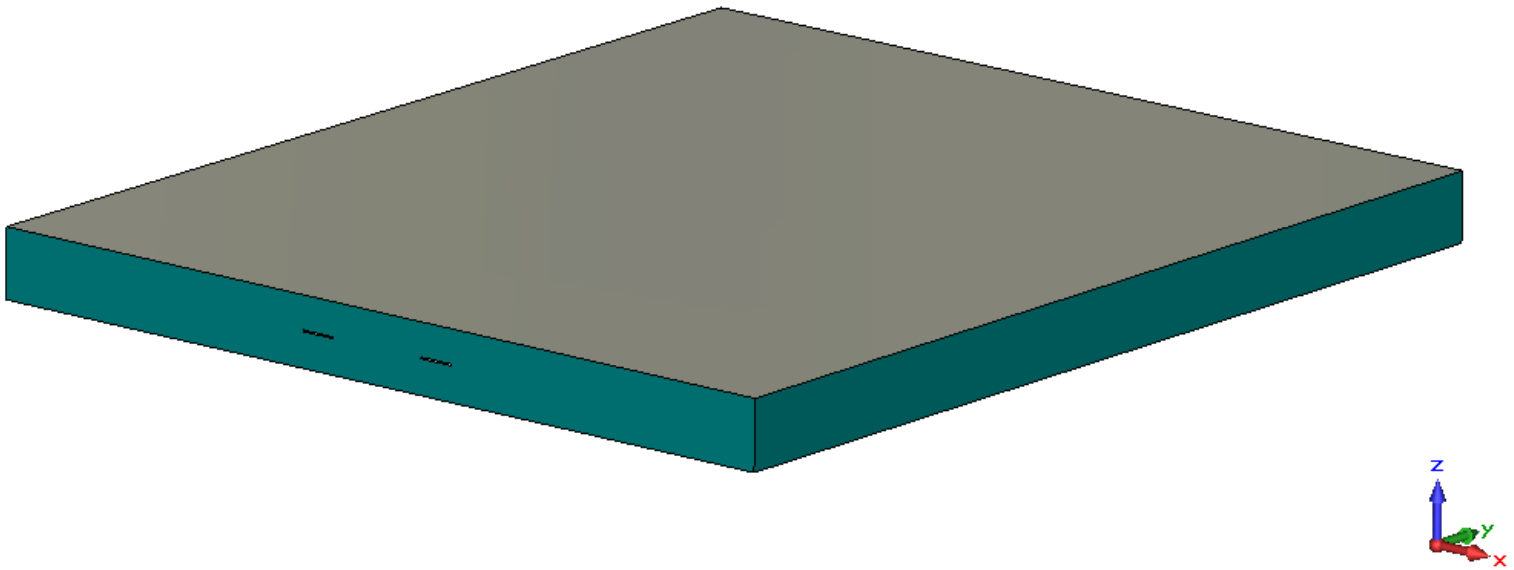




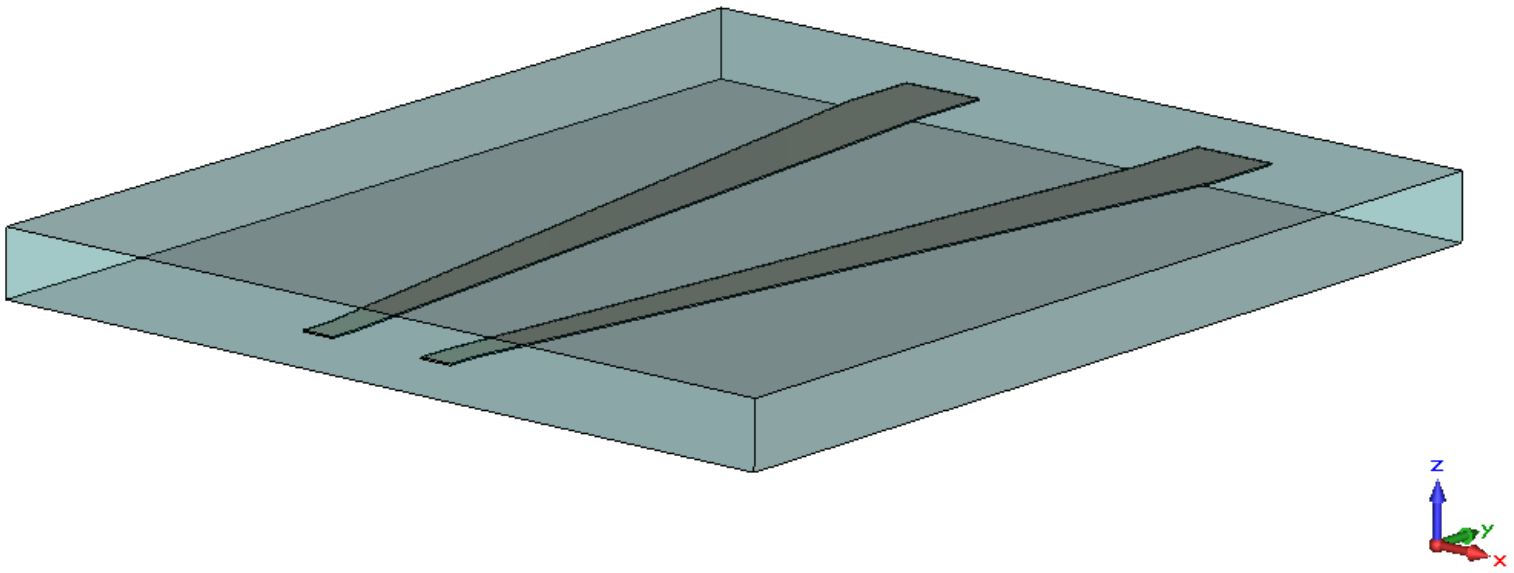


选择平面器件，微带线模板，默认单位为GHz，mm。
仿真频带范围0-8GHz。

模型

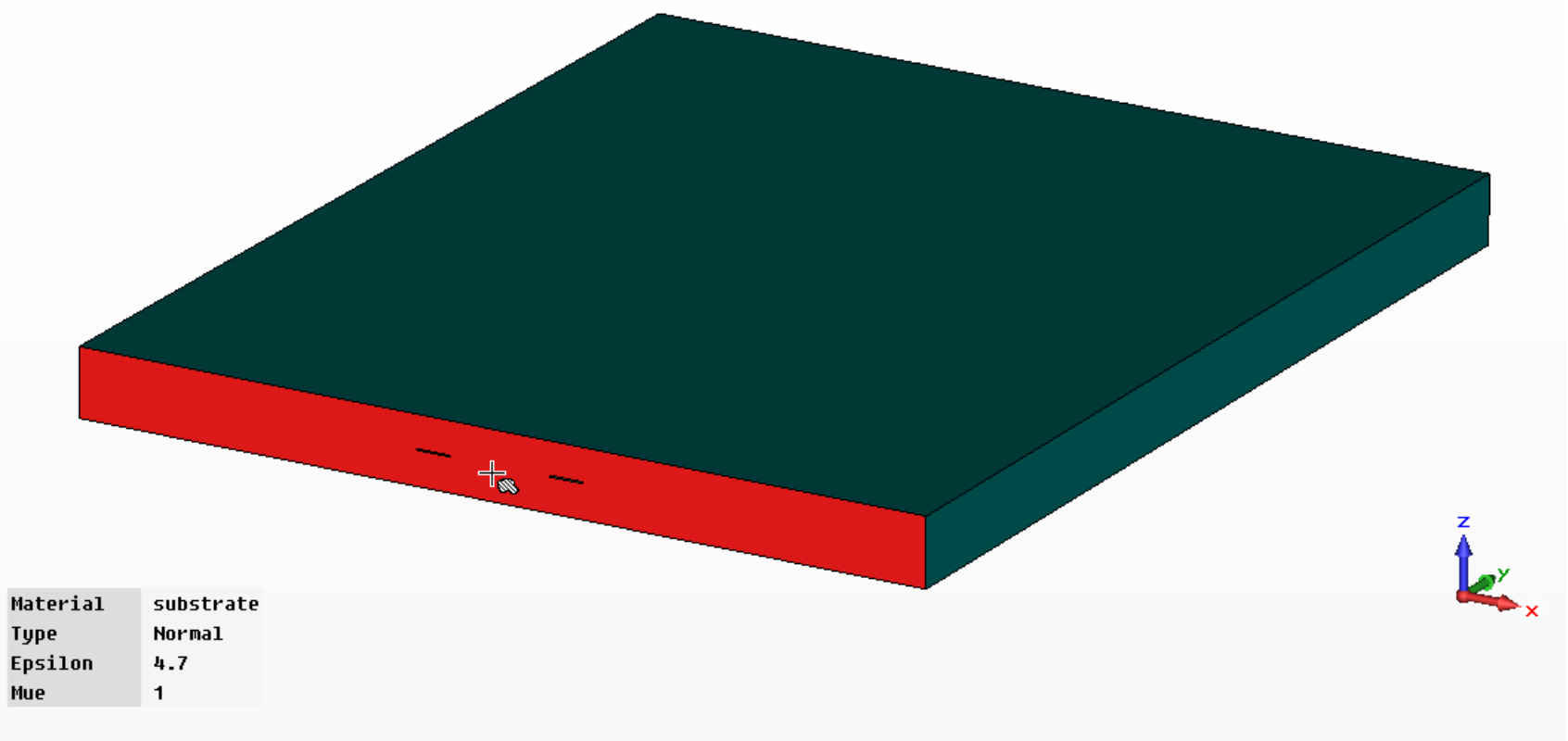


渐变带状线

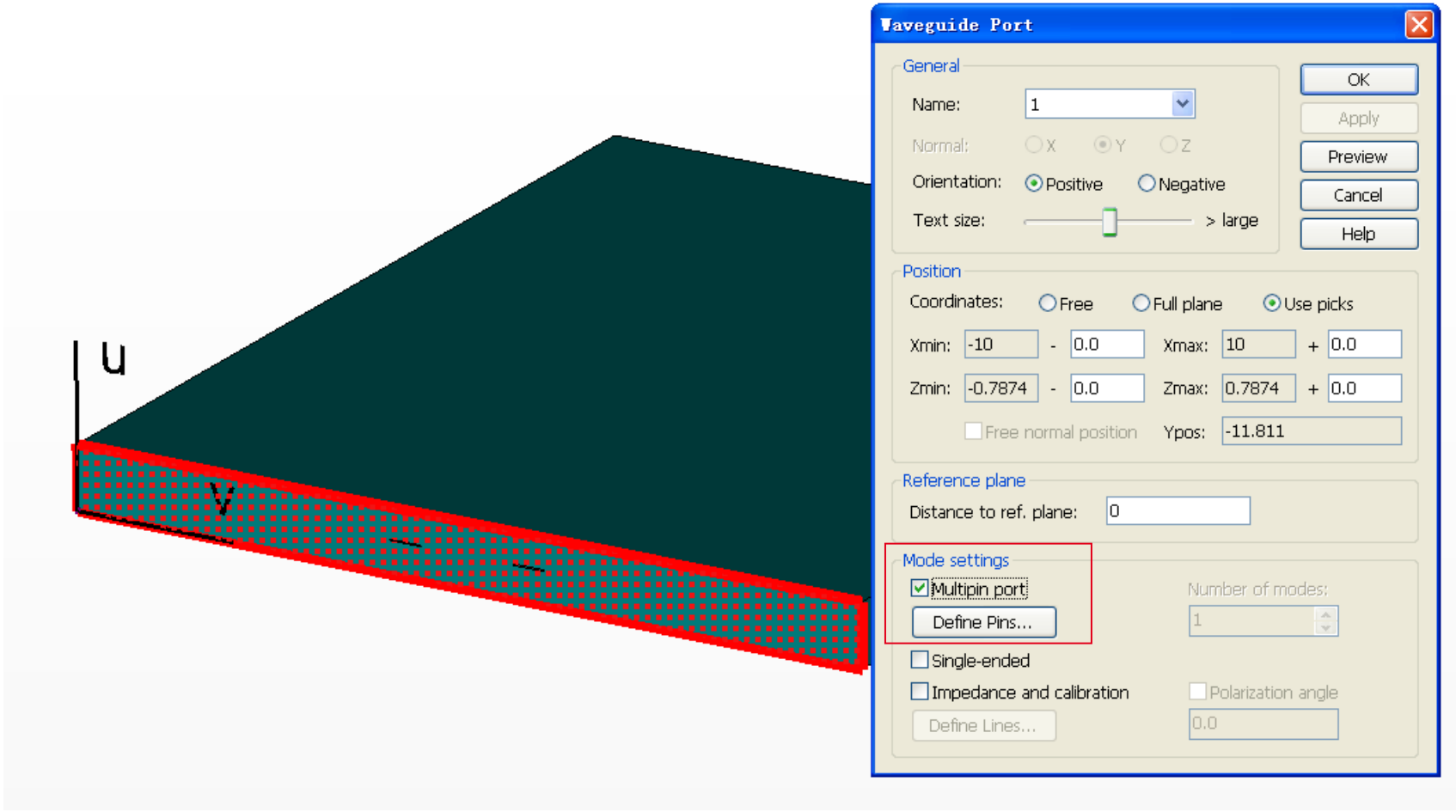


选中面

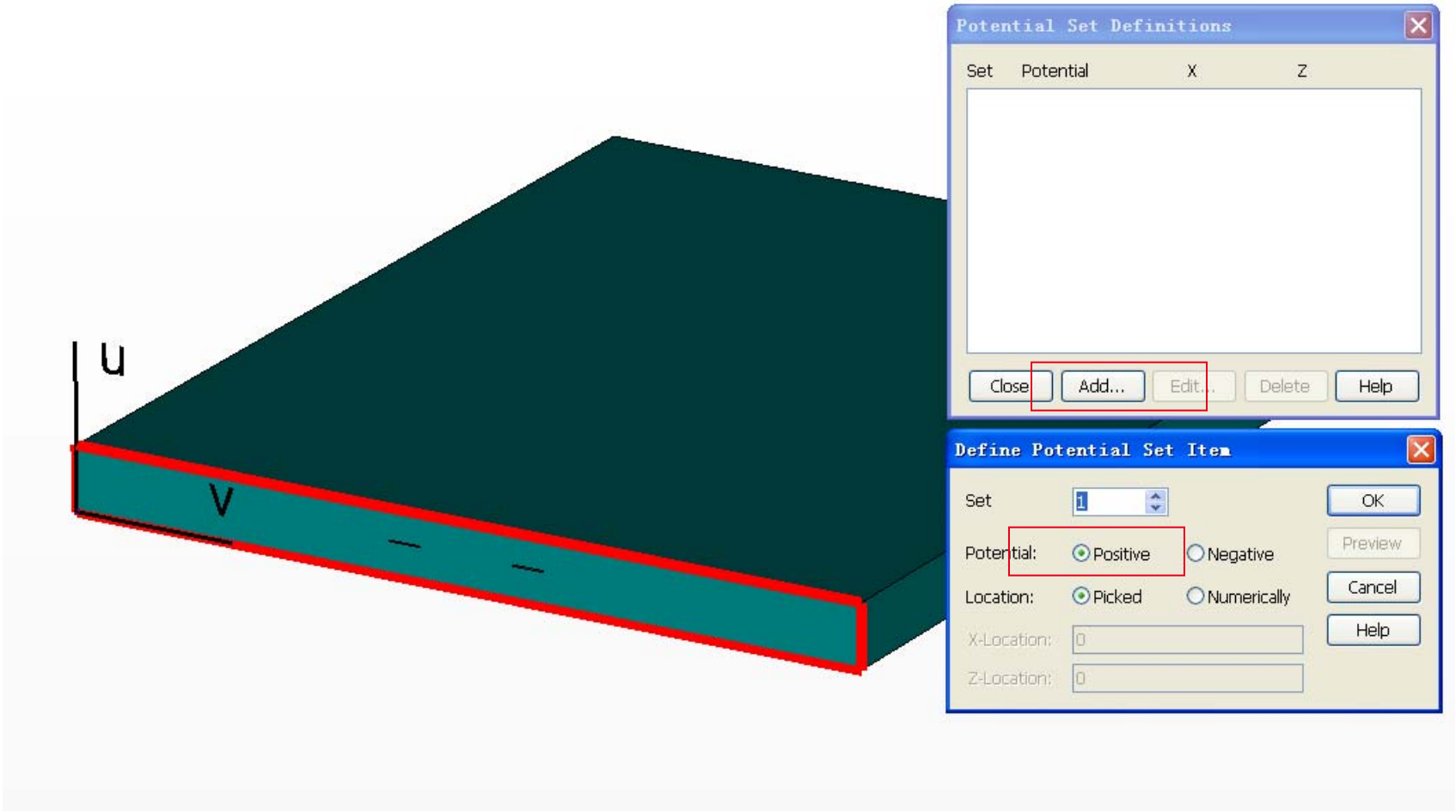
Select face in main view (Press RETURN or ESC to leave this mode)



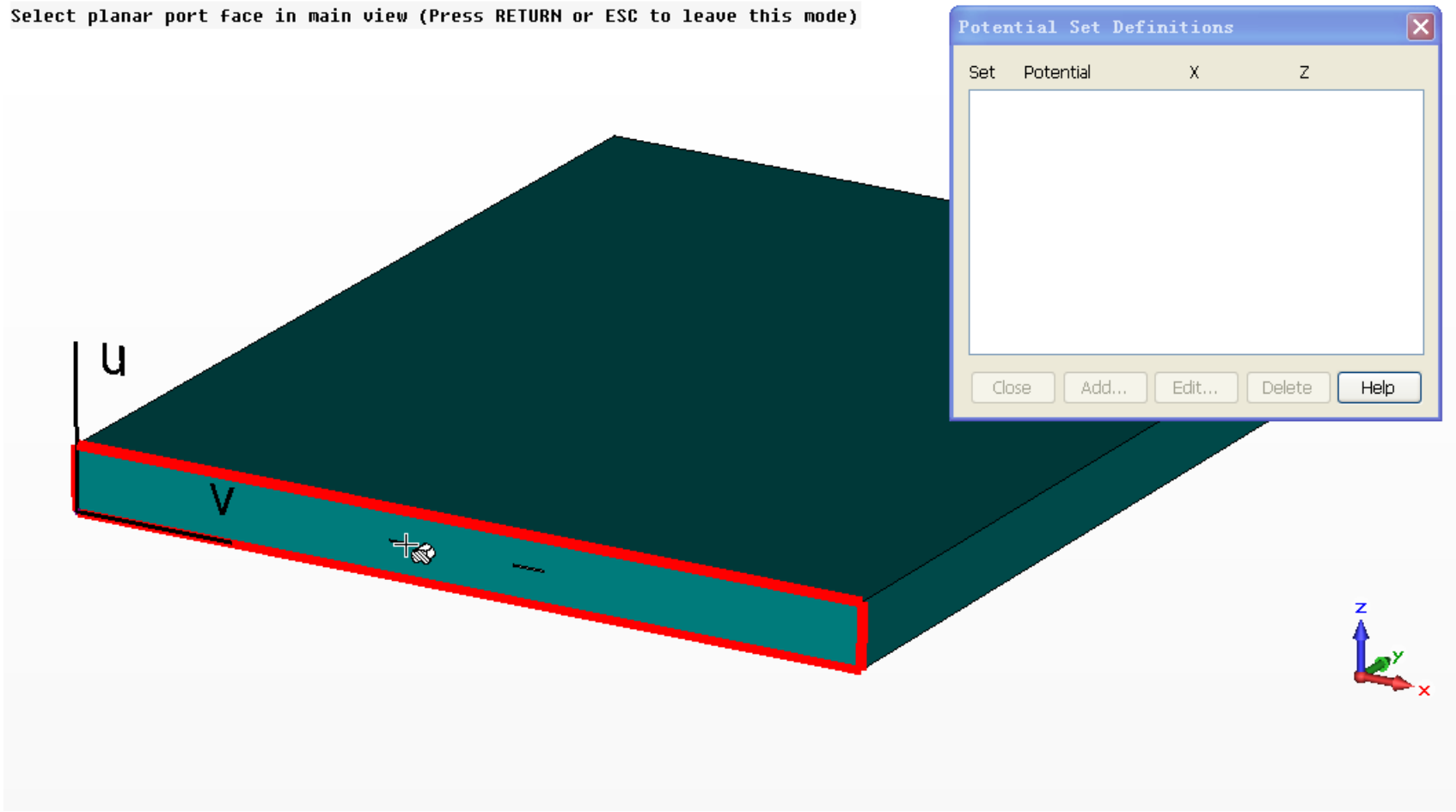
激活多芯端口按钮


u
v
Waveguide Port
General
Name:
1
Normal:
X
Y
Z
Orientation:
Positive
Negative
Text size:
> large
OK
Apply
Preview
Cancel
Help
Position
Coordinates:
Free
Full plane
Use picks
Xmin:
-10
0.0
Xmax:
10
0.0
Zmin:
-0.7874
0.0
Zmax:
0.7874
0.0
Free normal position
Ypos:
-11.811
Reference plane
Distance to ref. plane:
0
Mode settings
Multipin port
Define Pins...
Number of modes:
1
Single-ended
Impedance and calibration
Define Lines...
Polarization angle
0.0

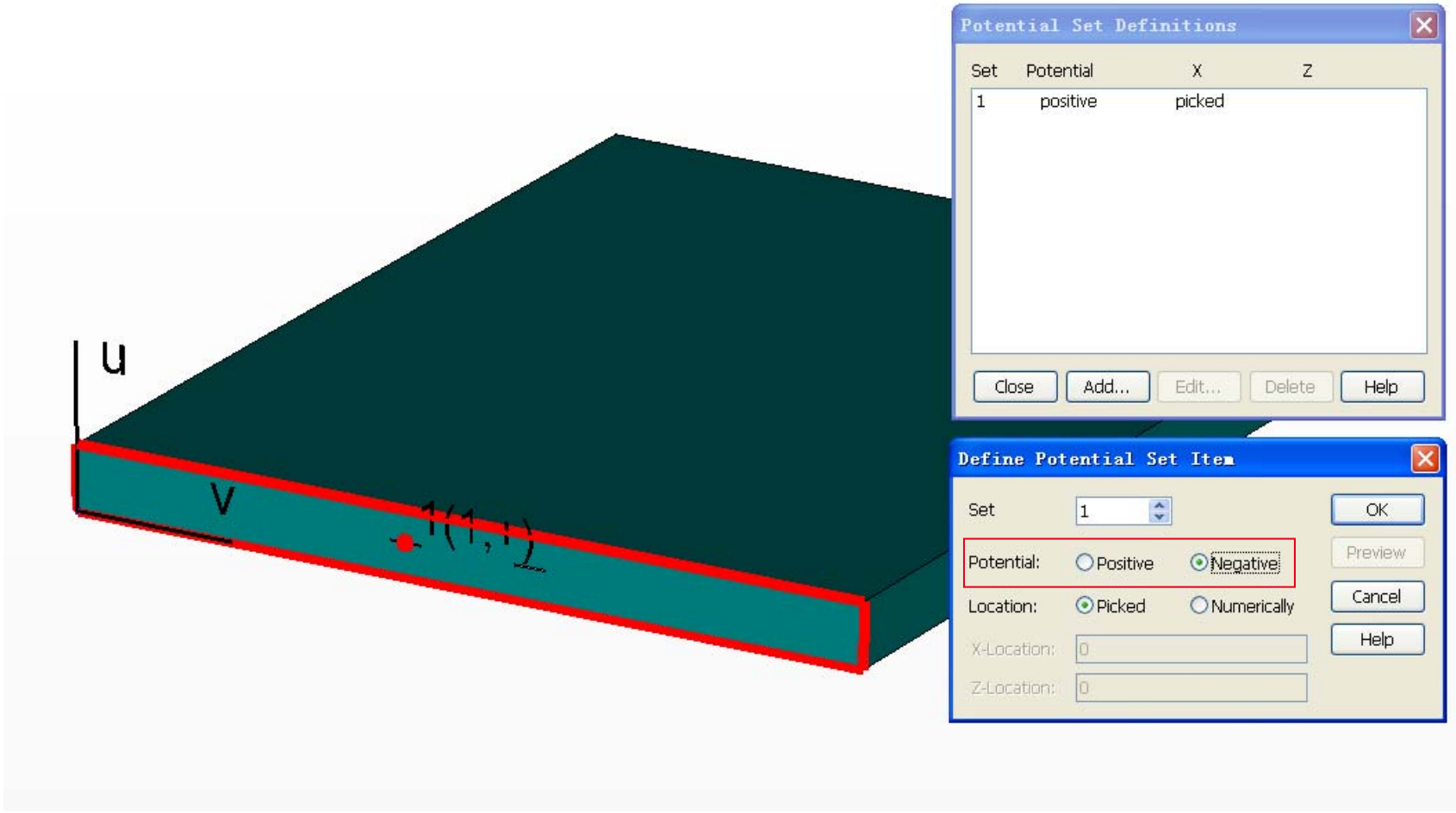
定义多芯端口——指定模式场



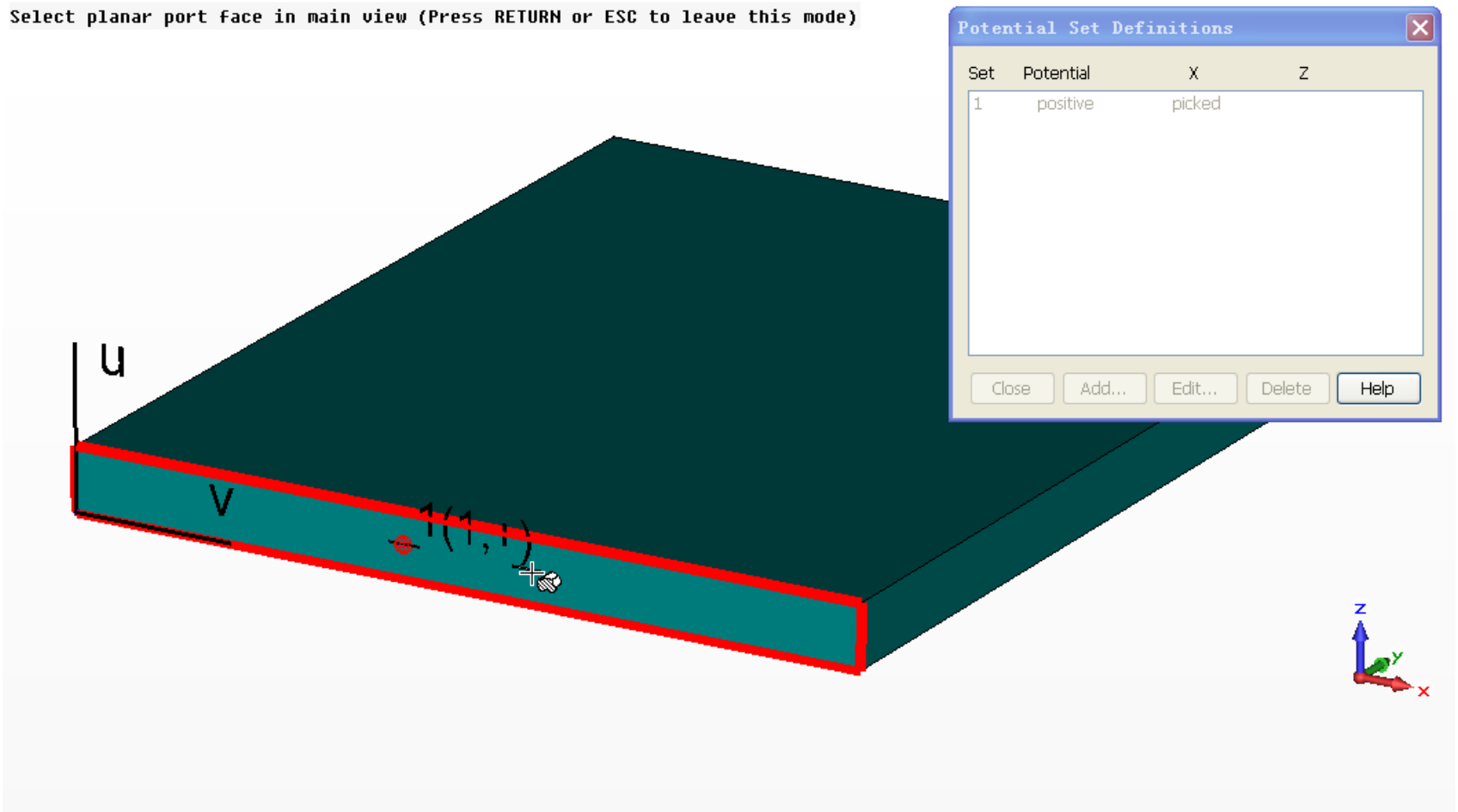
定义模式


Select planar port face in main view (Press RETURN or ESC to leave this mode)
Potential Set Definitions
Set
Potential
X
Z
Close
Add...
Edit...
Delete
Help
u
v
z
y
x

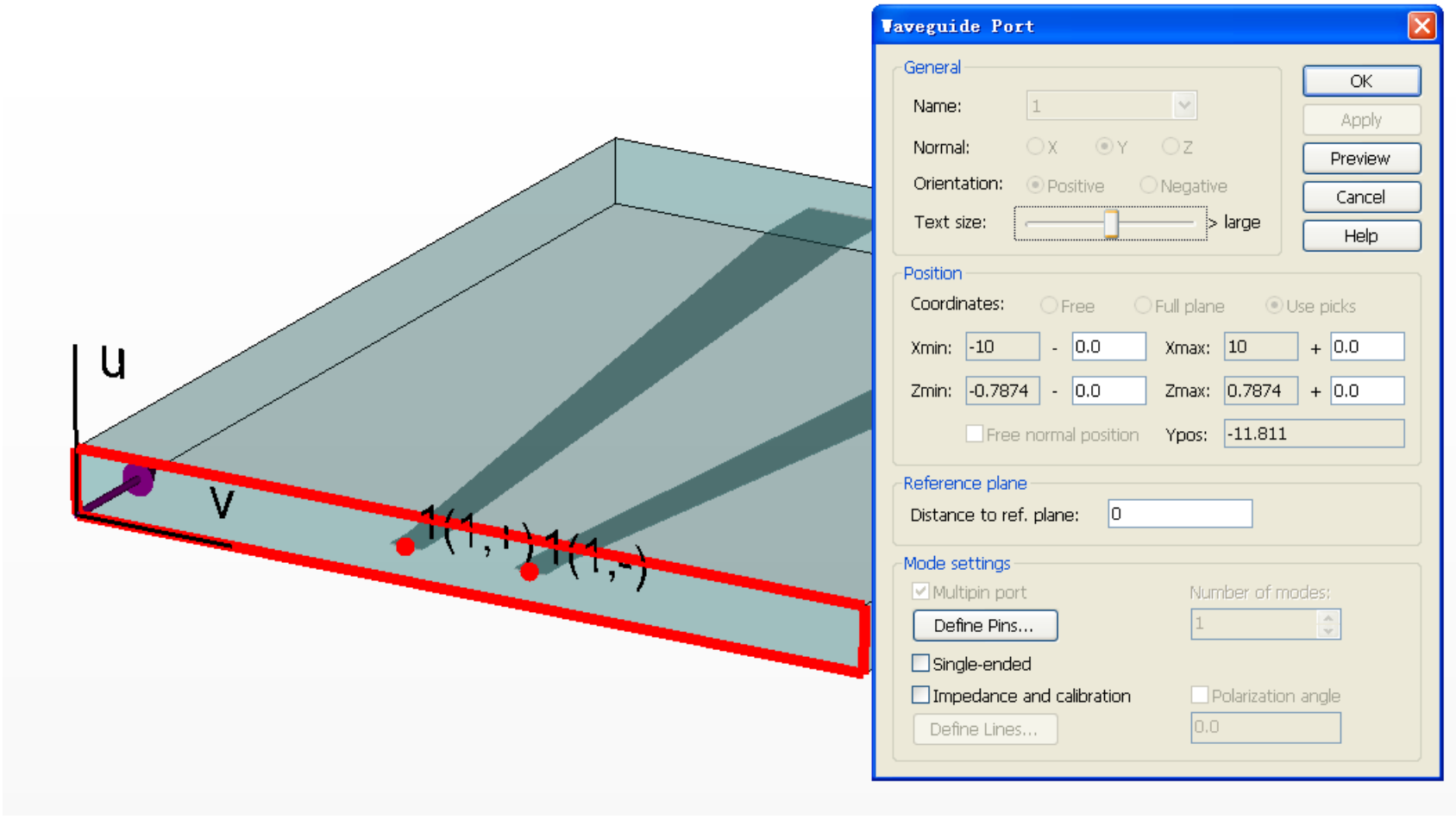
定义指定模式的另一极


u
V
1(1, 1)
Potential Set Definitions
Set
Potential
X
Z
1
positive
picked
Close
Add...
Edit...
Delete
Help
Define Potential Set Item
Set
1
OK
Potential:
Positive
Negative
Preview
Location:
Picked
Numerically
Cancel
X-Location:
0
Help
Z-Location:
0

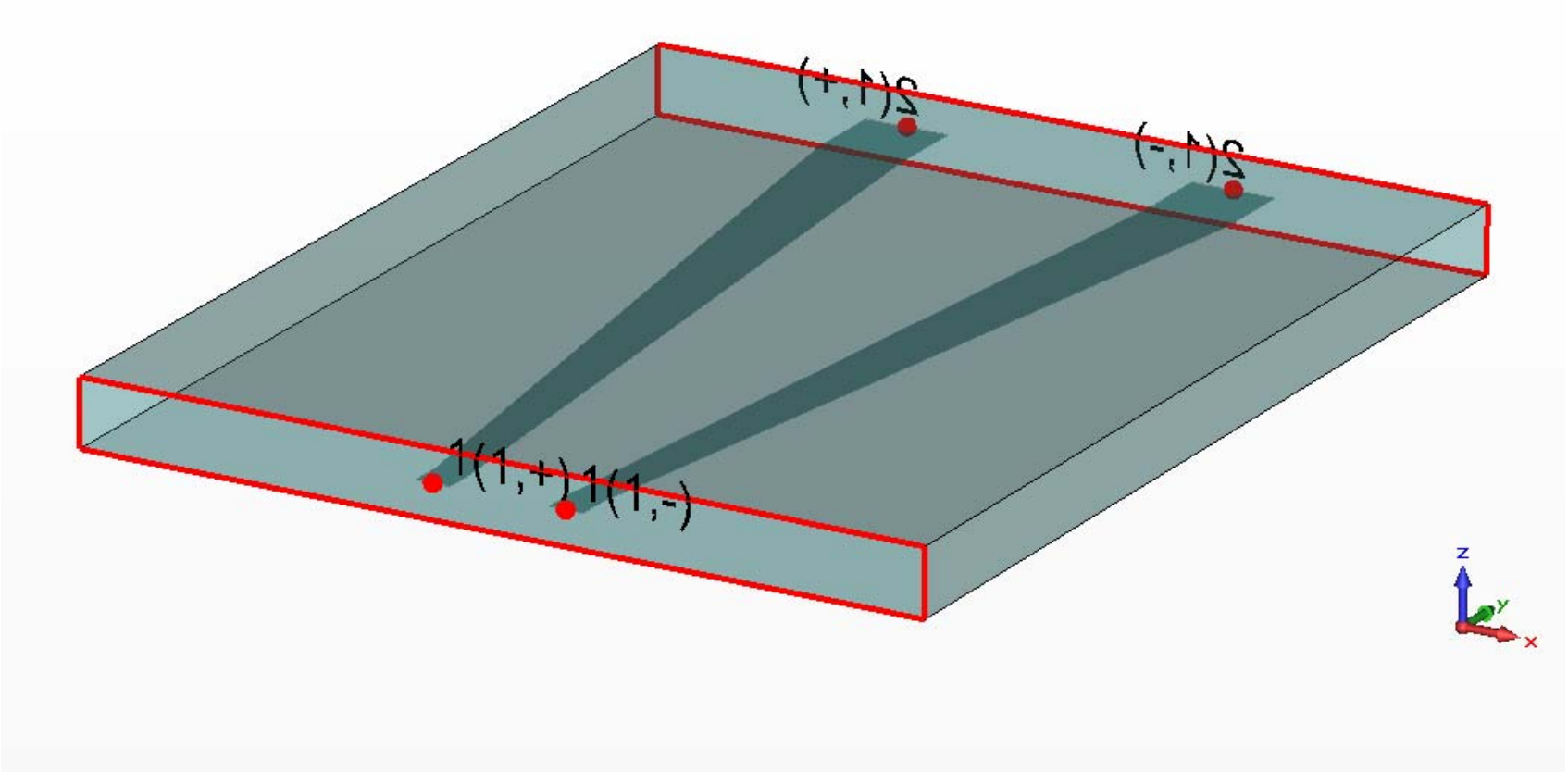
选择差分线的另一端


Select planar port face in main view (Press RETURN or ESC to leave this mode)
Potential Set Definitions
Set
Potential
X
Z
1
positive
picked
Close
Add...
Edit...
Delete
Help
u
v
1(1, 1)
z
y
x

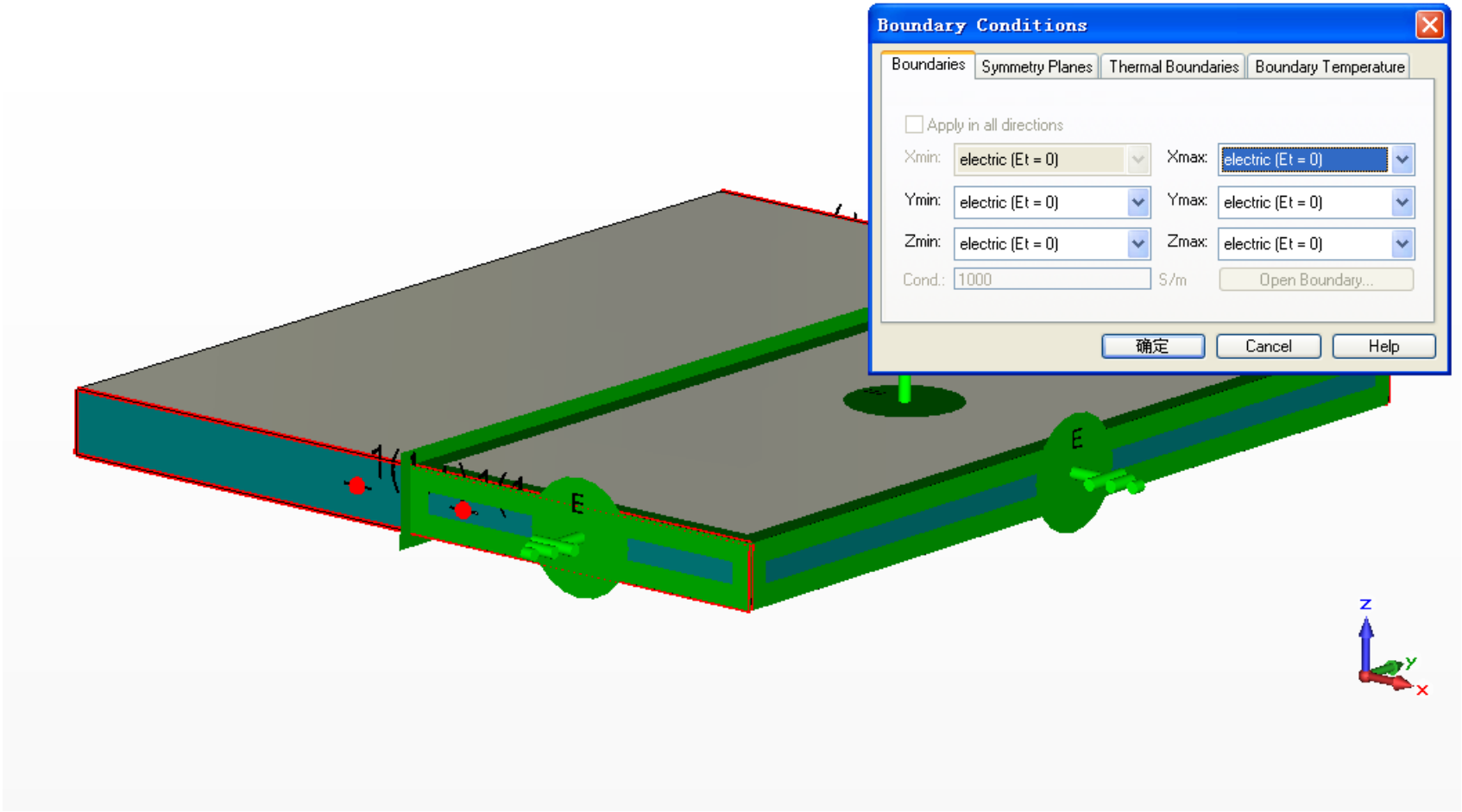
差分线1端口模式定义完成



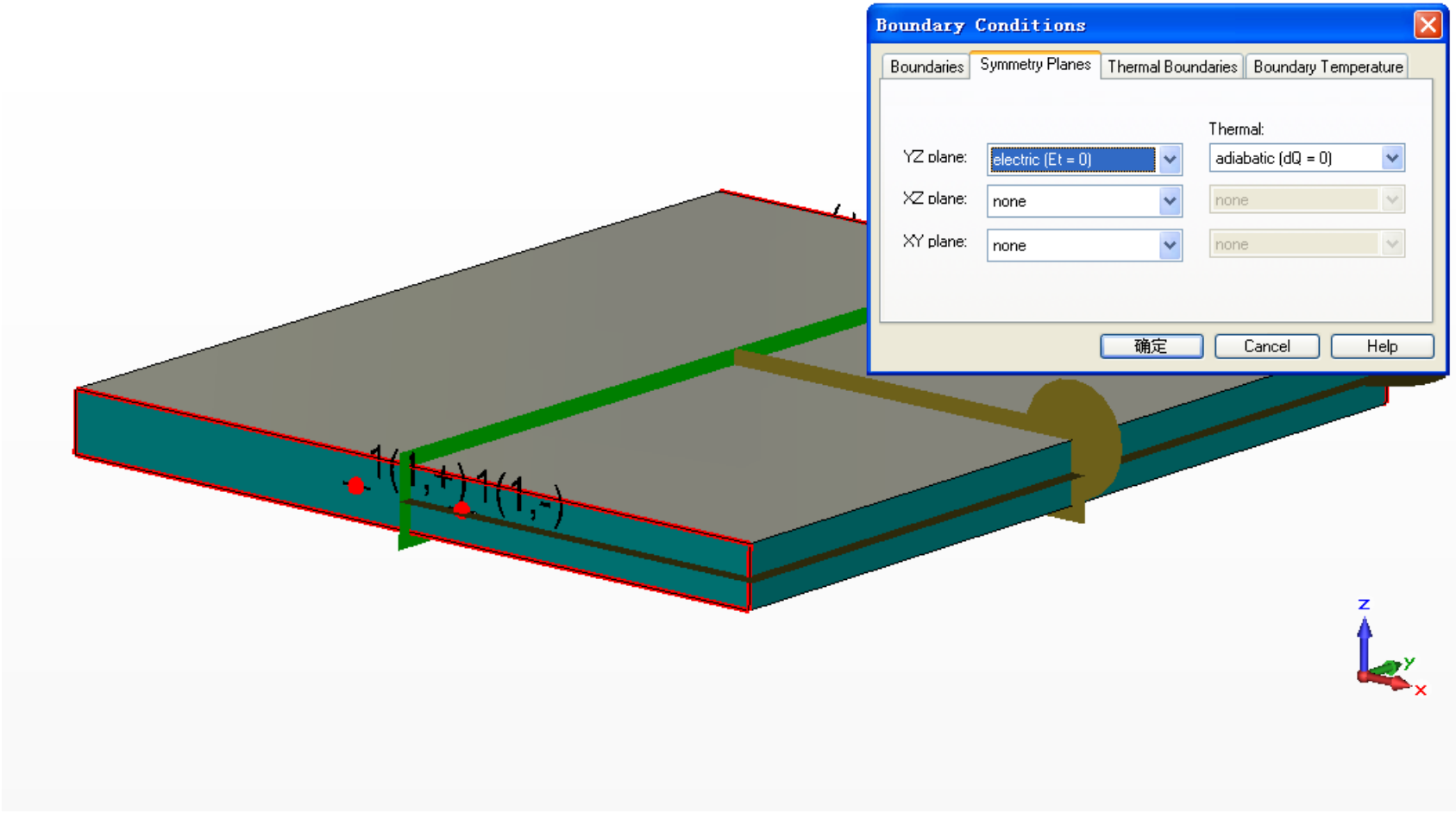
定义多芯端口2



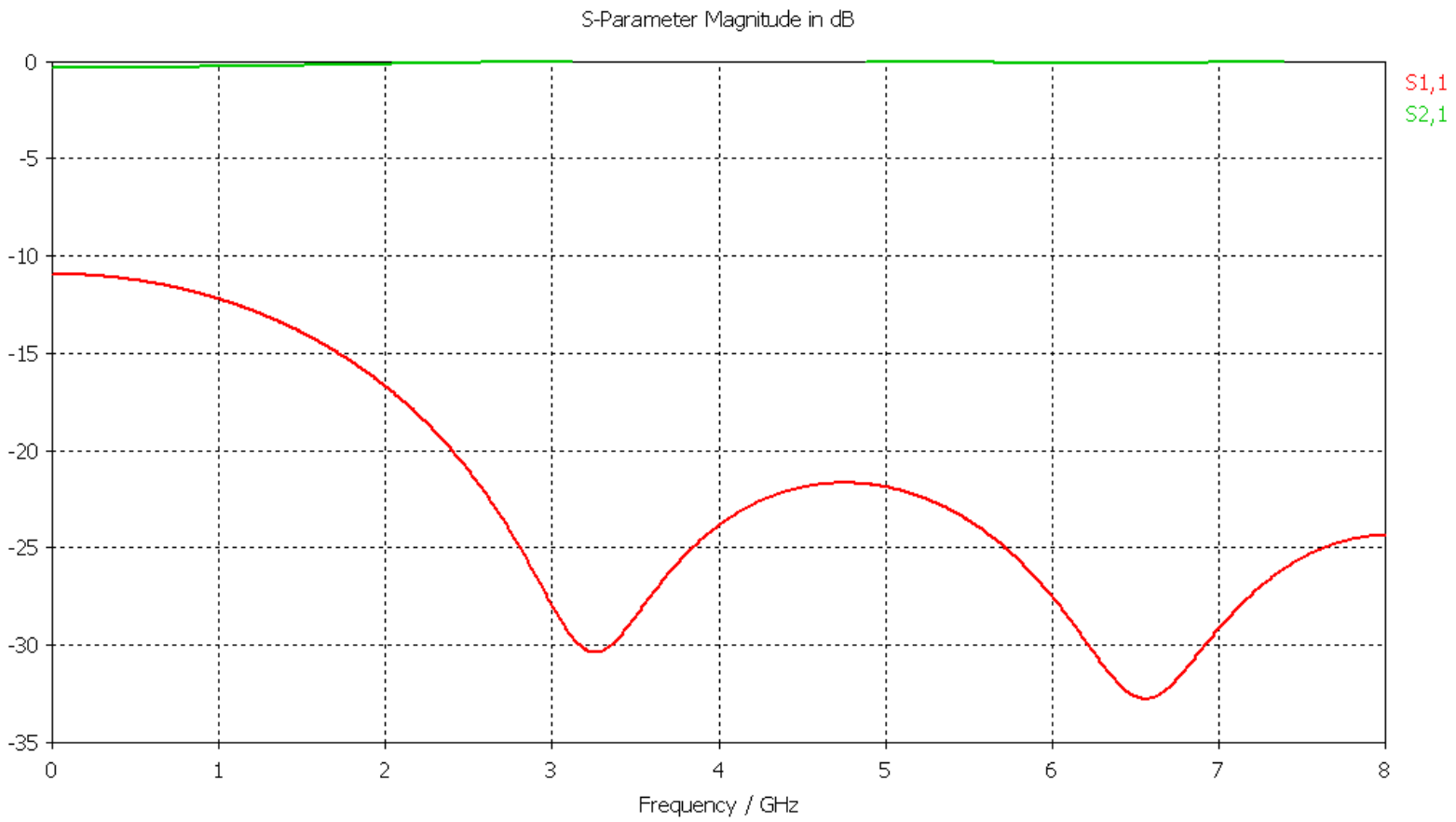
边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Xmin:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help
E
E
z
y
x

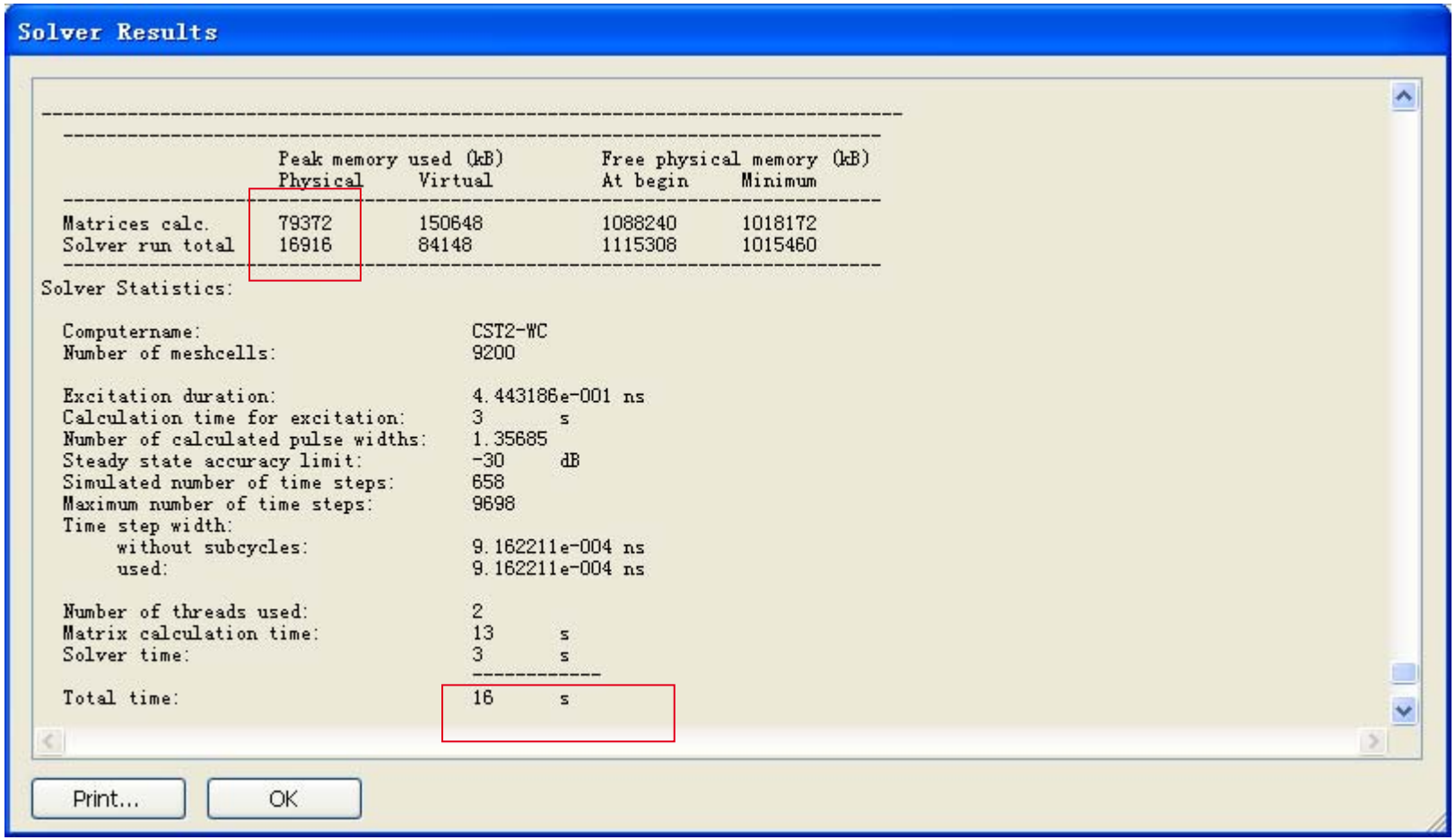
定义对称面，节省仿真时间



S参量 dB

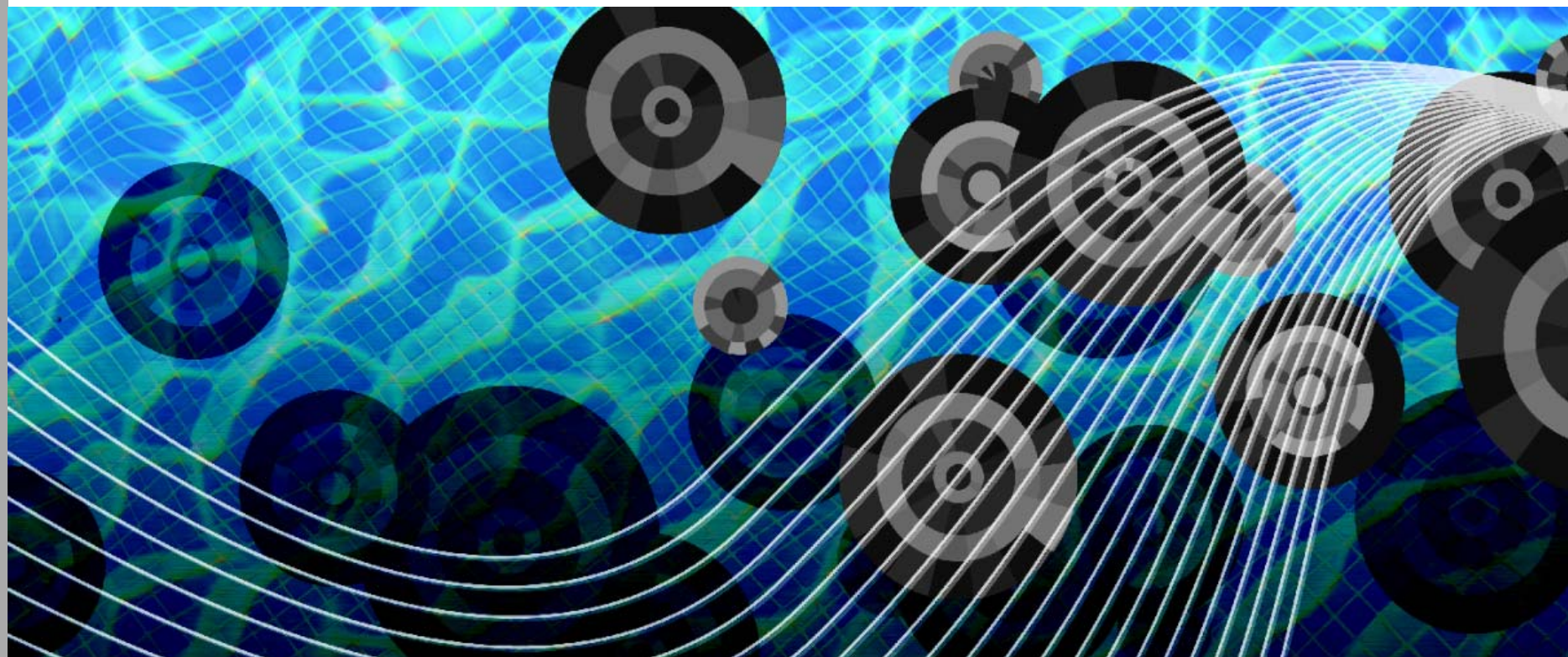

S-Parameter Magnitude in dB
0
-5
-10
-15
-20
-25
-30
-35
0
1
2
3
4
5
6
7
8
Frequency / GHz
S1,1
S2,1

占用系统资源

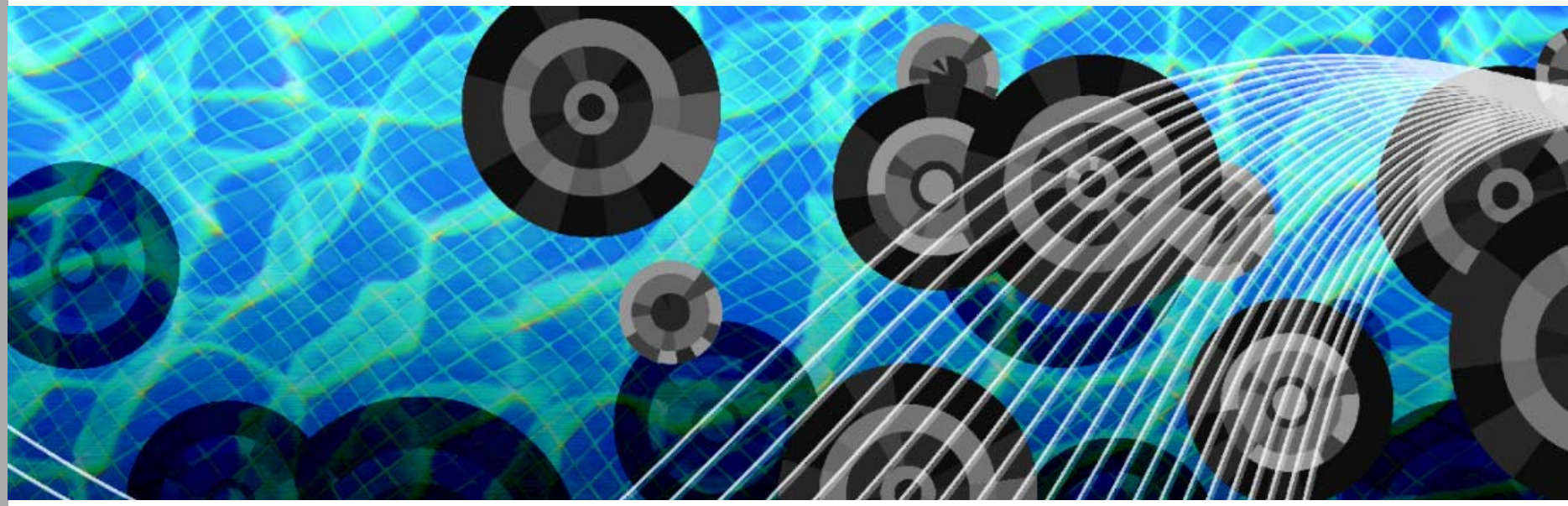


建立矩阵占用内存79M，求解占用内存17M，求解器计算时间16s。

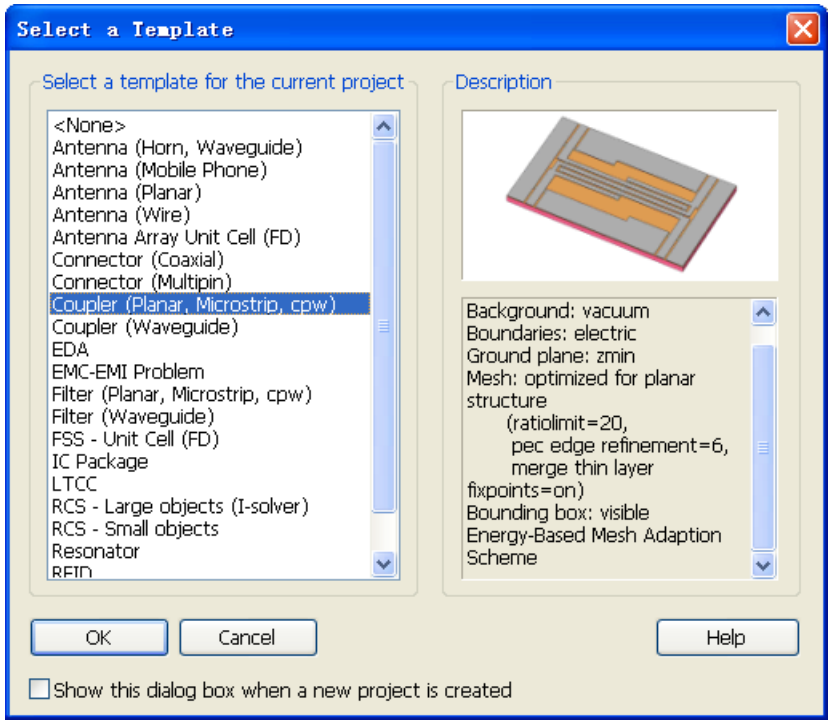
微带线

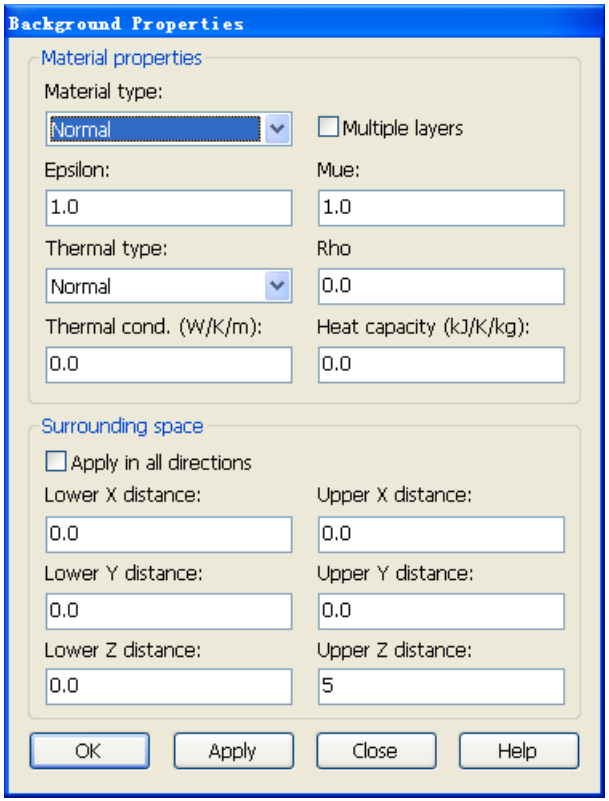


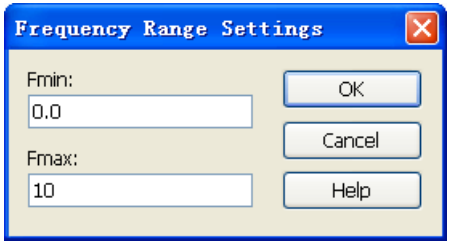
开放微带线



选择模板、背景条件、频带范围

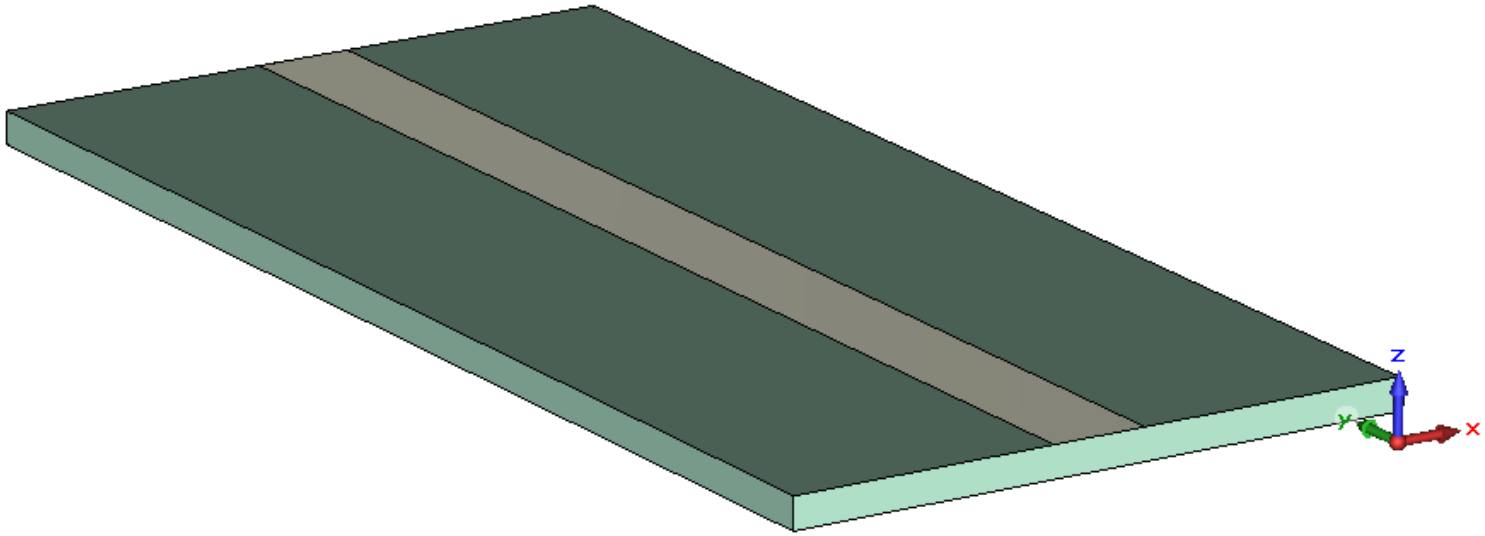




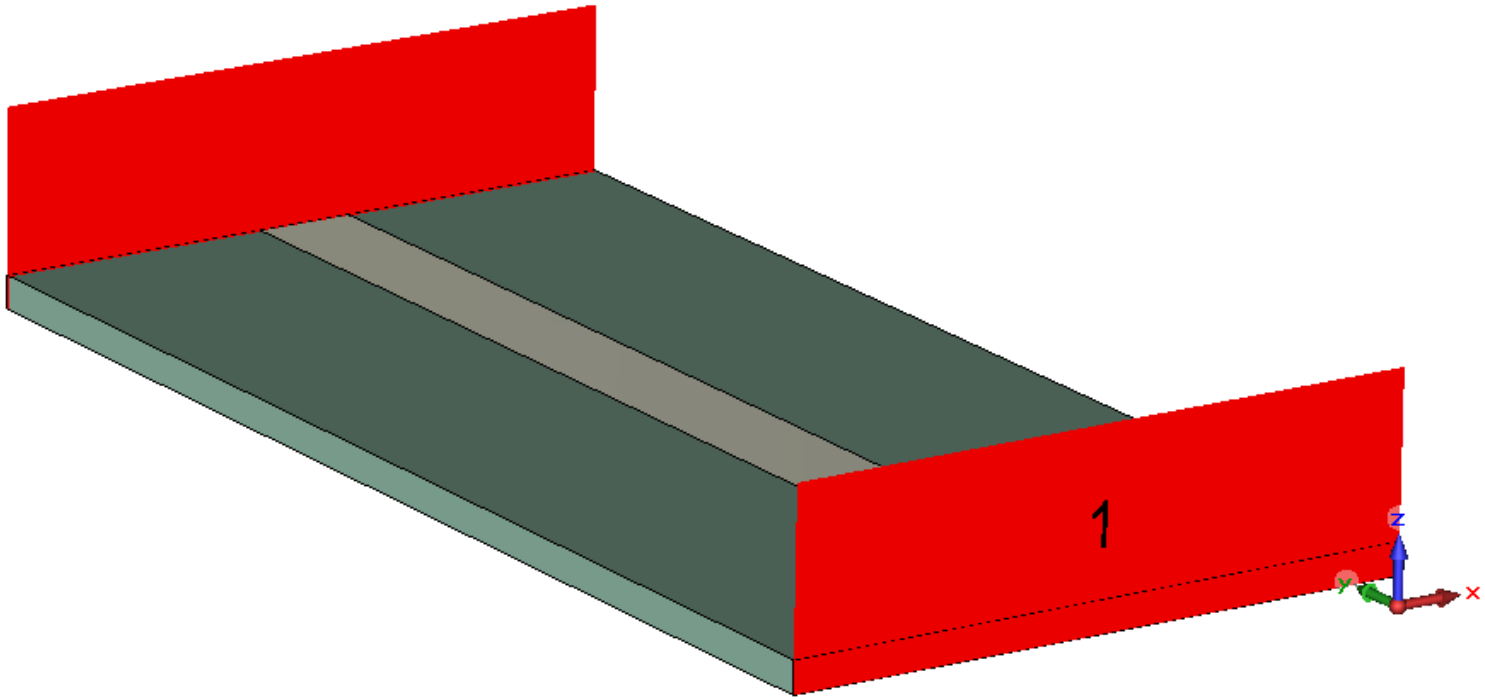


选好模板之后，默认单位为GHz，mm。

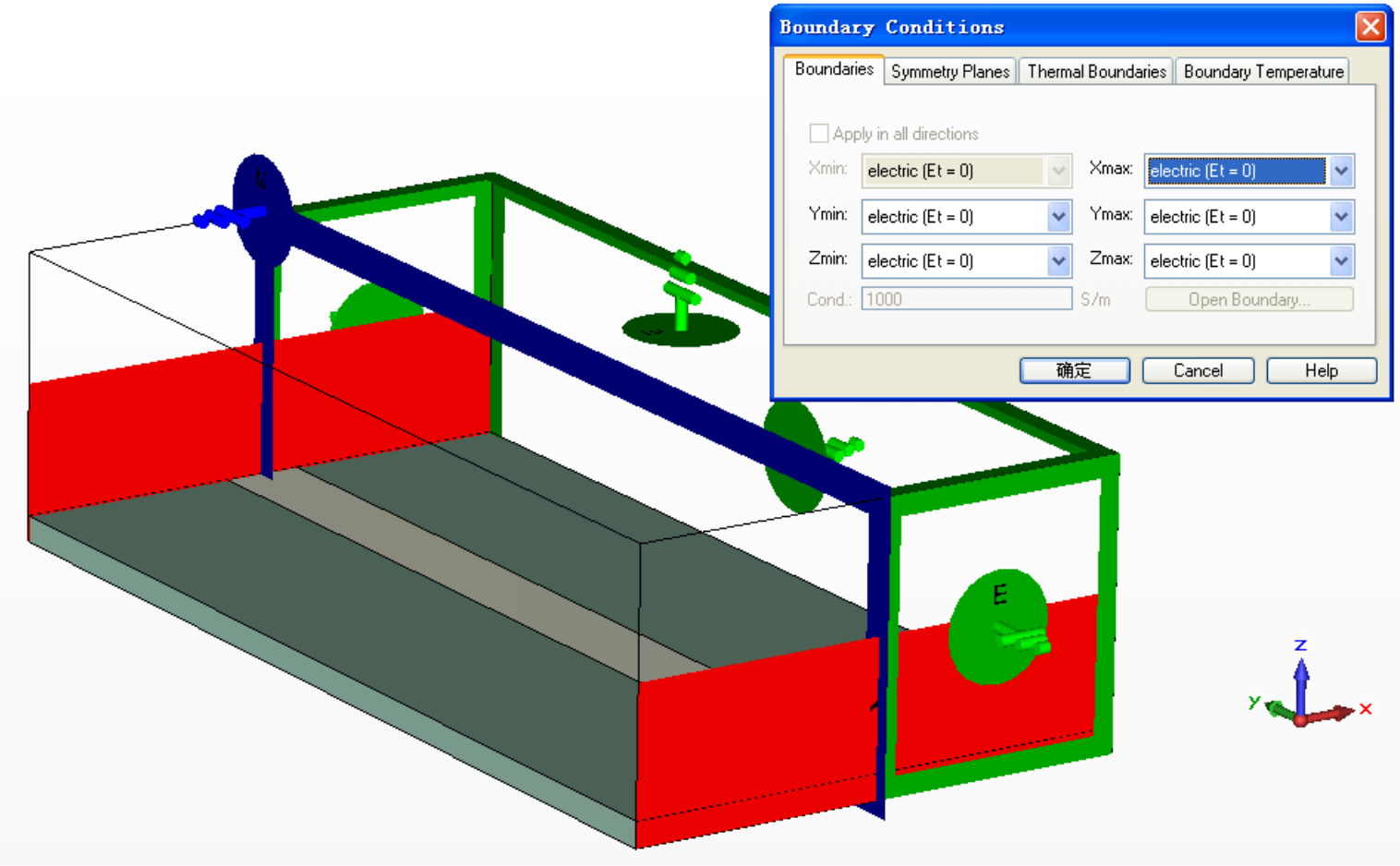
开放微带线模型



激励端口

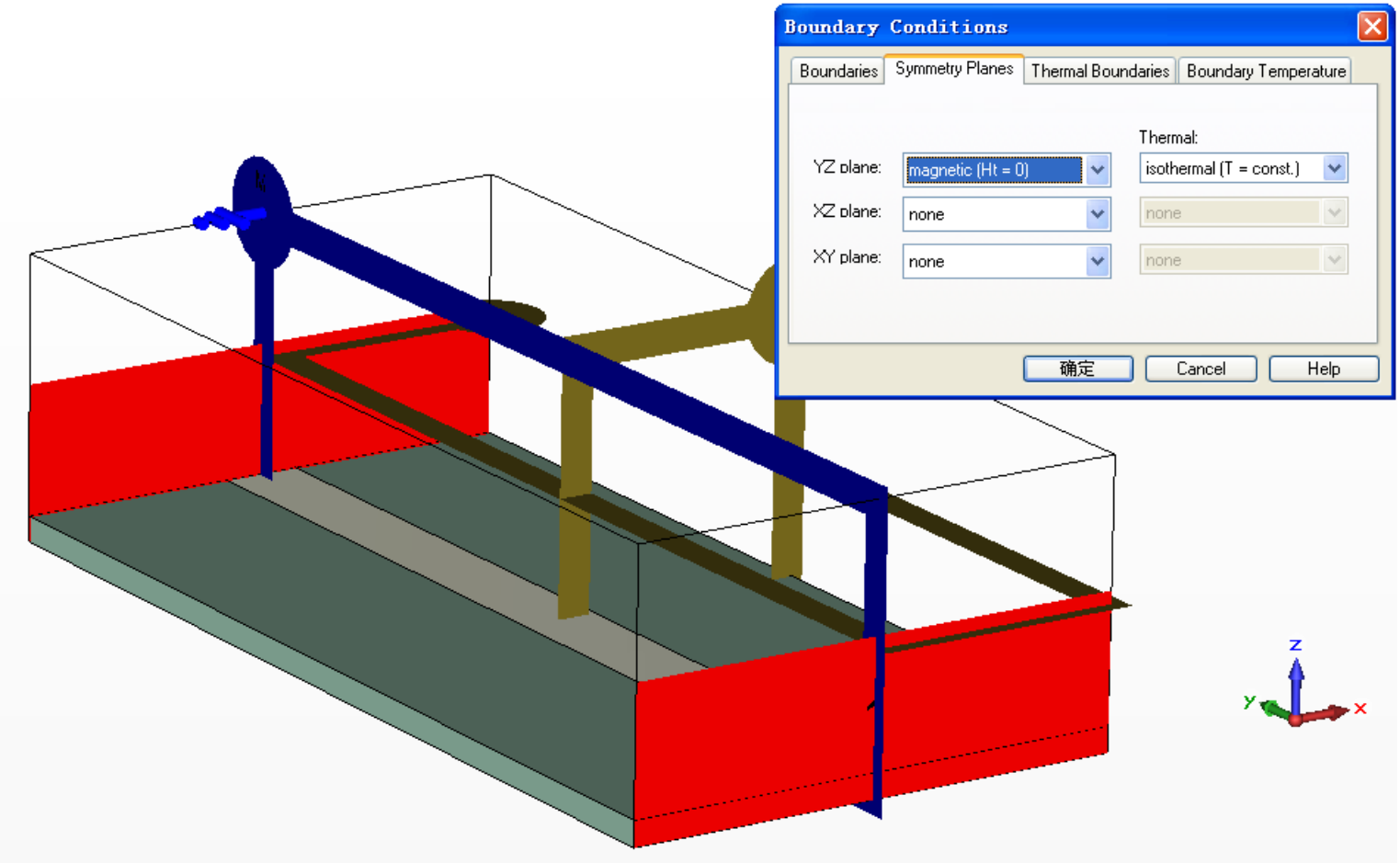


边界条件

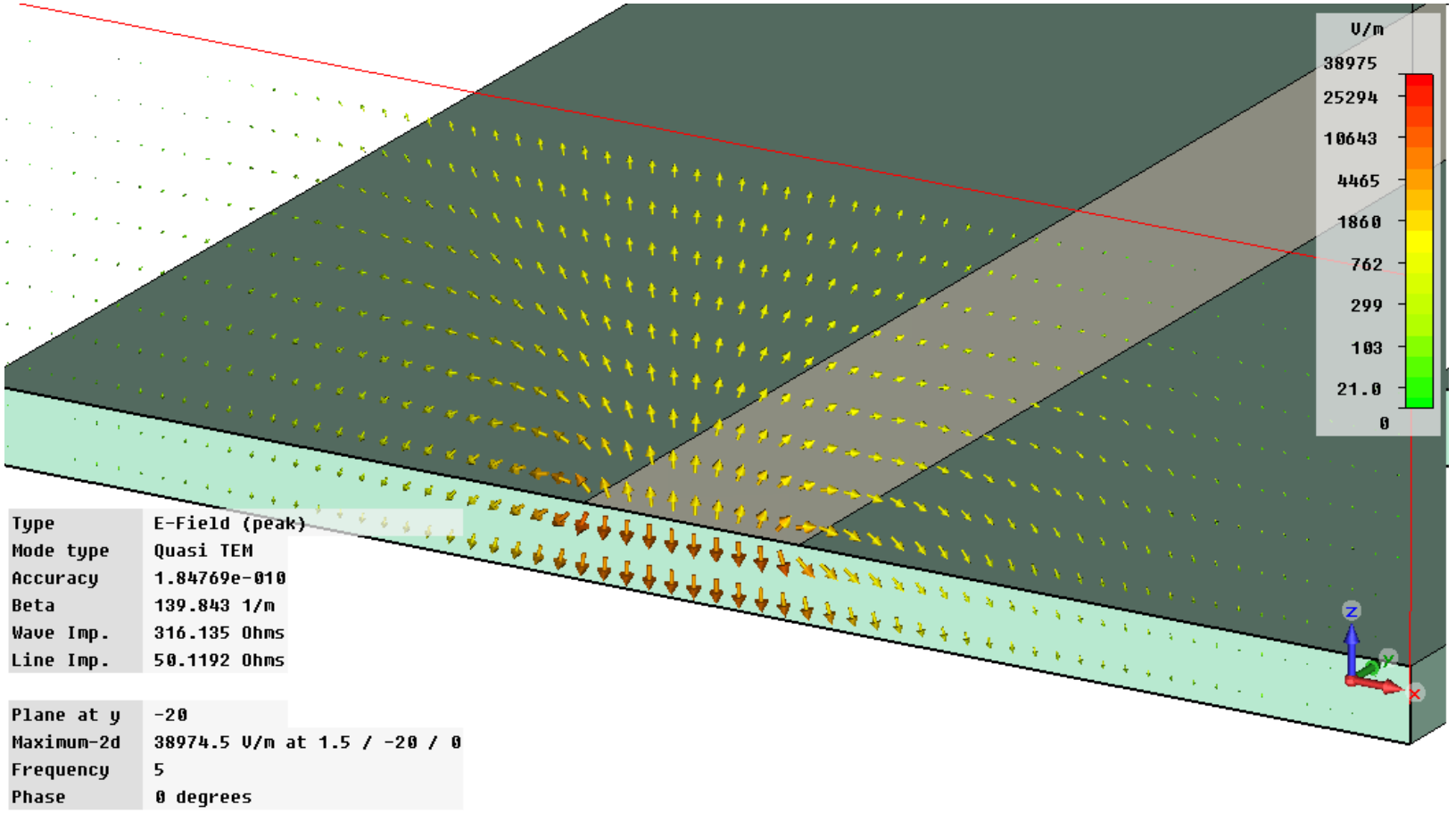


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使用磁对称面，减少仿真时间

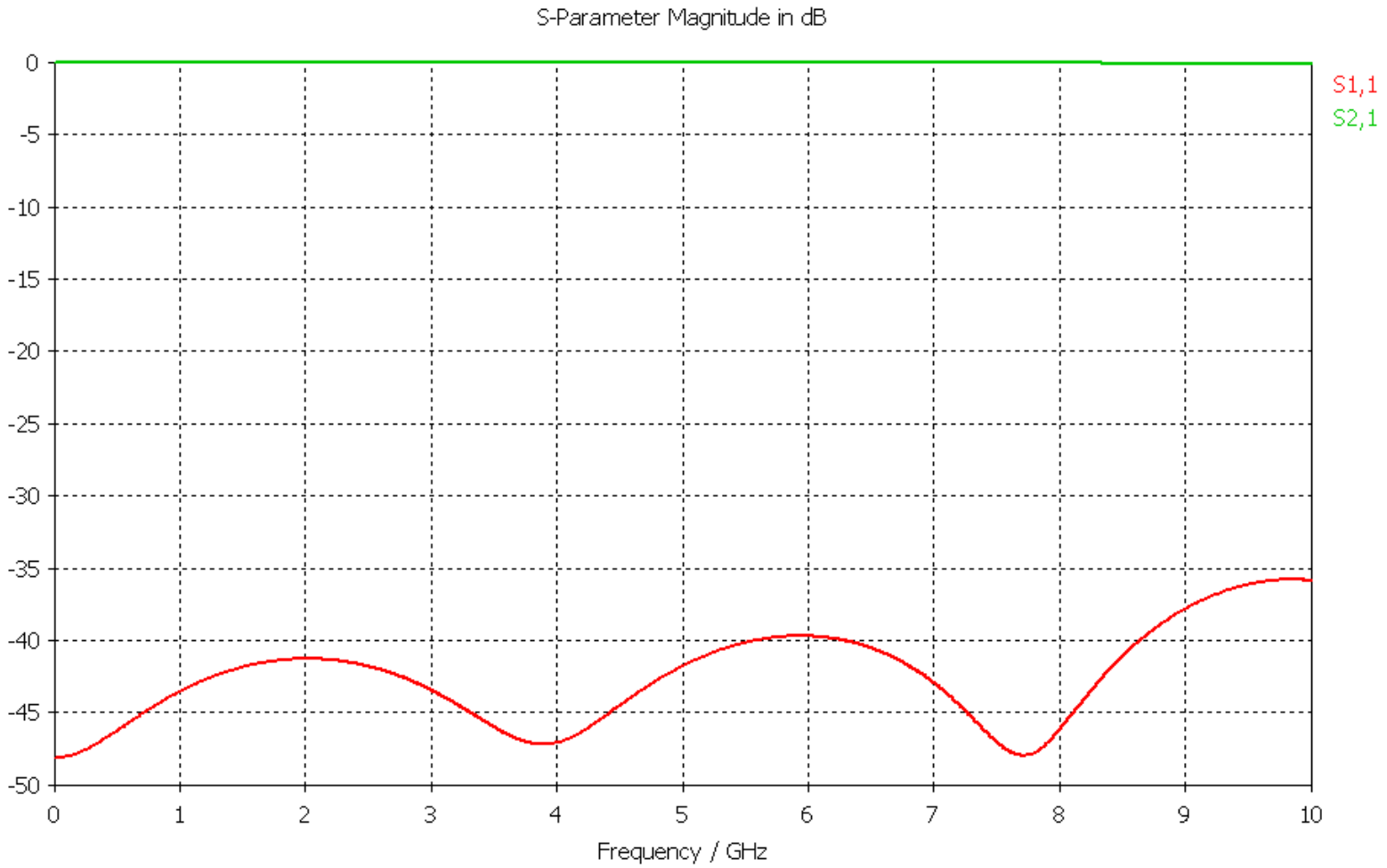


电场端口模式

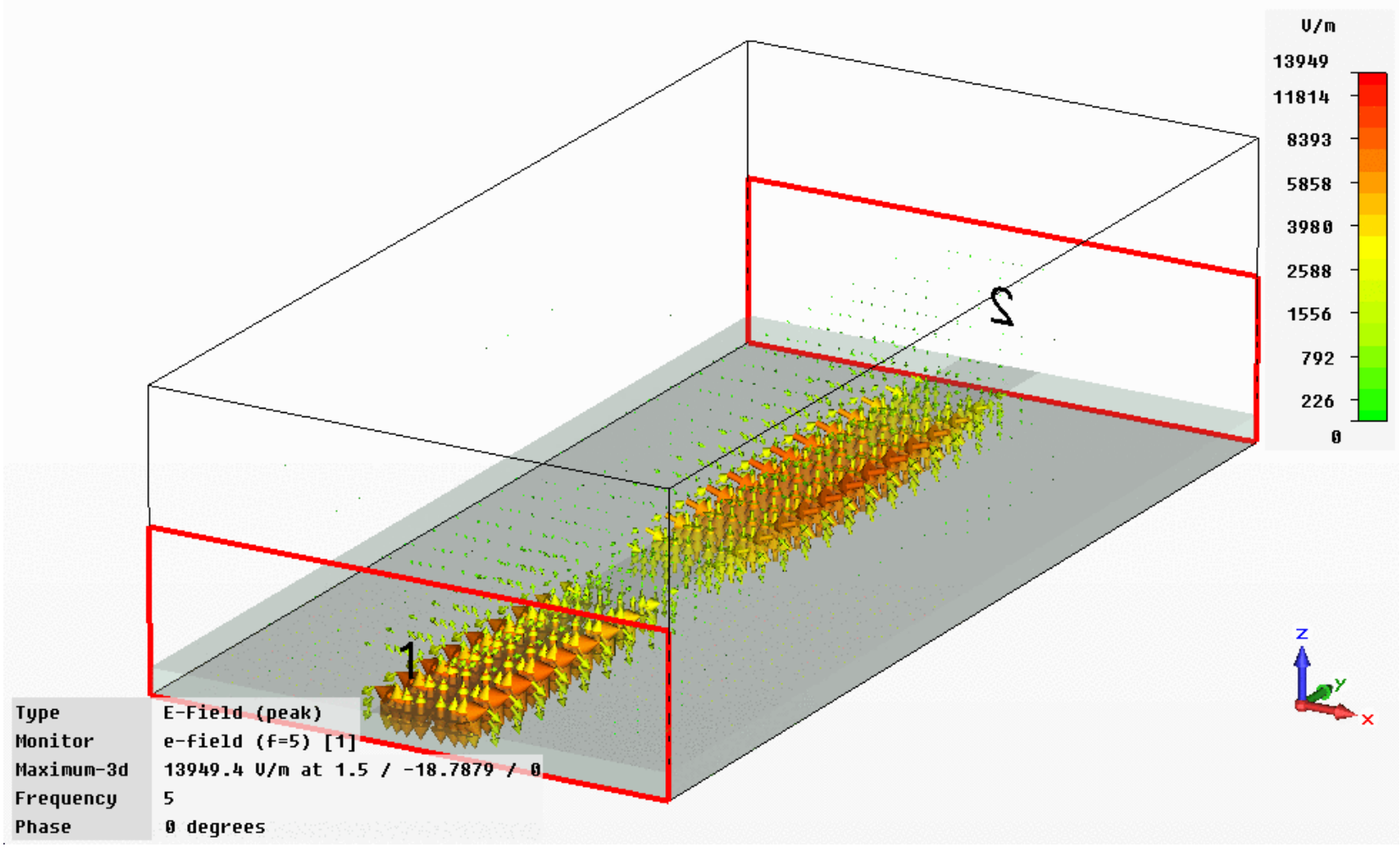


端口阻抗50欧姆

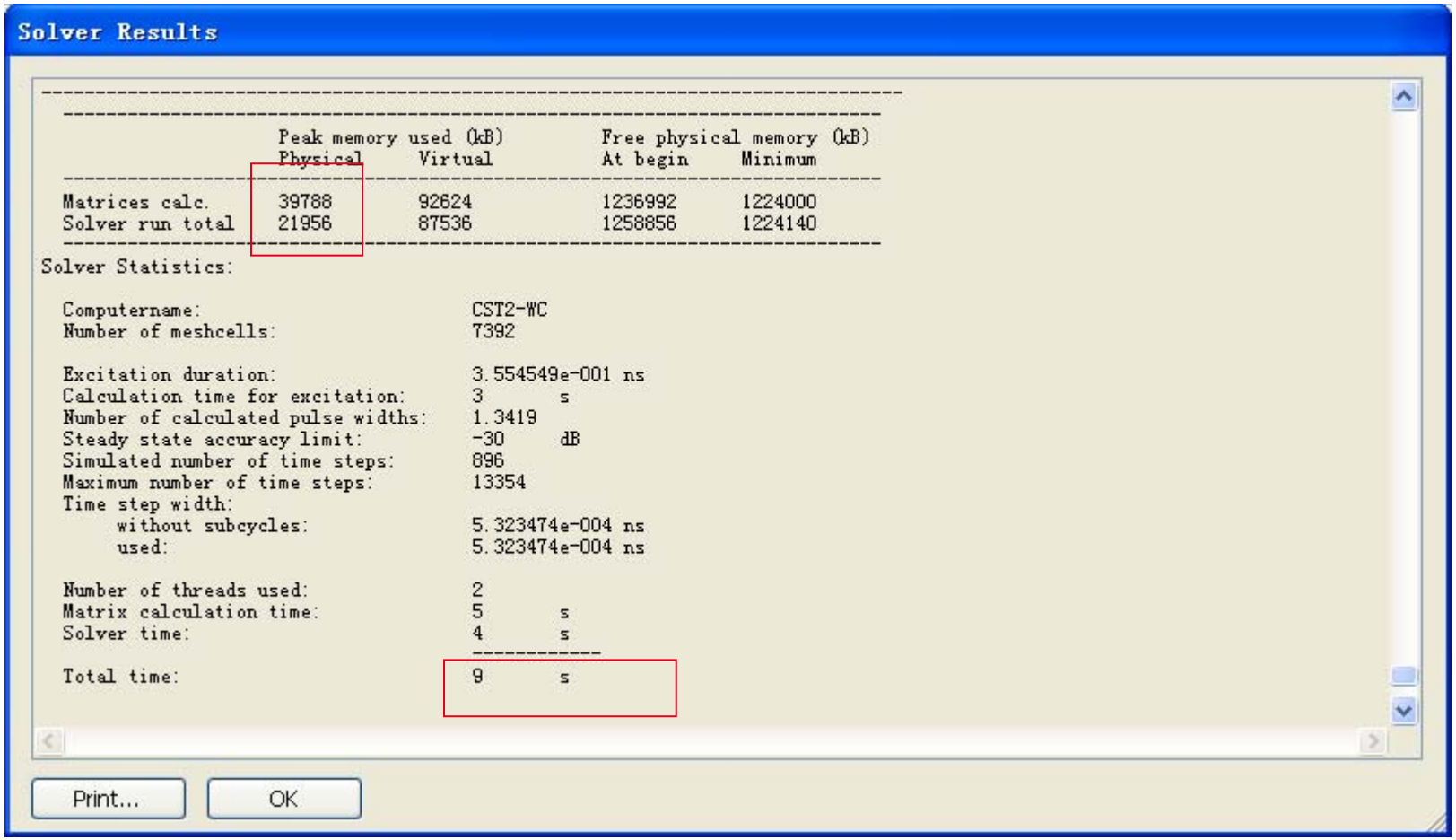
S参量


S-Parameter Magnitude in dB
0
-5
-10
-15
-20
-25
-30
-35
-40
-45
-50
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1
S2,1

f=5GHz 电场分布图

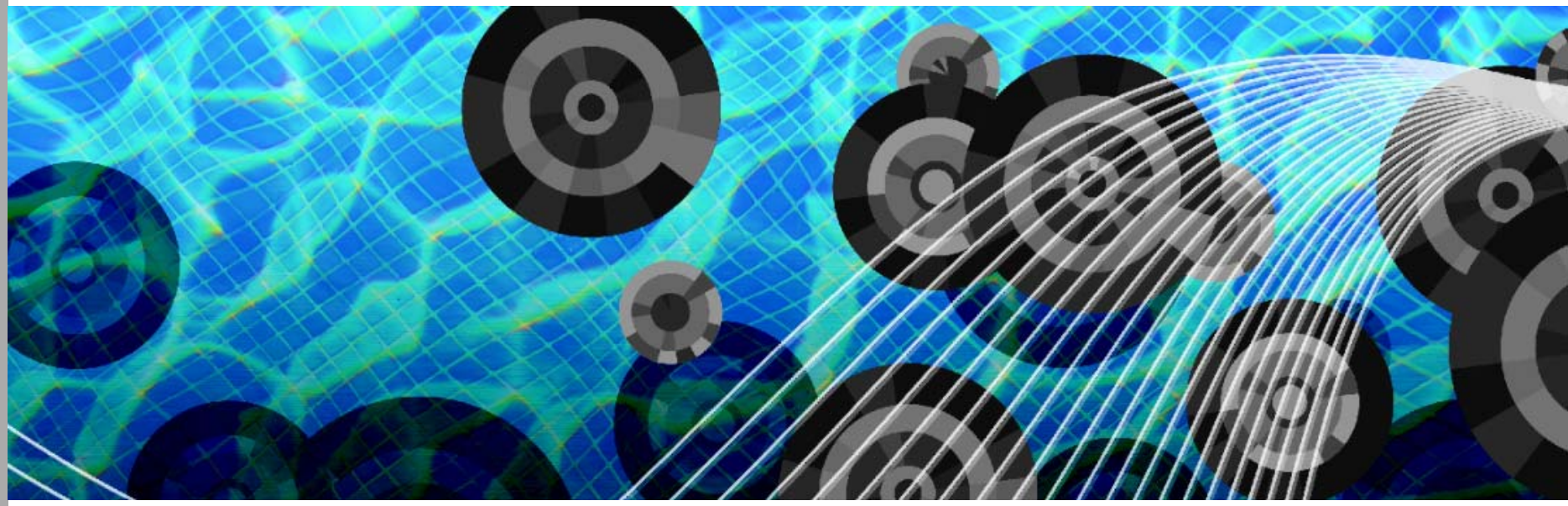

V/m
13949
11814
8393
5858
3980
2588
1556
792
226
0
2
1
z
y
x
Type E-Field (peak)
Monitor e-field (f=5) [1]
Maximum-3d 13949.4 V/m at 1.5 / -18.7879 / 0
Frequency 5
Phase 0 degrees

占用系统资源

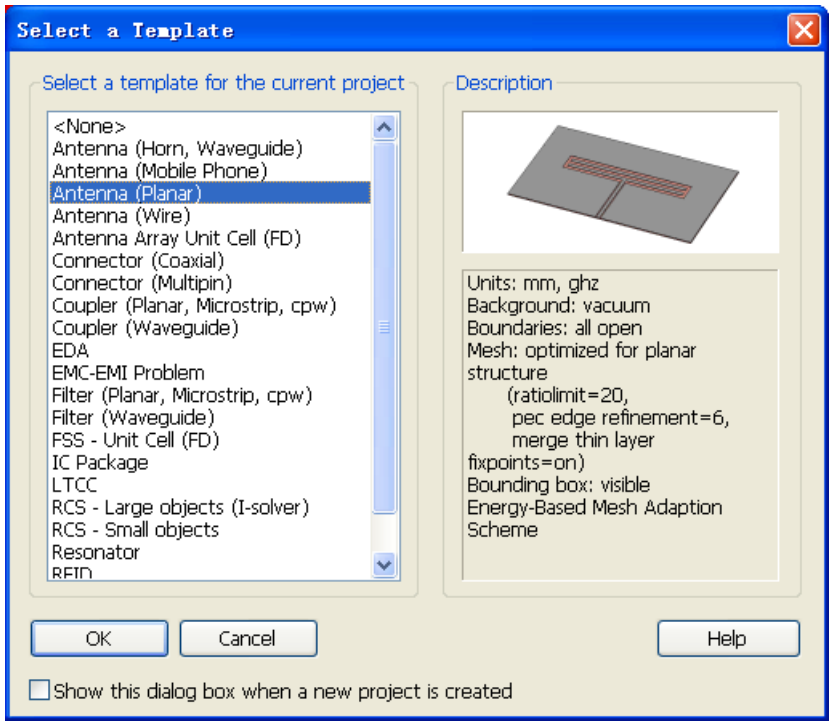


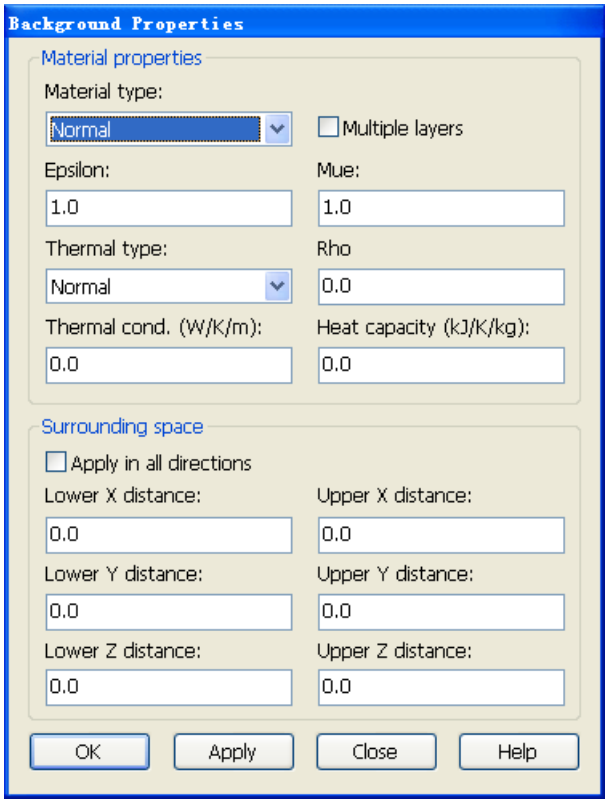
建立矩阵占用内存39M，求解占用内存22M，计算时间9s。

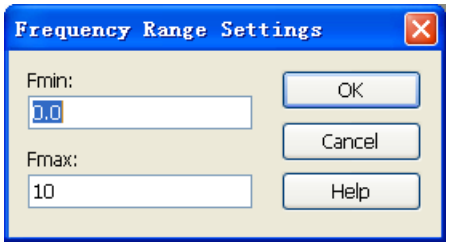
屏蔽微带线



选择模板、背景条件、频带范围

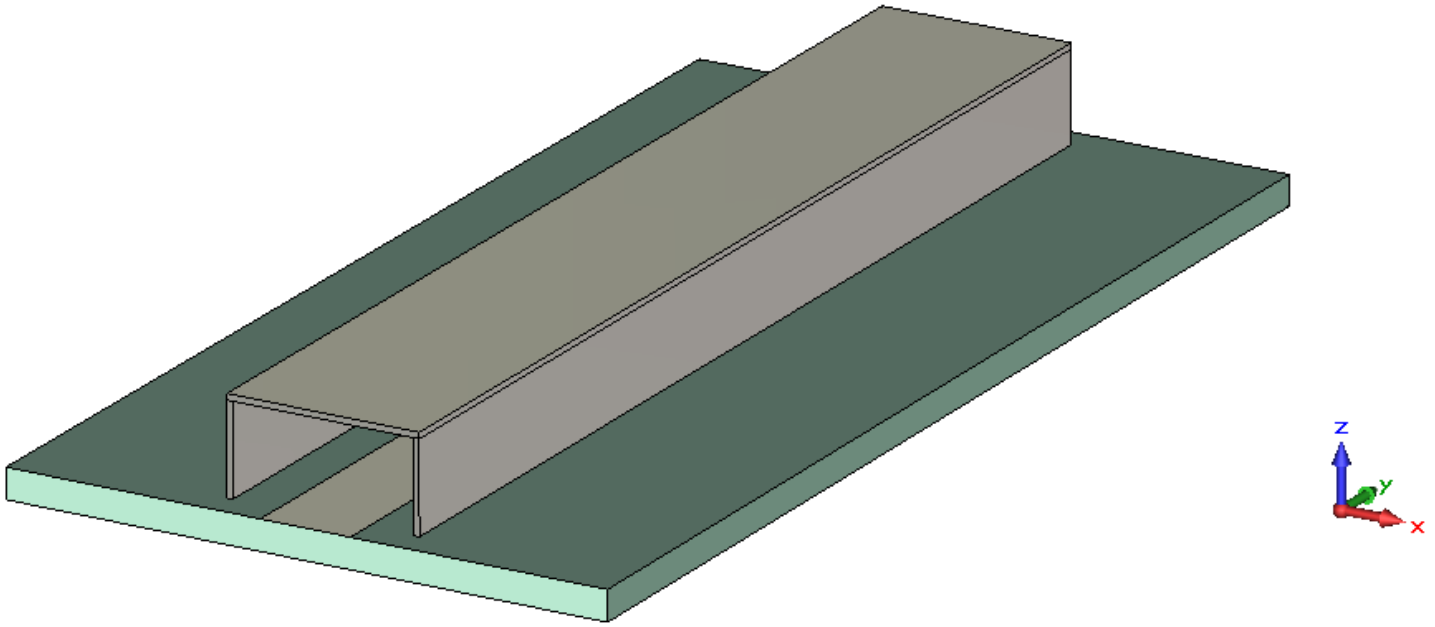




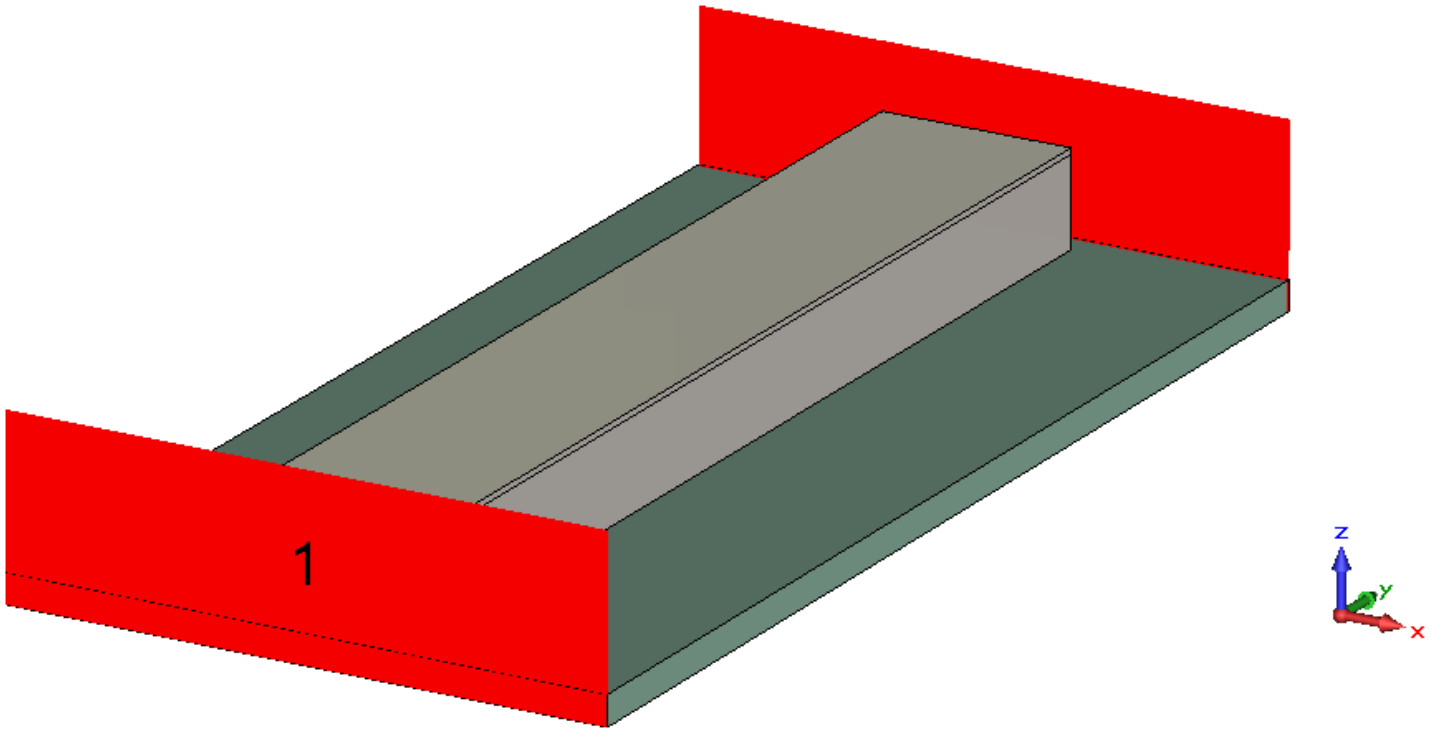


选好模板之后，默认单位为GHz，mm。

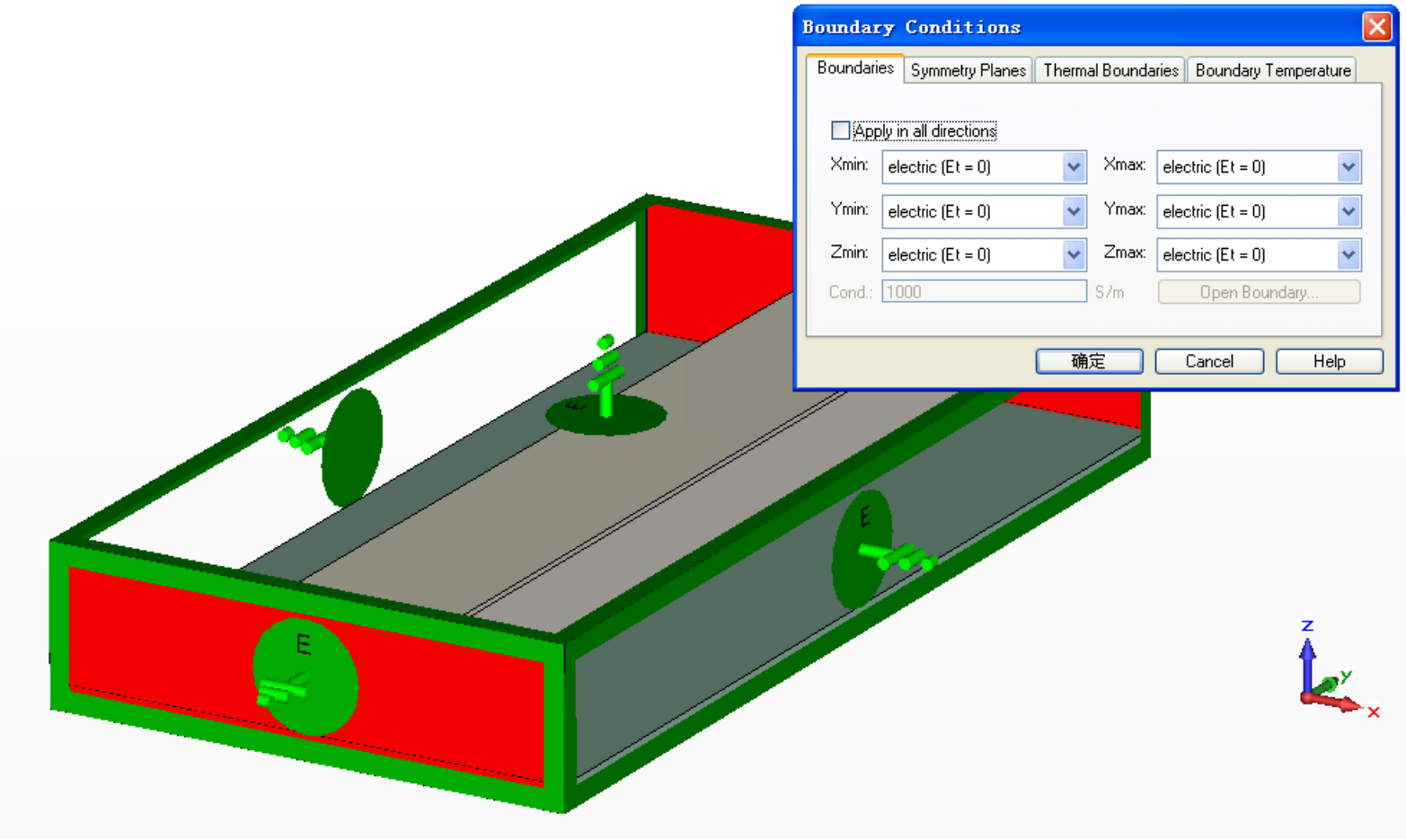
屏蔽微带线模型



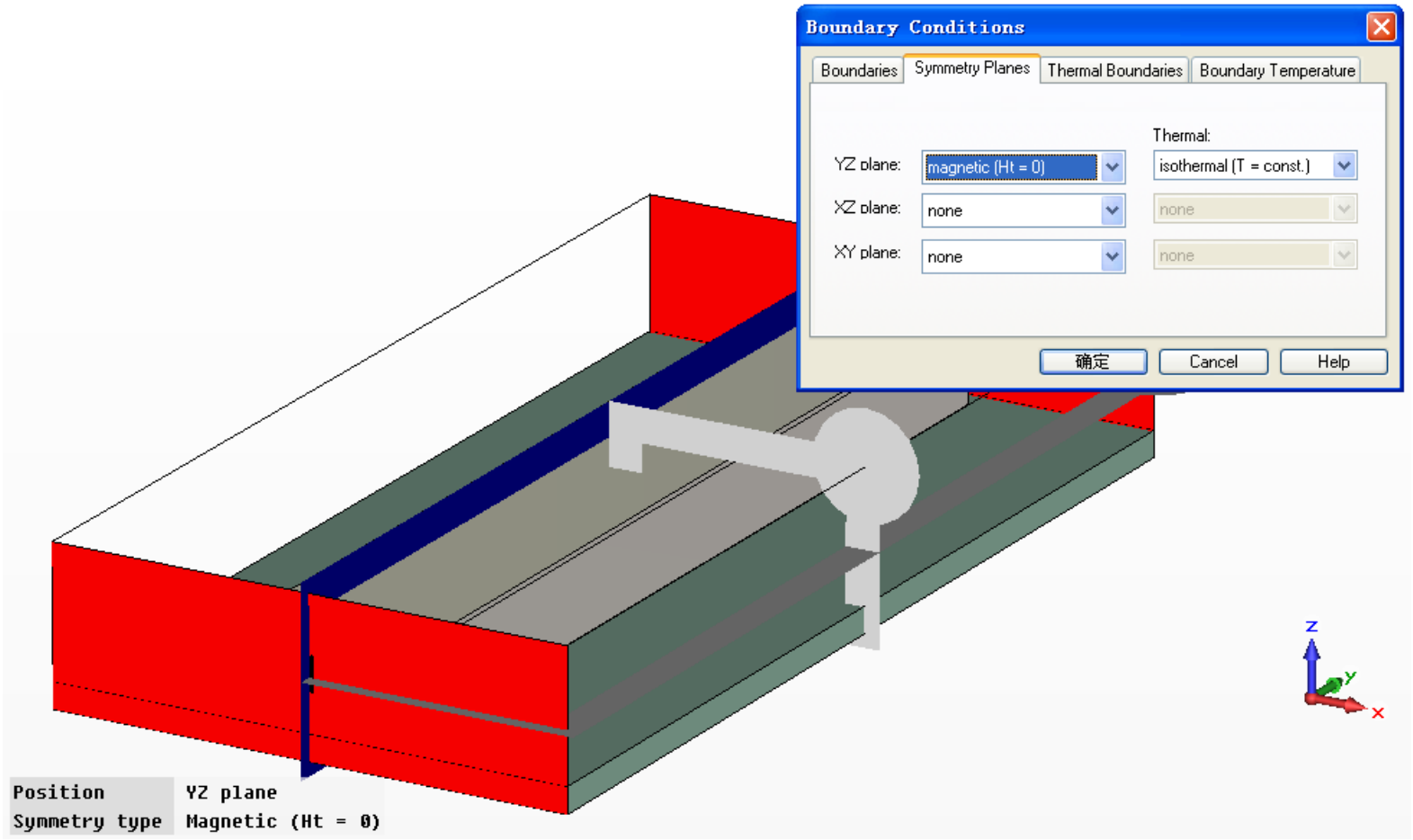
加载端口



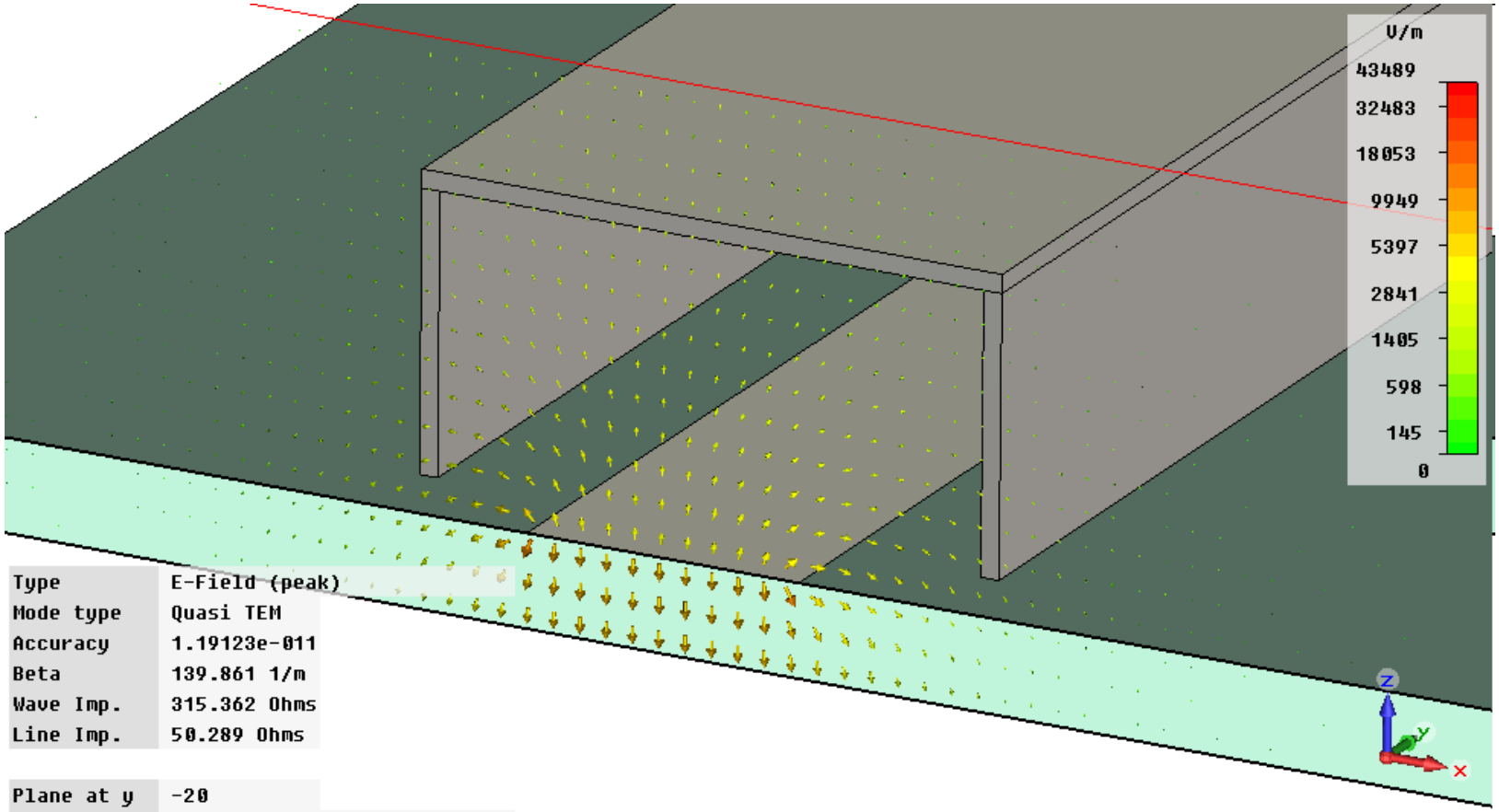
边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Xmin:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help
E
E
z
y
x

设置对称面


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Thermal:
YZ plane:
magnetic (Ht = 0)
isothermal (T = const.)
XZ plane:
none
none
XY plane:
none
none
确定
Cancel
Help
Z
Y
X
Position
YZ plane
Symmetry type
Magnetic (Ht = 0)

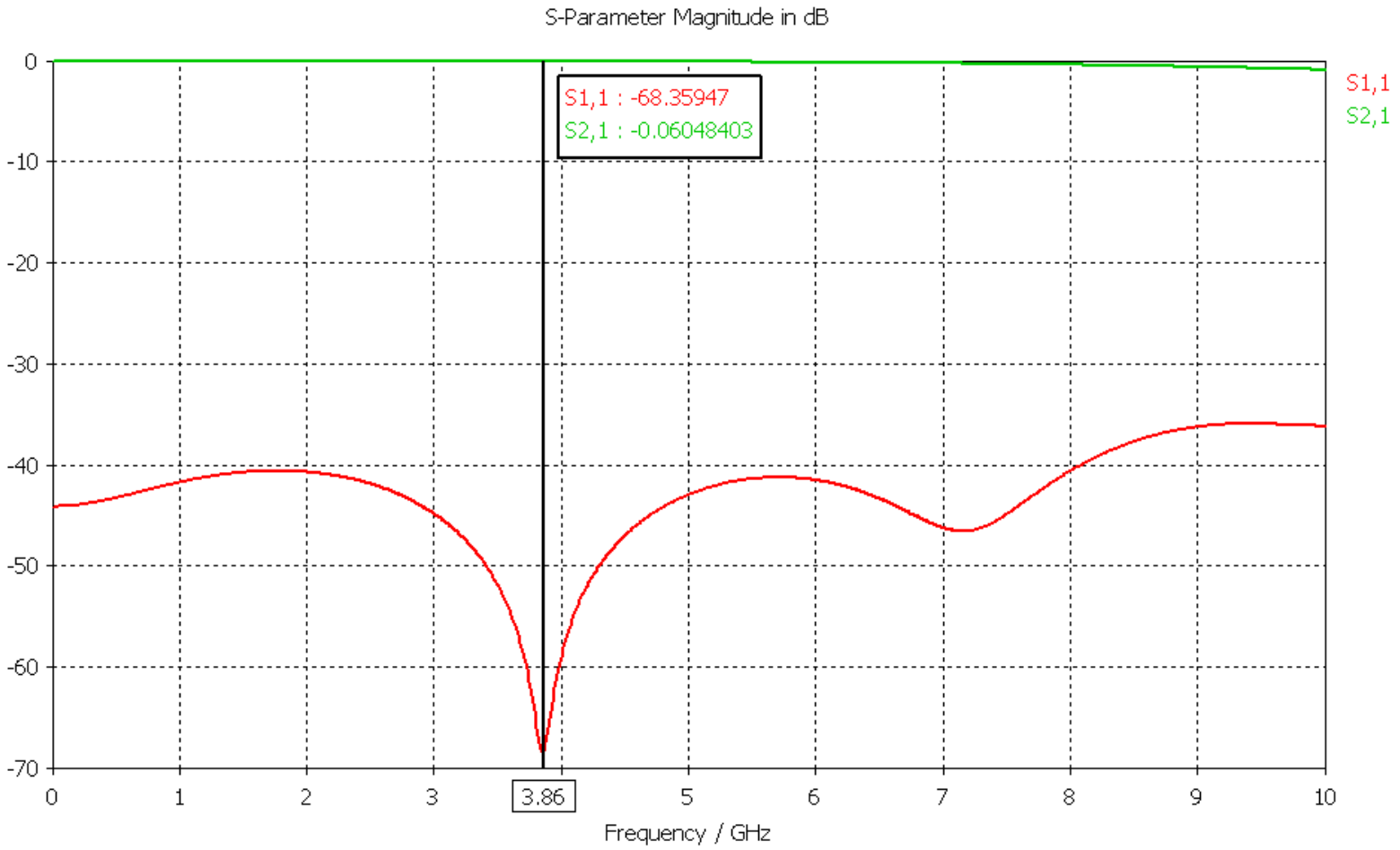
Port Modes > Port1 > e1



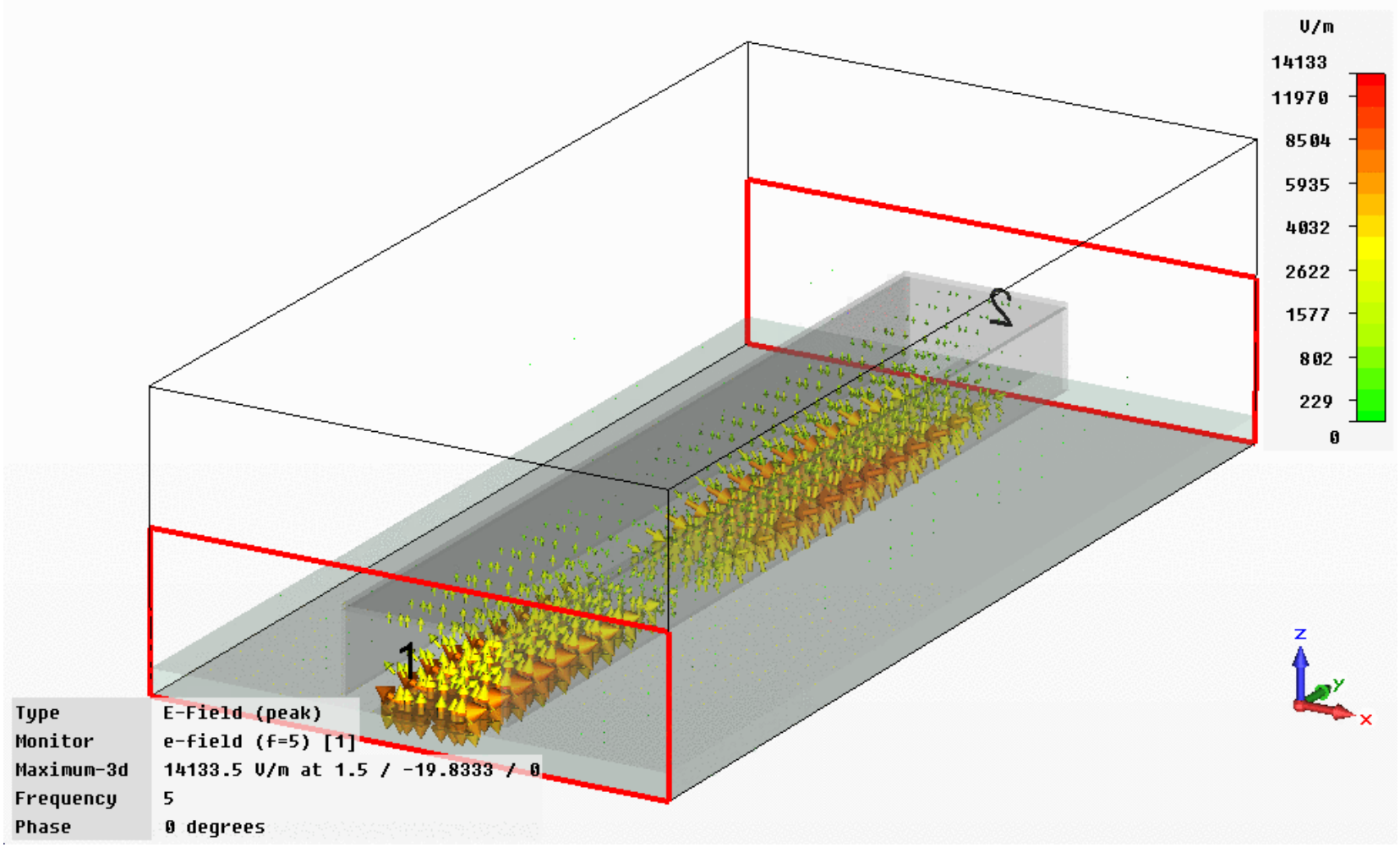
Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	1.19123e-011
Beta	139.861 1/m
Wave Imp.	315.362 Ohms
Line Imp.	50.289 Ohms

Plane at y	-20
Maximum-2d	43488.6 V/m at 1.5 / -20 / 0
Frequency	5
Phase	0 degrees

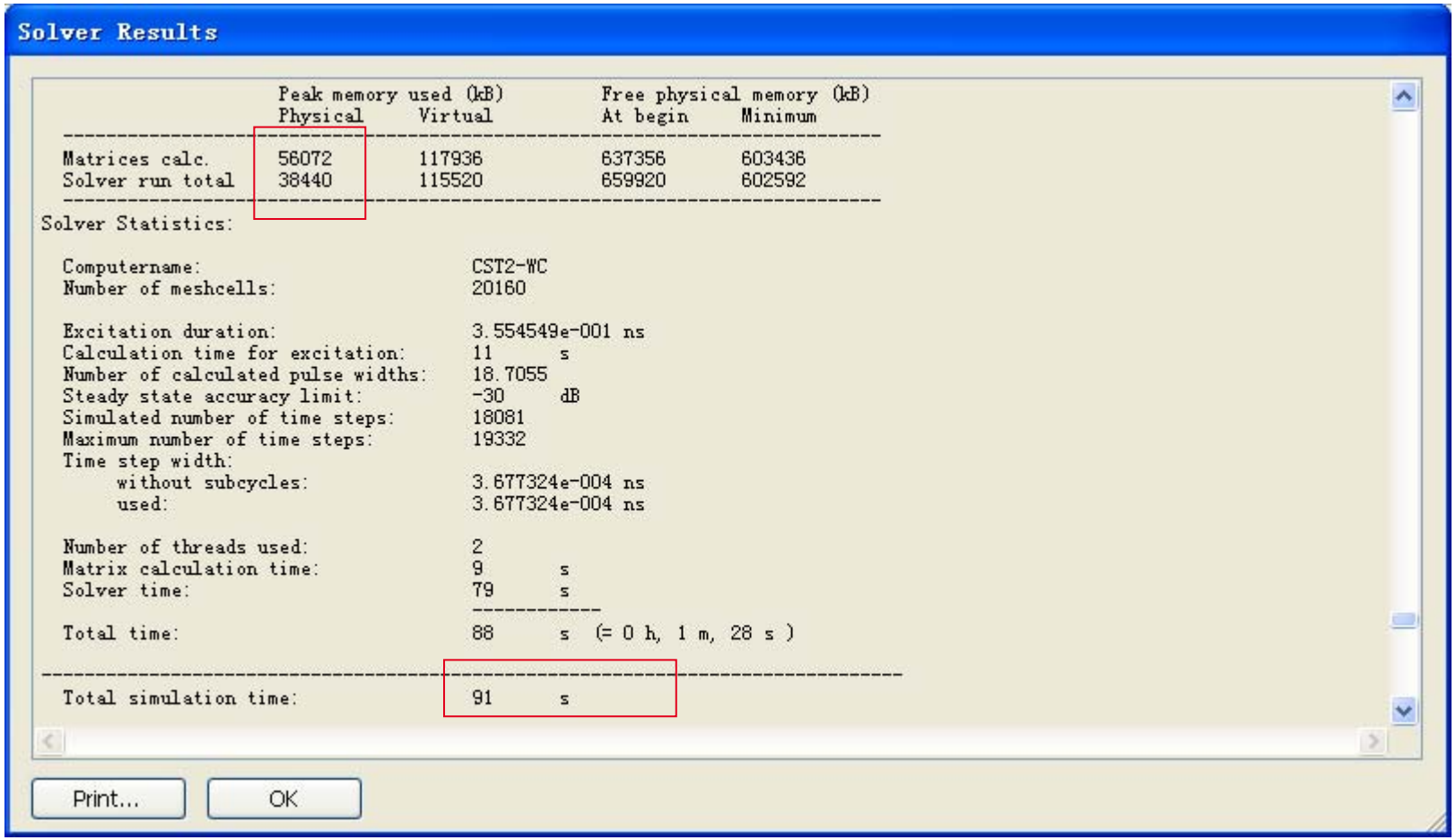
S11、S21


S-Parameter Magnitude in dB
S1,1 : -68.35947
S2,1 : -0.06048403
S1,1
S2,1
0
-10
-20
-30
-40
-50
-60
-70
0
1
2
3
3.86
5
6
7
8
9
10
Frequency / GHz

f=5GHz 电场分布图

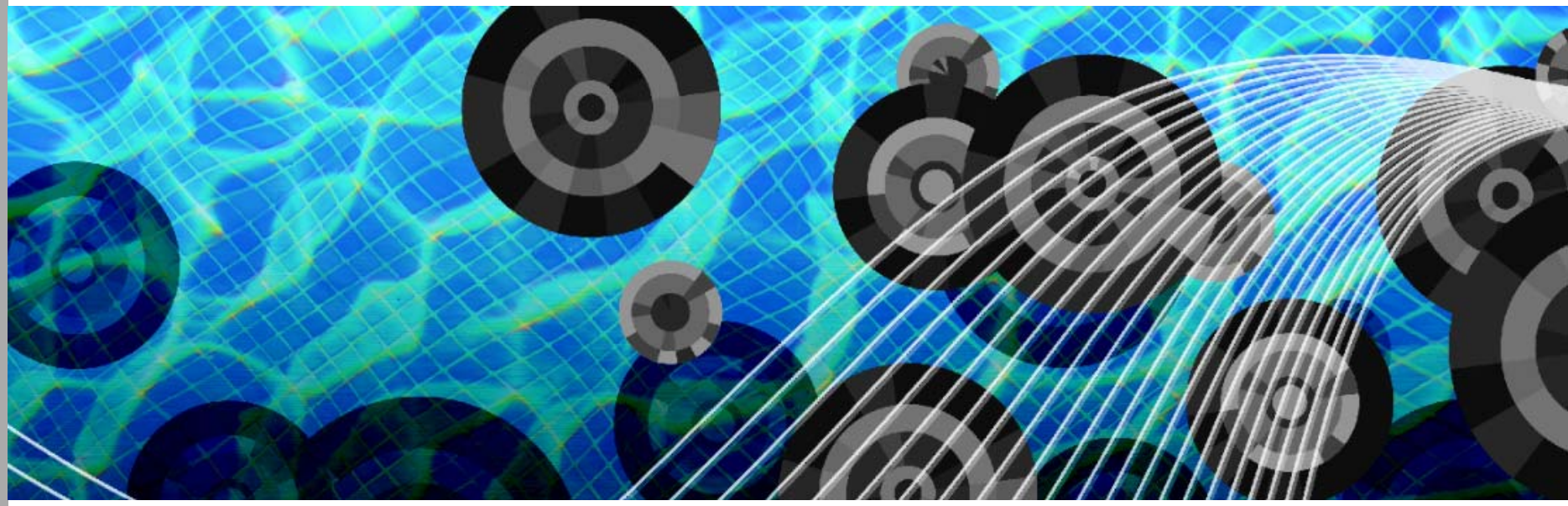

V/m
14133
11970
8504
5935
4032
2622
1577
802
229
0
1
2
Z
Y
X
Type E-Field (peak)
Monitor e-field (f=5) [1]
Maximum-3d 14133.5 V/m at 1.5 / -19.8333 / 0
Frequency 5
Phase 0 degrees

占用系统资源

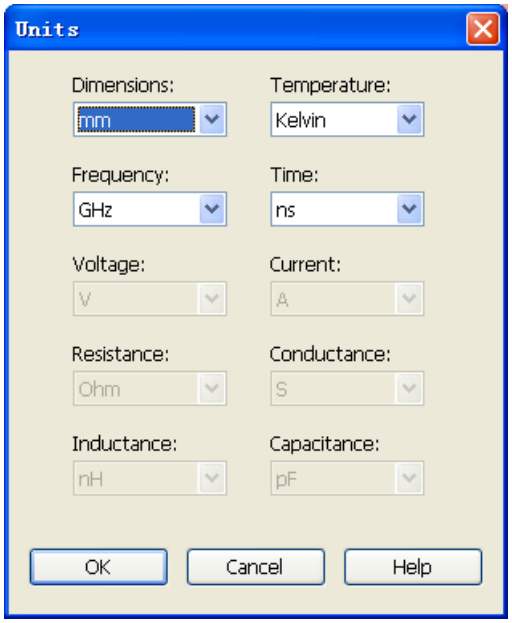


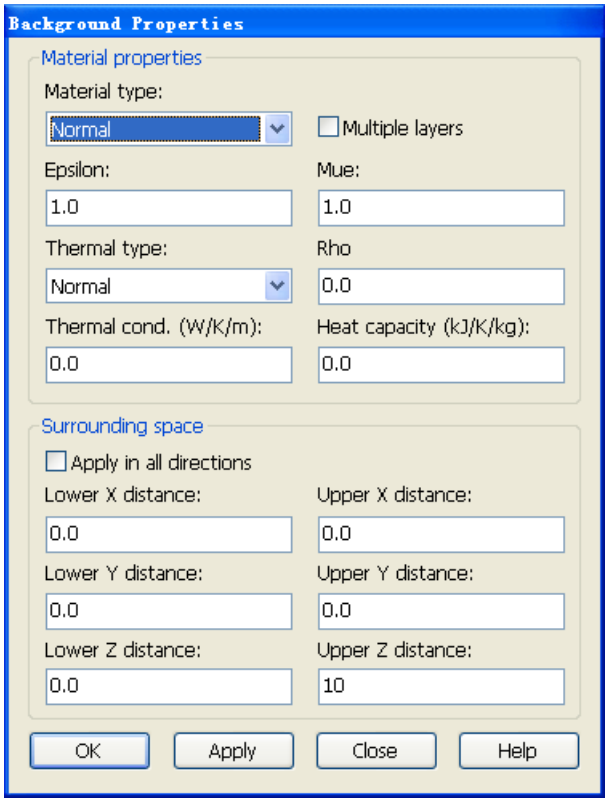
建立矩阵占用内存56M，求解占用内存38M，求解器计算时间91s。

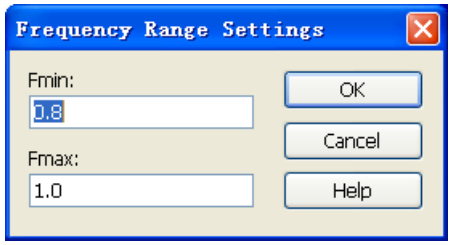
耦合微带线



选择模板、背景条件、频带范围

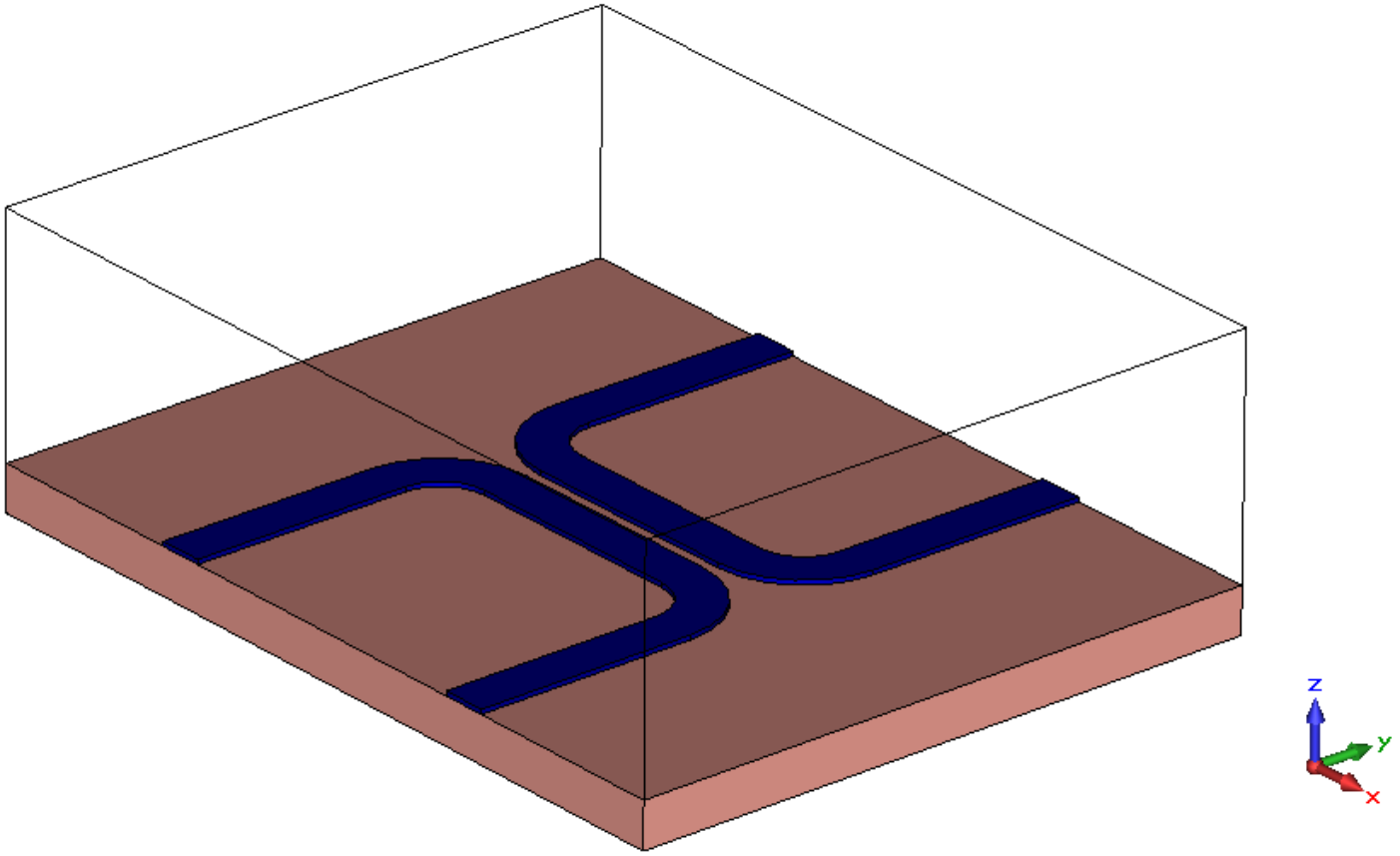




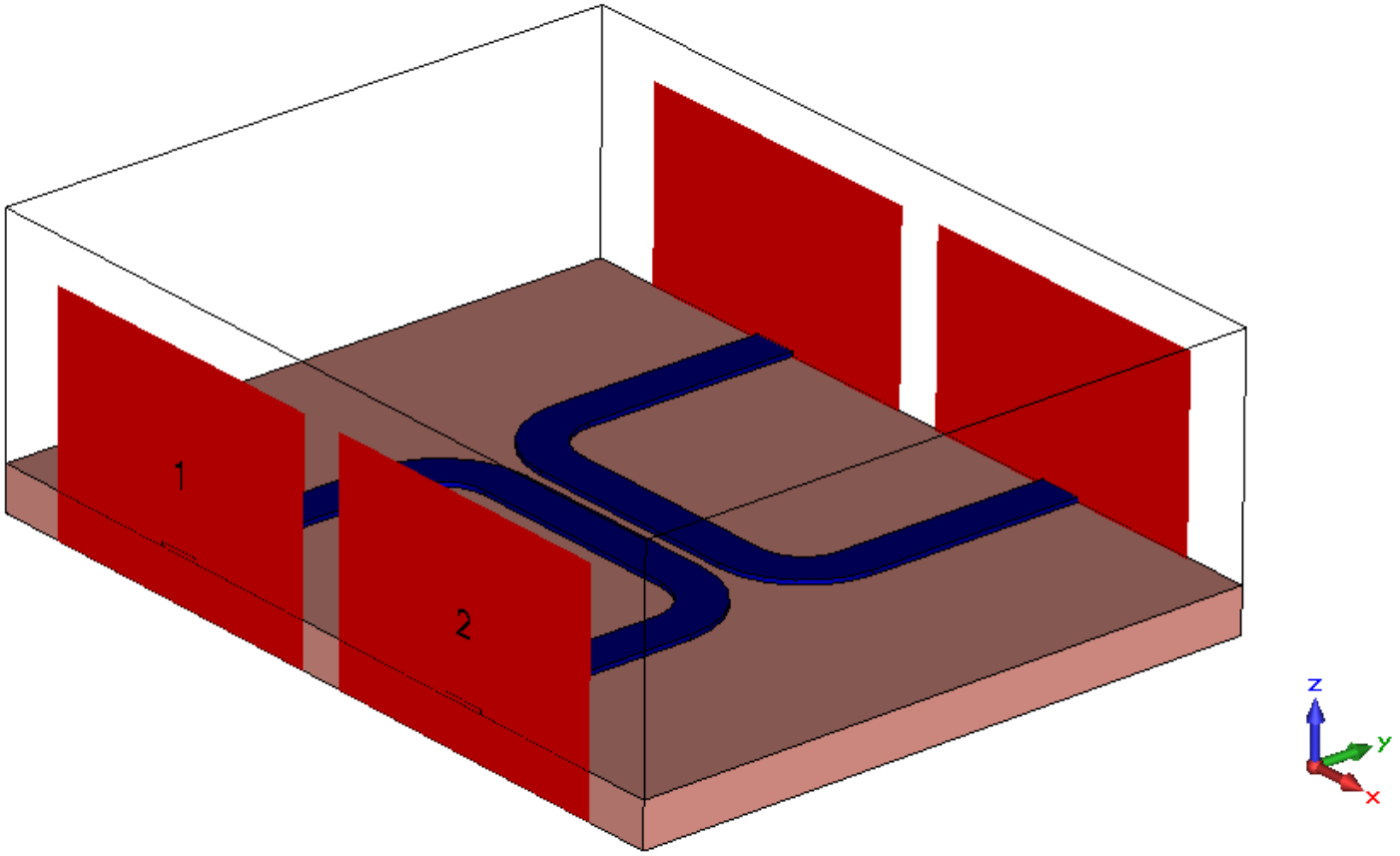


选择单位，背景材料，以及仿真的频带范围。

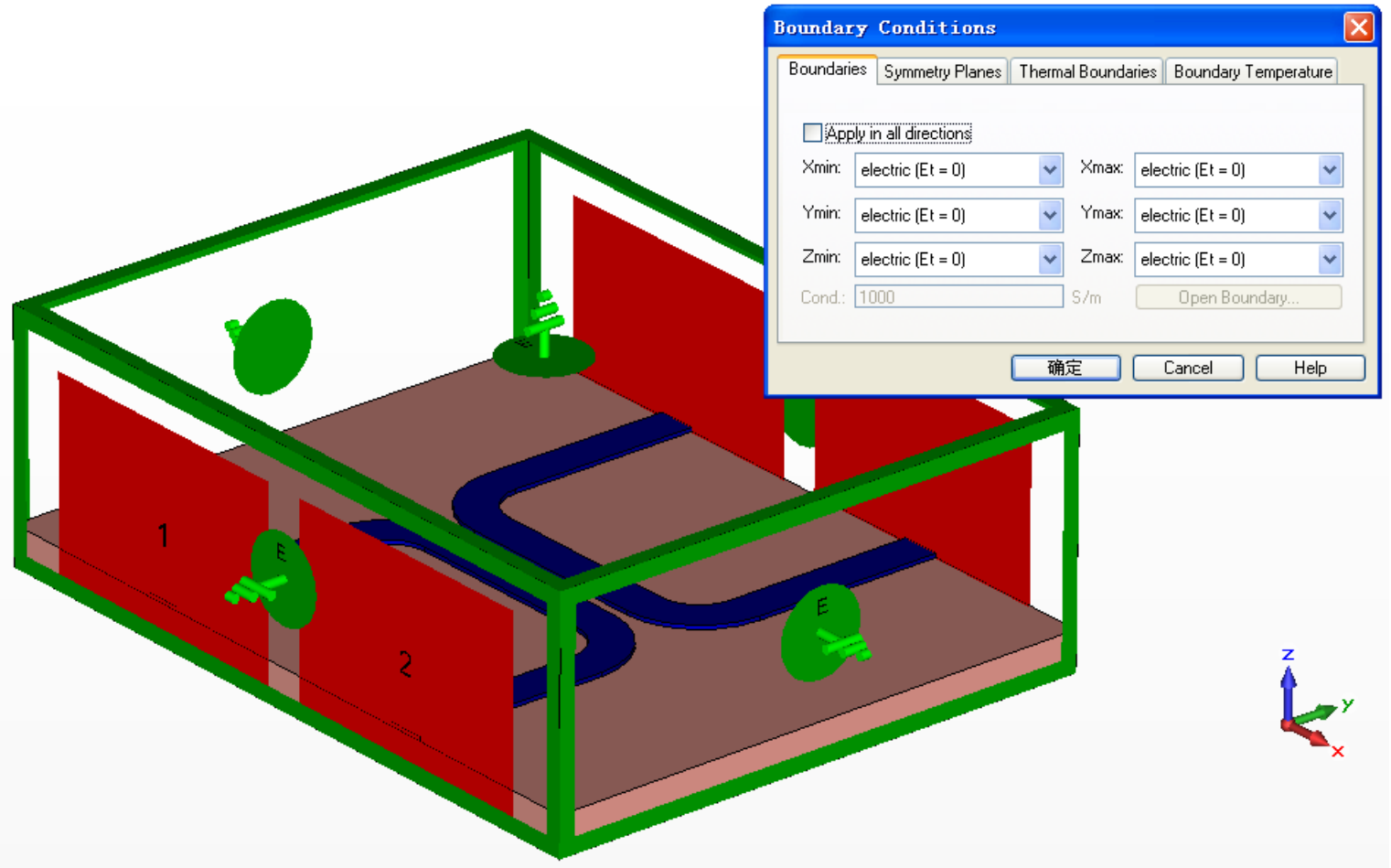
耦合微带线模型



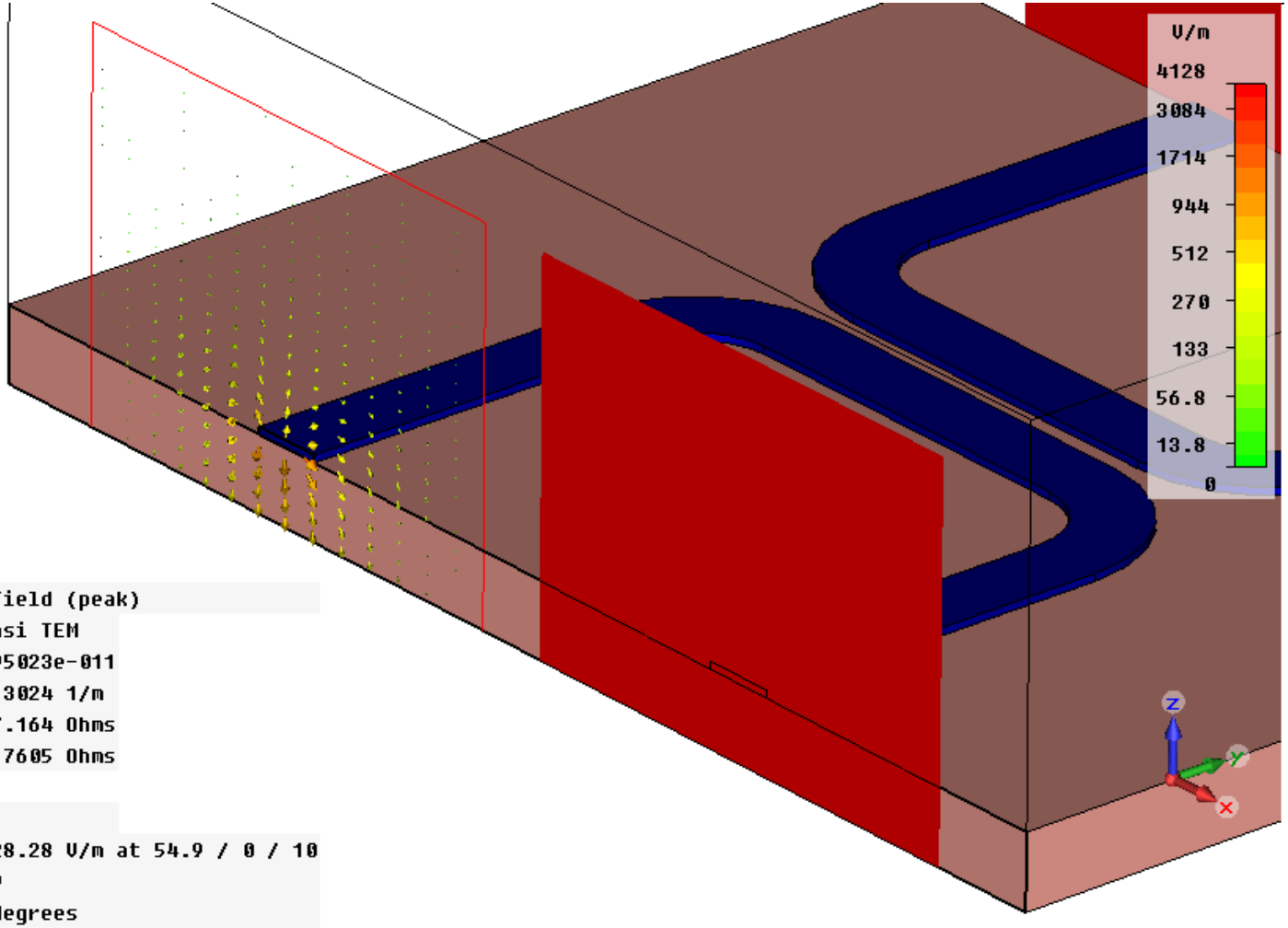
四端口激励



边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Xmin: electric (Et = 0)
Xmax: electric (Et = 0)
Ymin: electric (Et = 0)
Ymax: electric (Et = 0)
Zmin: electric (Et = 0)
Zmax: electric (Et = 0)
Cond.: 1000
S/m
Open Boundary...
确定
Cancel
Help
1
2
E
E
E
z
y
x

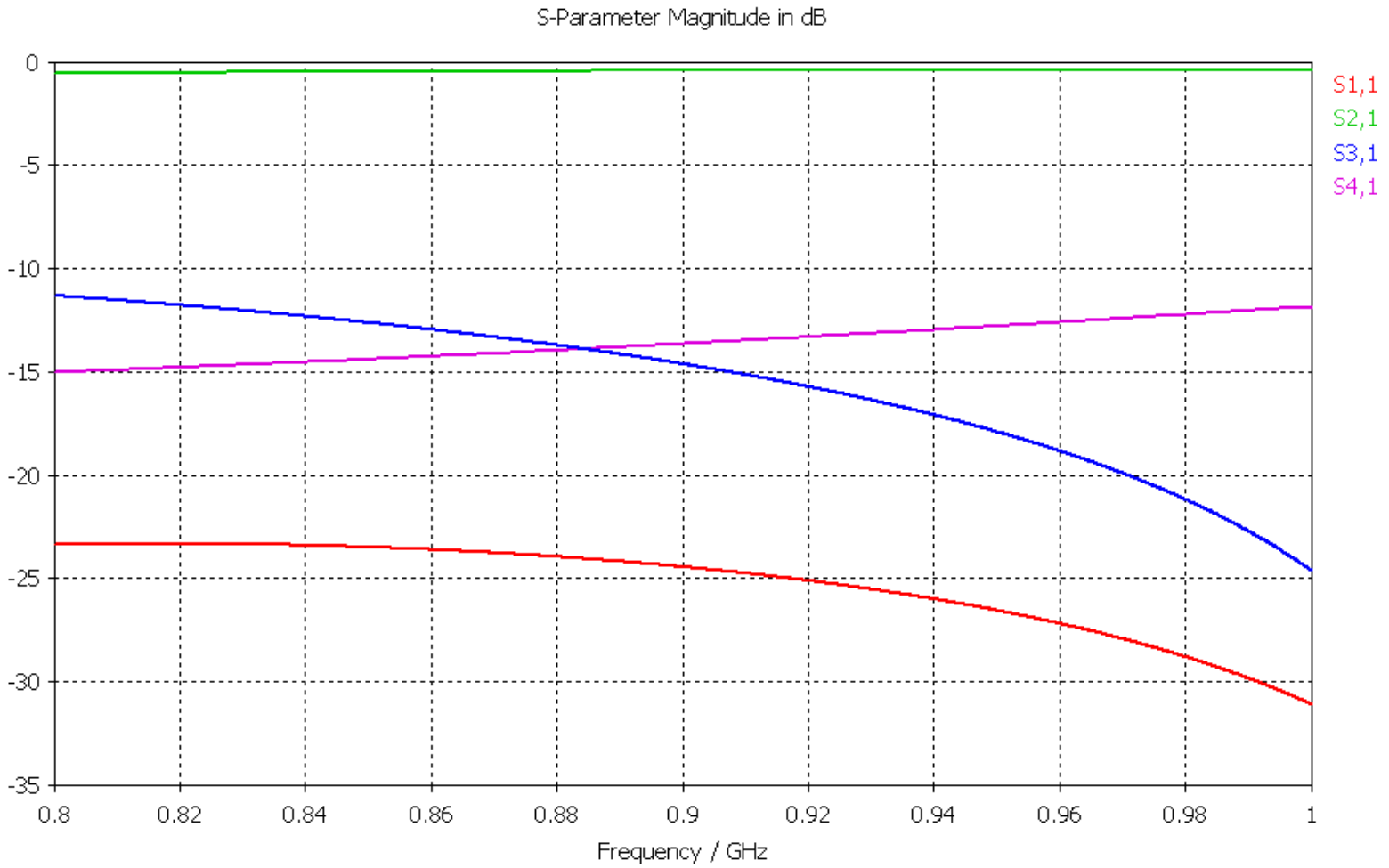
1端口处端口模式



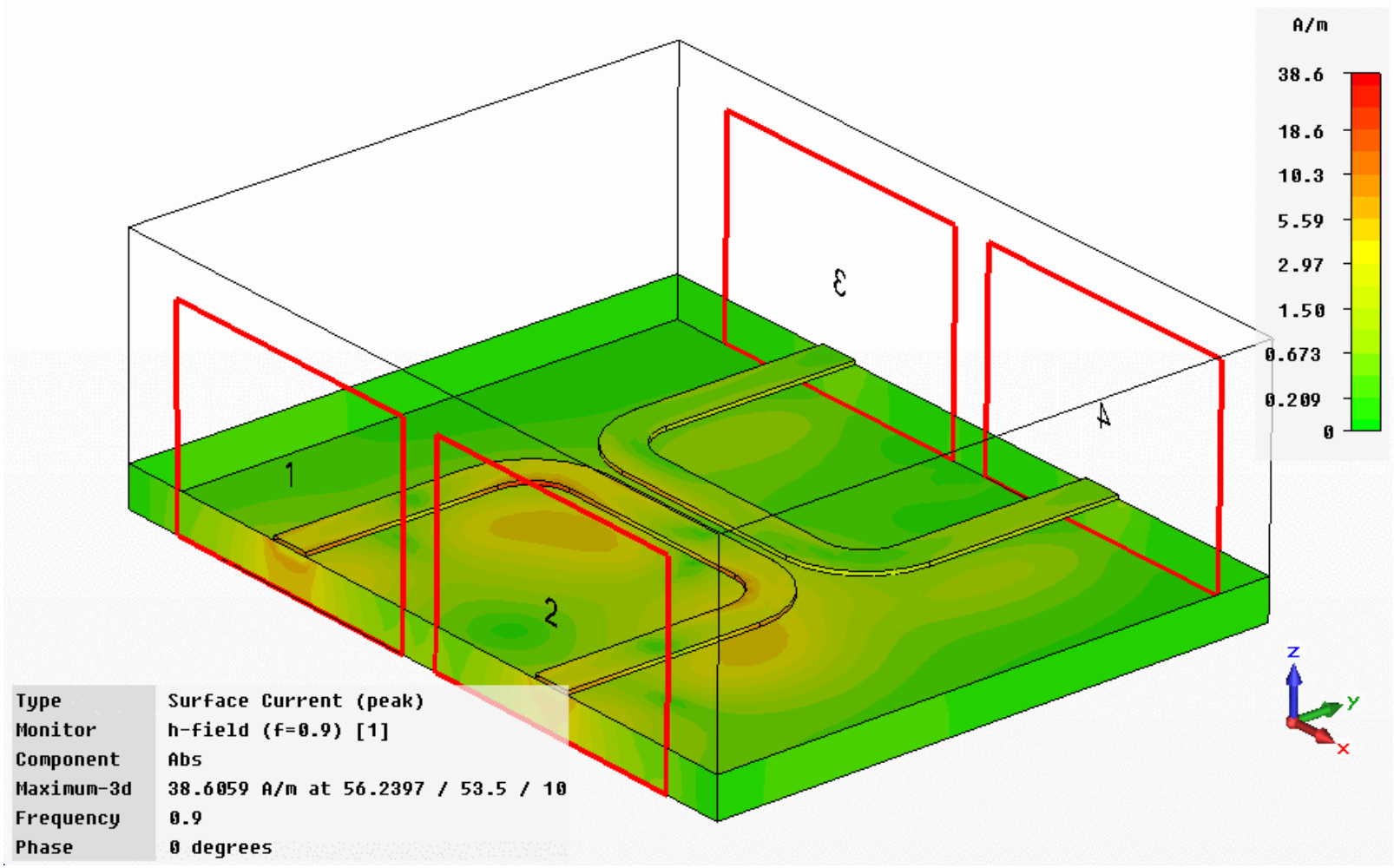
Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	8.95023e-011
Beta	47.3024 1/m
Wave Imp.	257.164 Ohms
Line Imp.	46.7605 Ohms

Plane at y	0
Maximum-2d	4128.28 V/m at 54.9 / 0 / 10
Frequency	0.9
Phase	0 degrees

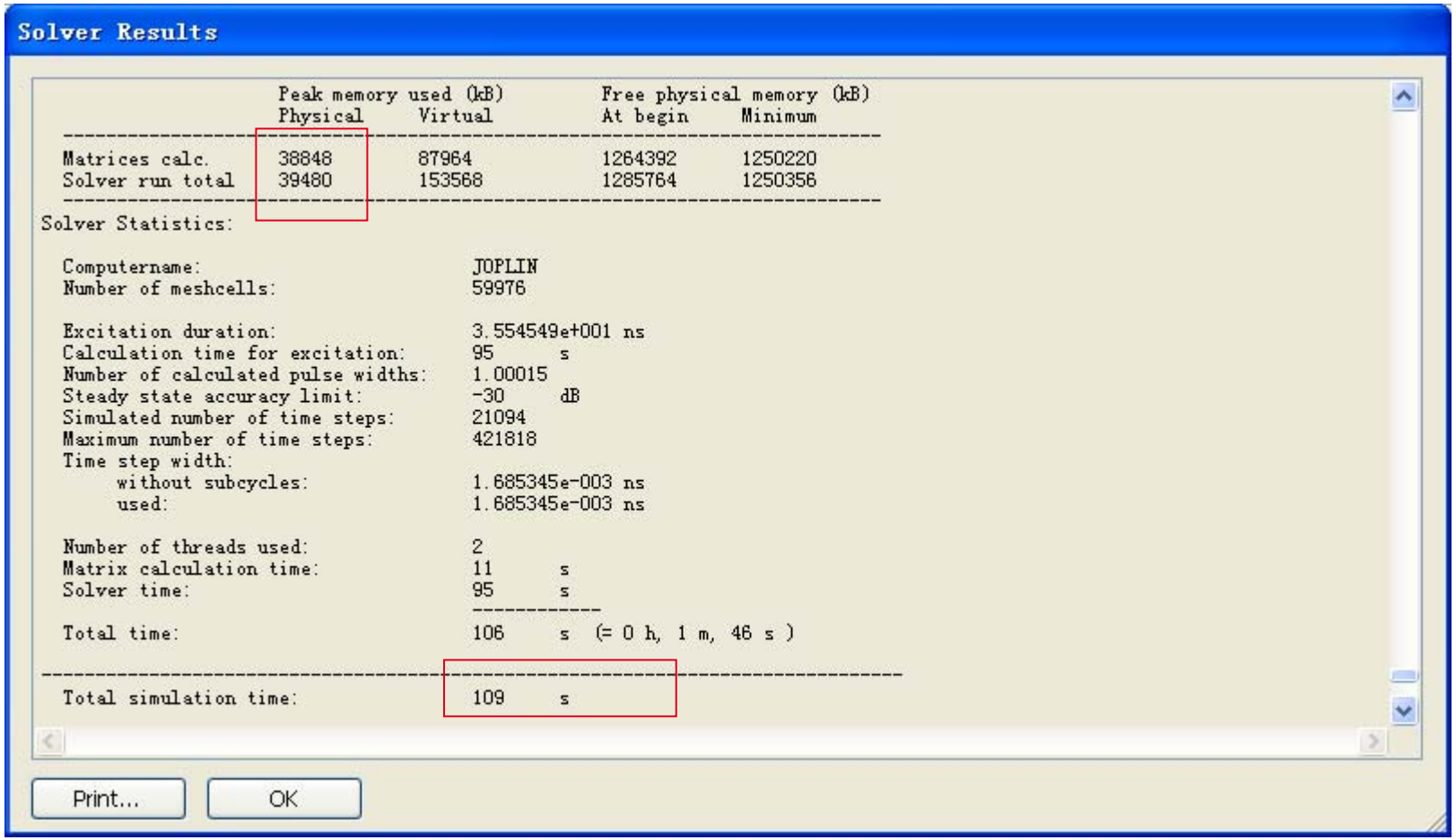
1端口与其他端口的耦合度曲线



表面电流

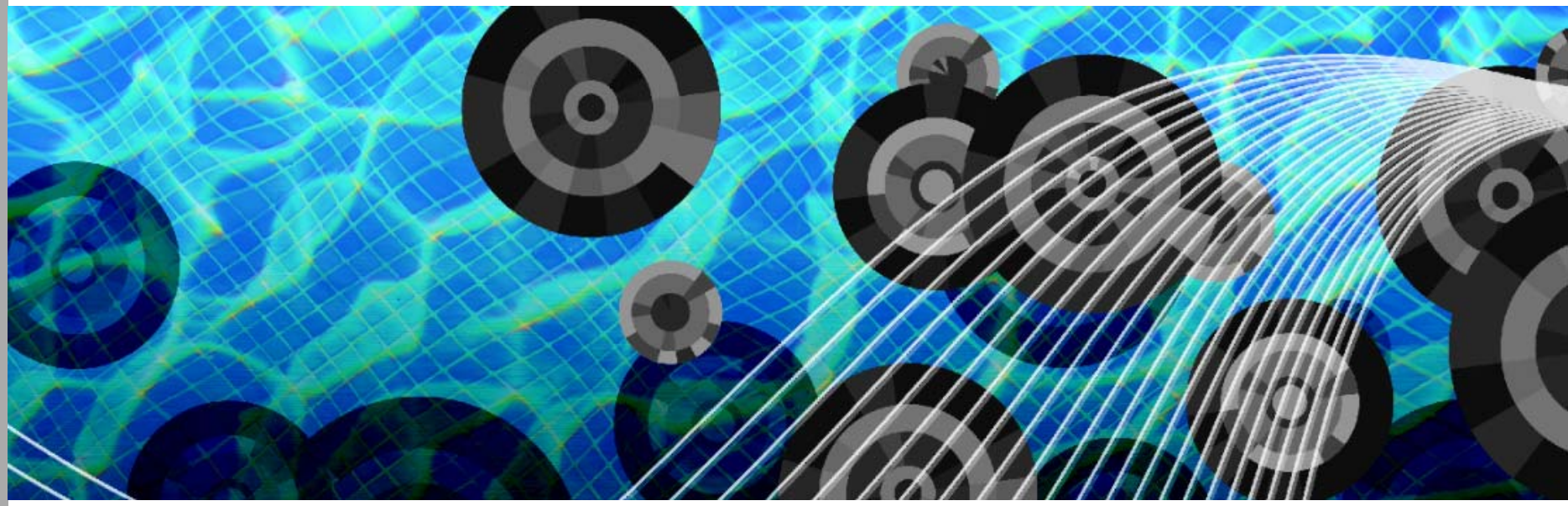

A/m
38.6
18.6
10.3
5.59
2.97
1.50
0.673
0.209
0
1
2
3
4
z
y
x
Type Surface Current (peak)
Monitor h-field (f=0.9) [1]
Component Abs
Maximum-3d 38.6059 A/m at 56.2397 / 53.5 / 10
Frequency 0.9
Phase 0 degrees

占用系统资源

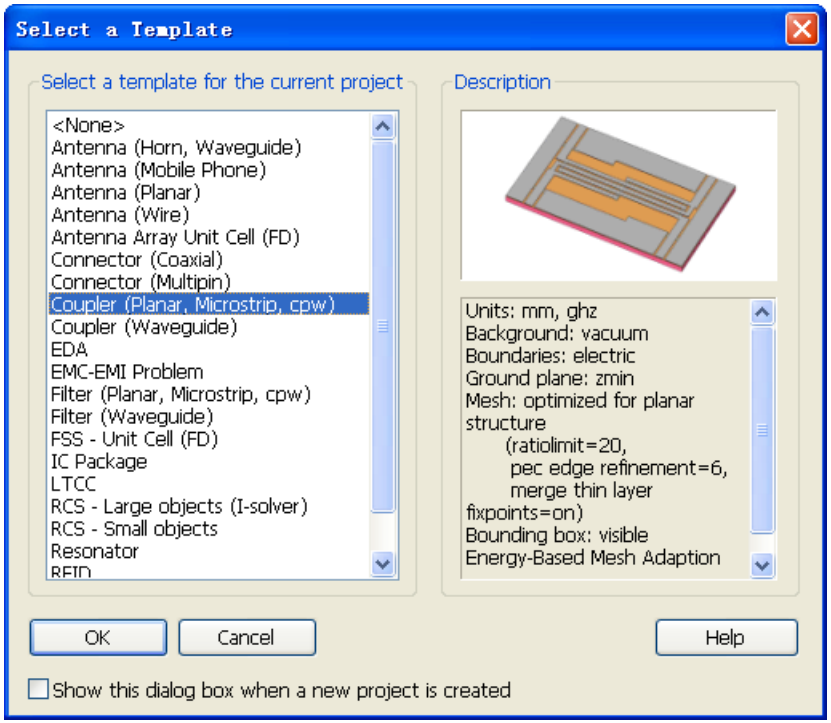


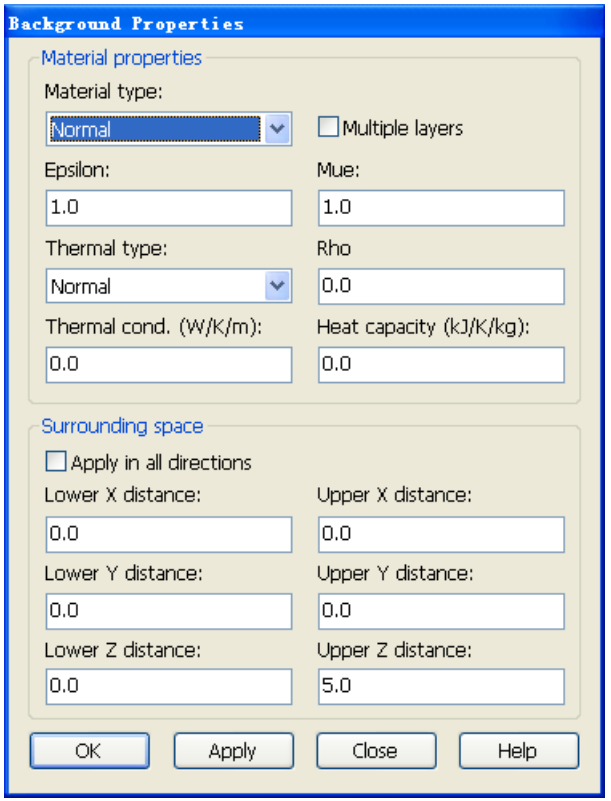
建立矩阵占用内存38M，求解占用内存39M，计算时间109s。

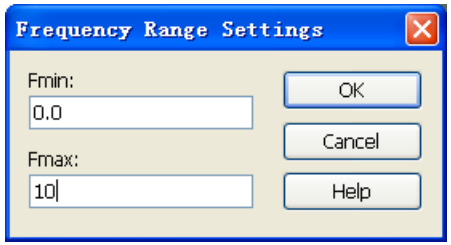
悬置微带线



选择模板、背景条件、频带范围

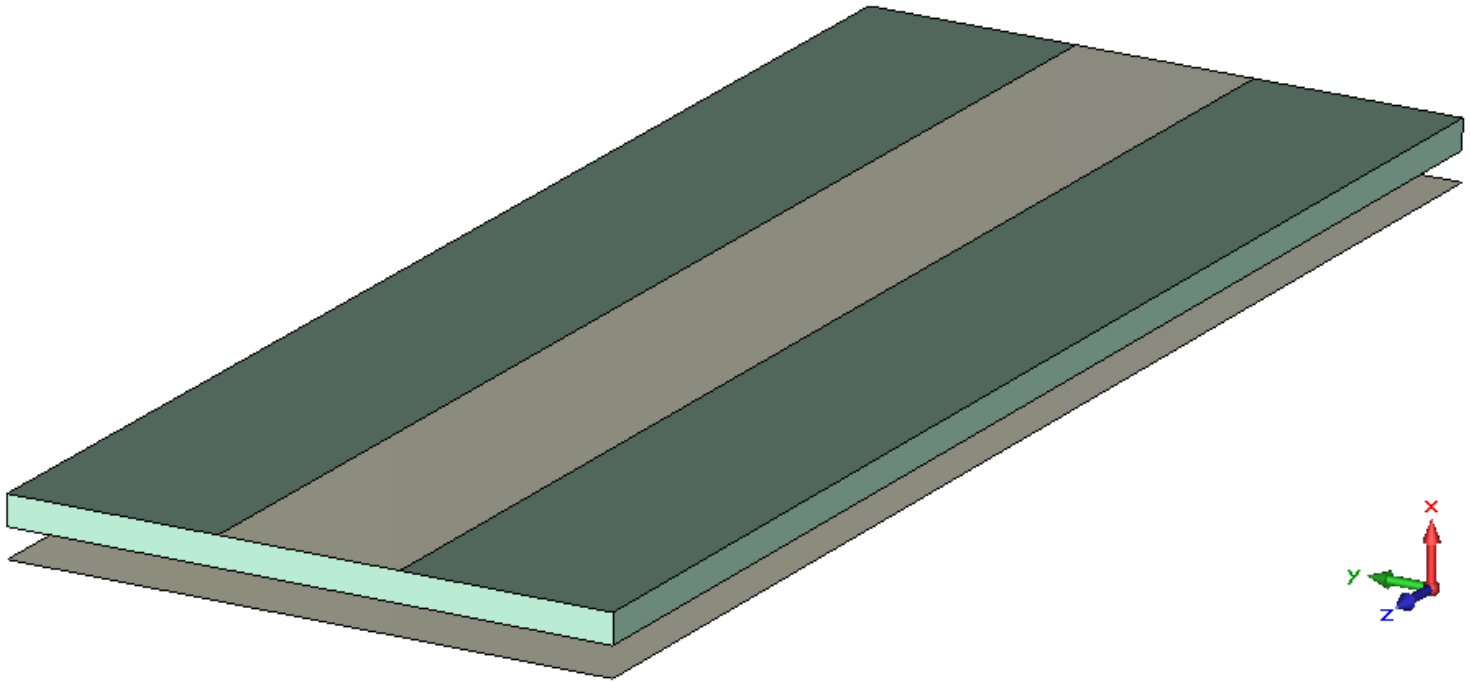




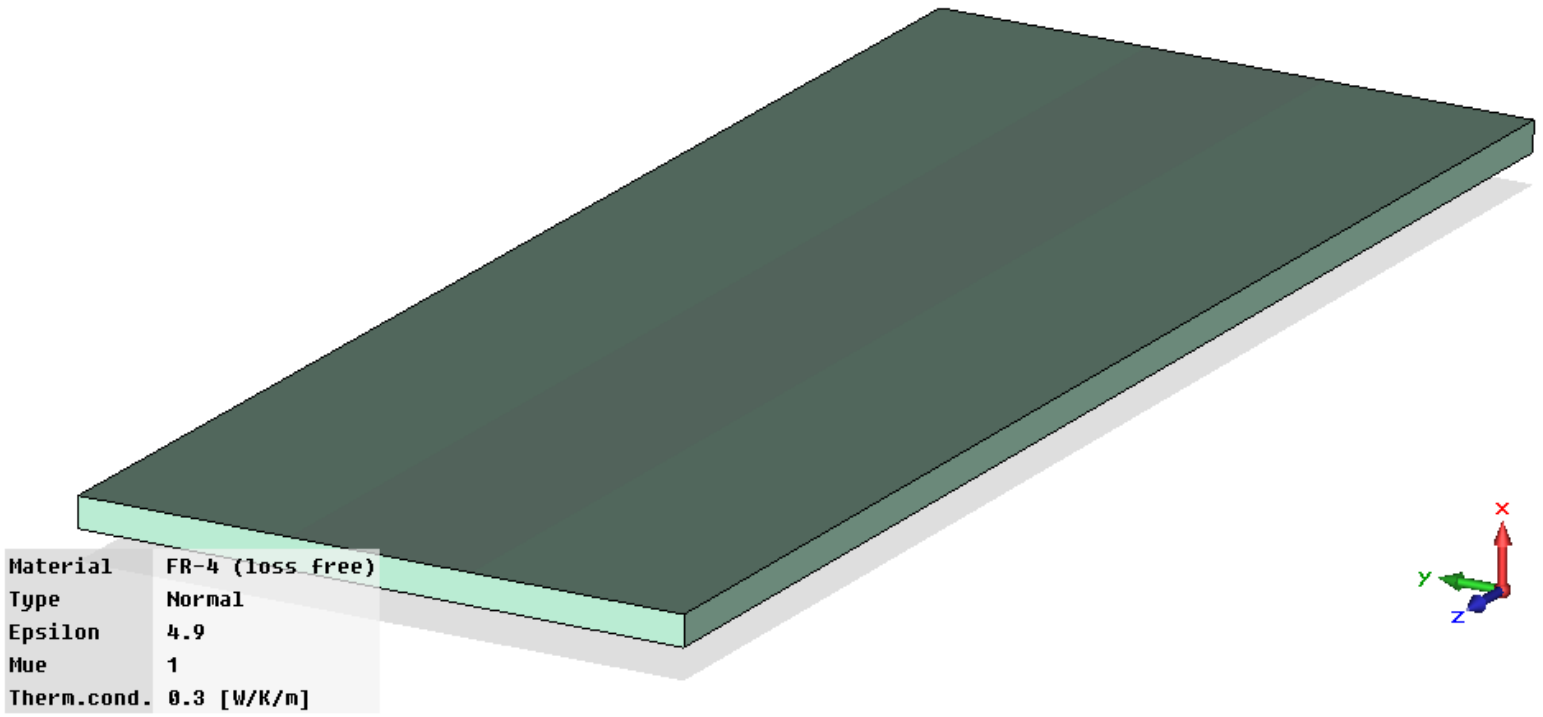


选择平面器件，微带线模板，默认单位为GHz，mm。

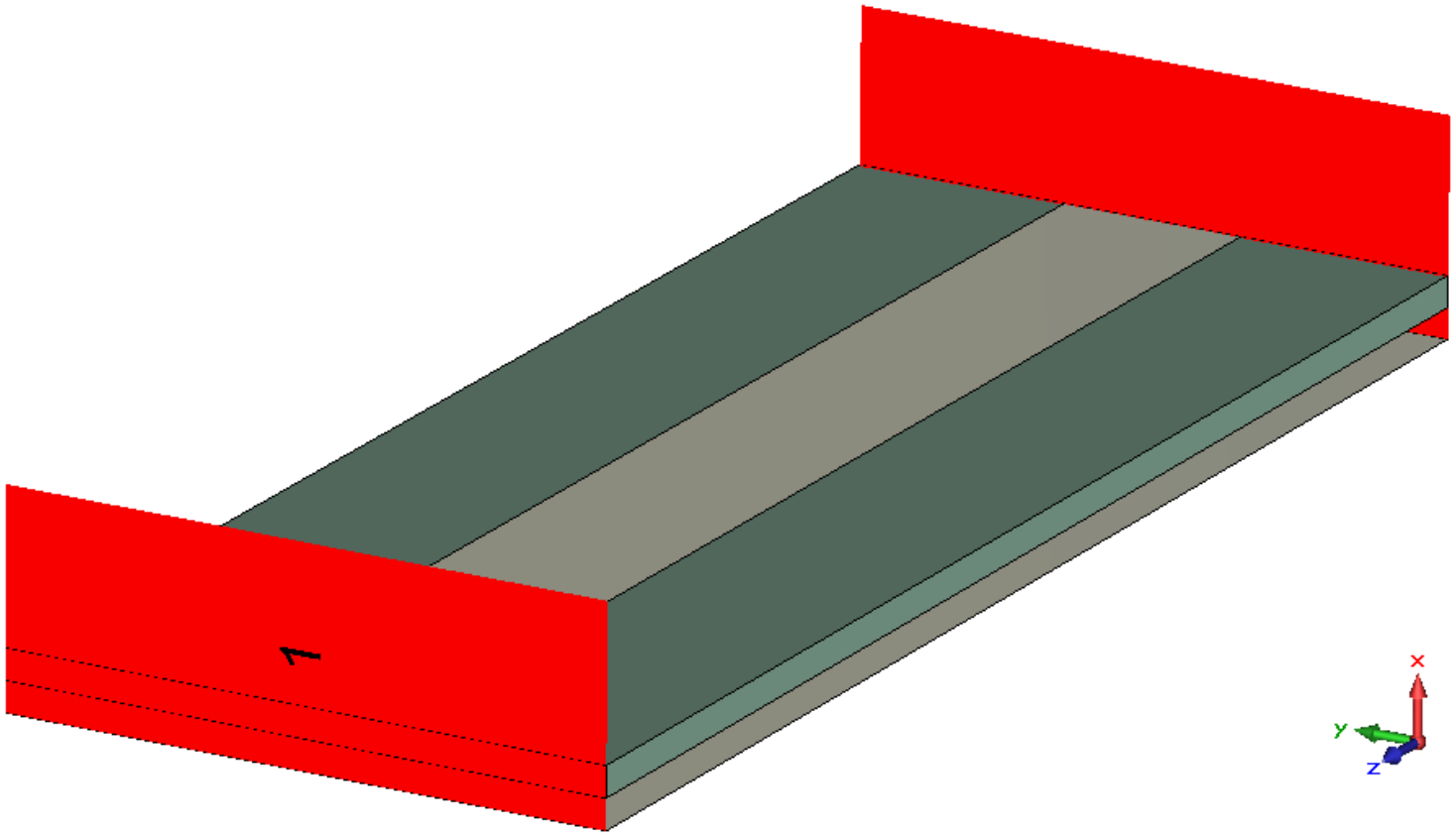
悬置微带线模型



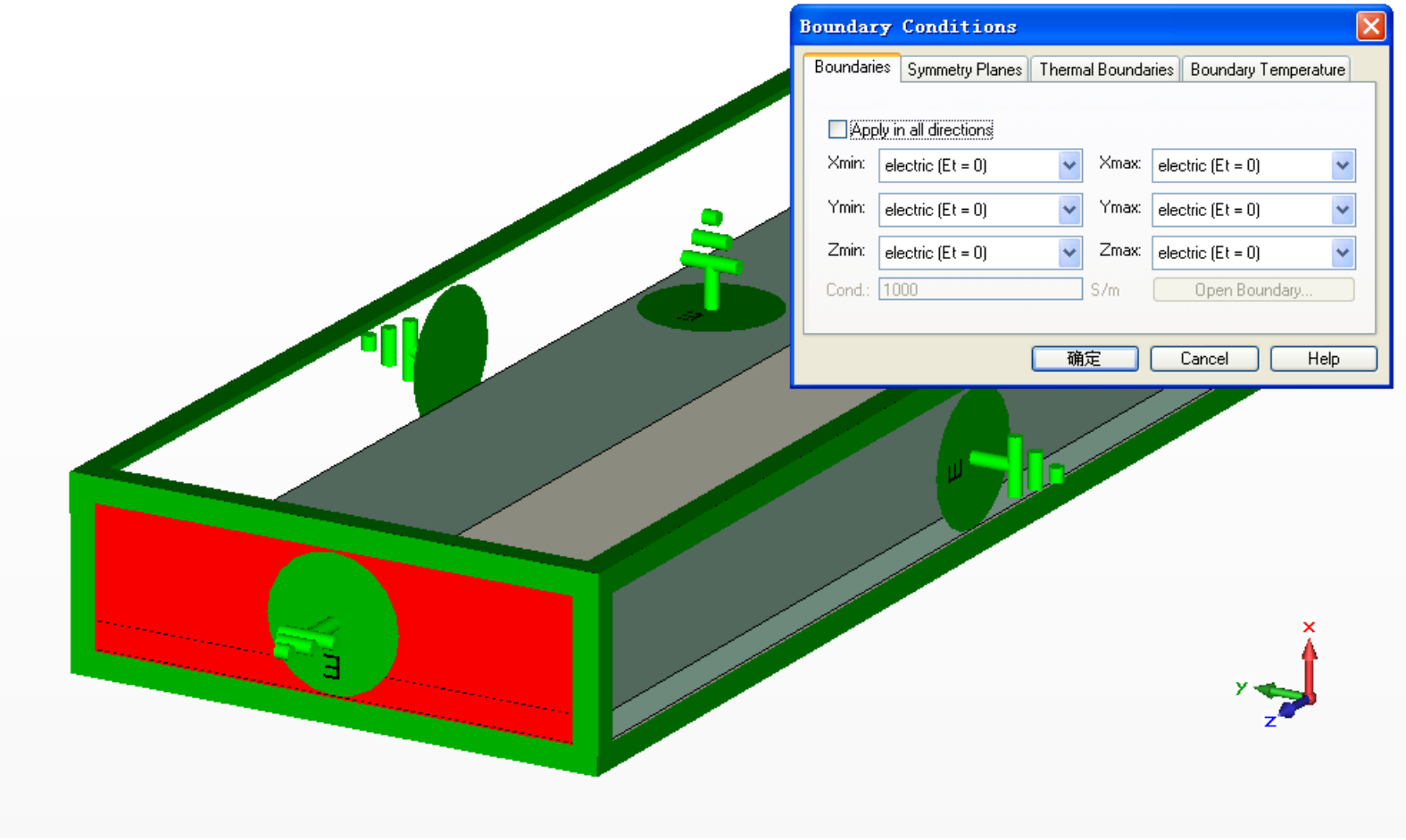
中间介质板材料


Material FR-4 (loss free)
Type Normal
Epsilon 4.9
Mue 1
Therm.cond. 0.3 [W/K/m]
x
y
z

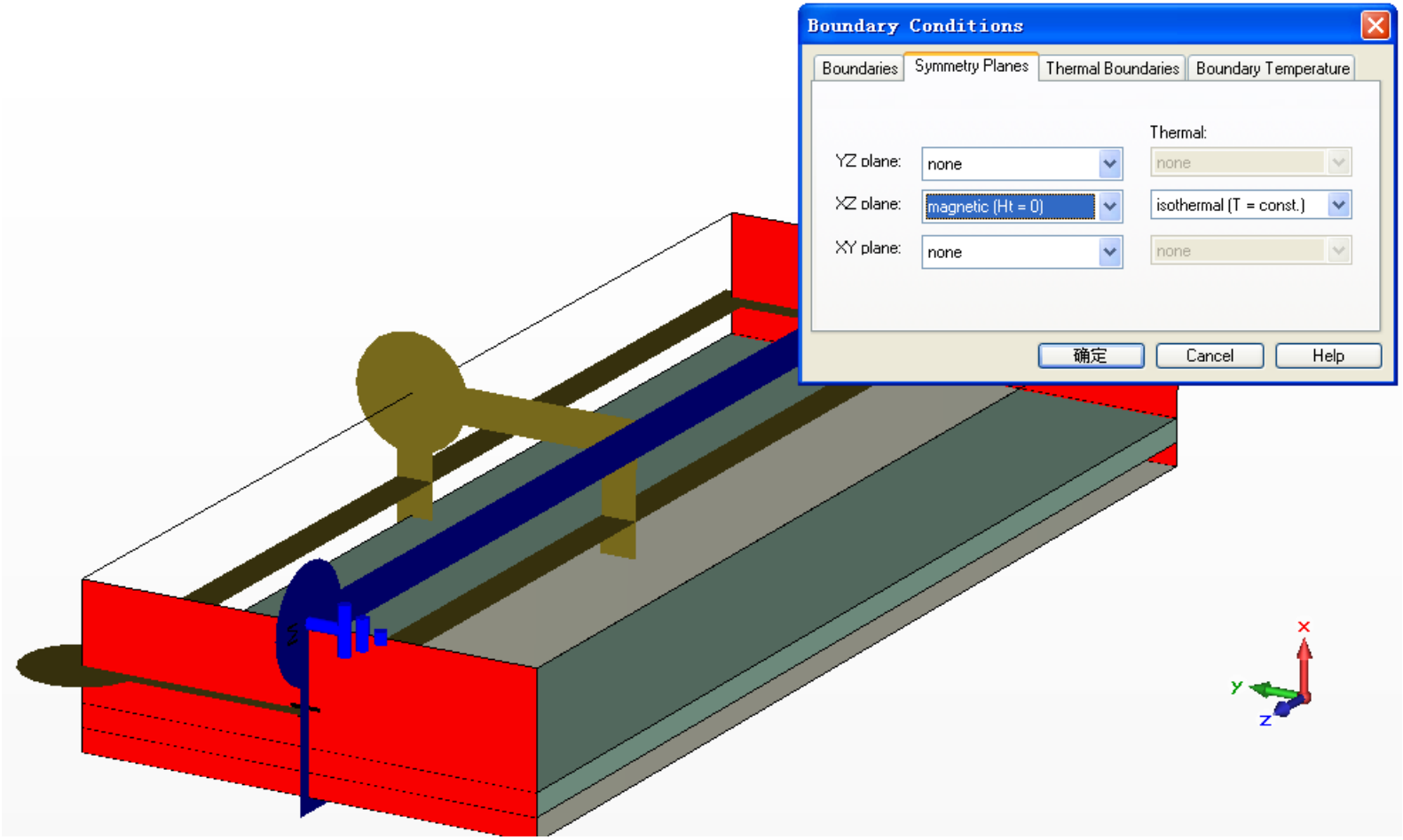
端口定义



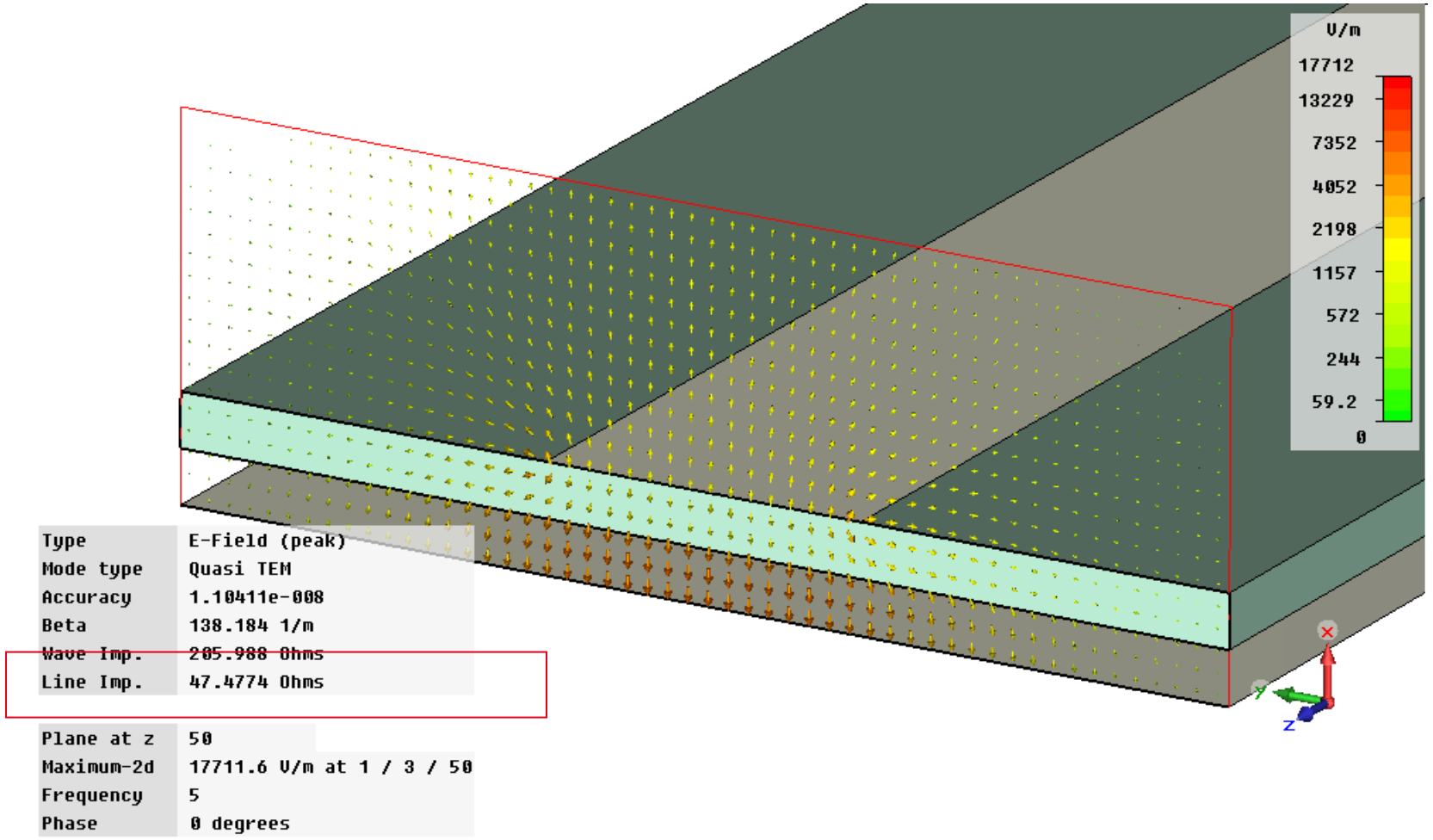
边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Xmin:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help
x
y
z

使用对称面加快仿真速度

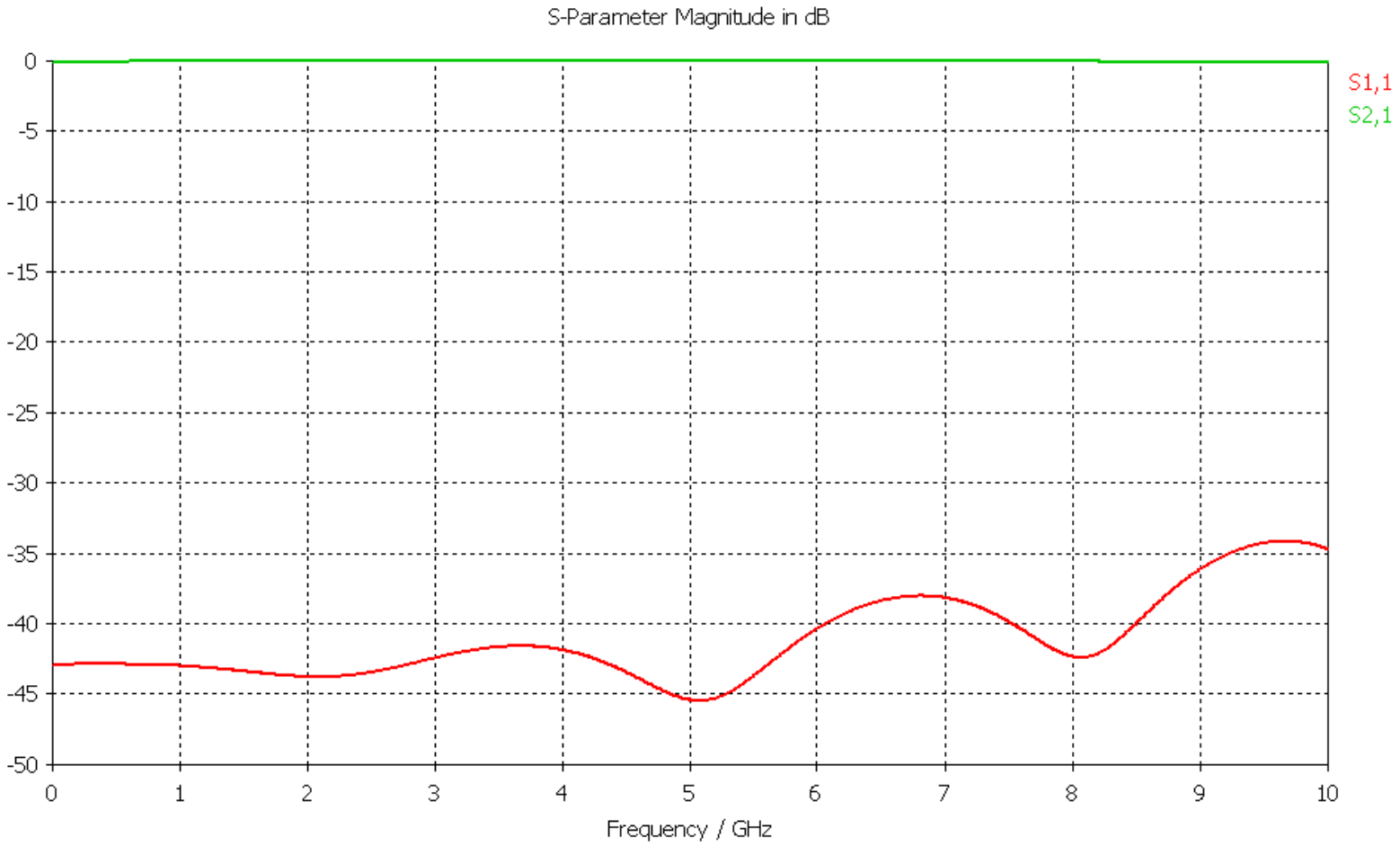


1端口电场端口模式

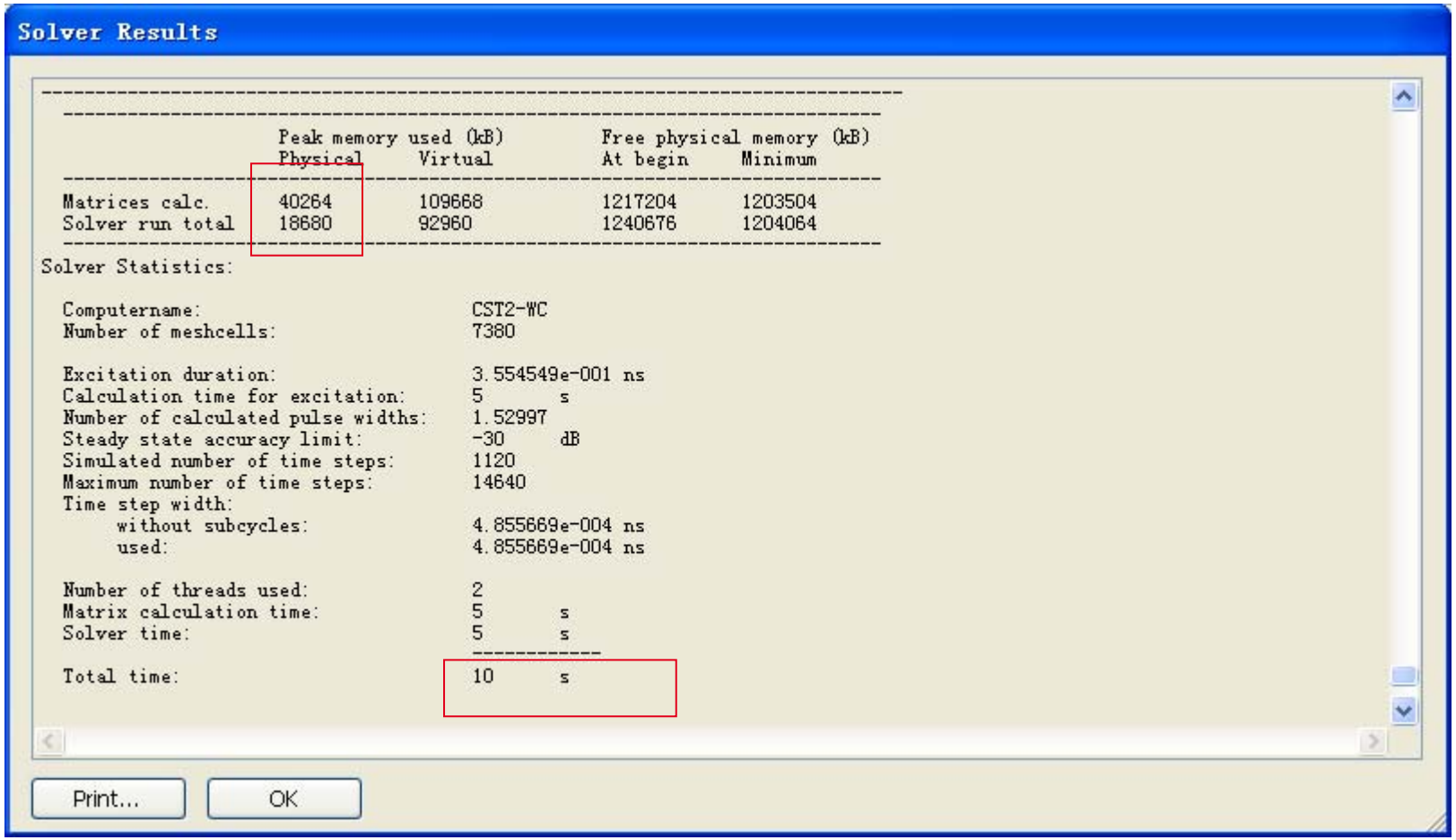


端口阻抗值为47欧姆。

S参量 dB

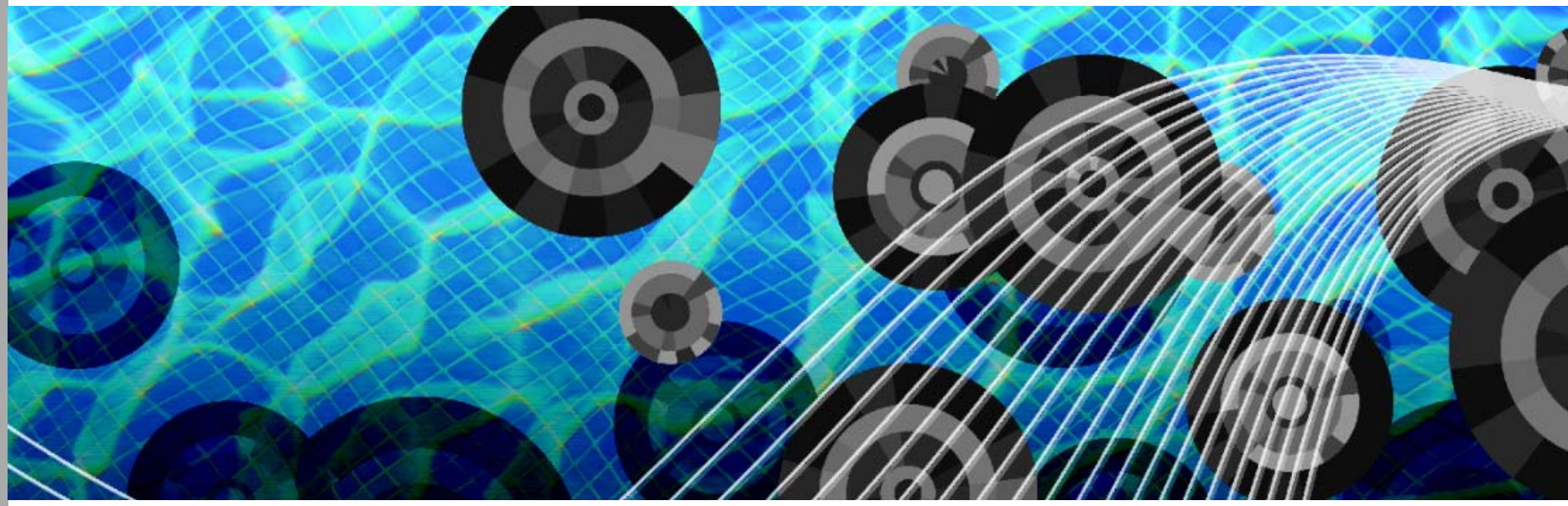

S-Parameter Magnitude in dB
0
-5
-10
-15
-20
-25
-30
-35
-40
-45
-50
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1
S2,1

占用系统资源

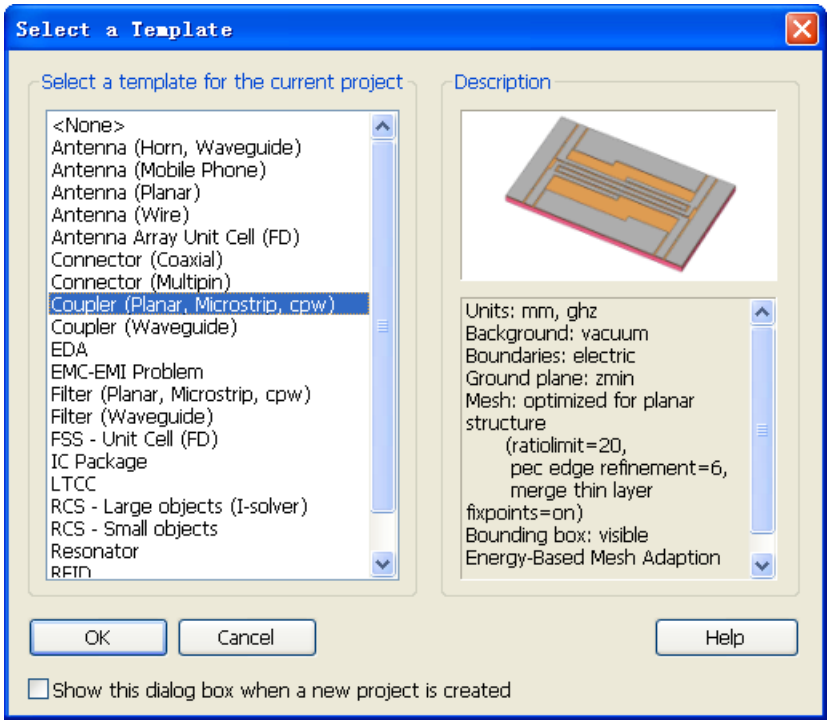


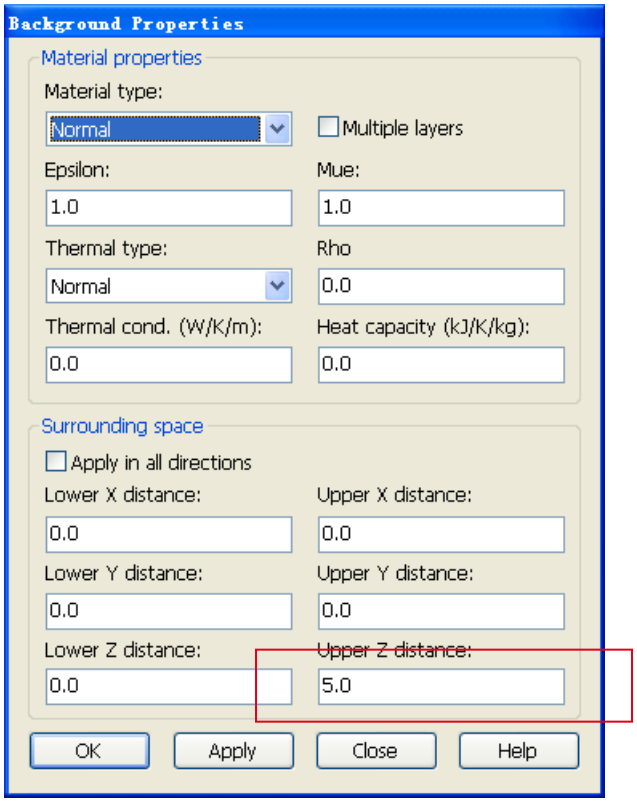
建立矩阵占用内存40M，求解占用内存19M，求解器计算时间10s。

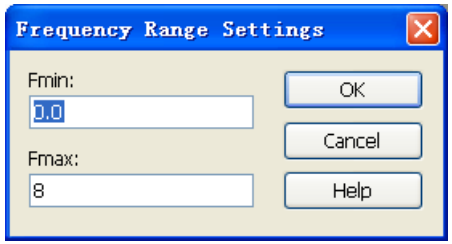
渐变微带线



选择模板、背景条件、频带范围



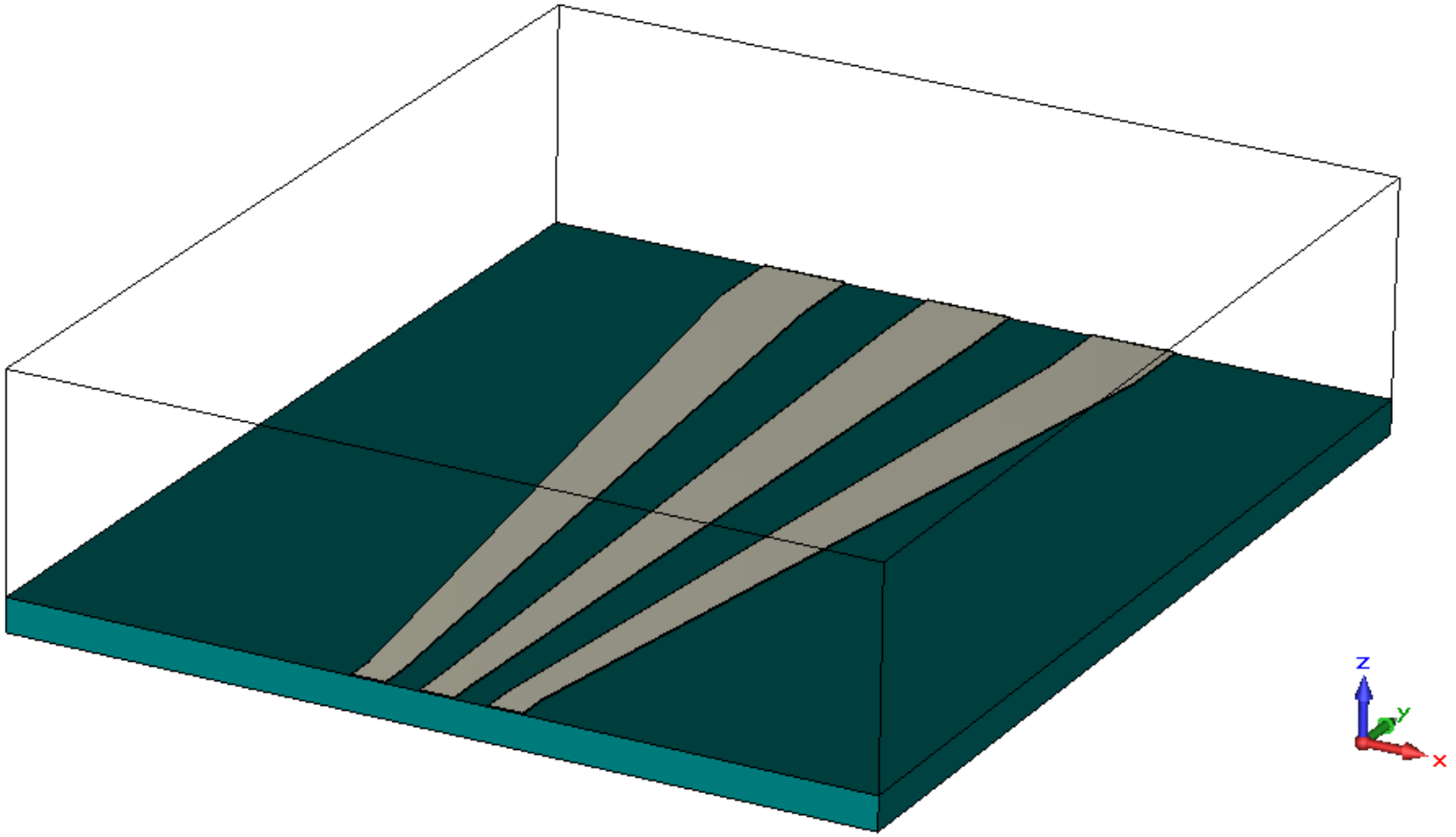




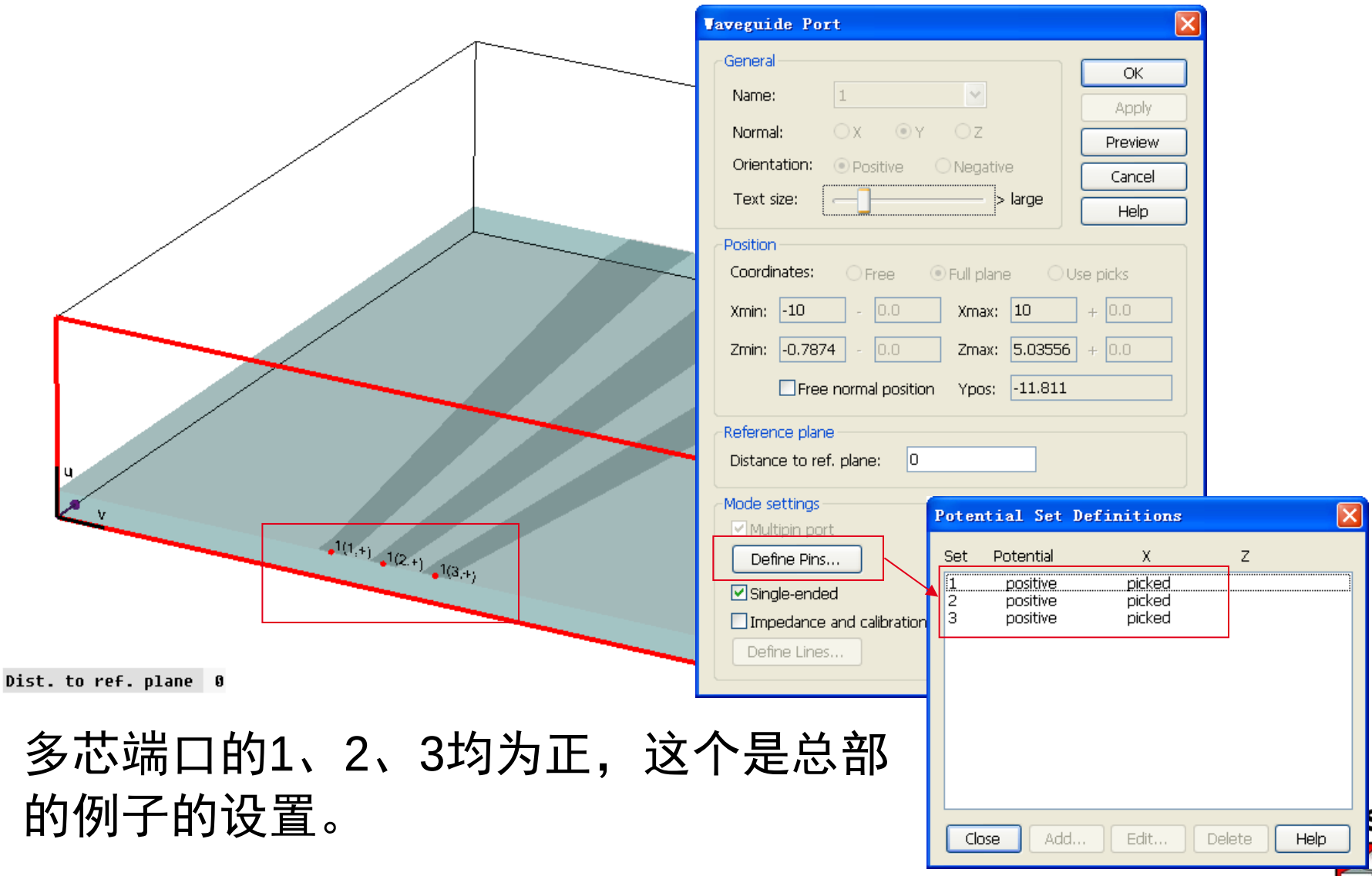
选择微带线模板，默认单位为GHz，mm。

由于模型在空间上分布了一定电磁场，因此需要在Zmax方向定义一定空间。

渐变微带线

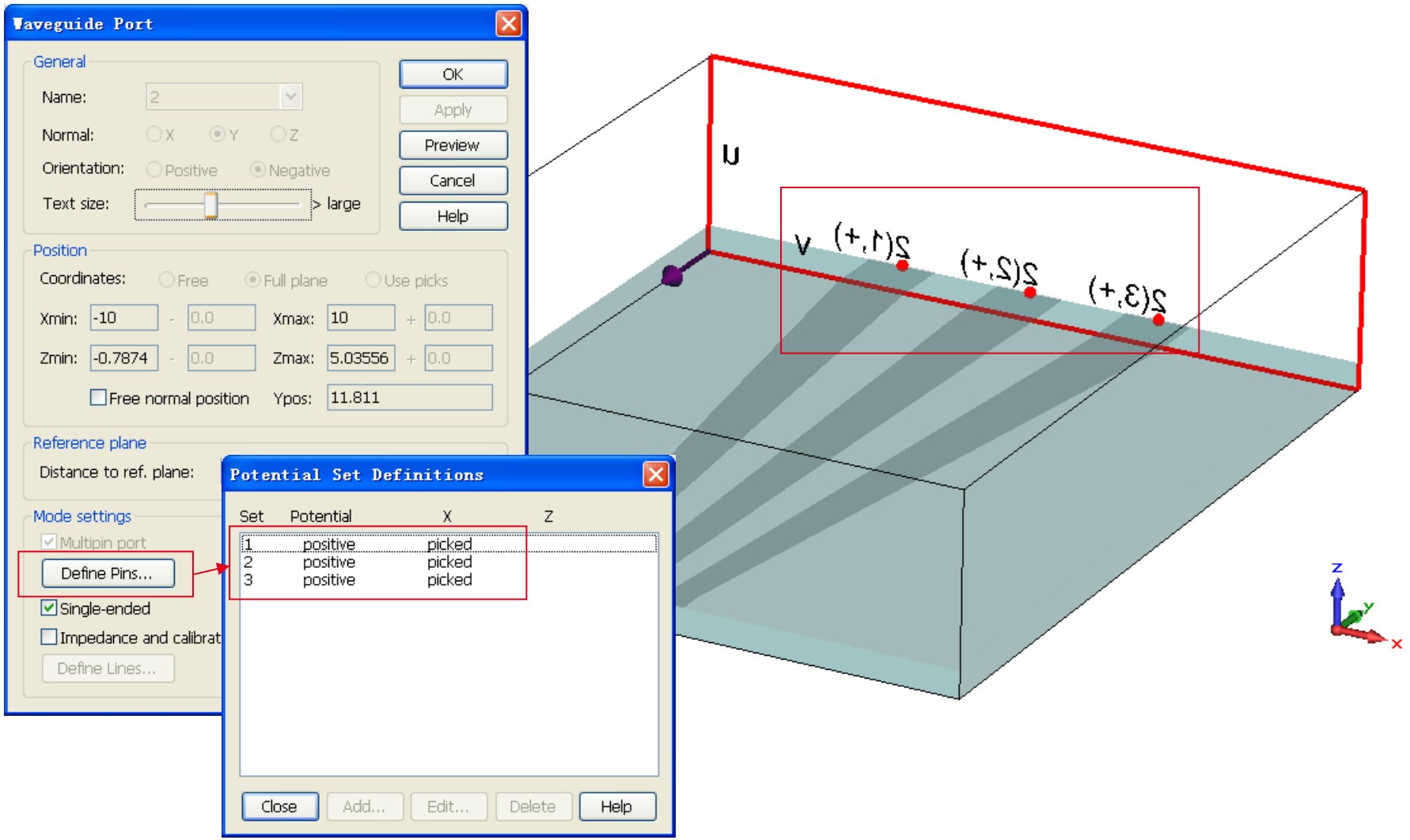


1端口处的设置

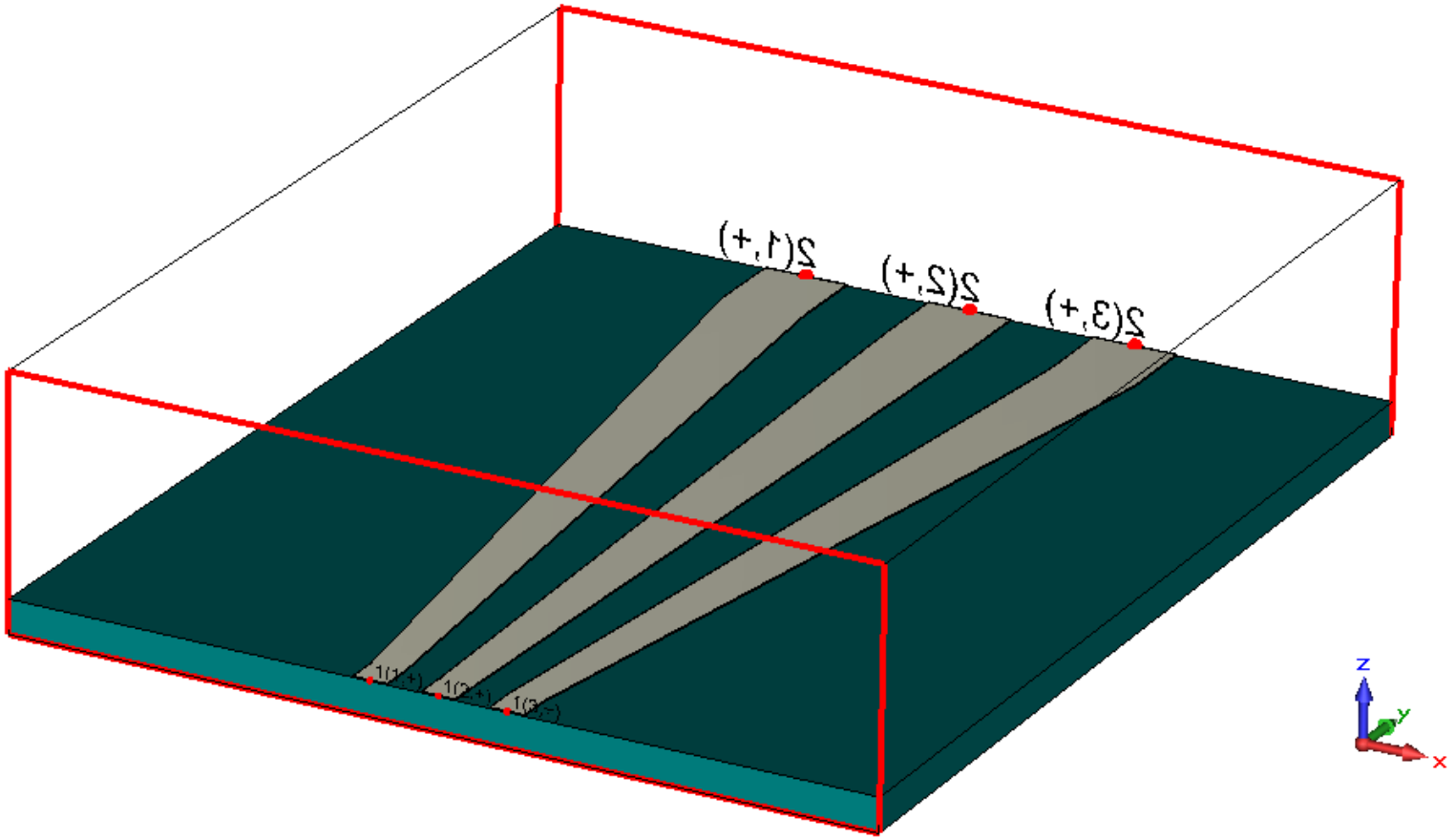


多芯端口的1、2、3均为正，这个是总部的例子的设置。

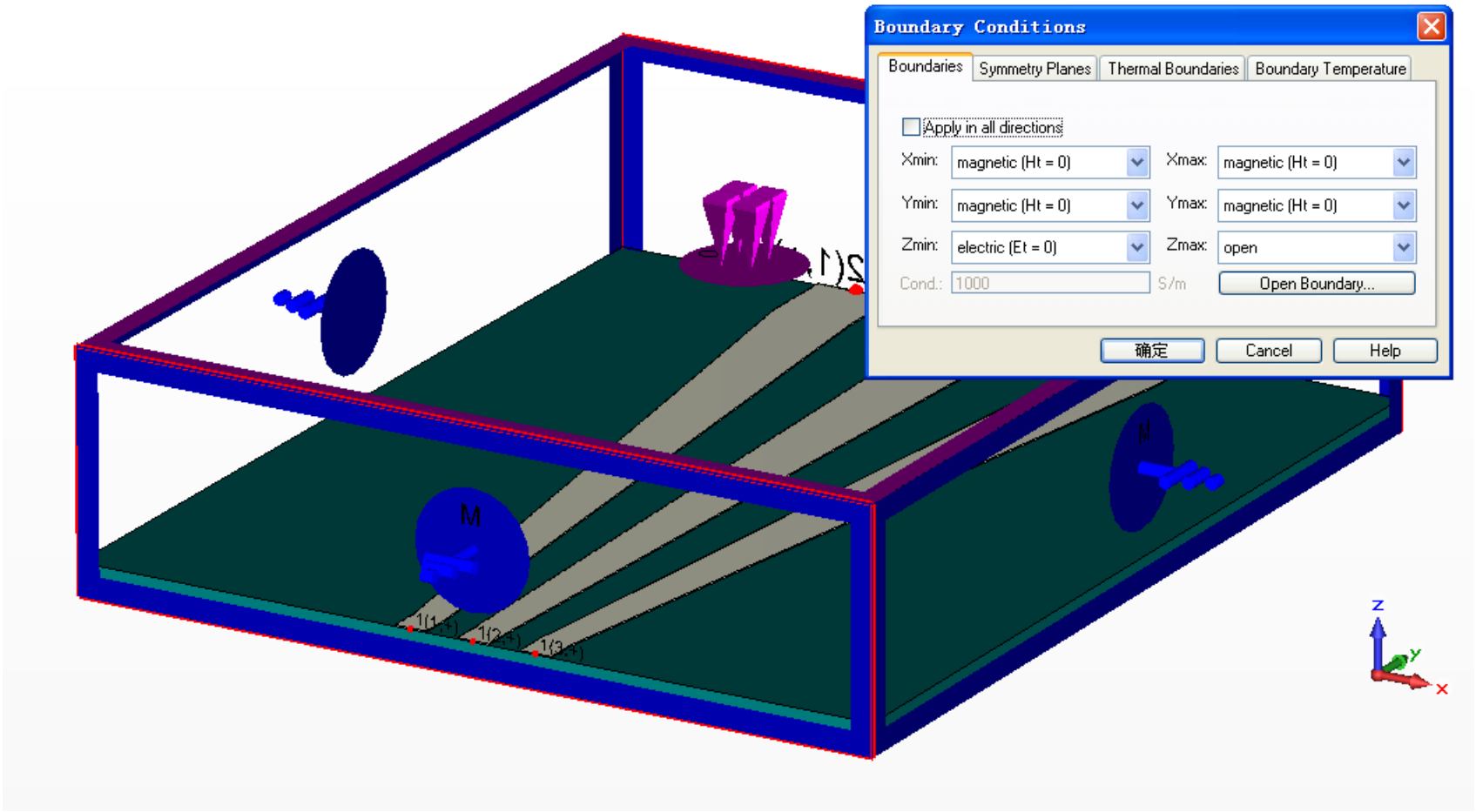
2端口处的设置


Waveguide Port
General
Name:
2
Normal:
X
Y
Z
Orientation:
Positive
Negative
Text size:
large
OK
Apply
Preview
Cancel
Help
Position
Coordinates:
Free
Full plane
Use picks
Xmin:
-10
0.0
Xmax:
10
0.0
Zmin:
-0.7874
0.0
Zmax:
5.03556
0.0
Free normal position
Ypos:
11.811
Reference plane
Distance to ref. plane:
Mode settings
Multipin port
Define Pins...
Single-ended
Impedance and calibrat
Define Lines...
Potential Set Definitions
Set
Potential
X
Z
1
positive
picked
2
positive
picked
3
positive
picked
Close
Add...
Edit...
Delete
Help
U
V
Z(1,+)
Z(2,+)
Z(3,+)
Z
Y
X

模型+端口

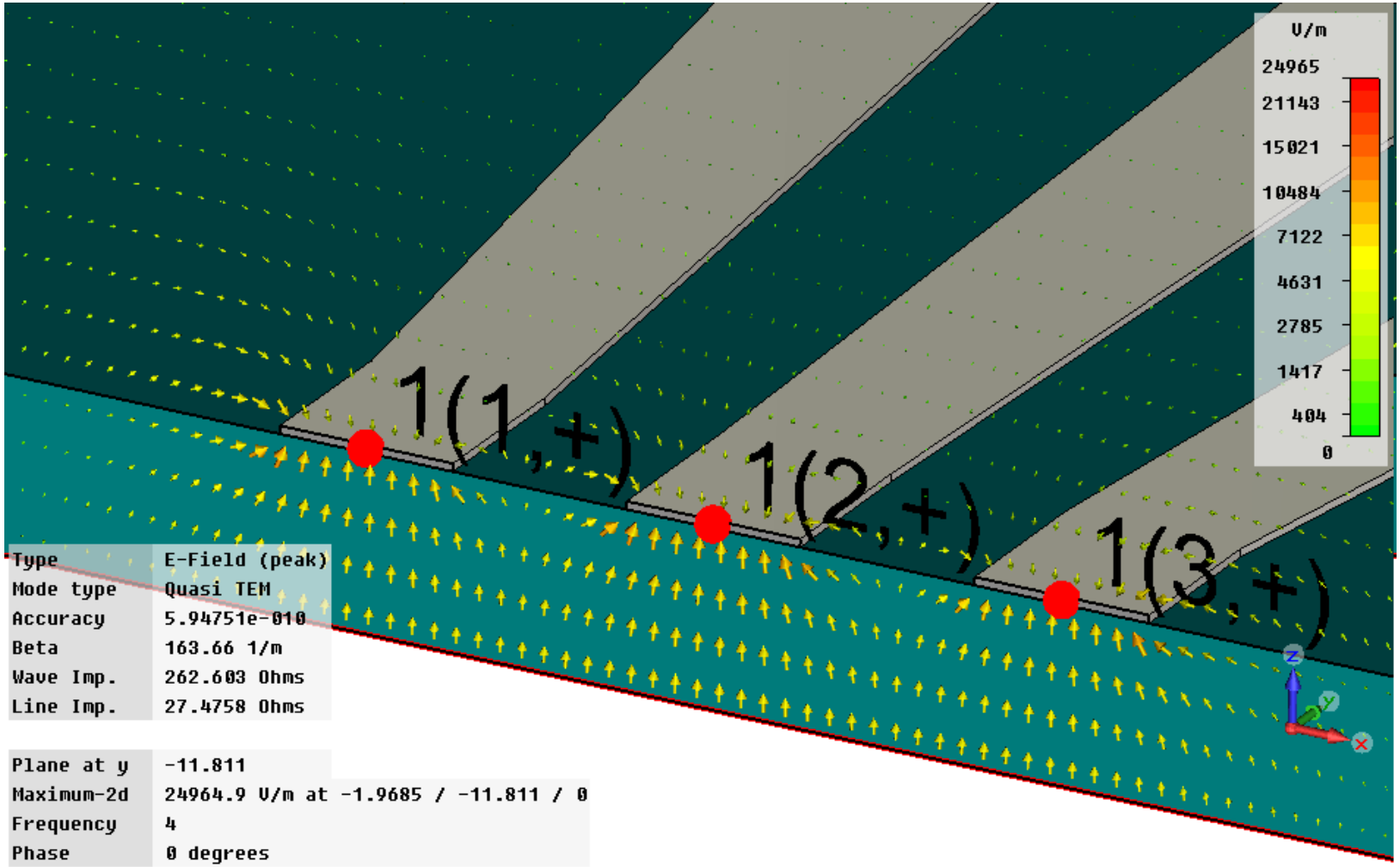


边界条件

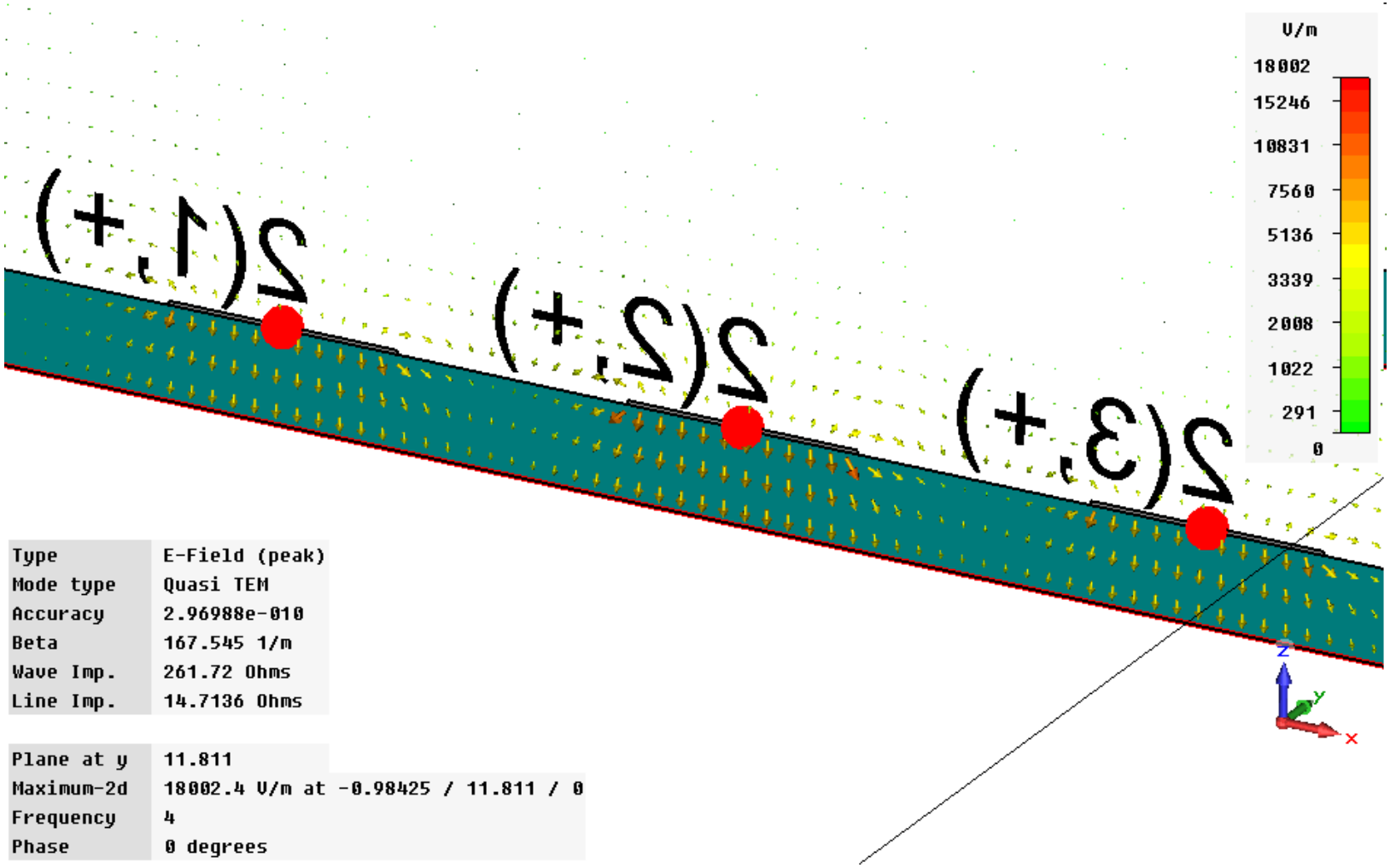


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1端口处端口模式1


V/m
24965
21143
15021
10484
7122
4631
2785
1417
404
0
1(1,+)
1(2,+)
1(3,+)
Type E-Field (peak)
Mode type Quasi TEM
Accuracy 5.94751e-010
Beta 163.66 1/m
Wave Imp. 262.603 Ohms
Line Imp. 27.4758 Ohms
Plane at y -11.811
Maximum-2d 24964.9 V/m at -1.9685 / -11.811 / 0
Frequency 4
Phase 0 degrees

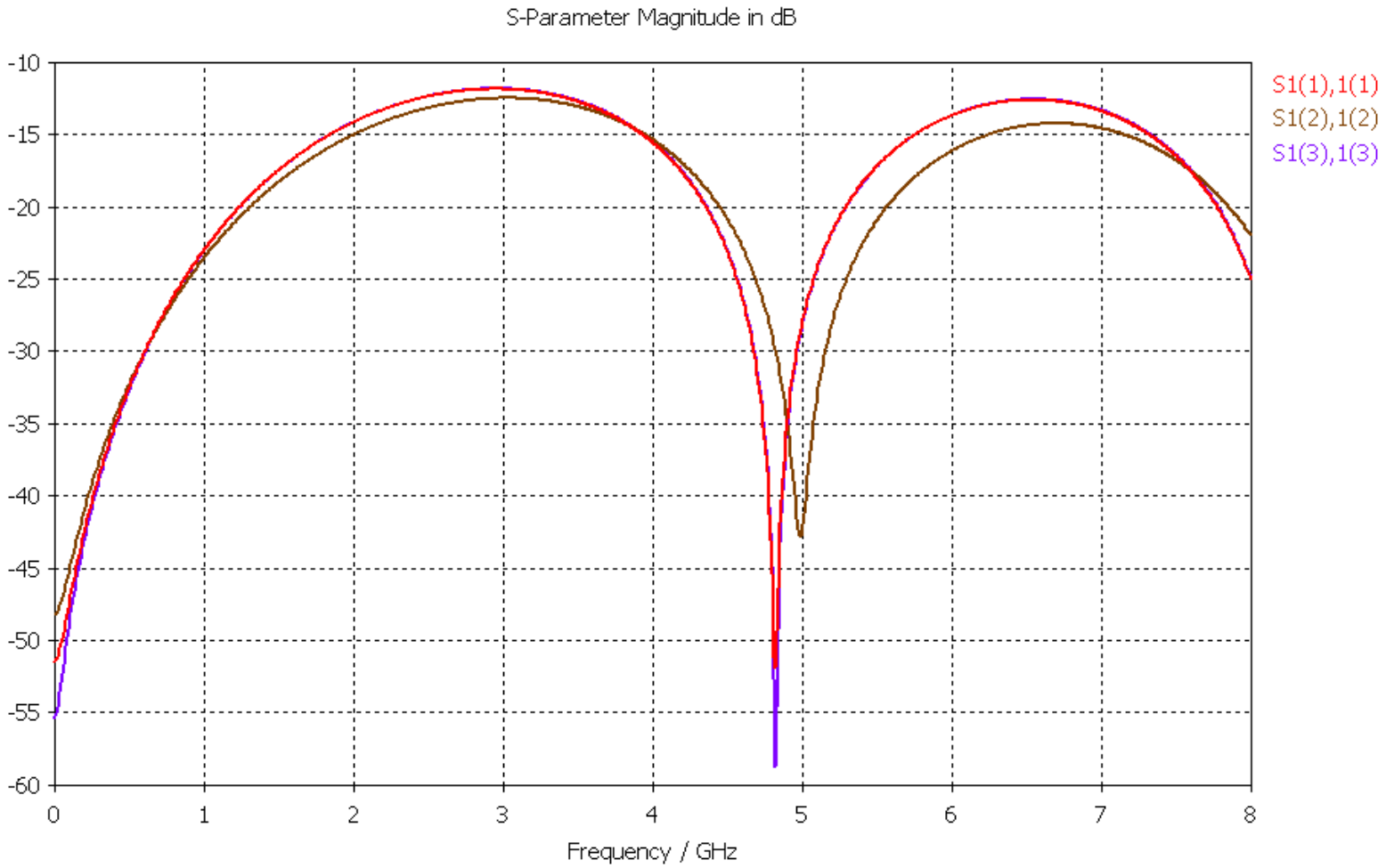
2端口处端口模式1



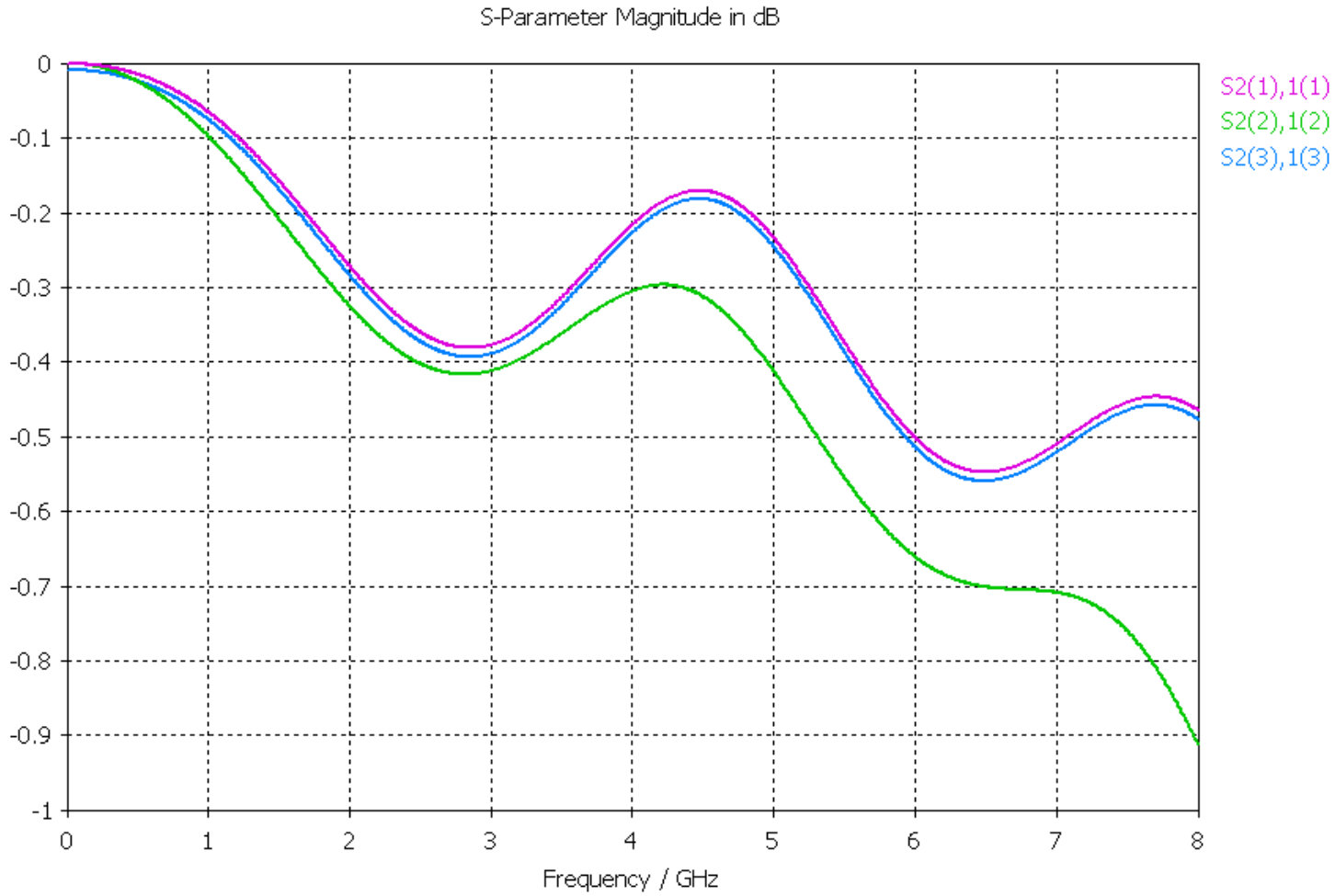
Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	2.96988e-010
Beta	167.545 1/m
Wave Imp.	261.72 Ohms
Line Imp.	14.7136 Ohms

Plane at y	11.811
Maximum-2d	18002.4 V/m at -0.98425 / 11.811 / 0
Frequency	4
Phase	0 degrees

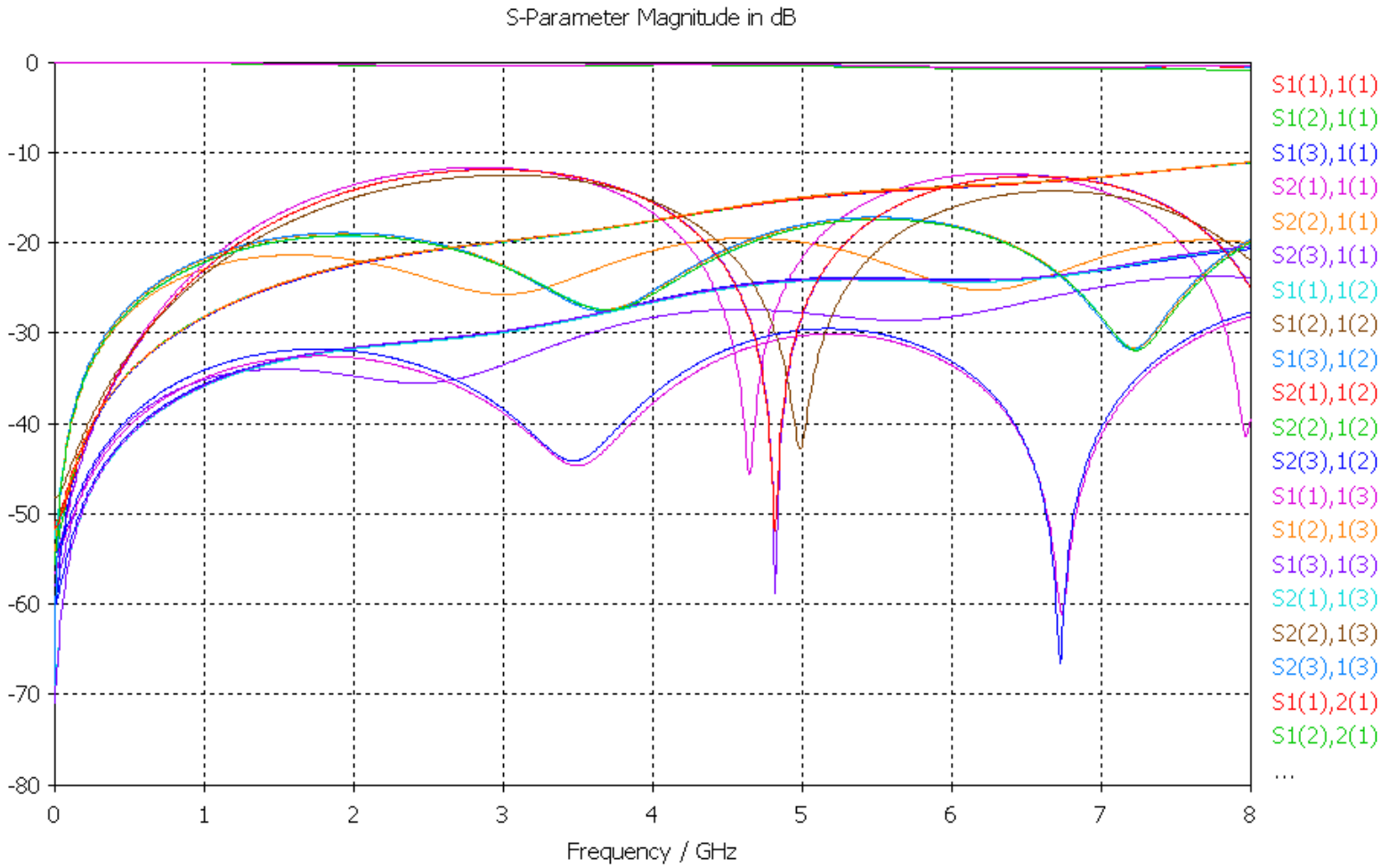
各条带线的回波损耗


S-Parameter Magnitude in dB
-10
-15
-20
-25
-30
-35
-40
-45
-50
-55
-60
0
1
2
3
4
5
6
7
8
Frequency / GHz
S1(1),1(1)
S1(2),1(2)
S1(3),1(3)

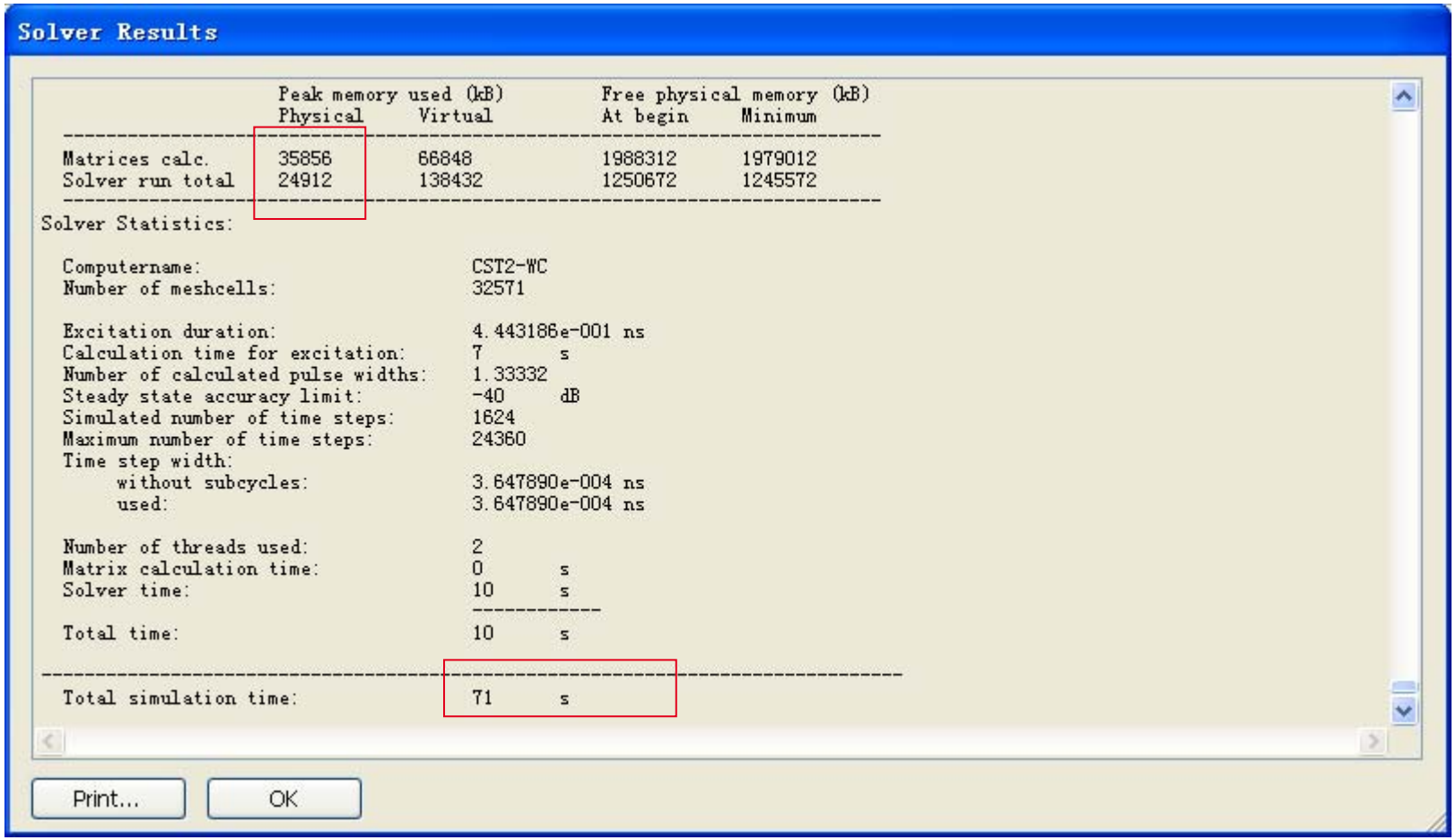
S21


S-Parameter Magnitude in dB
0
-0.1
-0.2
-0.3
-0.4
-0.5
-0.6
-0.7
-0.8
-0.9
-1
0
1
2
3
4
5
6
7
8
Frequency / GHz
S2(1),1(1)
S2(2),1(2)
S2(3),1(3)

所有模式下的S参量

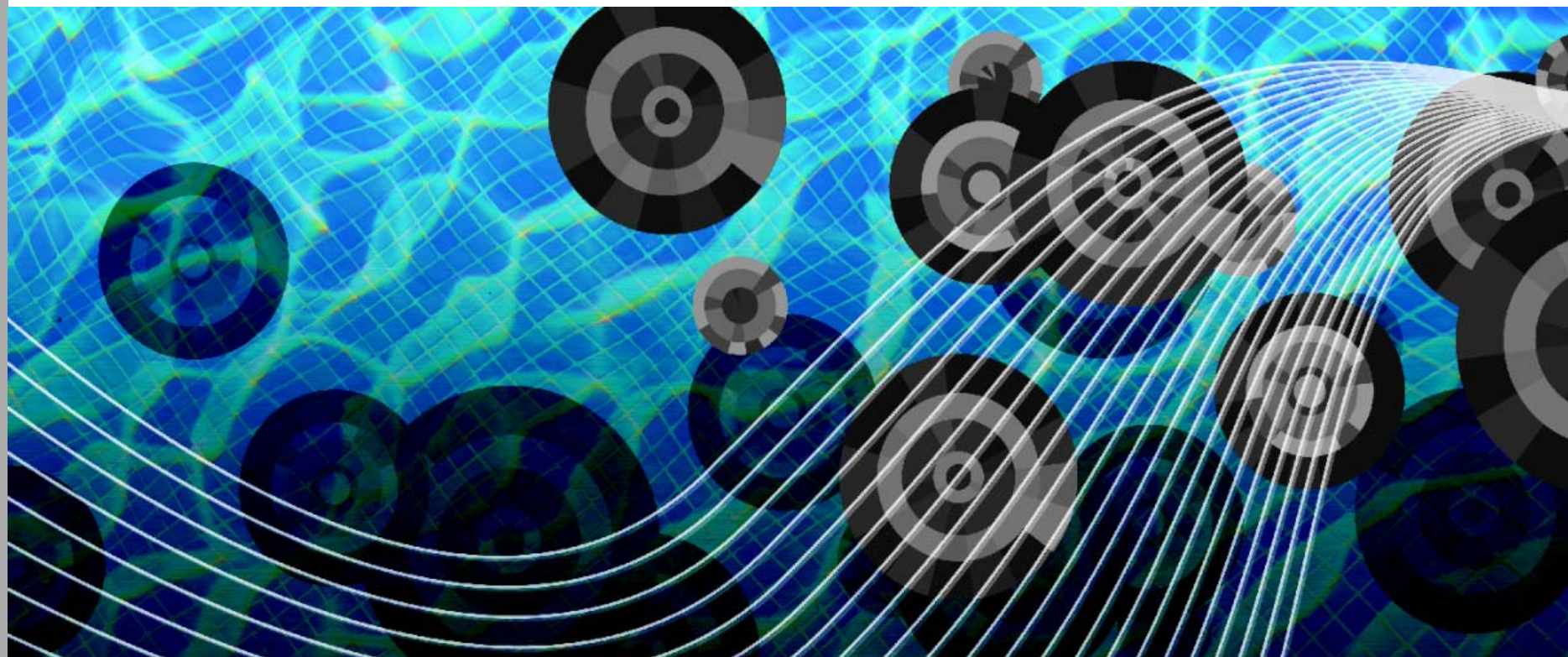

S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
-70
-80
0
1
2
3
4
5
6
7
8
Frequency / GHz
S1(1),1(1)
S1(2),1(1)
S1(3),1(1)
S2(1),1(1)
S2(2),1(1)
S2(3),1(1)
S1(1),1(2)
S1(2),1(2)
S1(3),1(2)
S2(1),1(2)
S2(2),1(2)
S2(3),1(2)
S1(1),1(3)
S1(2),1(3)
S1(3),1(3)
S2(1),1(3)
S2(2),1(3)
S2(3),1(3)
S1(1),2(1)
S1(2),2(1)
...

占用系统资源

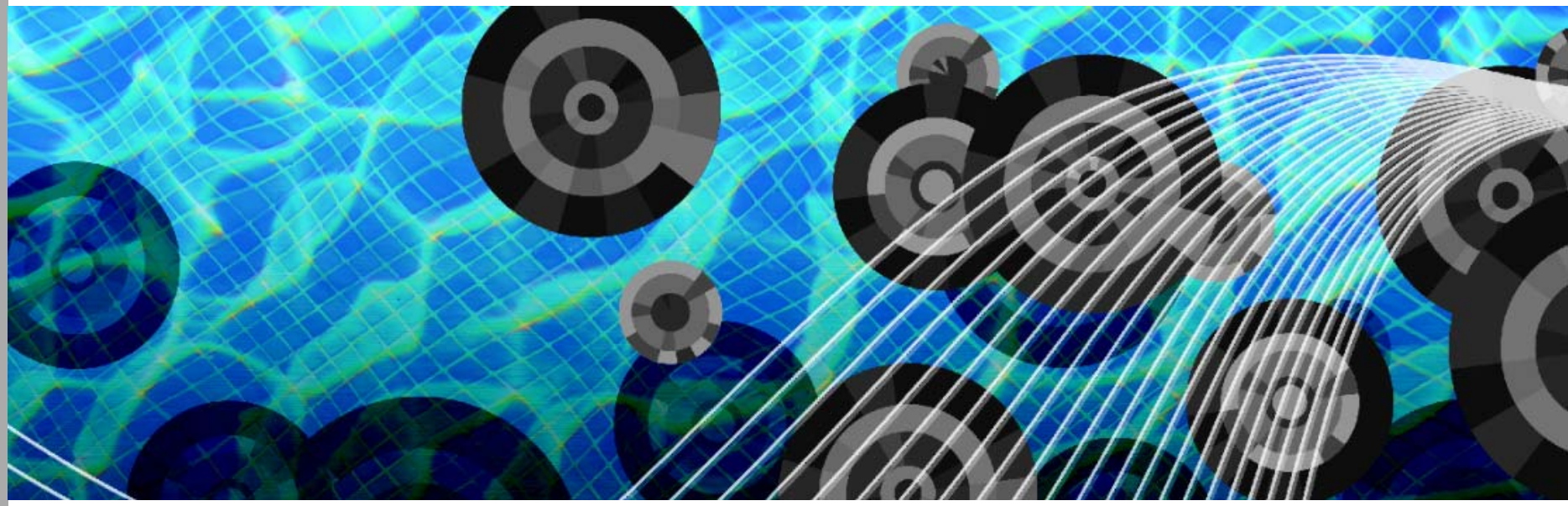


建立矩阵占用内存36M，求解占用内存25M，计算时间1m, 11s。

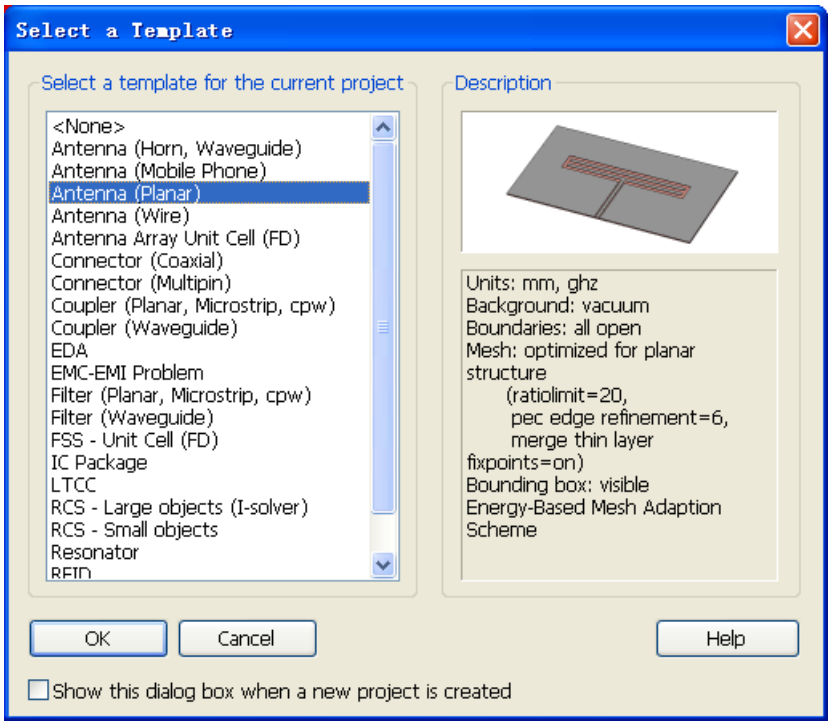
槽线

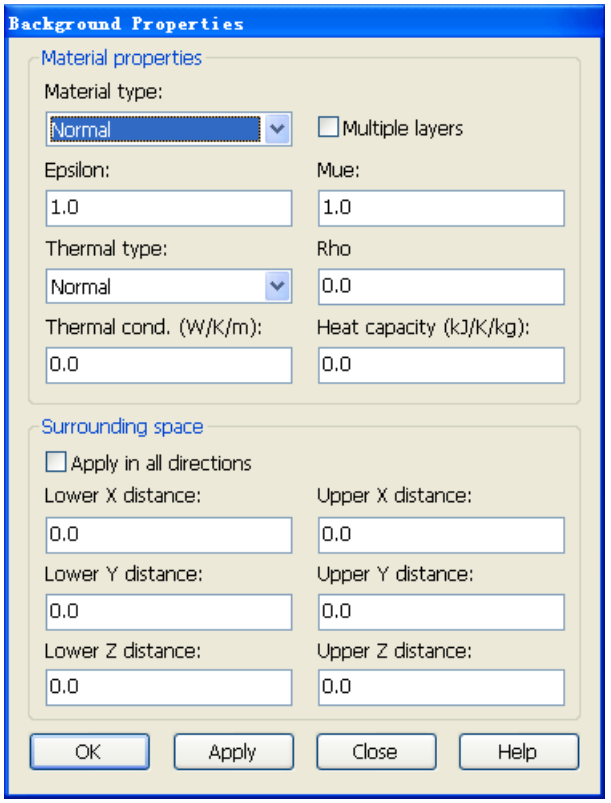


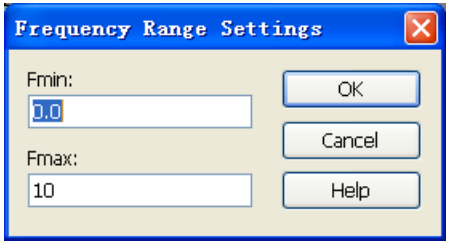
不带接地板槽线



选择模板、背景条件、频带范围

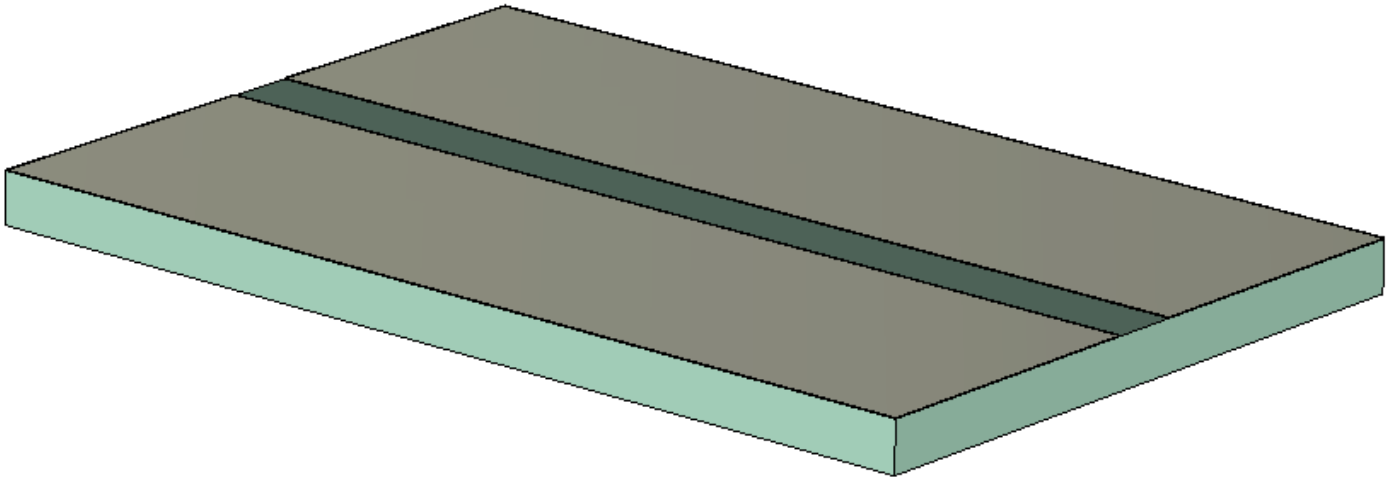




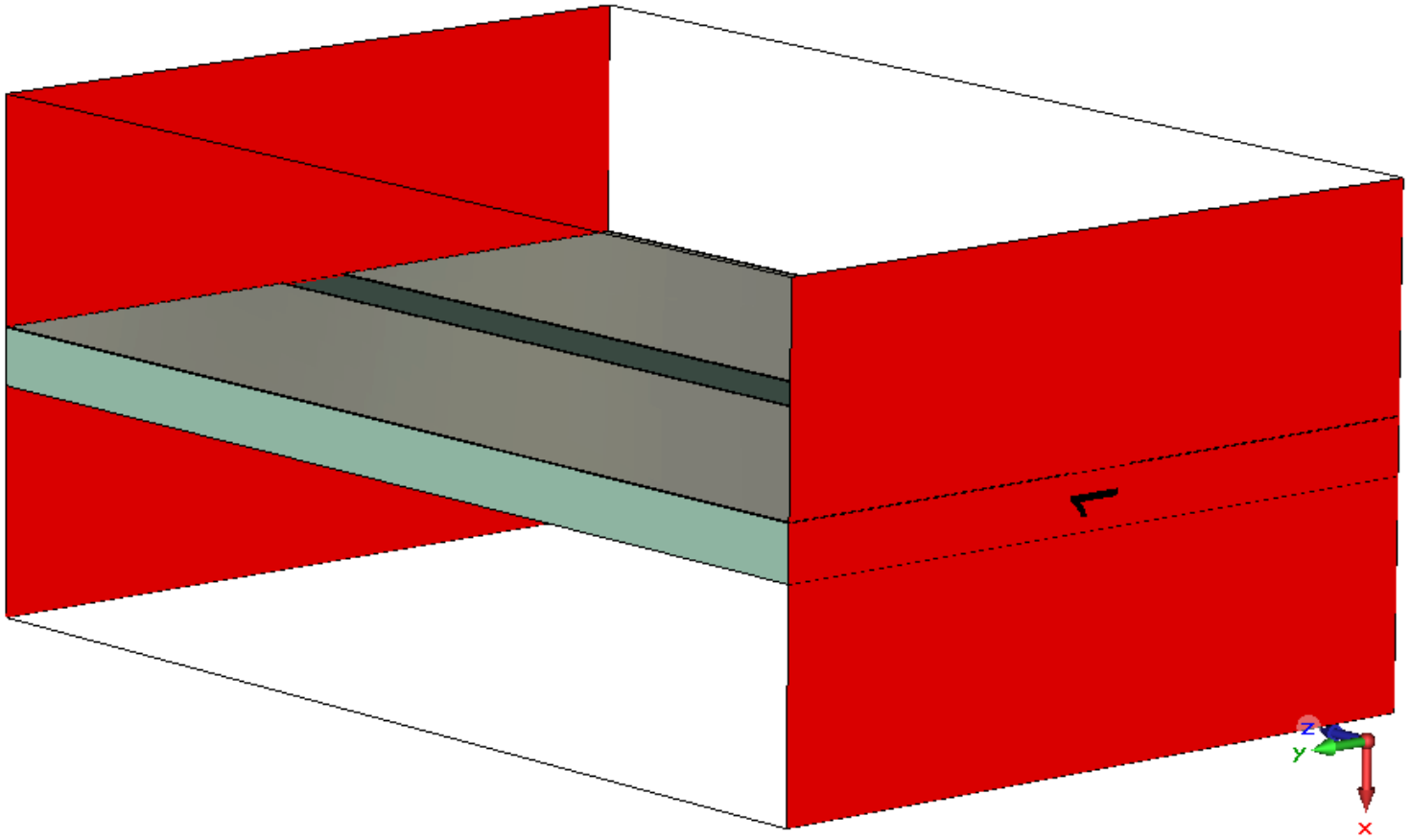


选好模板之后，默认单位为GHz，mm。

模型

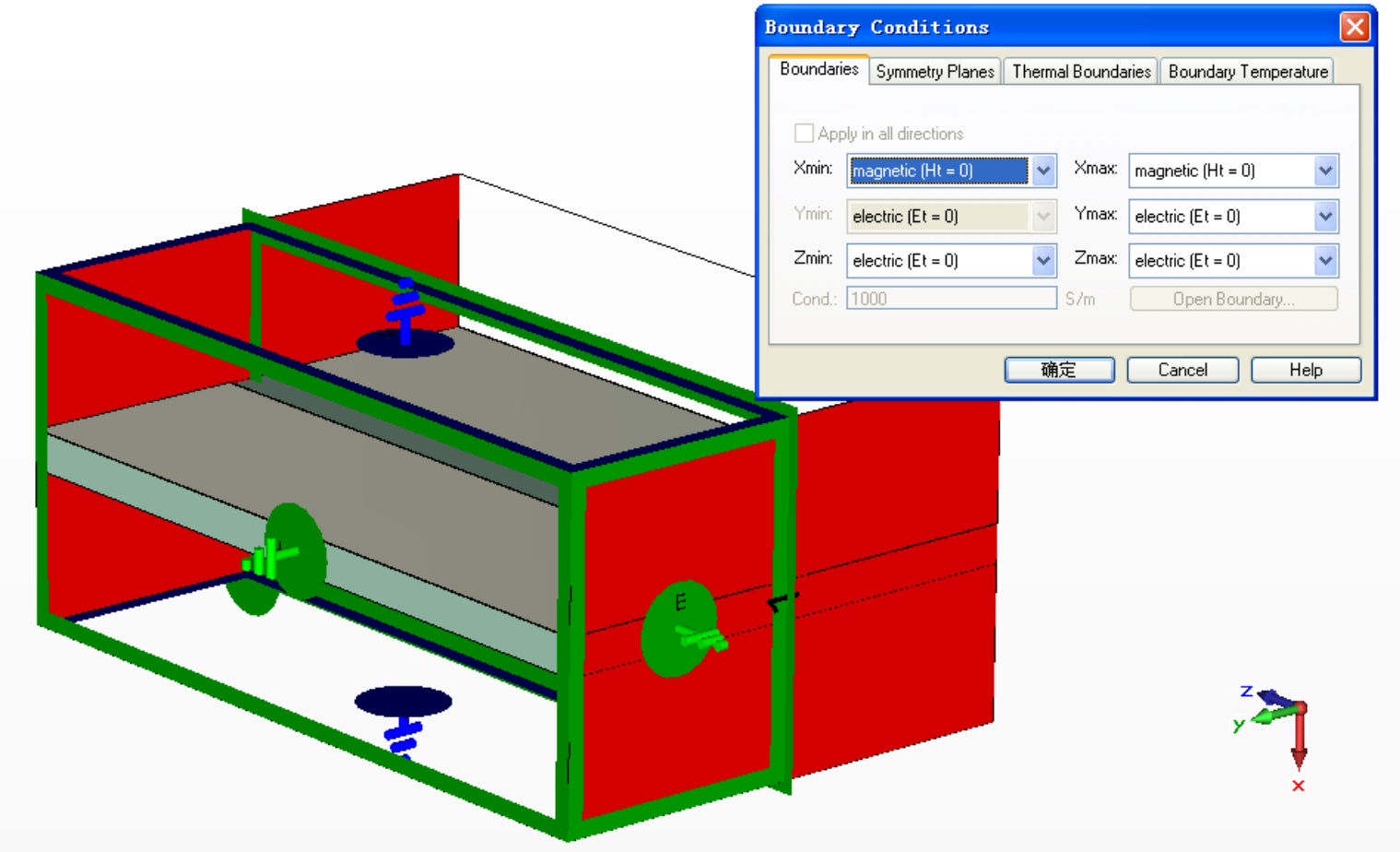


激励端口



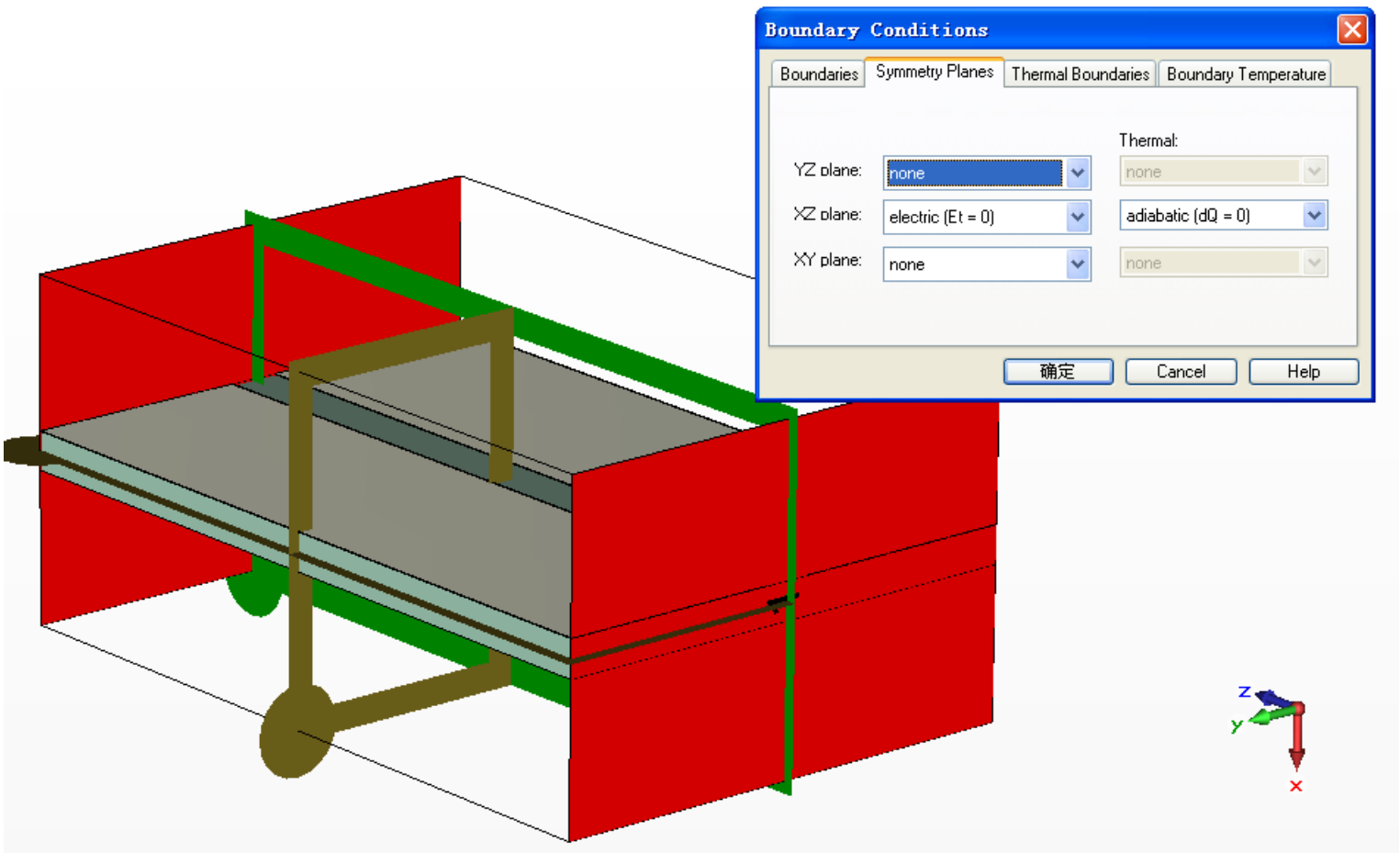
将端口覆盖整个平面即可，上下各留出一定的高度。

边界条件



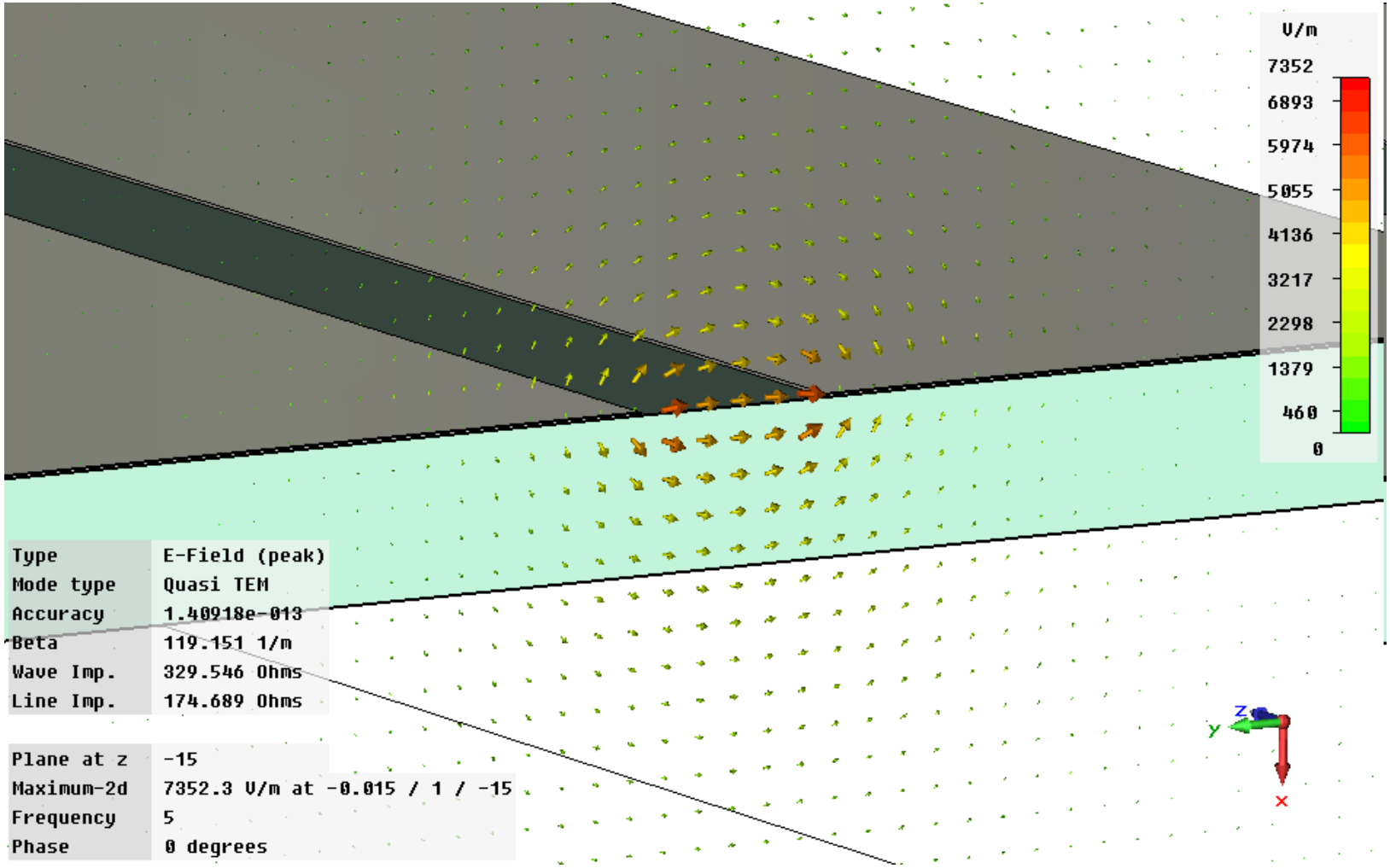
请将Xmin与Xmax设为磁边界。

设置对称面

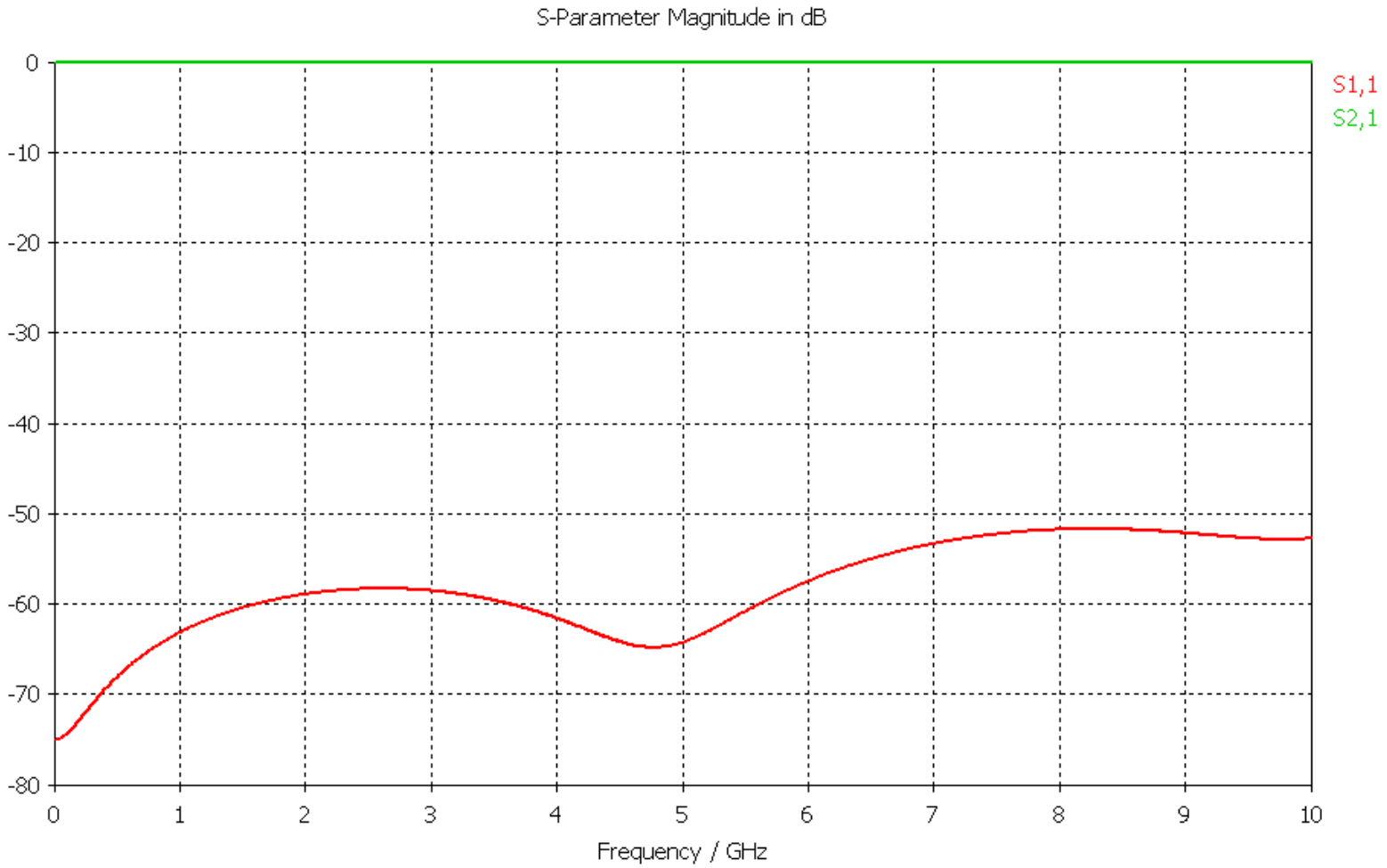


由于模型对称，且XOZ面与电场垂直，因此可以将XOZ面设为电对称面，用以加快仿真时间。

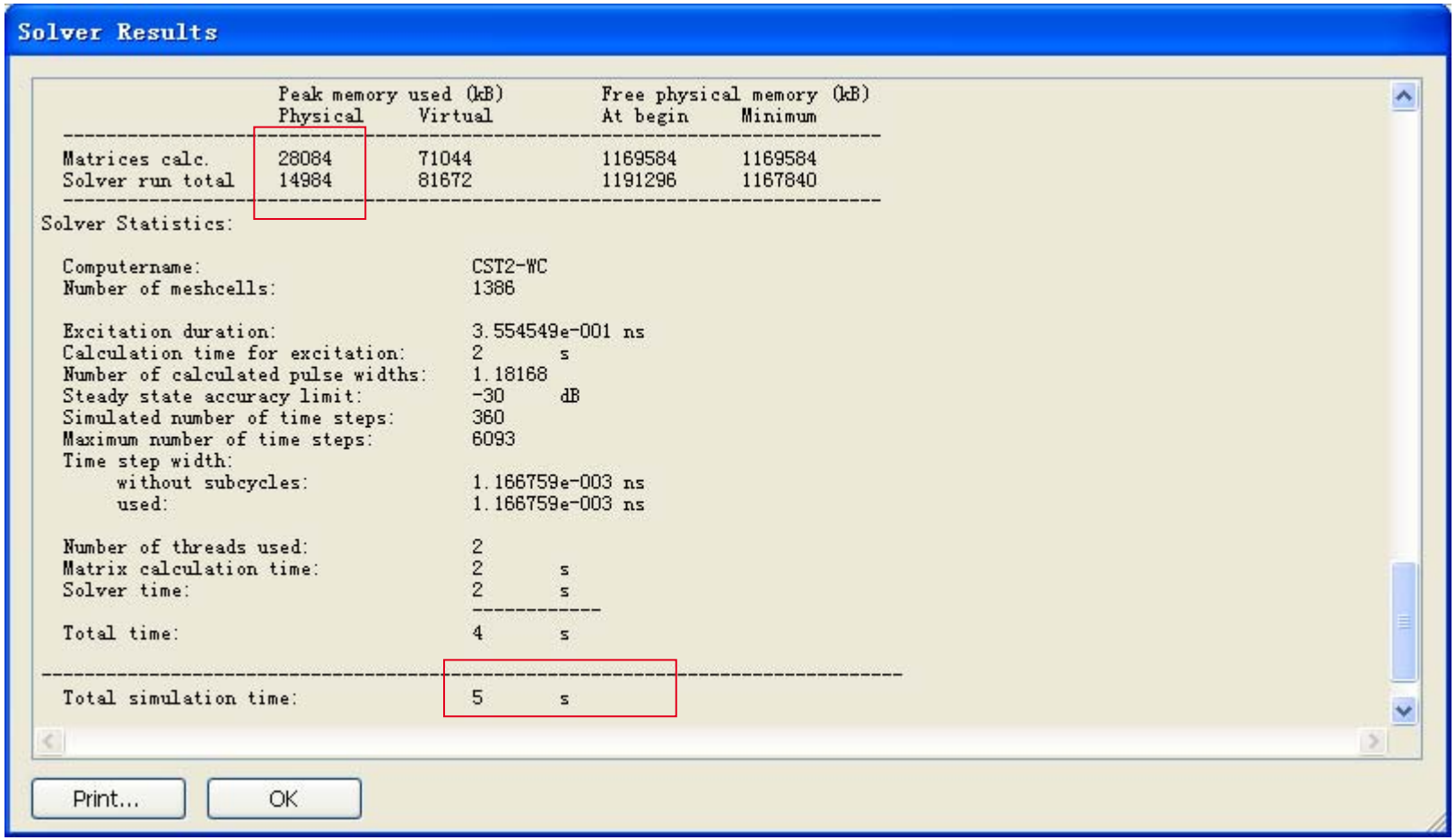
端口1处的端口模式


V/m
7352
6893
5974
5055
4136
3217
2298
1379
460
0
Type E-Field (peak)
Mode type Quasi TEM
Accuracy 1.40918e-013
Beta 119.151 1/m
Wave Imp. 329.546 Ohms
Line Imp. 174.689 Ohms
Plane at z -15
Maximum-2d 7352.3 V/m at -0.015 / 1 / -15
Frequency 5
Phase 0 degrees
z
y
x

S参量 dB

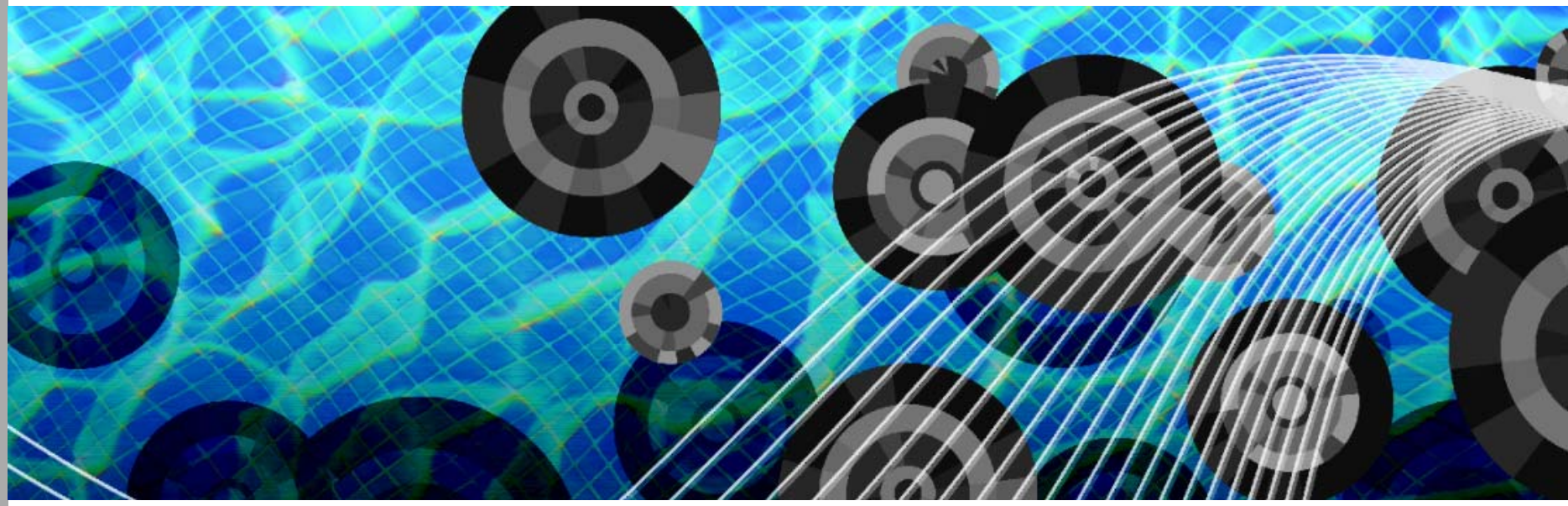

S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
-70
-80
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1
S2,1

占用系统资源

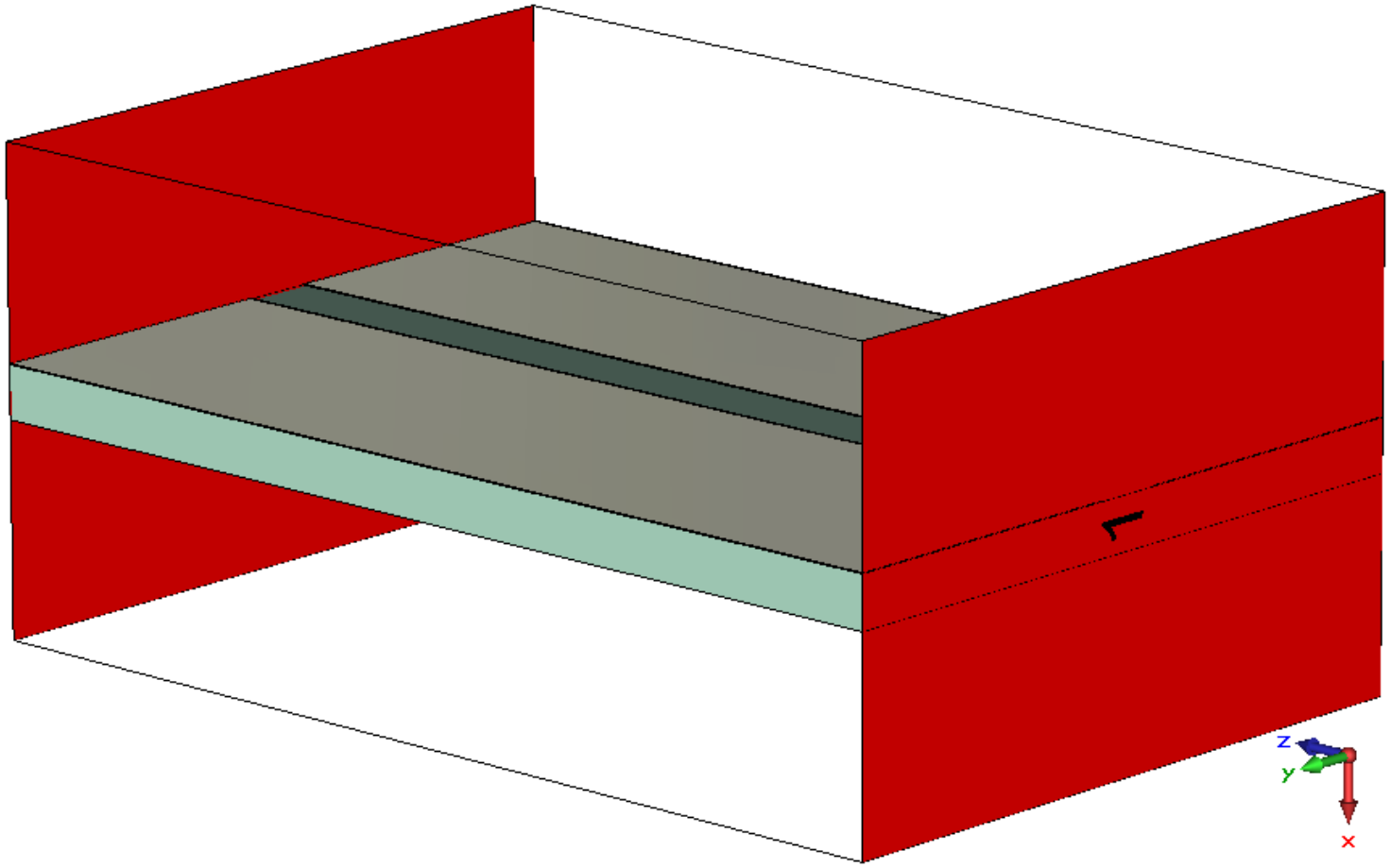


建立矩阵占用内存28M，求解占用内存15M，计算时间5s。

带接地板槽线

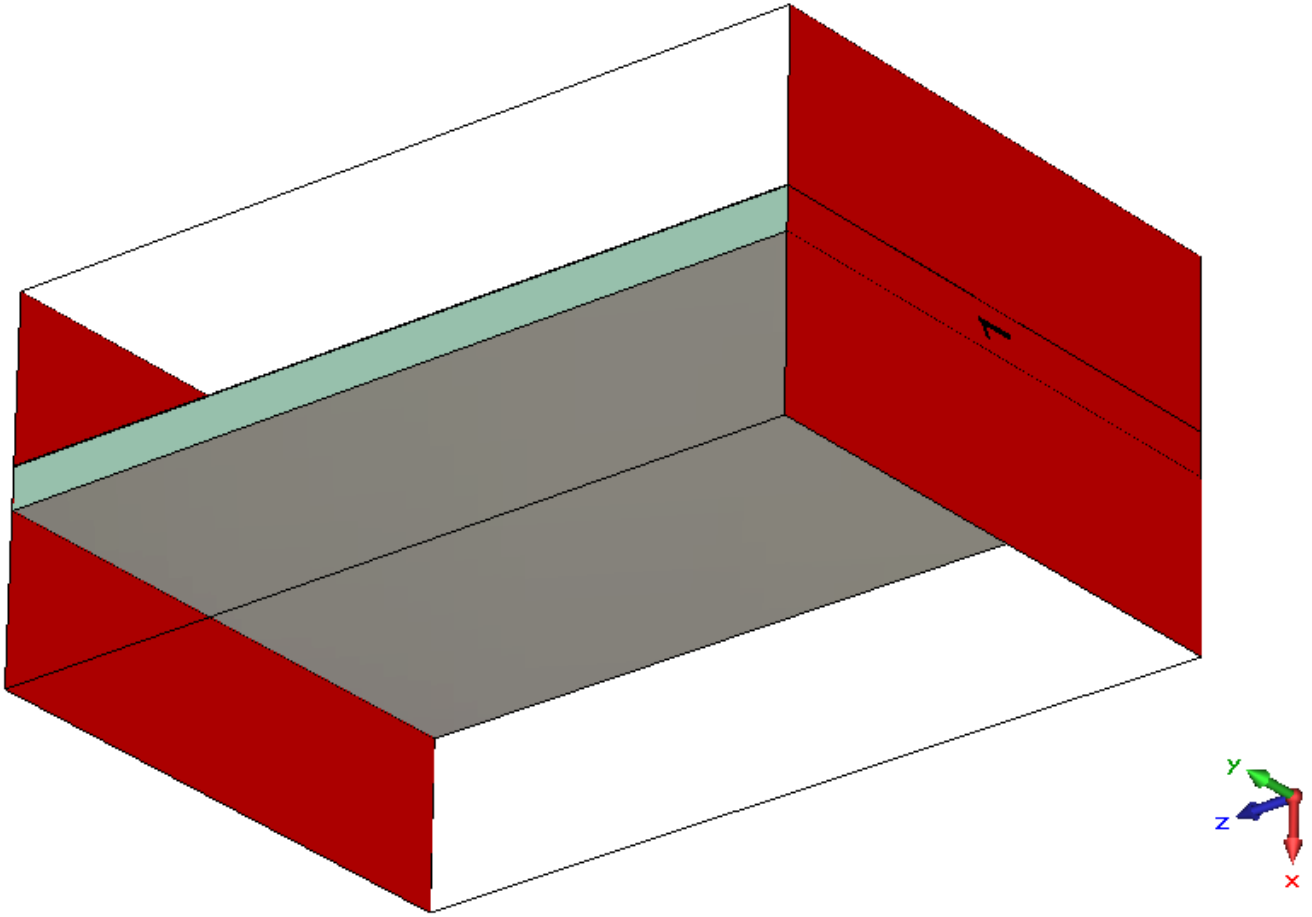


模型



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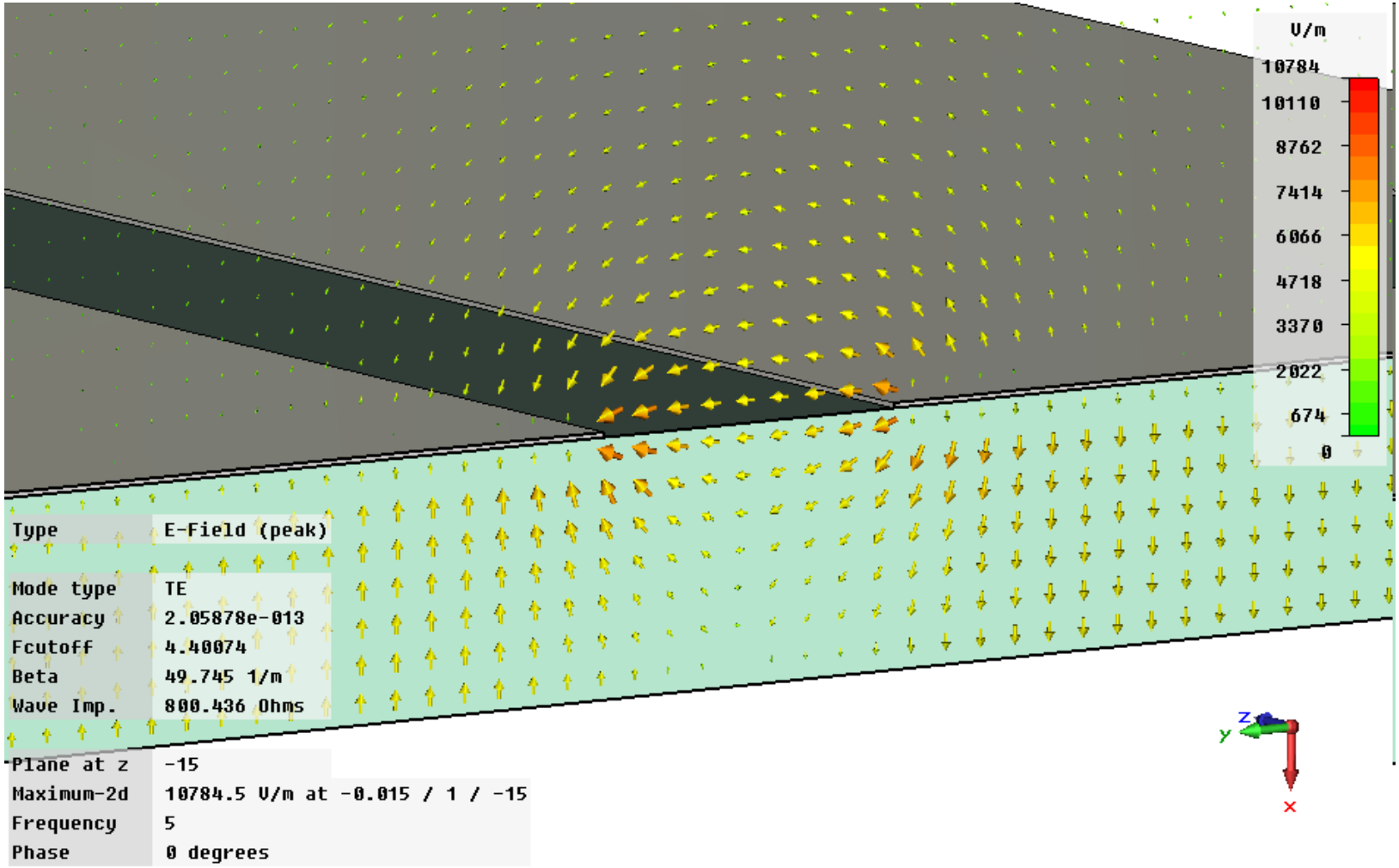
模型：底部为接地板



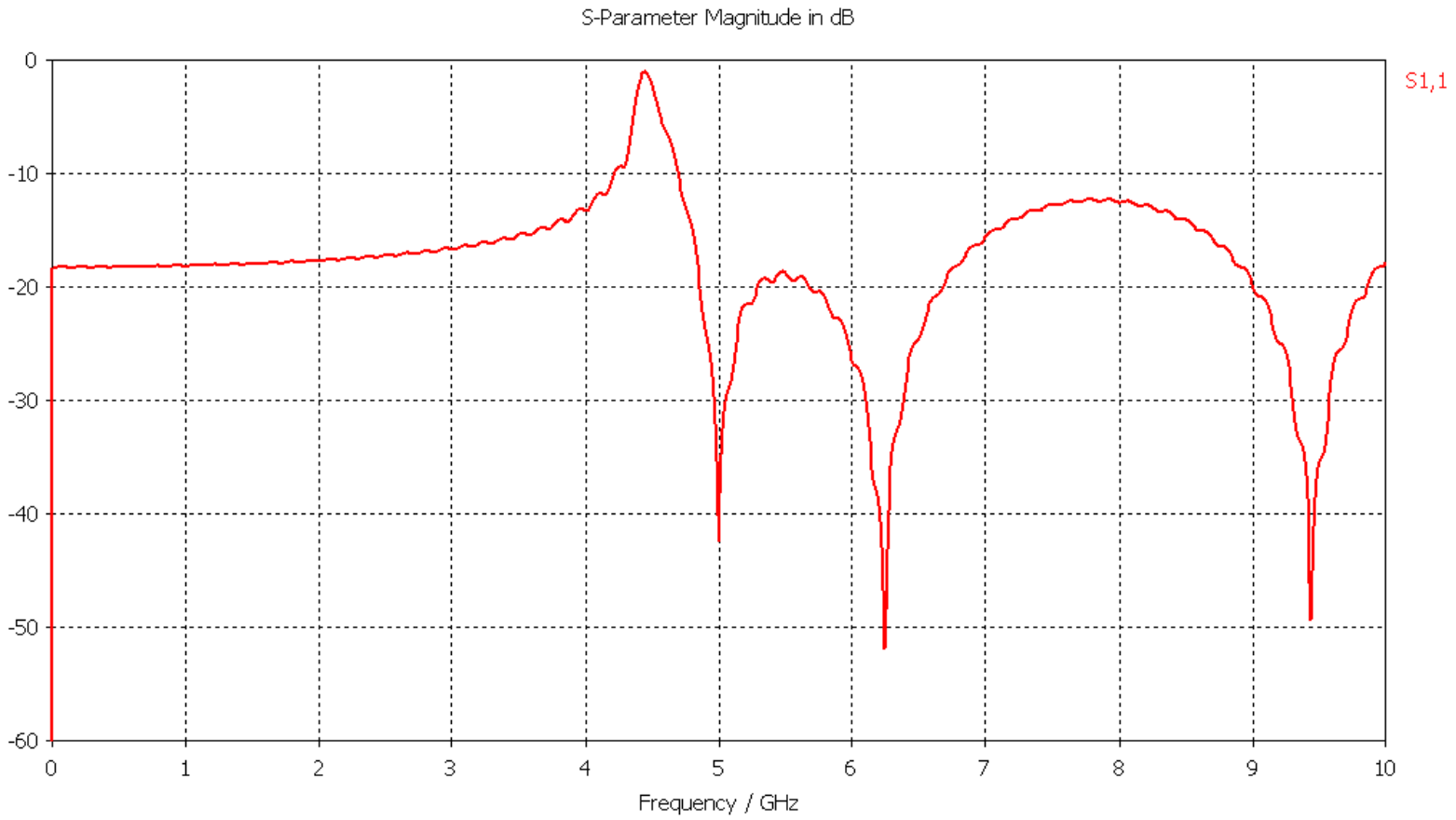
其余设置与非接地板槽线一样。

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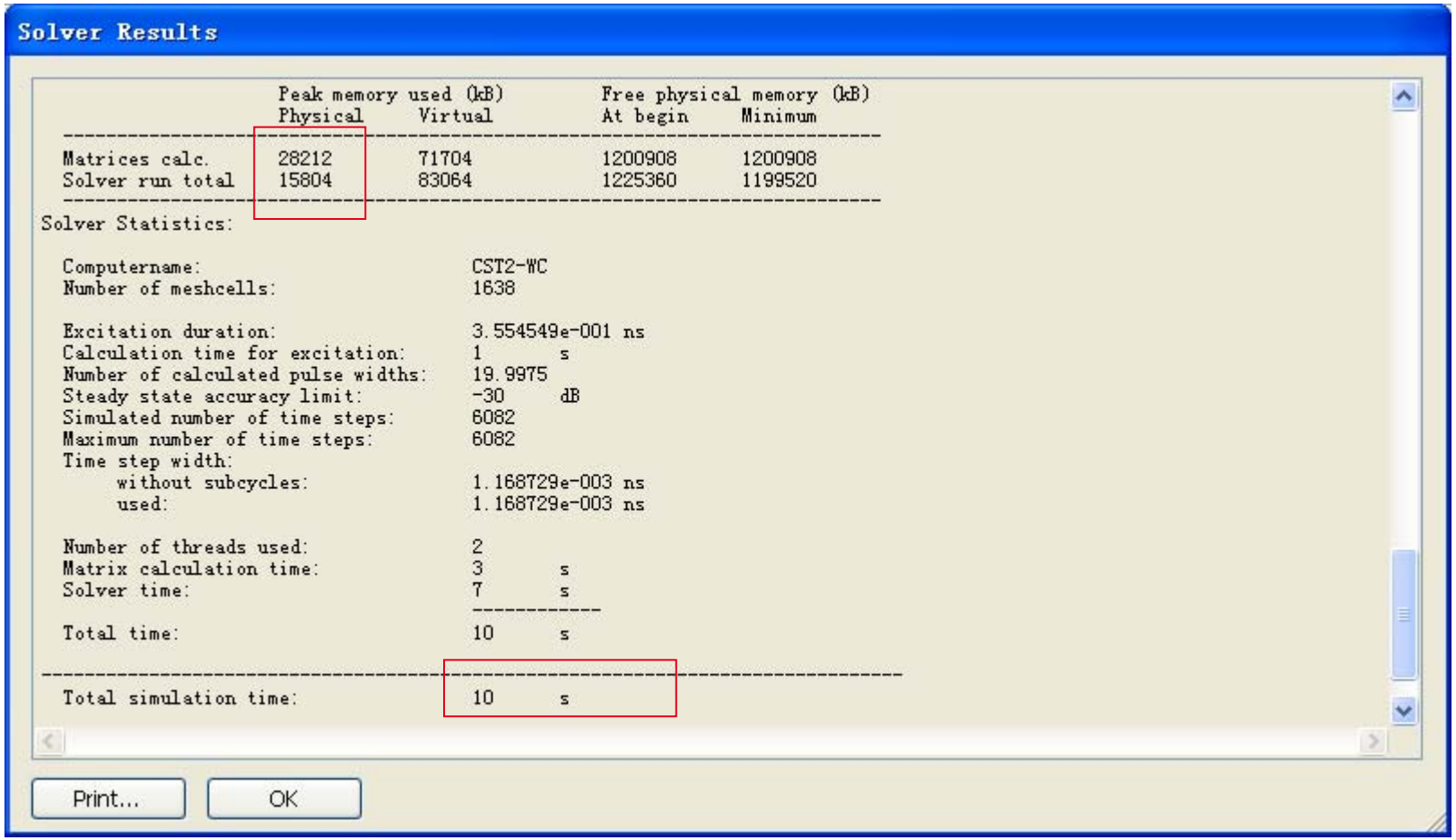
端口1处的端口模式


V/m
10784
10110
8762
7414
6066
4718
3370
2022
674
0
Type E-Field (peak)
Mode type TE
Accuracy 2.05878e-013
Fcutoff 4.40074
Beta 49.745 1/m
Wave Imp. 800.436 Ohms
Plane at z -15
Maximum-2d 10784.5 V/m at -0.015 / 1 / -15
Frequency 5
Phase 0 degrees
z
y
x

S11 dB

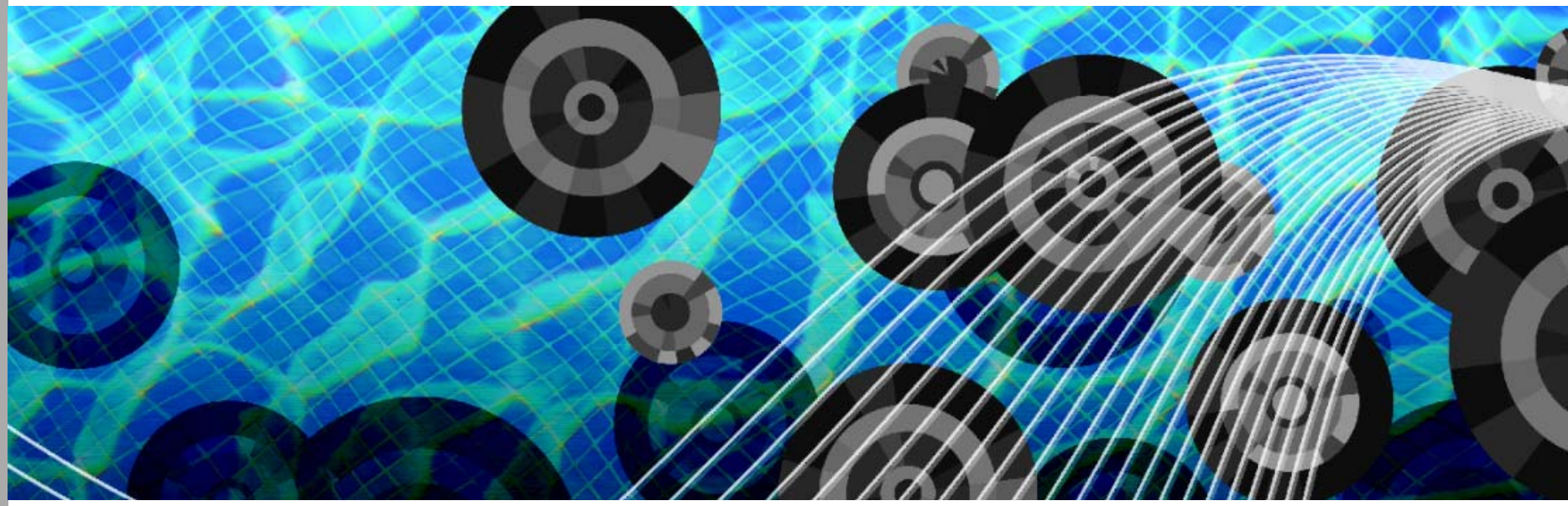

S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1

占用系统资源

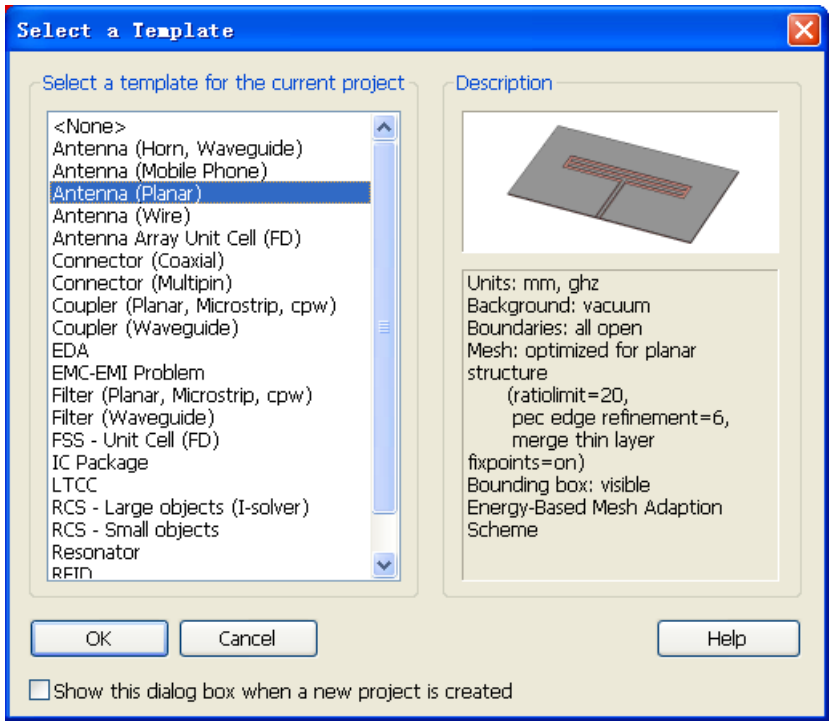


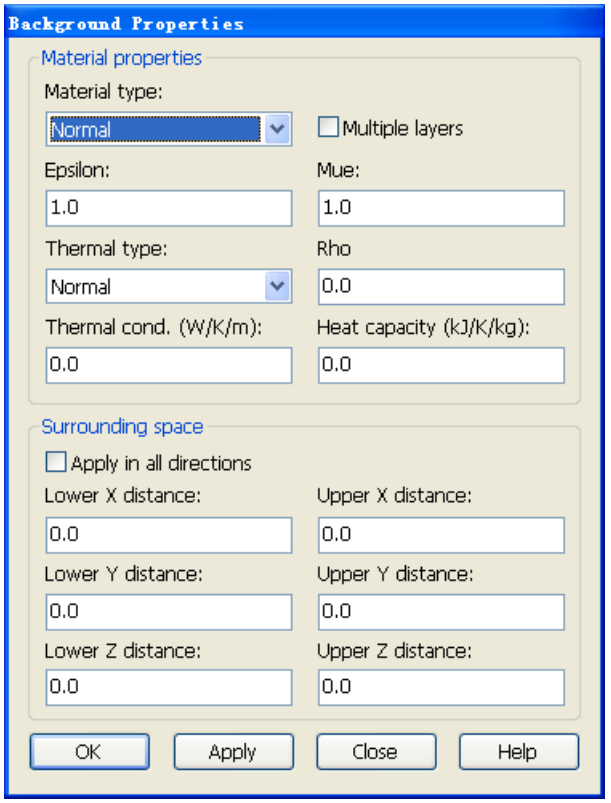
建立矩阵占用内存28M，求解占用内存16M，求解器计算时间10s。

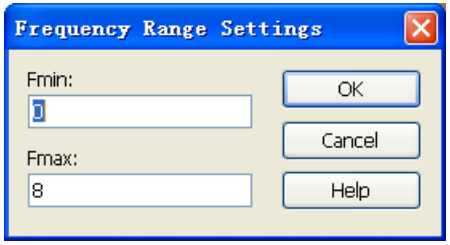
渐变槽线



选择模板、背景条件、频带范围

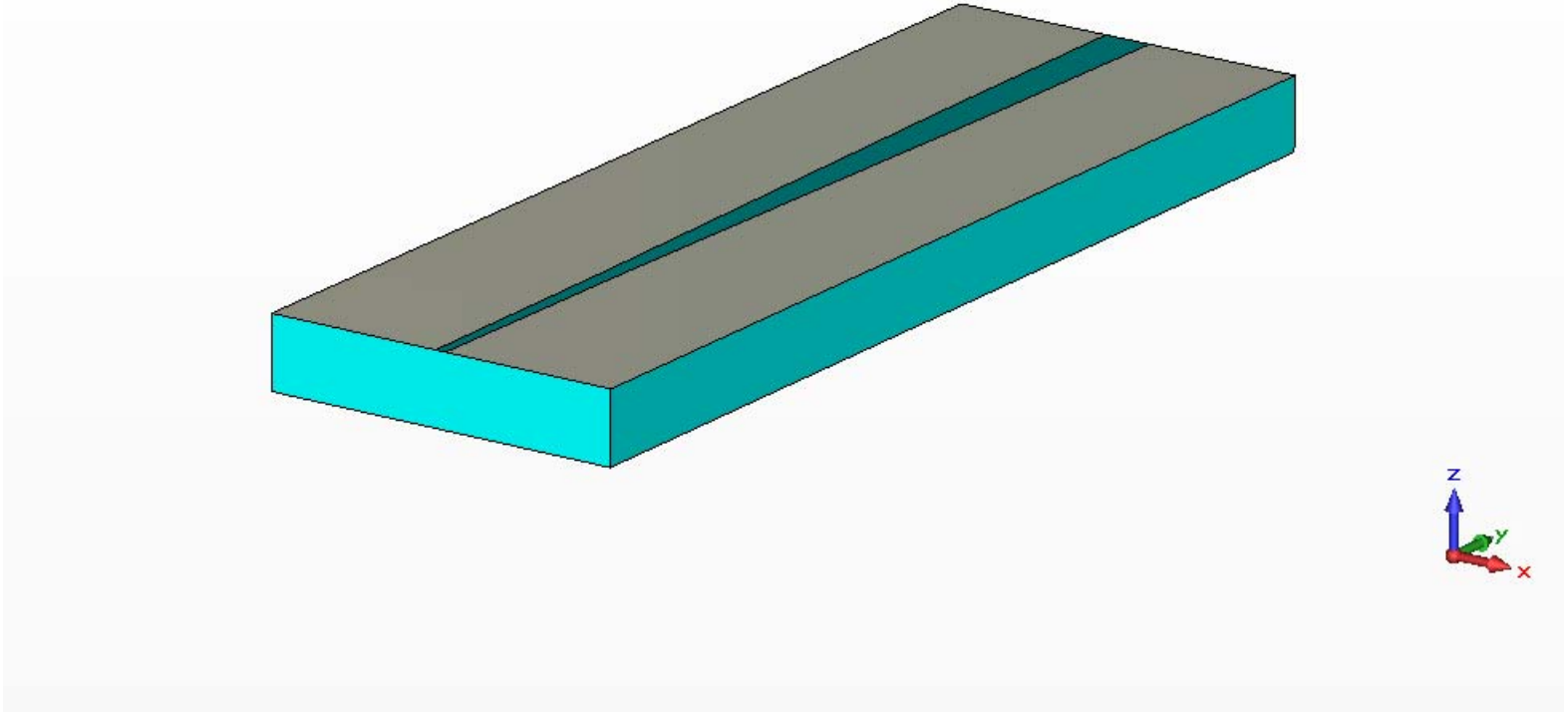




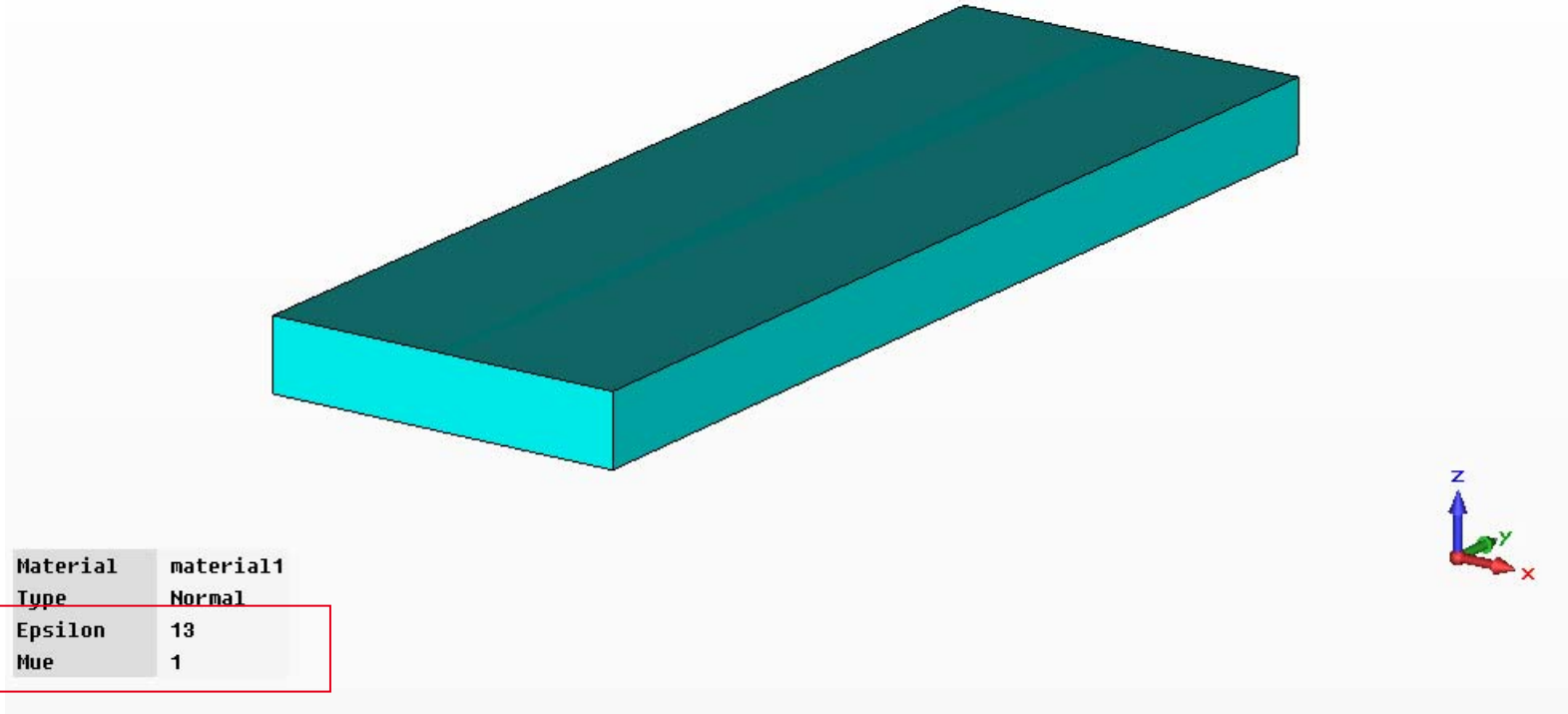


选好模板之后，默认单位为GHz，mm。

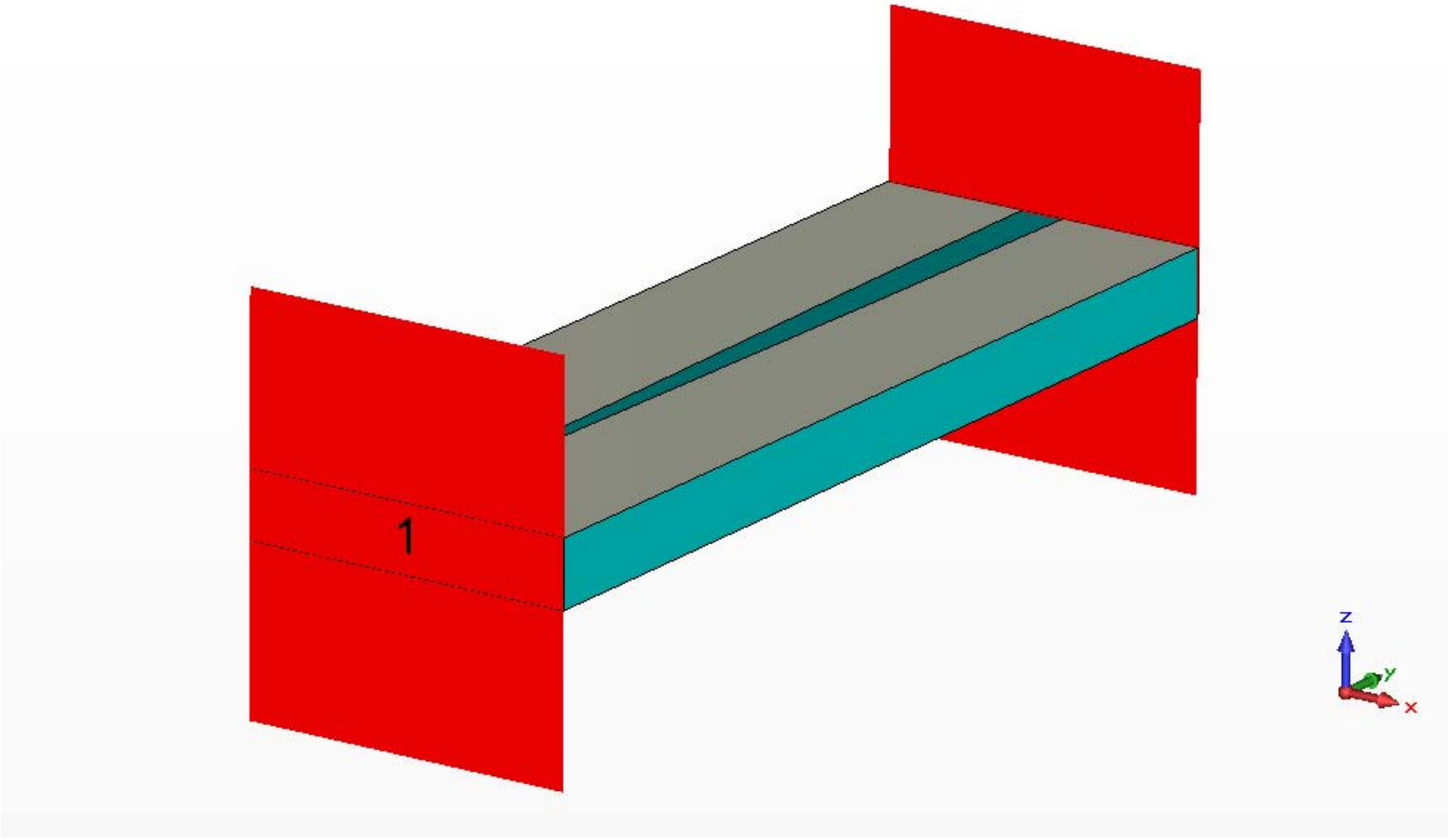
渐变槽线模型



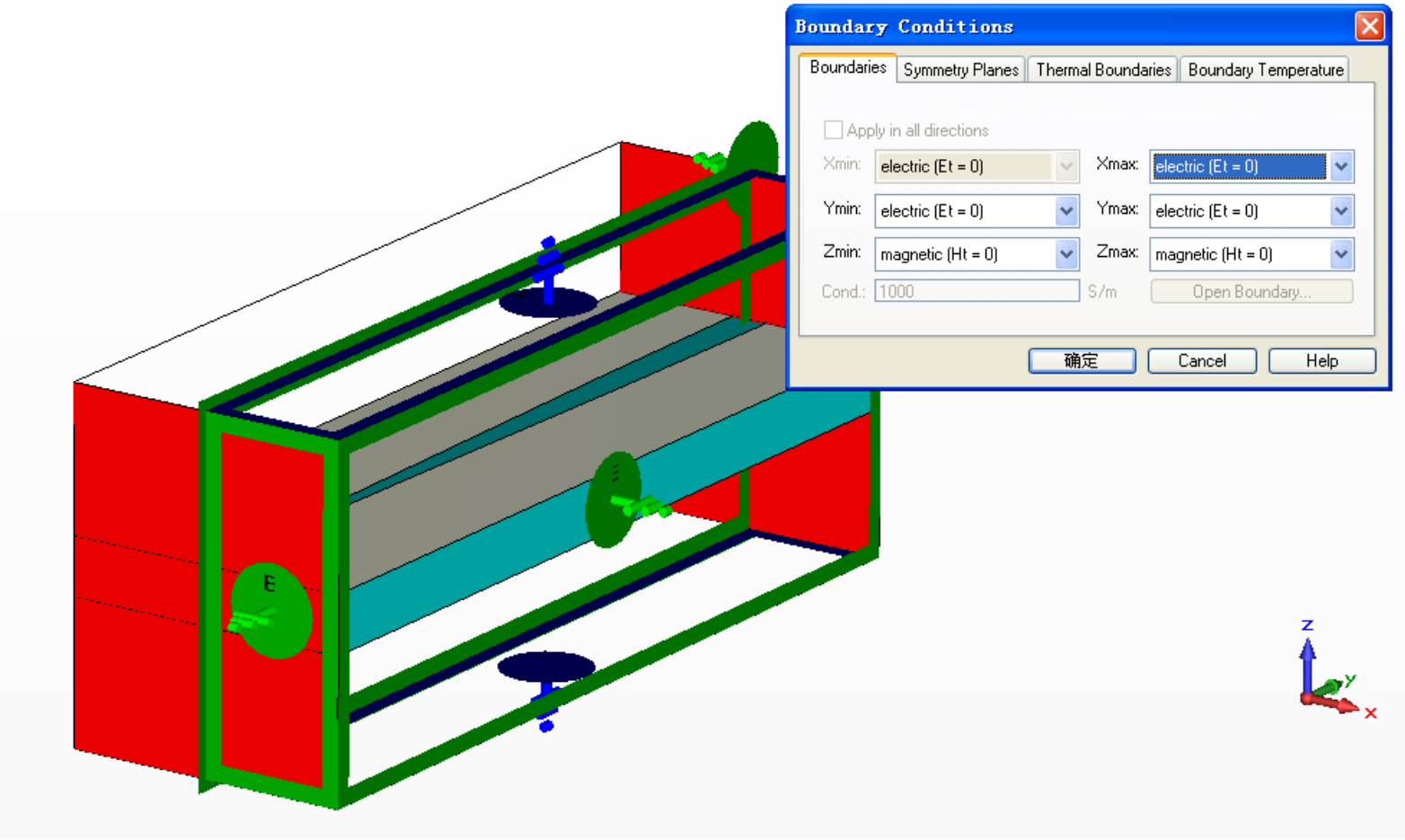
介质板材质



端口设置

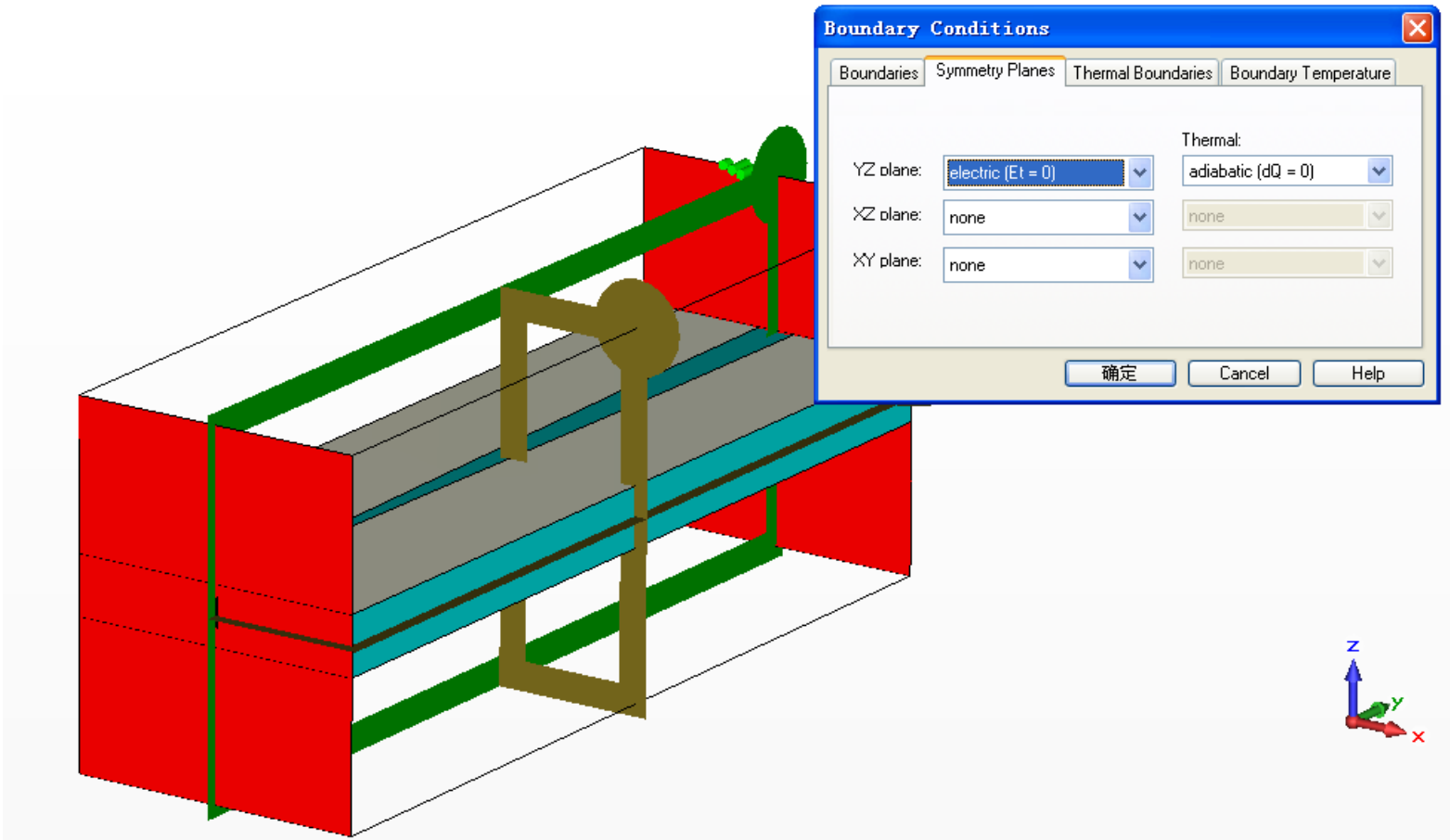


边界条件



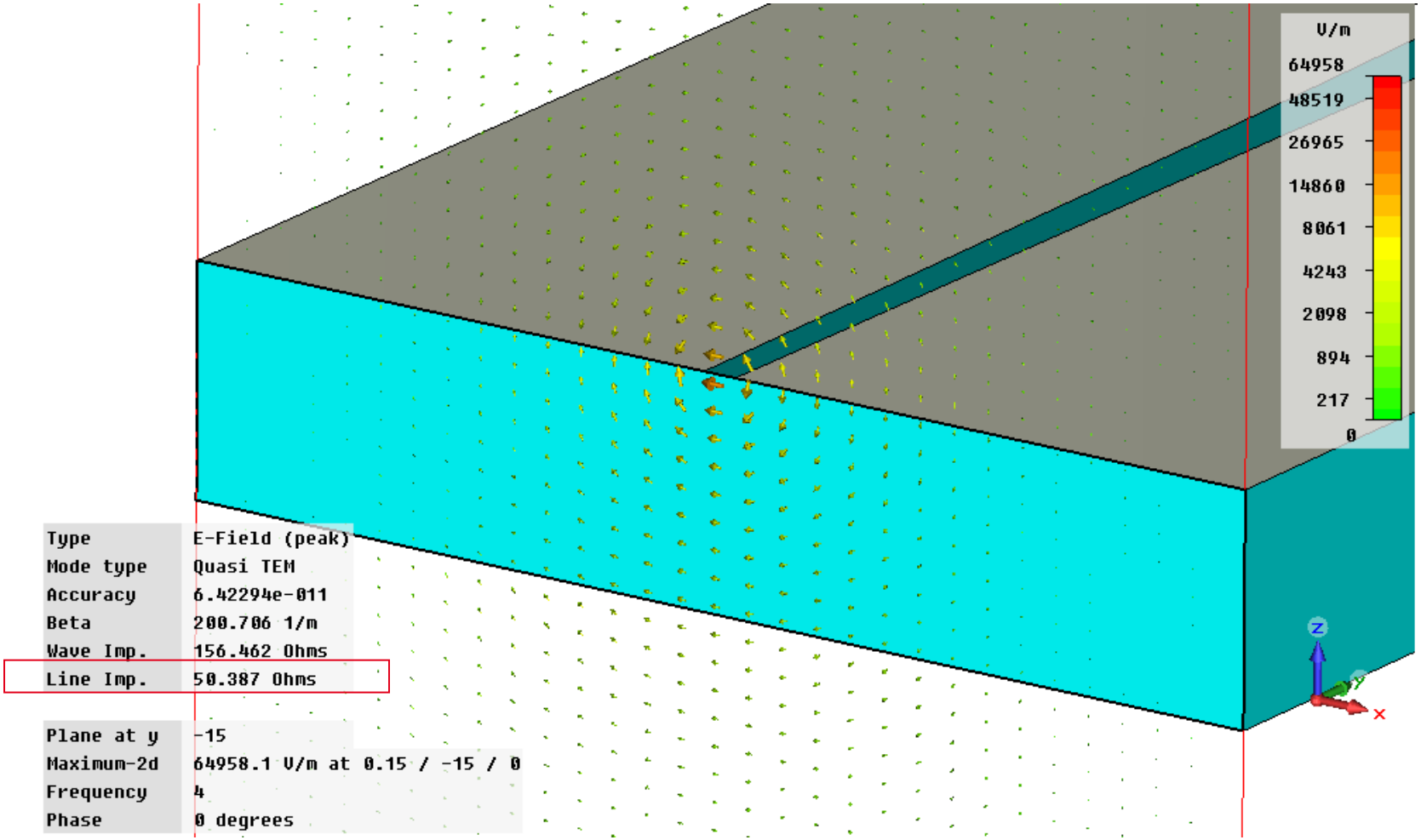
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设置对称面



由于端口处电场与YOZ面垂直，且模型关于YOZ面对称，因此可以讲YOZ面设为电对称面，以节省仿真时间。

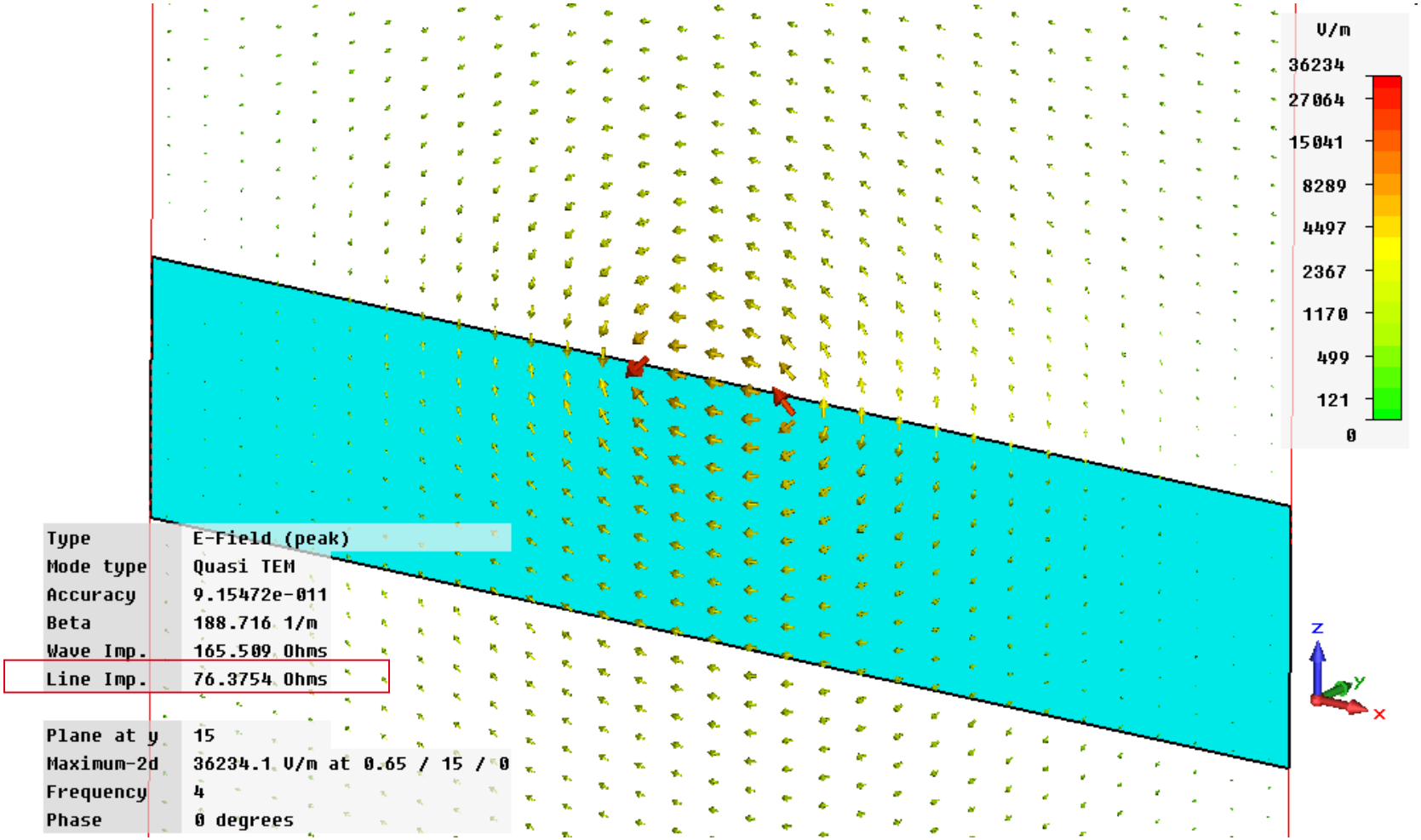
1端口模式



1端口端口阻抗50欧姆。

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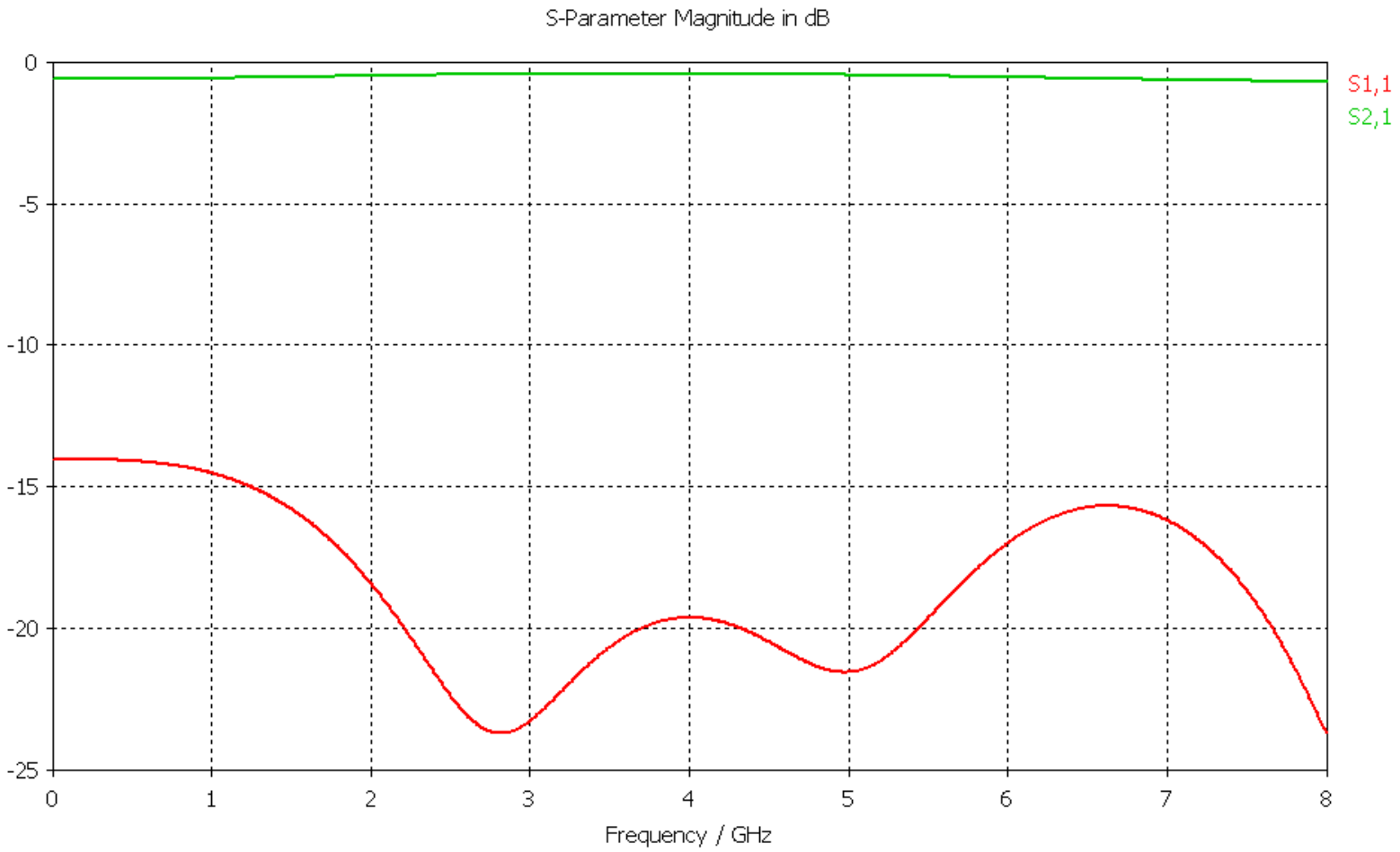
2端口模式场



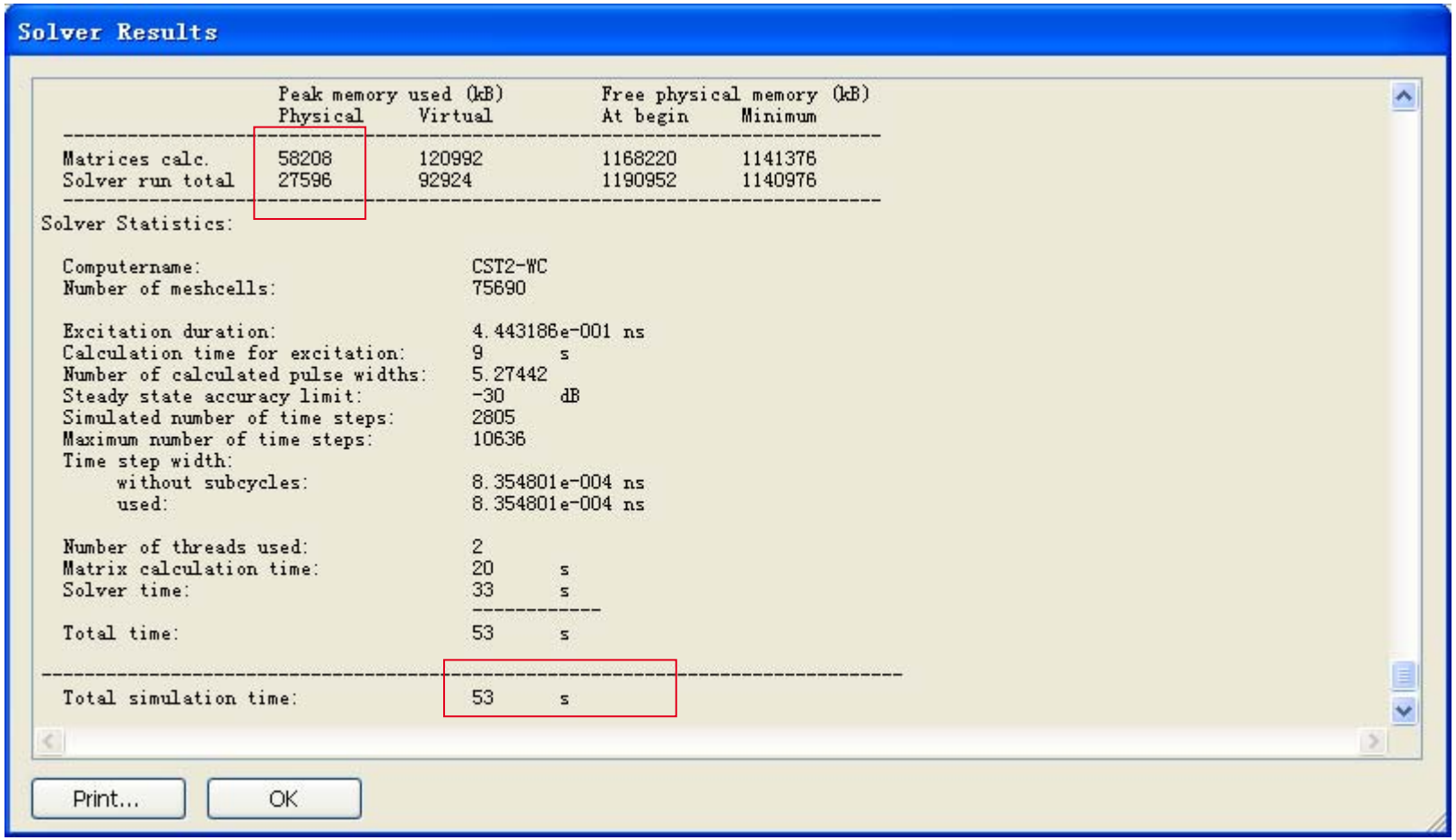
2端口端口阻抗75欧姆。

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S11 S21 dB

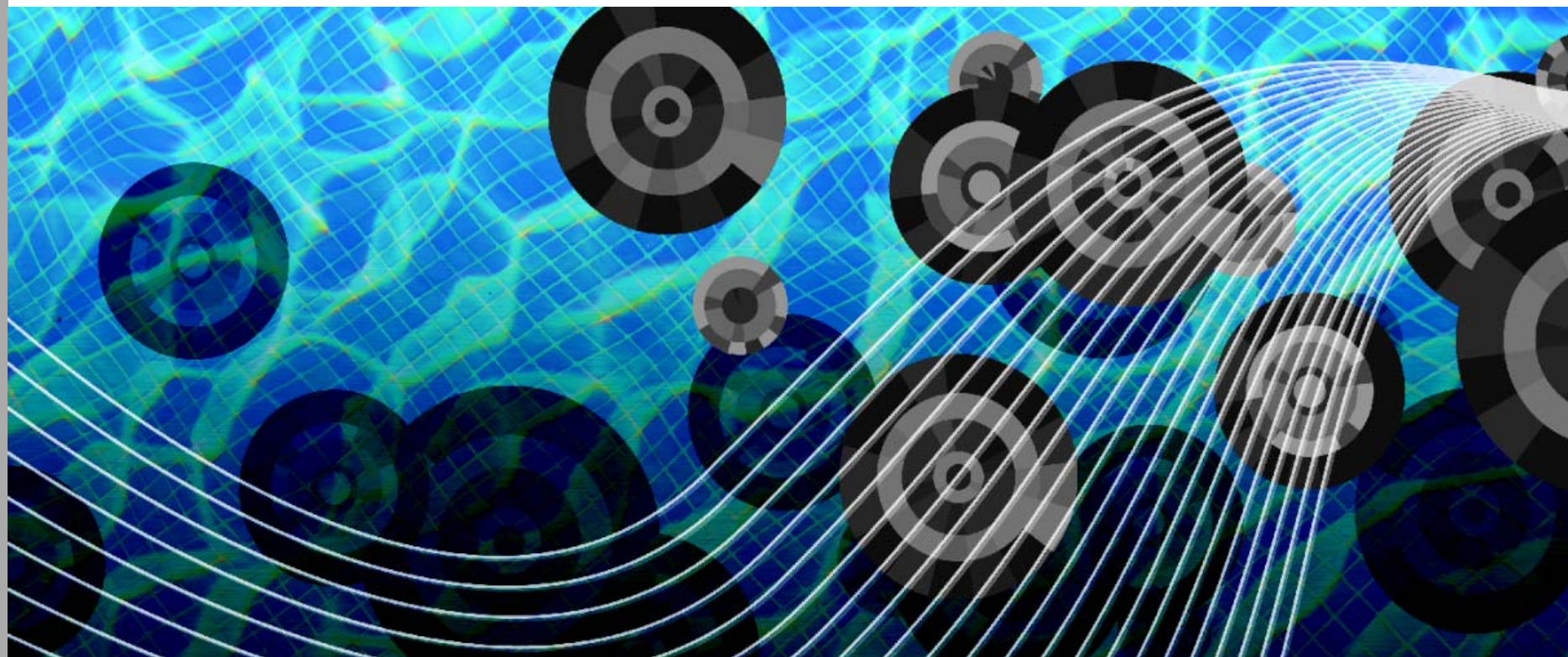

S-Parameter Magnitude in dB
0
-5
-10
-15
-20
-25
0
1
2
3
4
5
6
7
8
Frequency / GHz
S1,1
S2,1

占用系统资源

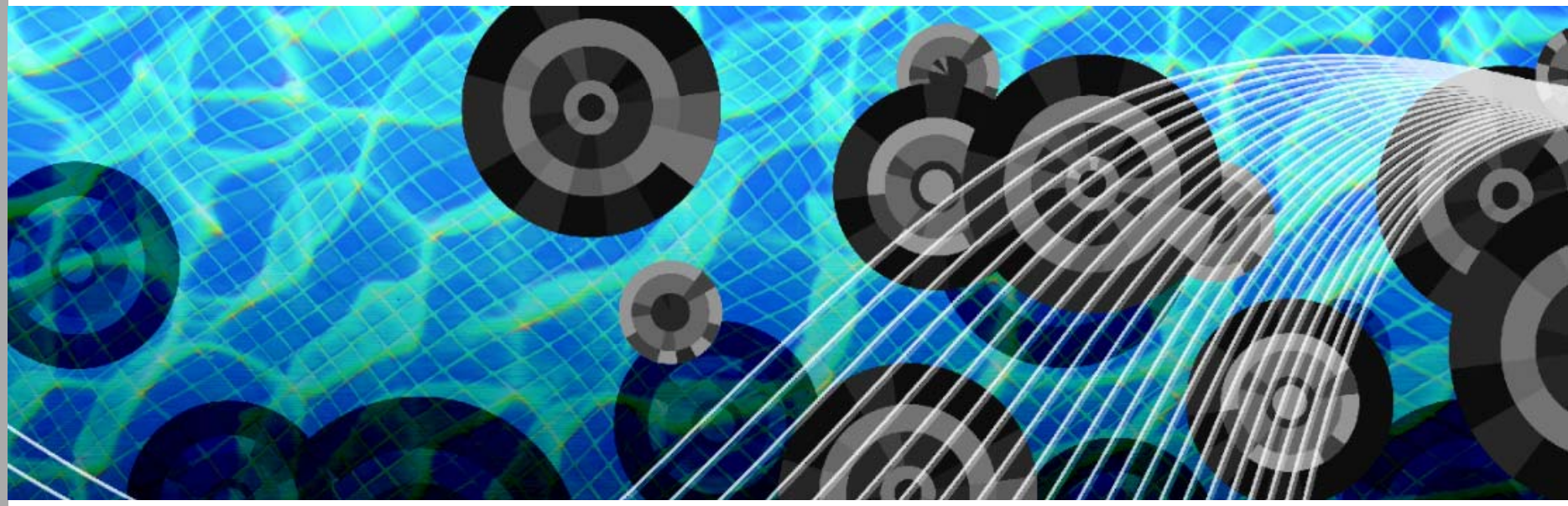


建立矩阵占用内存58M，求解占用内存27M，求解器计算时间53s。

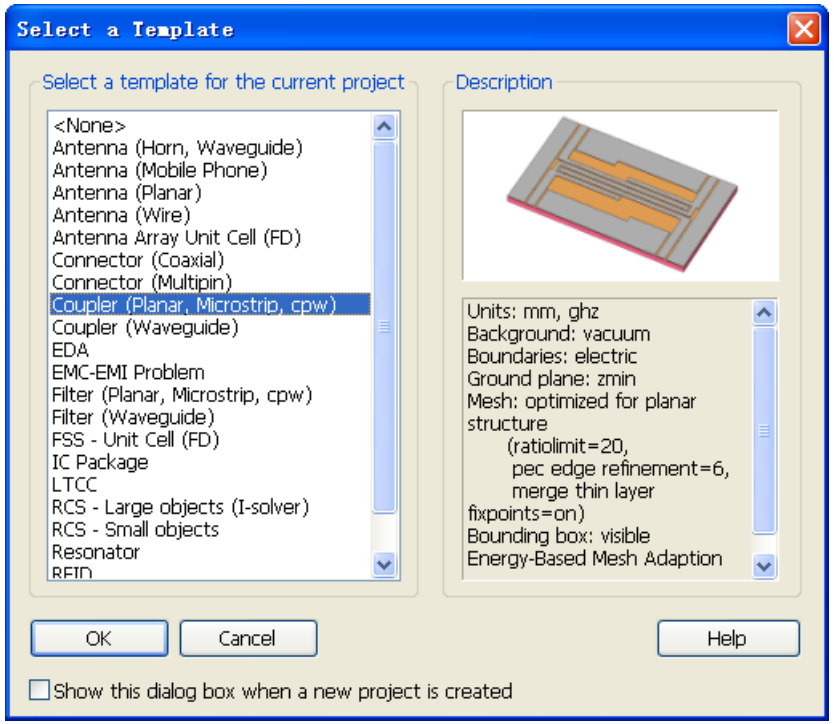
共面波导

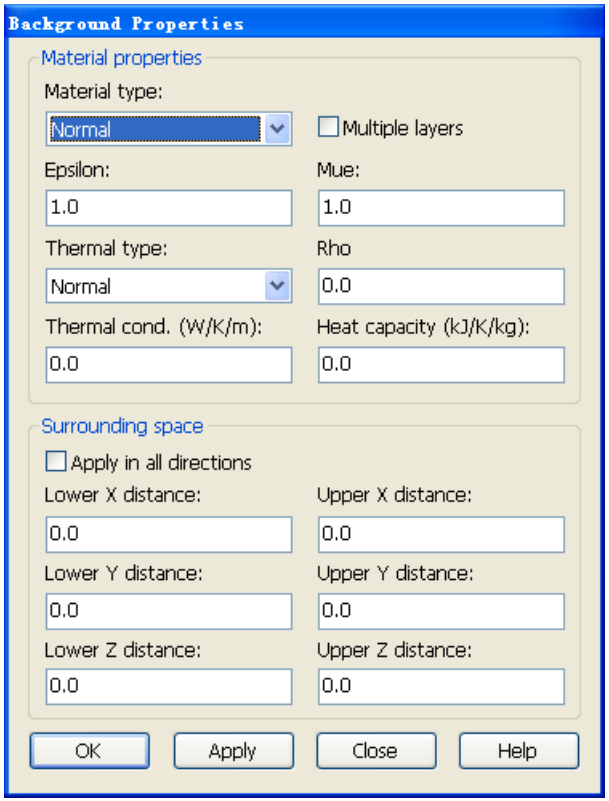


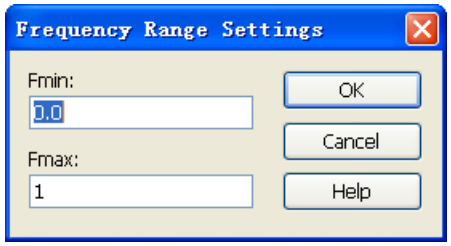
不带接地板的共面波导



单位、背景条件、频带范围设置

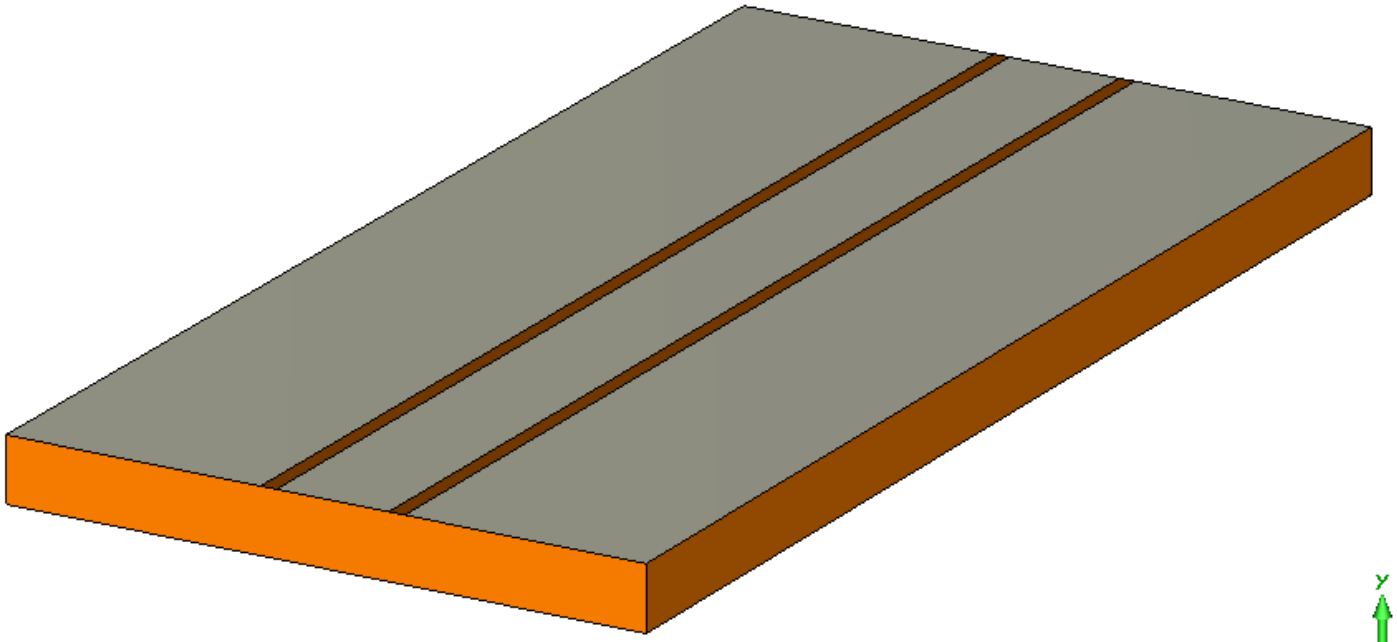




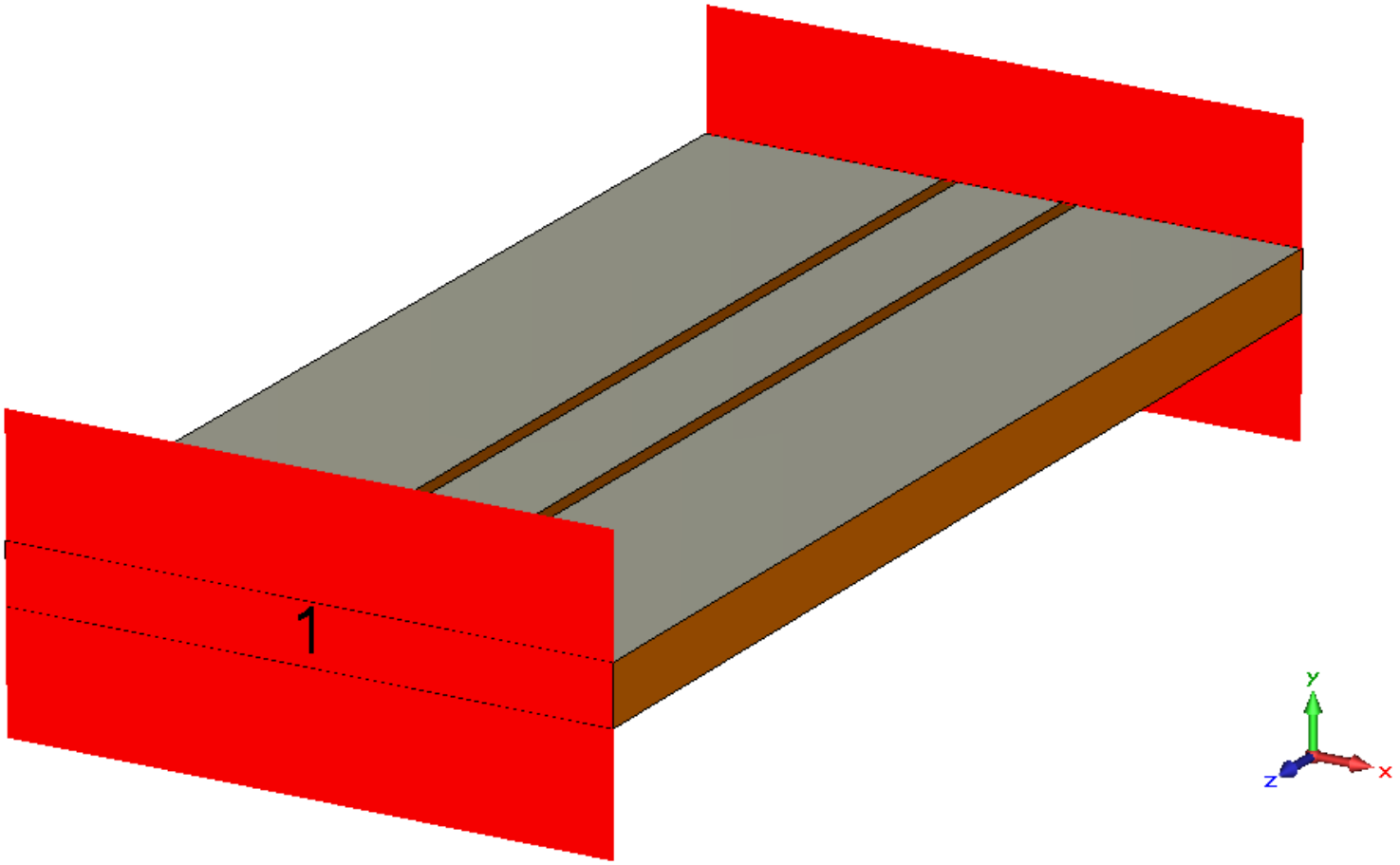


选好模板之后，默认单位为GHz，mm。

模型

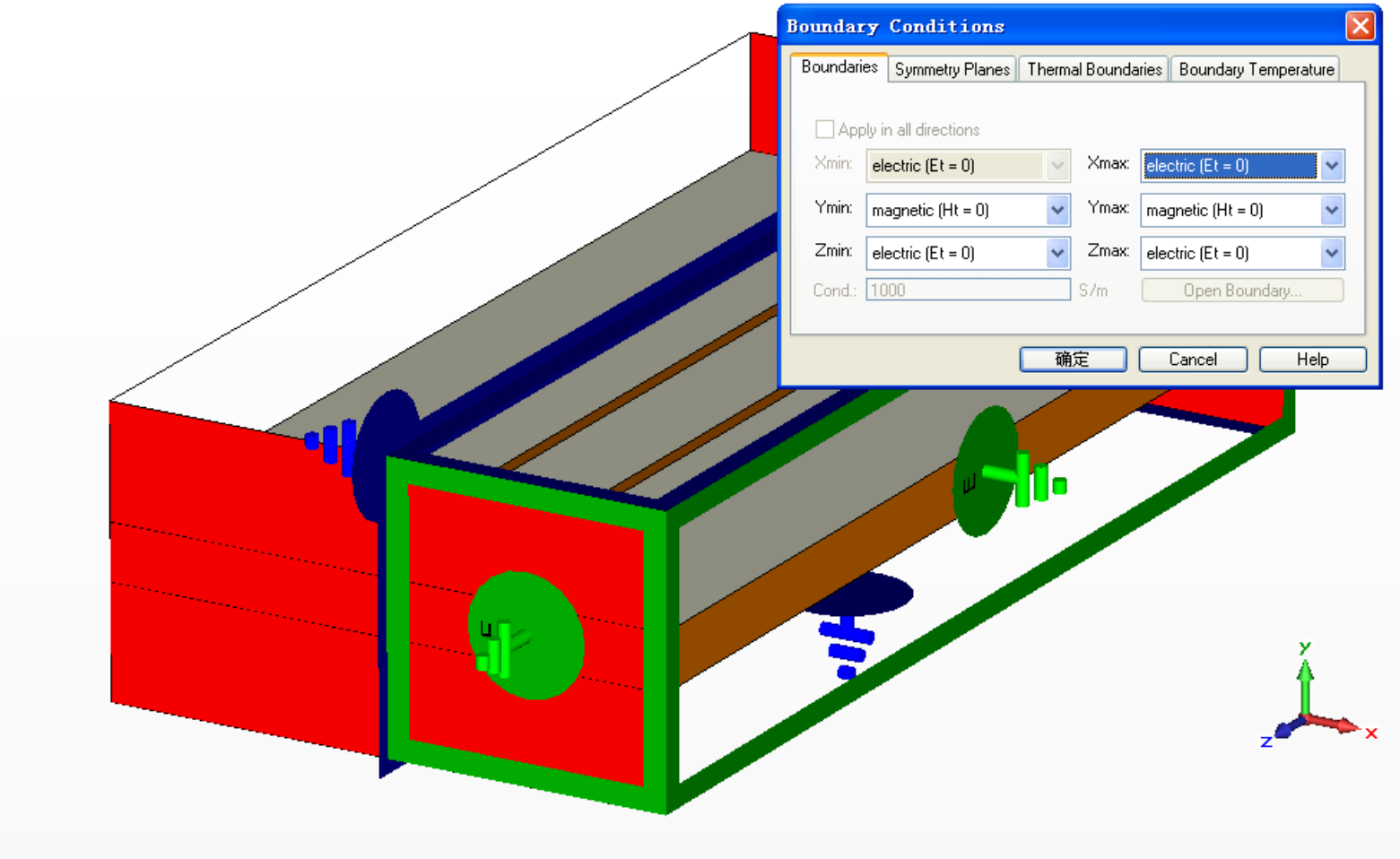


端口设置



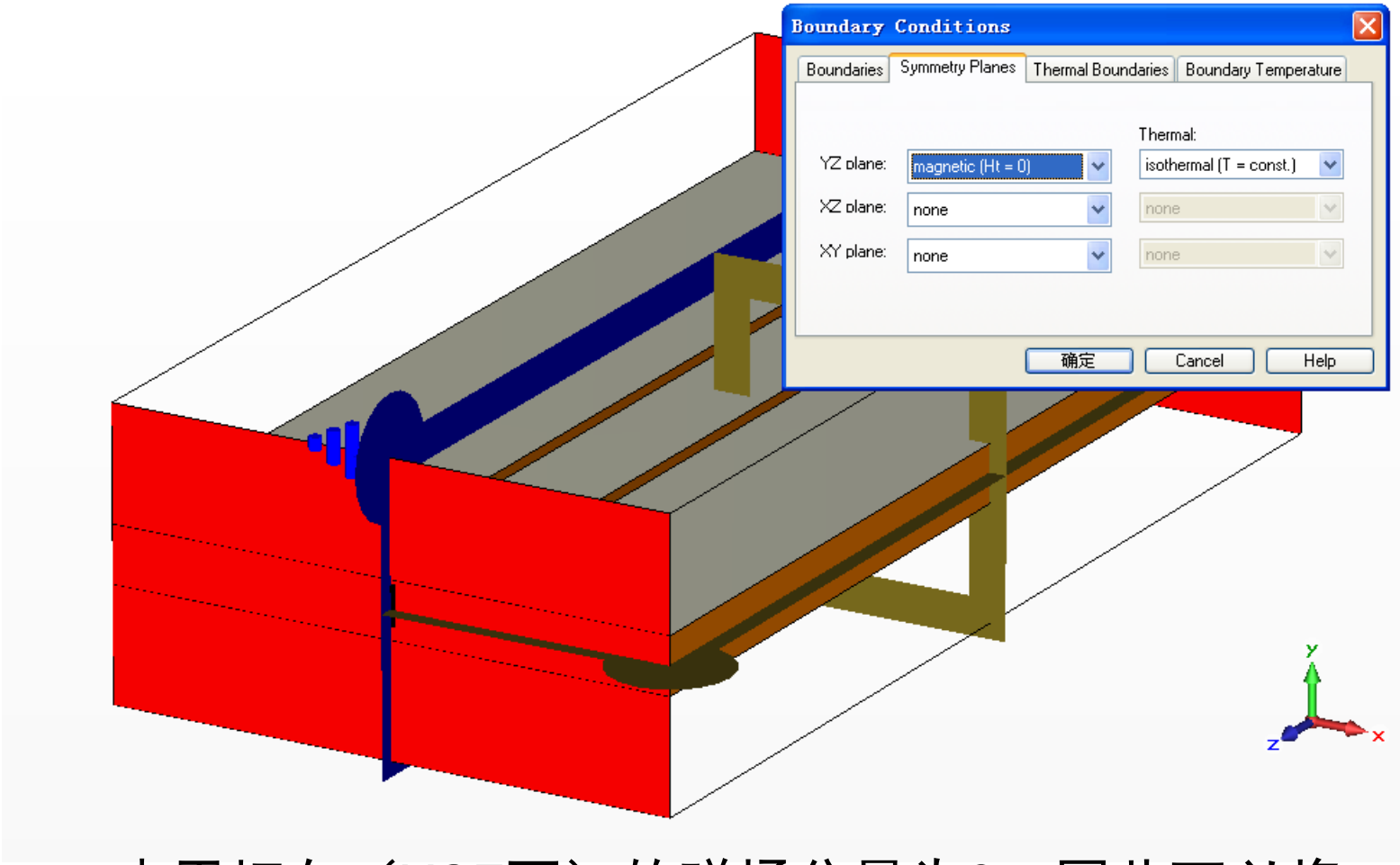
若中心带线宽度为W的话，则端口长度至少要保证2-3个W，高度至少为1个W。详情可见CST丛书3 P88-90。

边界条件



由于电壁相当于接地板，因此Ymin方向设置为磁壁。

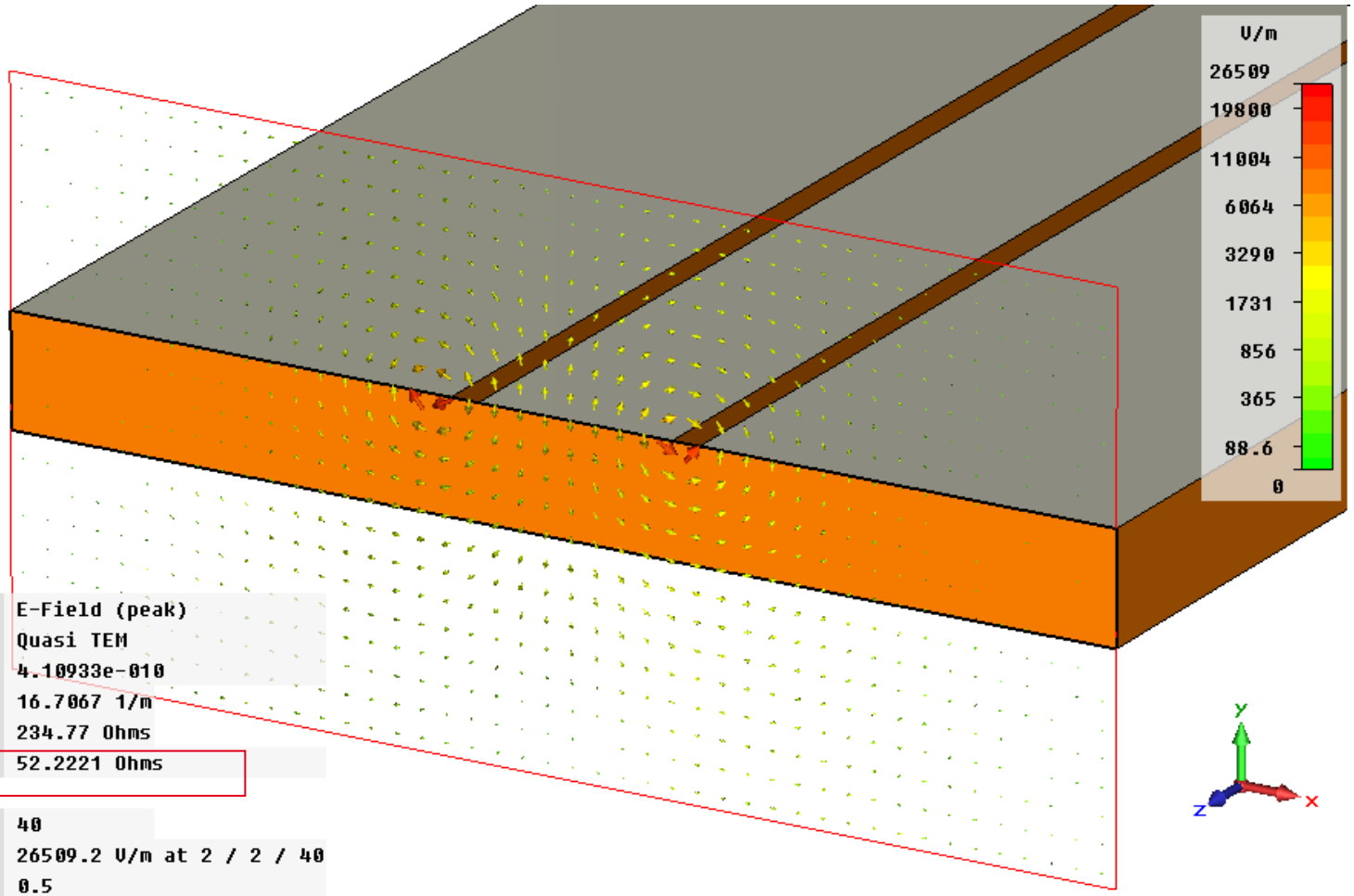
采用对称面减少仿真时间



由于切向（YOZ面）的磁场分量为0，因此可以将此平面设为对称面。

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端口模式

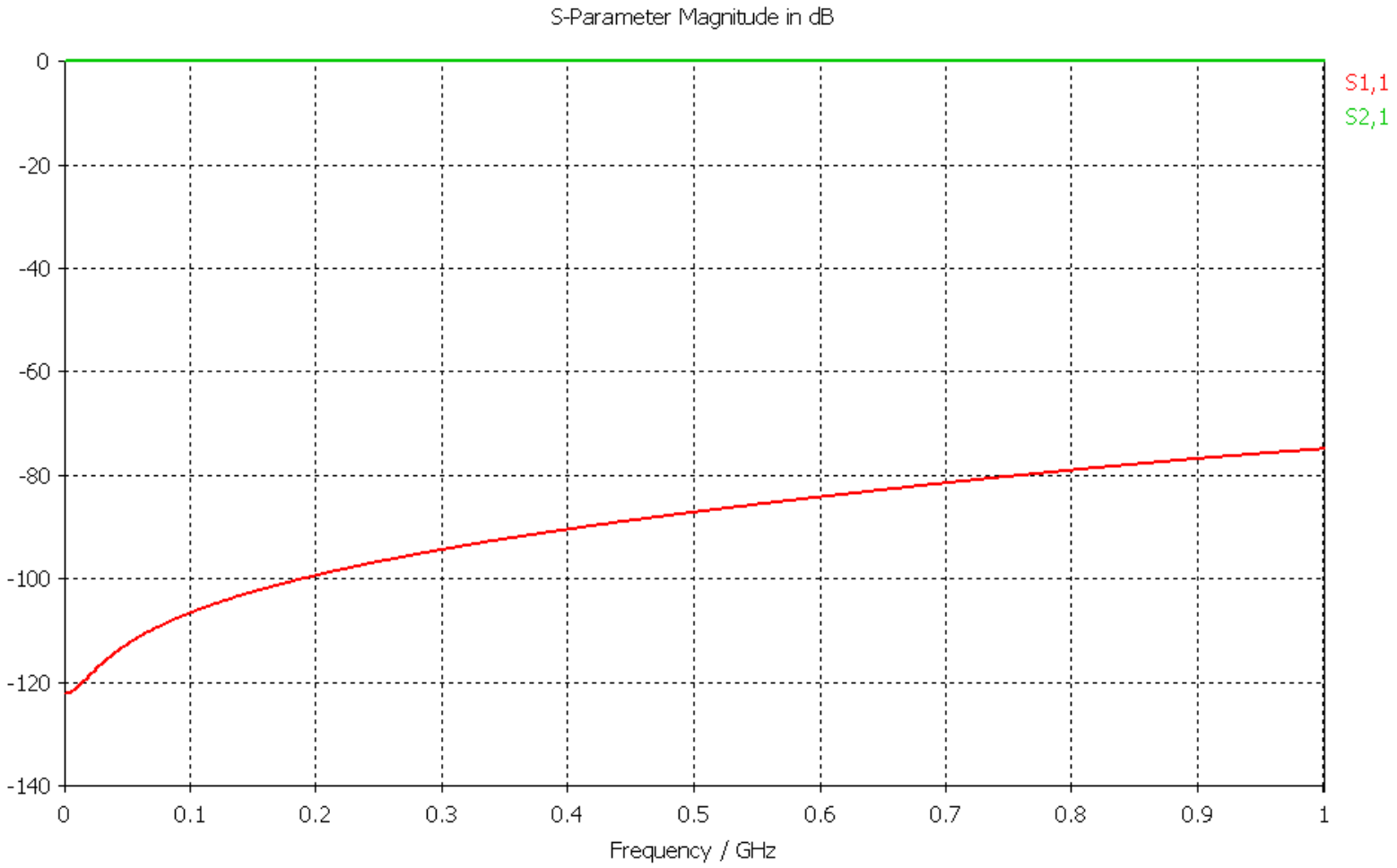


Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	4.10933e-010
Beta	16.7067 1/m
Wave Imp.	234.77 Ohms
Line Imp.	52.2221 Ohms

Plane at z	40
Maximum-2d	26509.2 V/m at 2 / 2 / 40
Frequency	0.5
Phase	0 degrees

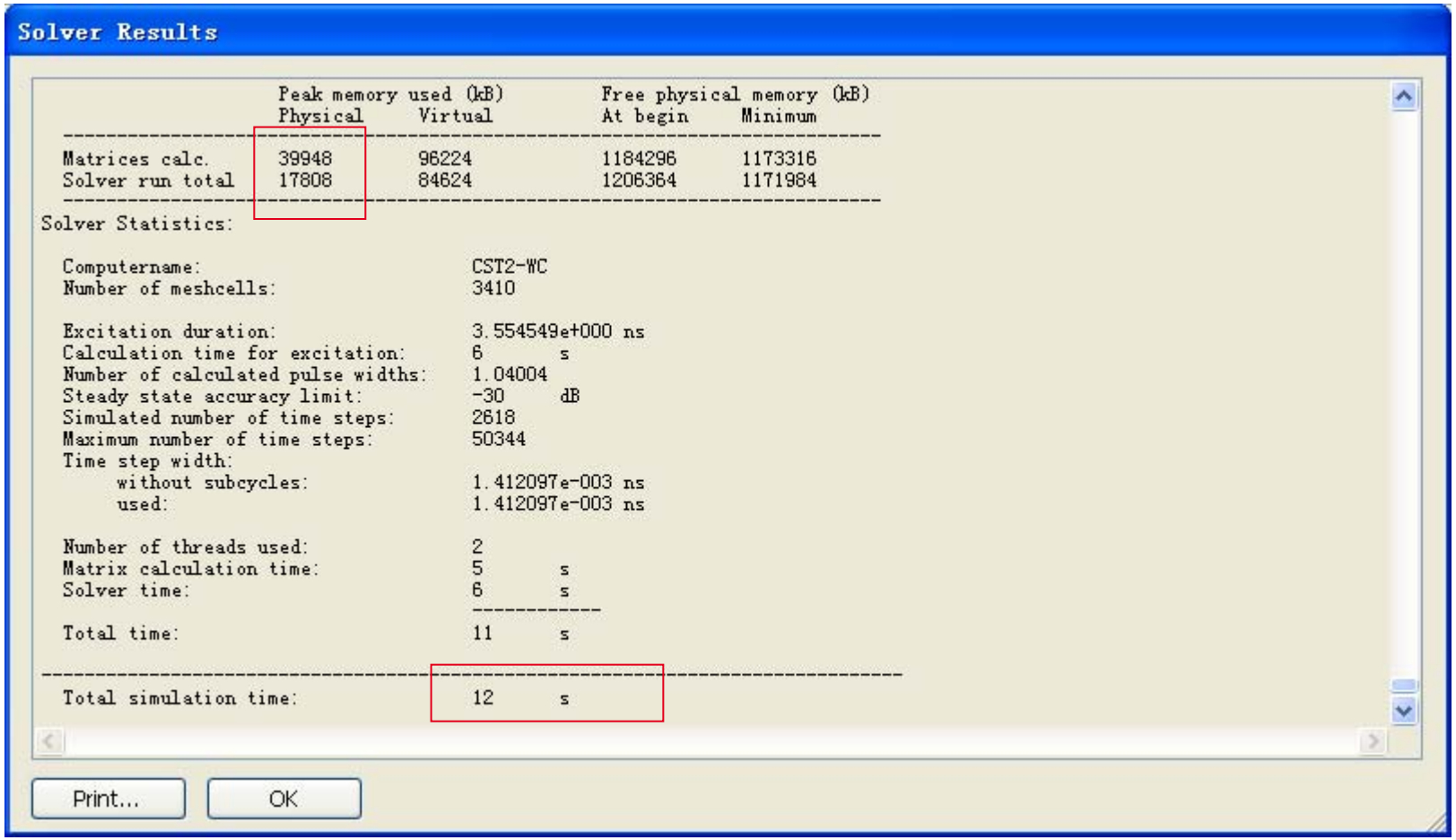
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S11、S21



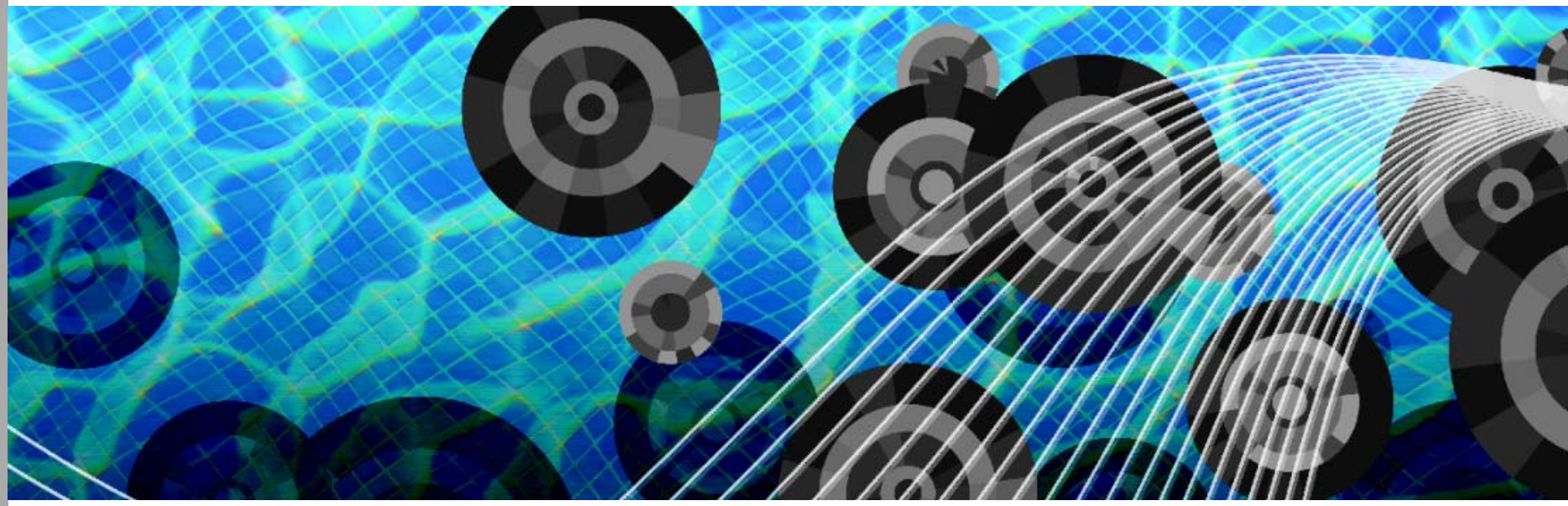
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占用系统资源

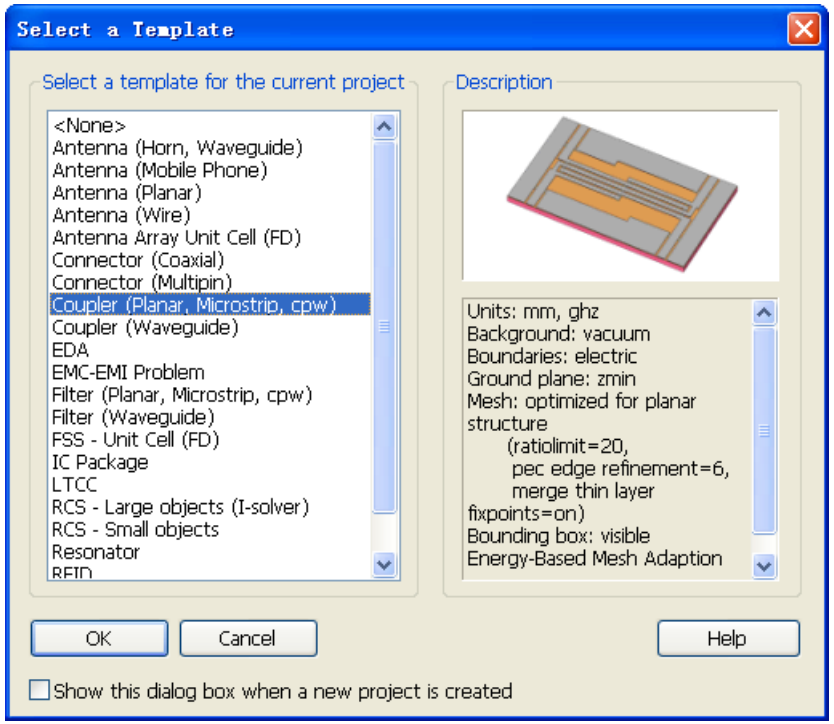


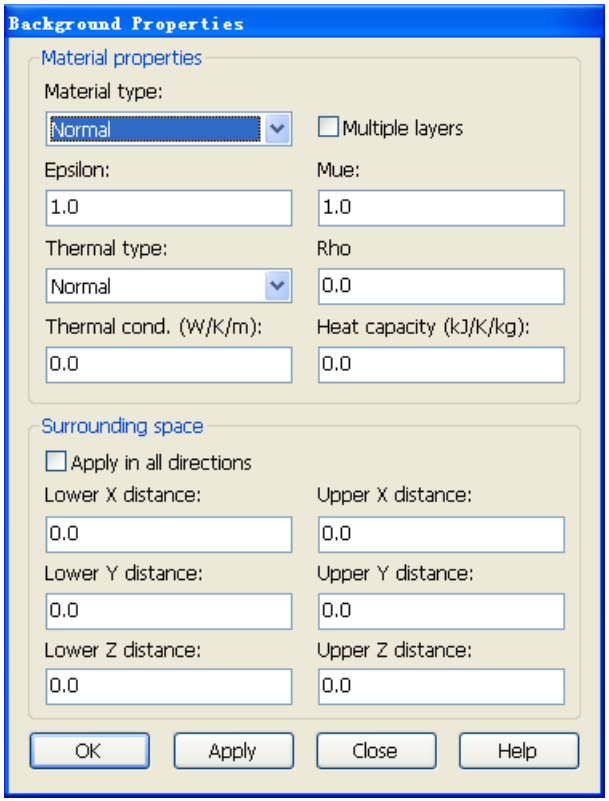
建立矩阵占用内存40M，求解占用内存18M，仿真求解计算时间12s。

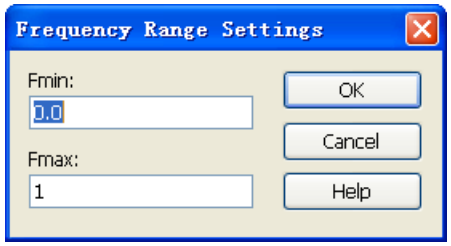
带接地板的共面波导



单位、背景条件、频带范围设置



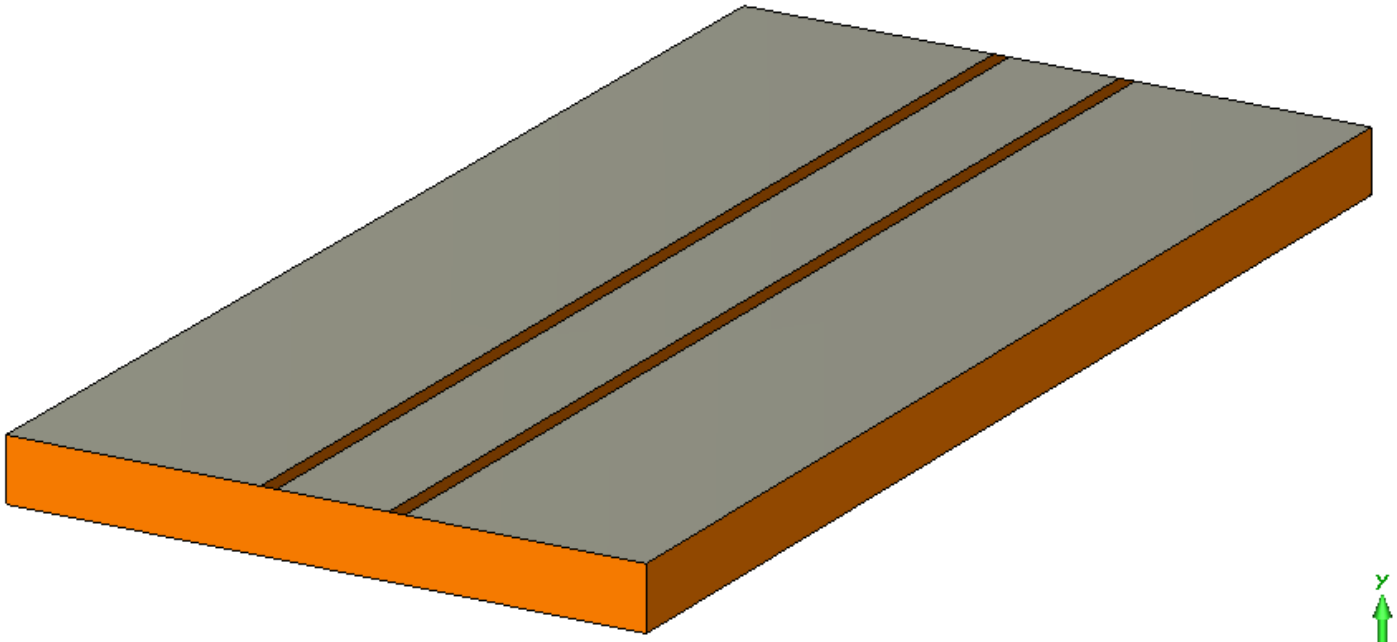




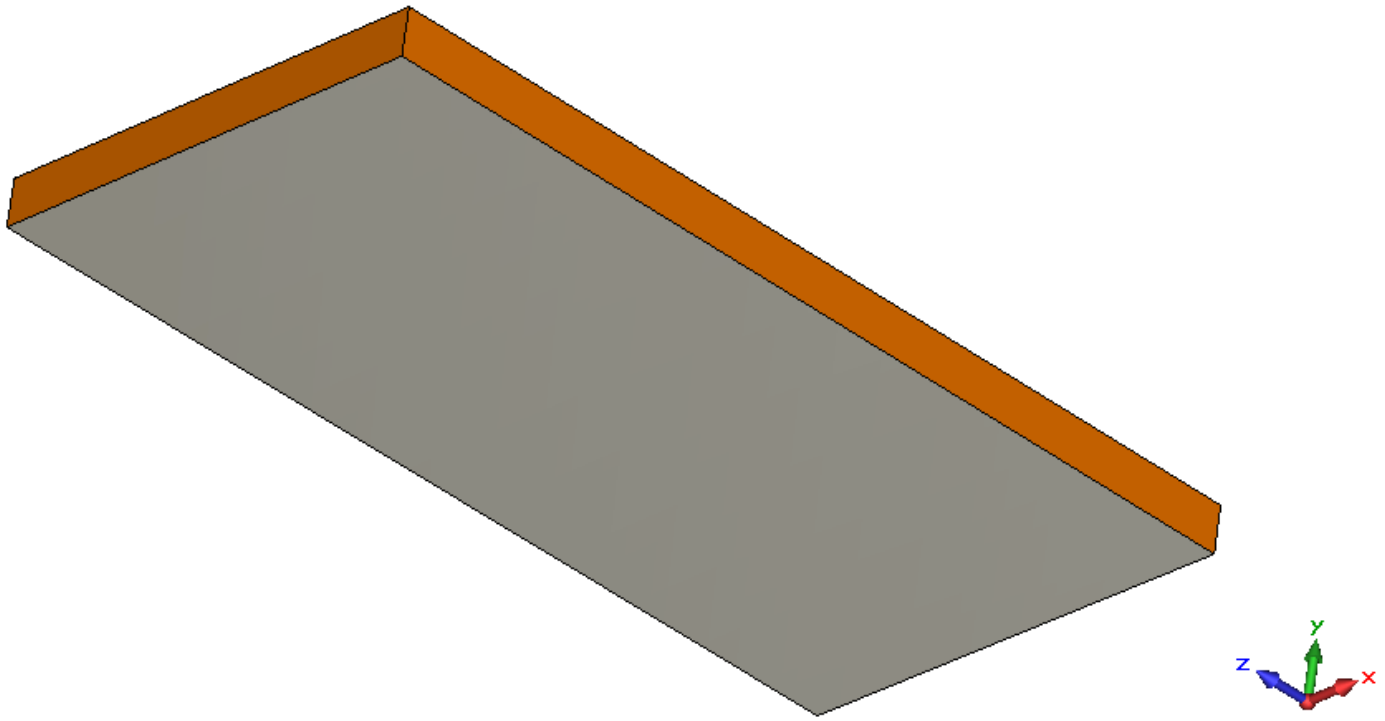
选好模板之后，默认单位为GHz，mm。

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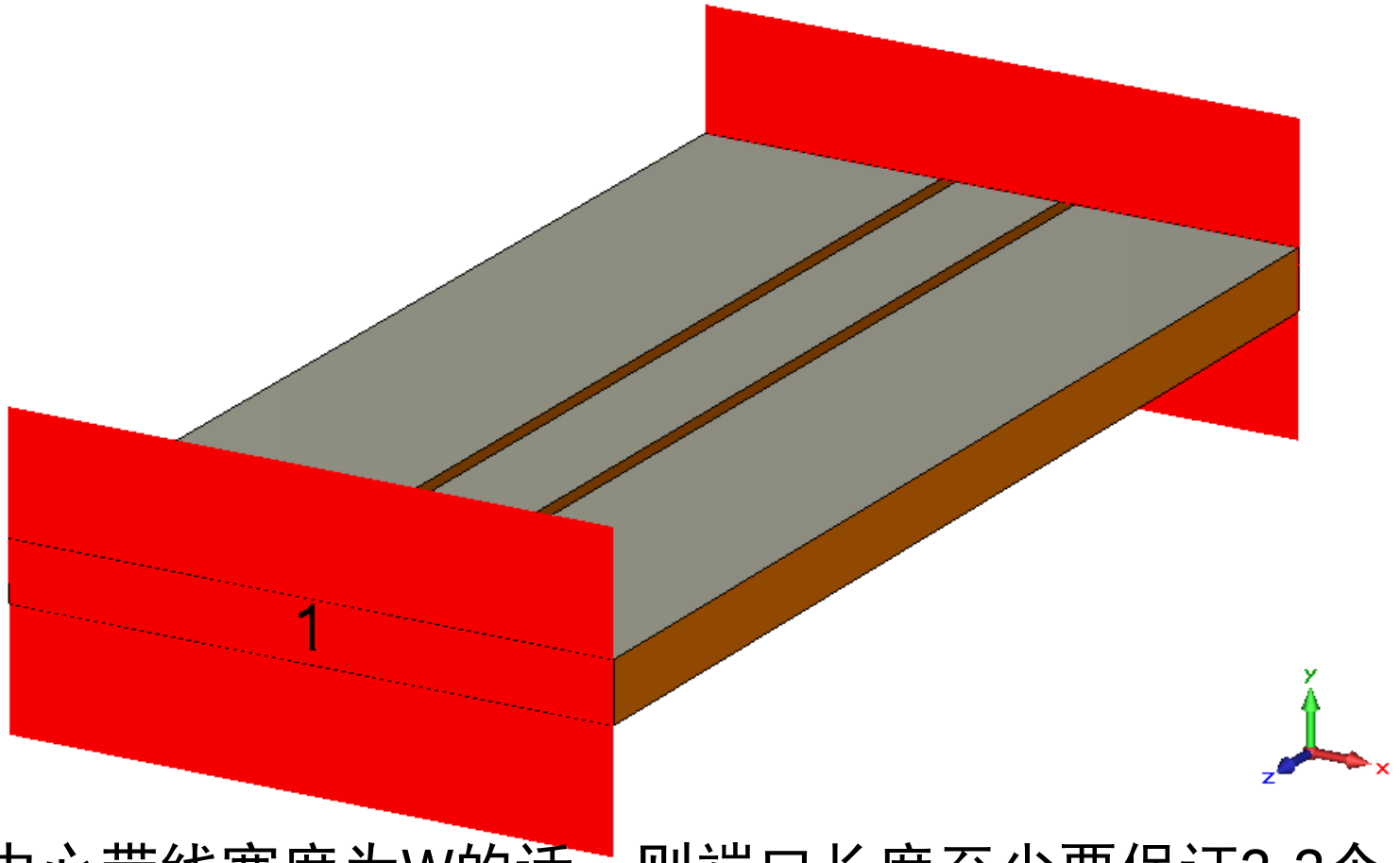
模型



带接地板

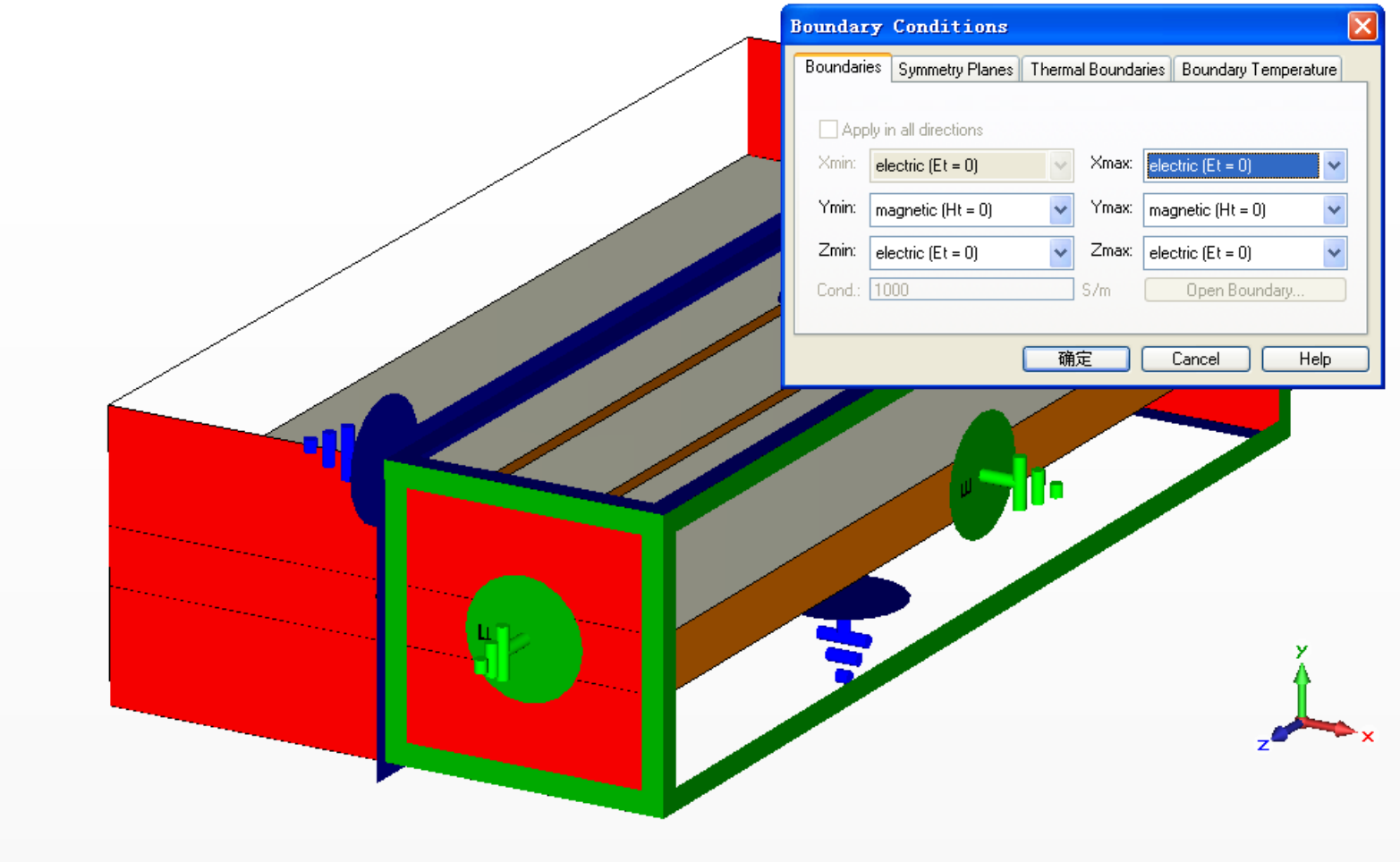


端口设置



若中心带线宽度为W的话，则端口长度至少要保证2-3个W，高度至少为1个W。详情可见CST丛书3 P88-90。

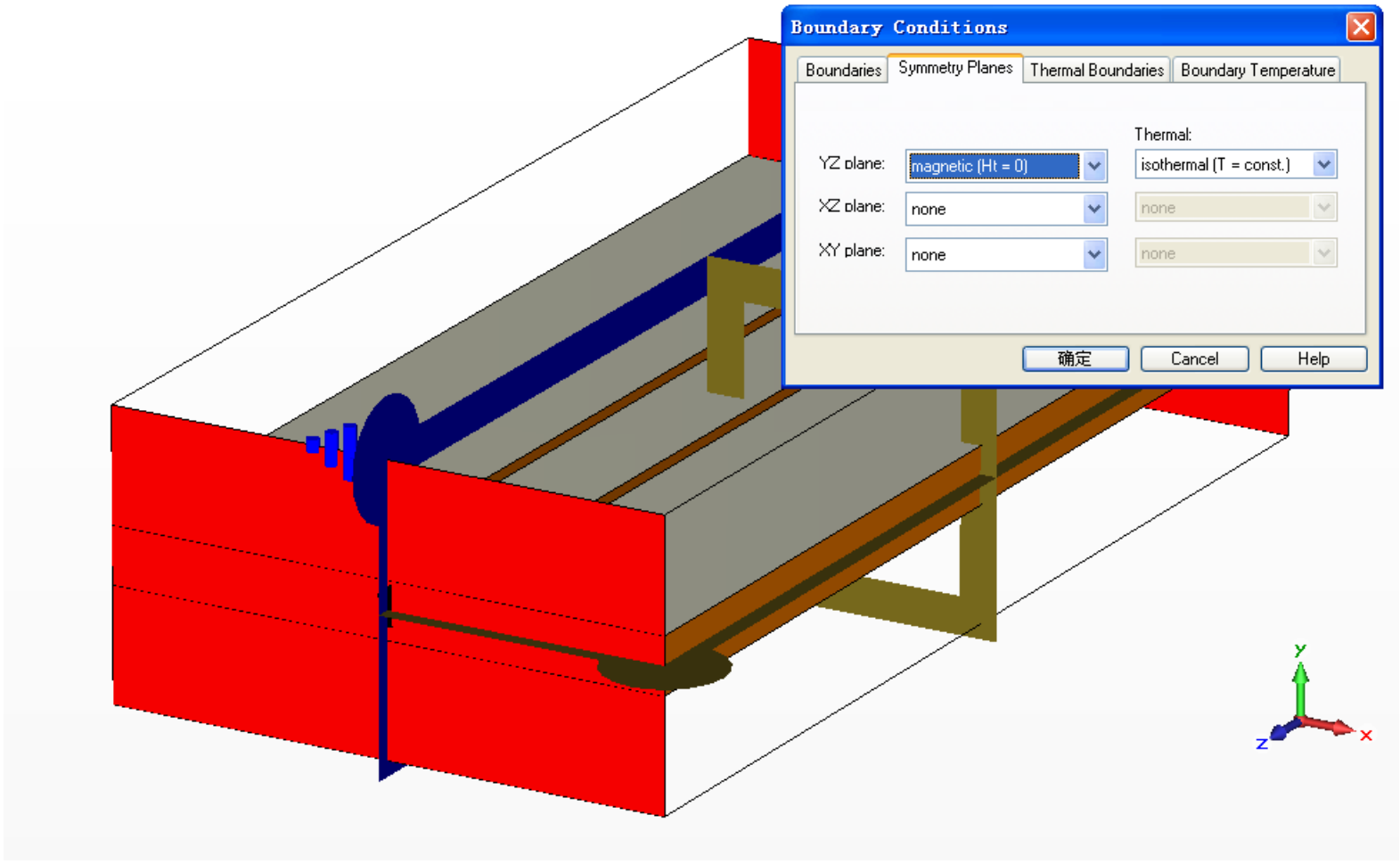
边界条件设置



边界条件设置与不带接地板的共面波导一样。

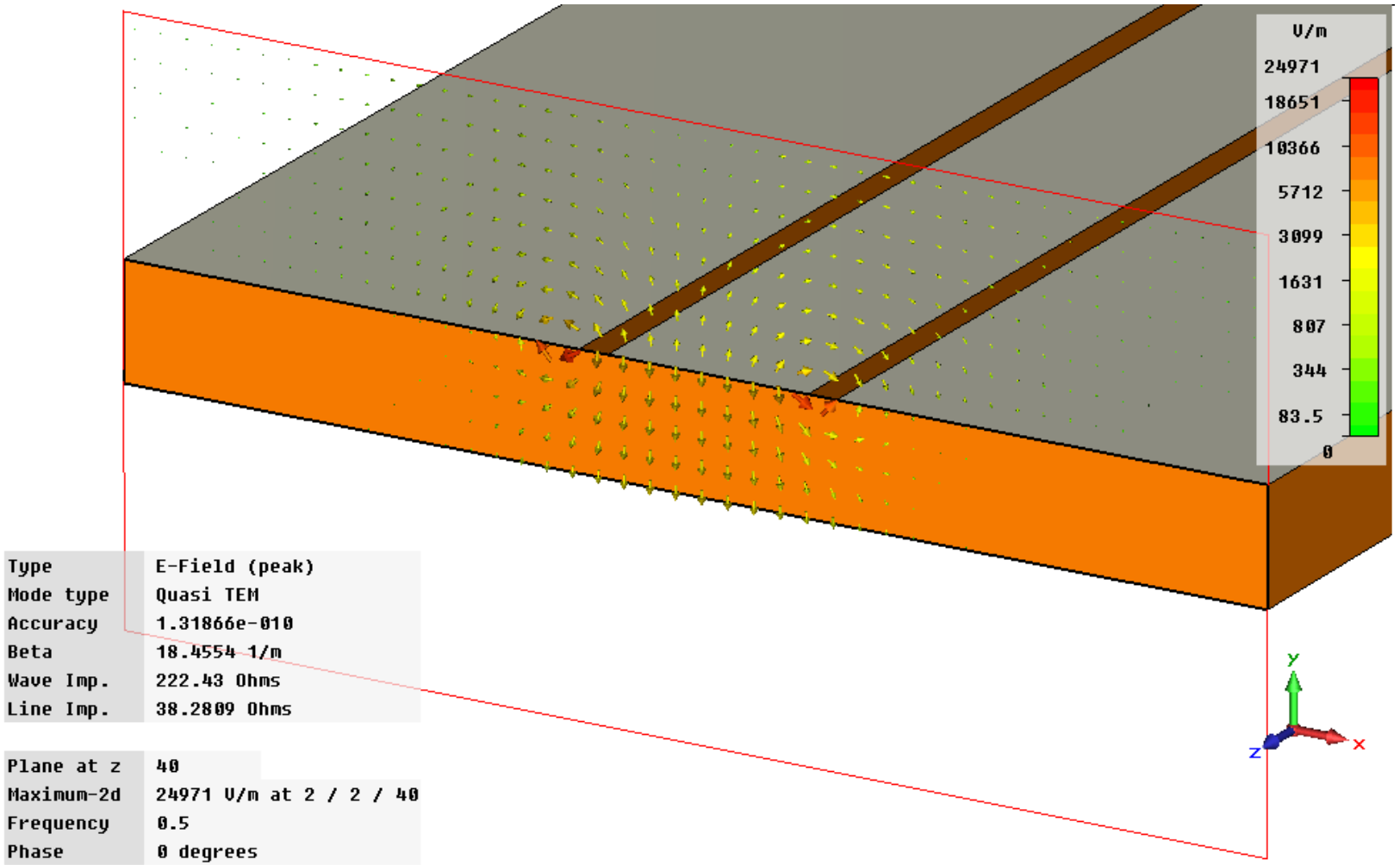
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采用对称面减少仿真时间



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端口模式



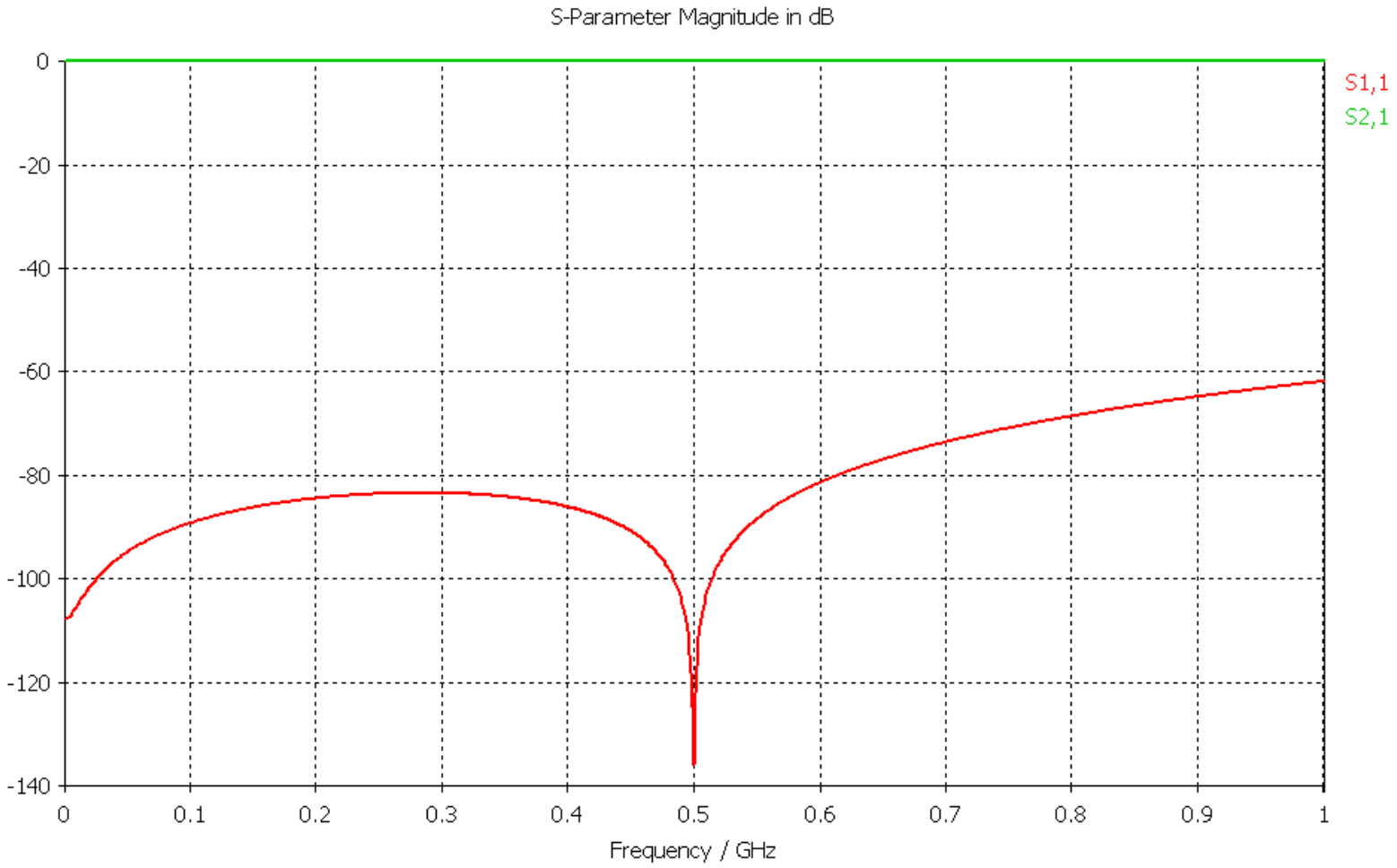
Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	1.31866e-010
Beta	18.4554 1/m
Wave Imp.	222.43 Ohms
Line Imp.	38.2809 Ohms

Plane at z	40
Maximum-2d	24971 V/m at 2 / 2 / 40
Frequency	0.5
Phase	0 degrees

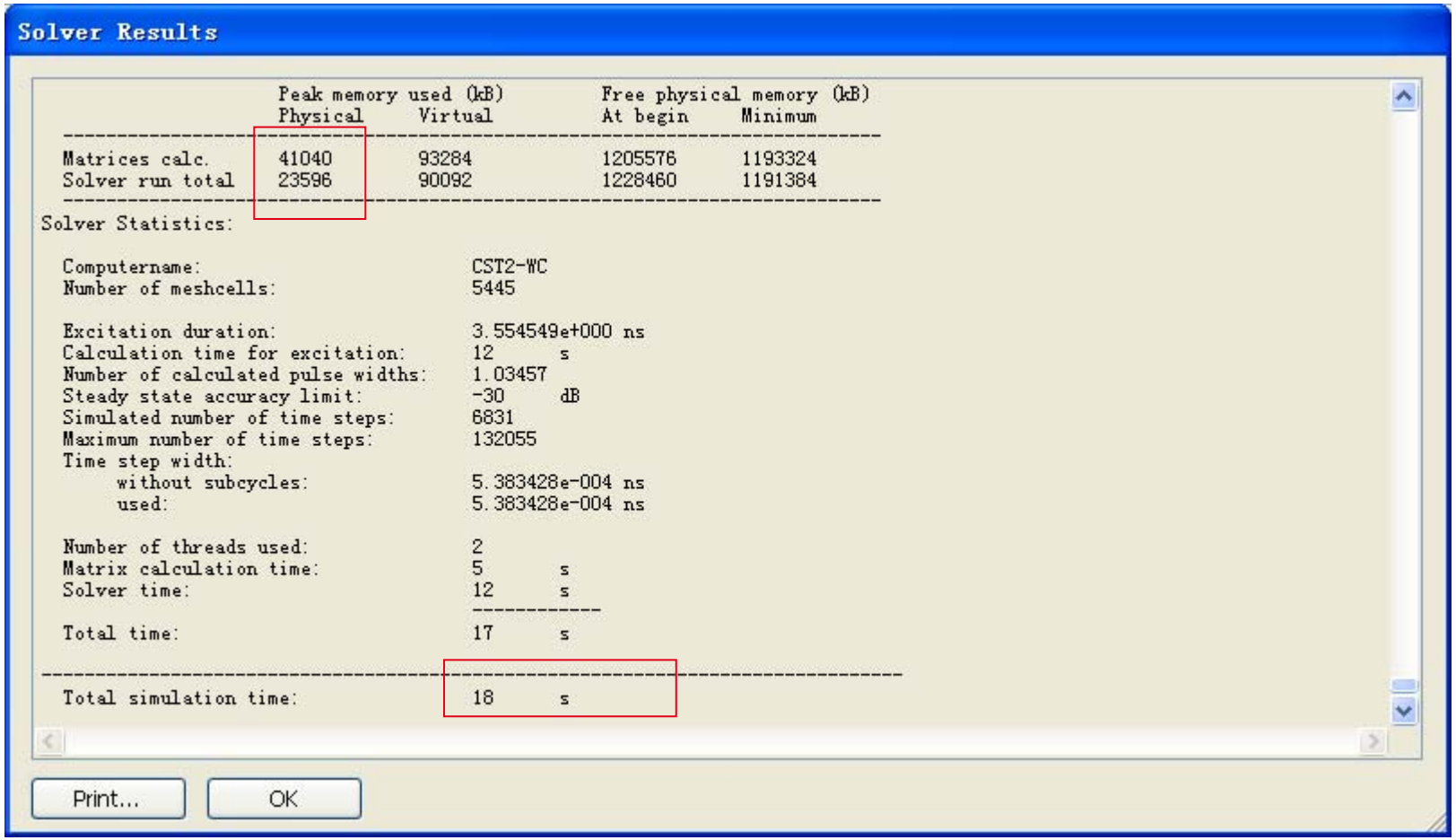
由于加上了接地板，因此端口模式场有所变化，端口线阻抗也随之变化。

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S11、S21

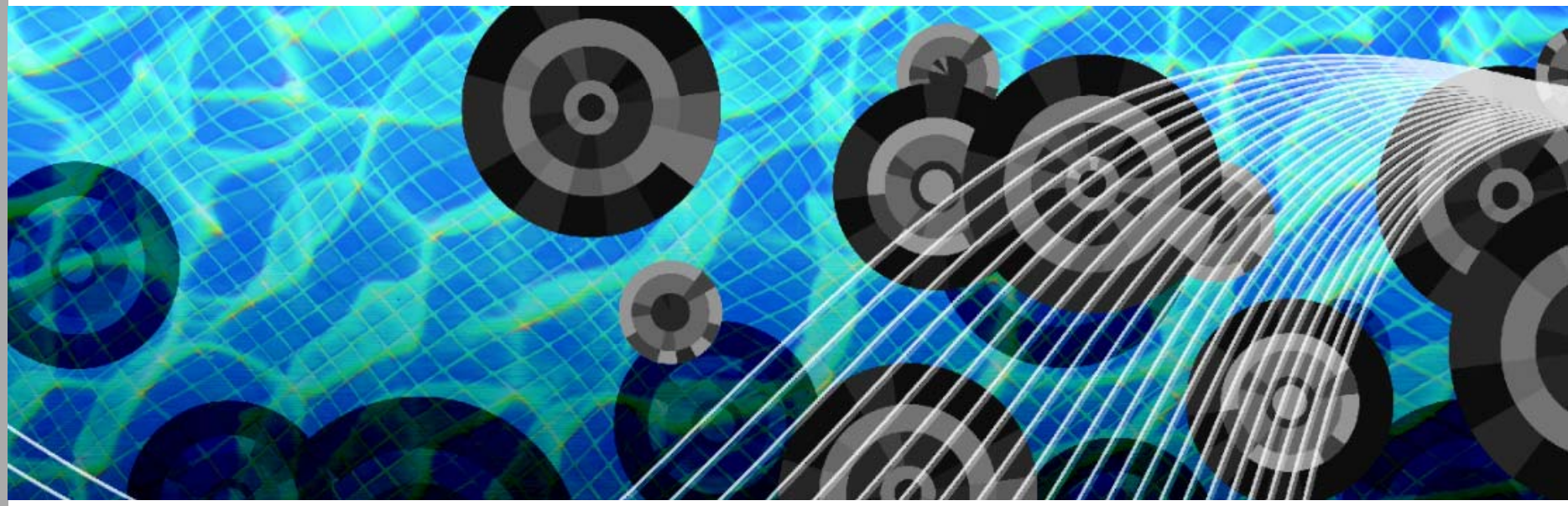

S-Parameter Magnitude in dB
0
-20
-40
-60
-80
-100
-120
-140
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Frequency / GHz
S1,1
S2,1

占用系统资源

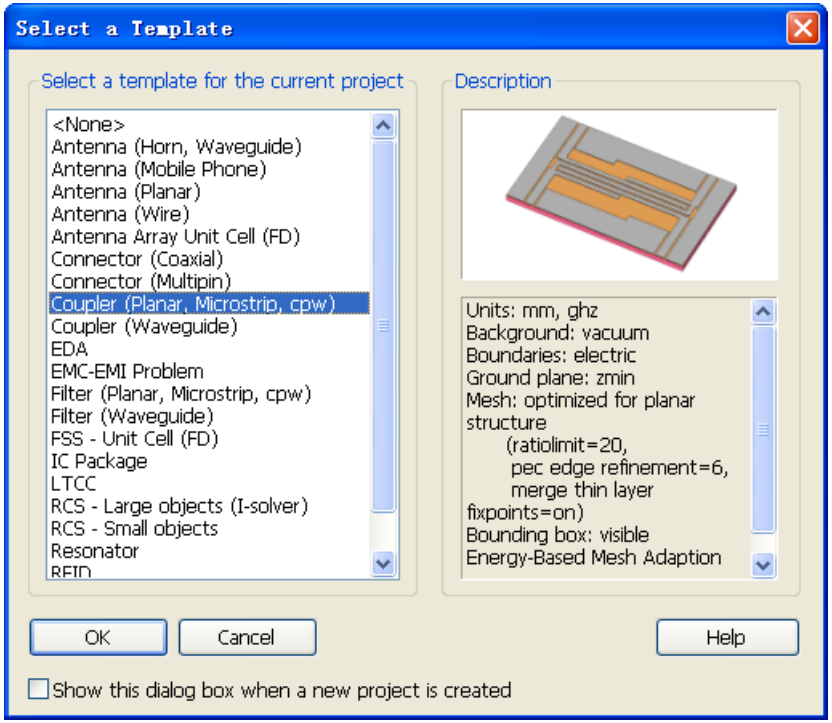


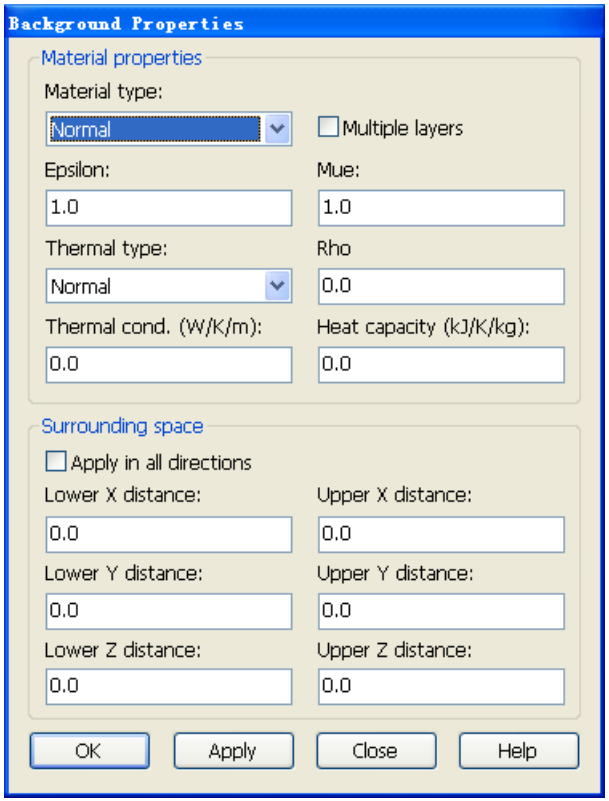
建立矩阵占用内存20M，求解占用内存40M，计算时间18s。

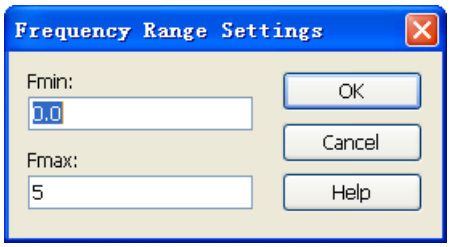
渐变耦合共面波导



单位、背景条件、频带范围设置



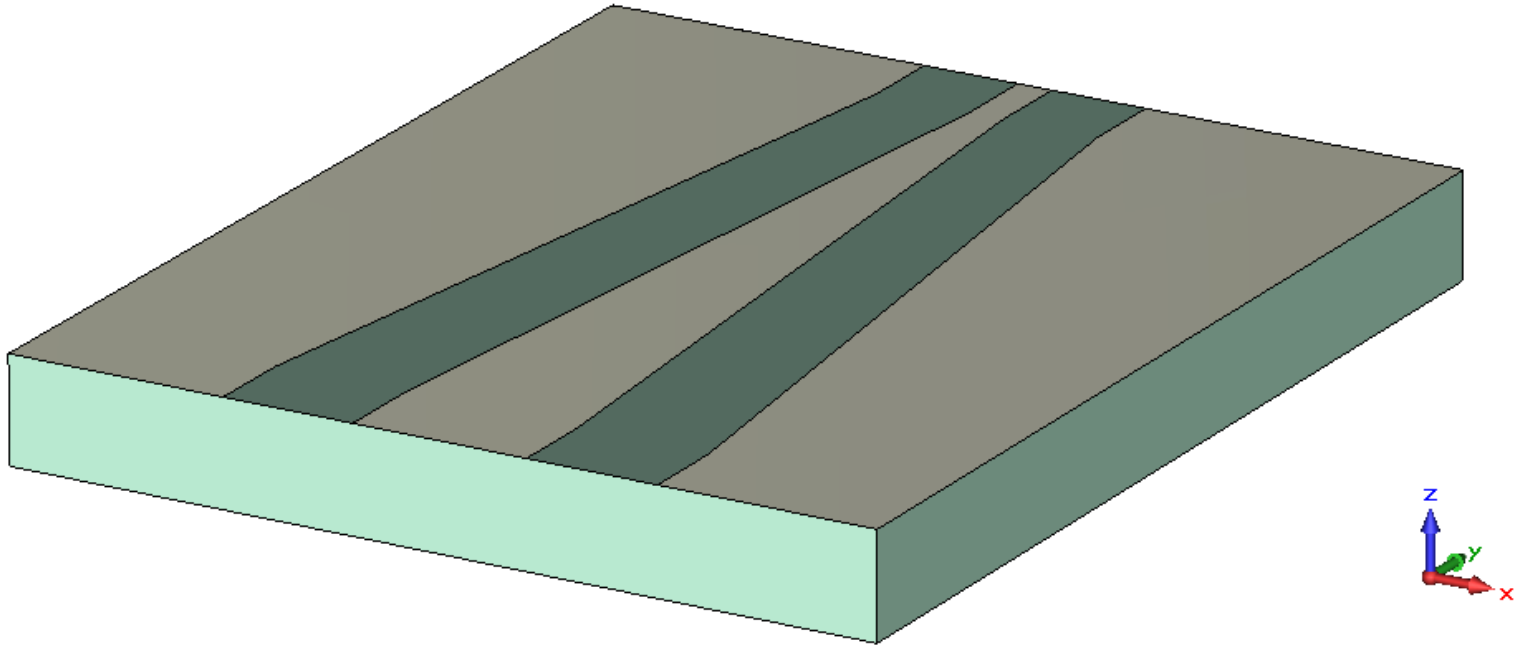




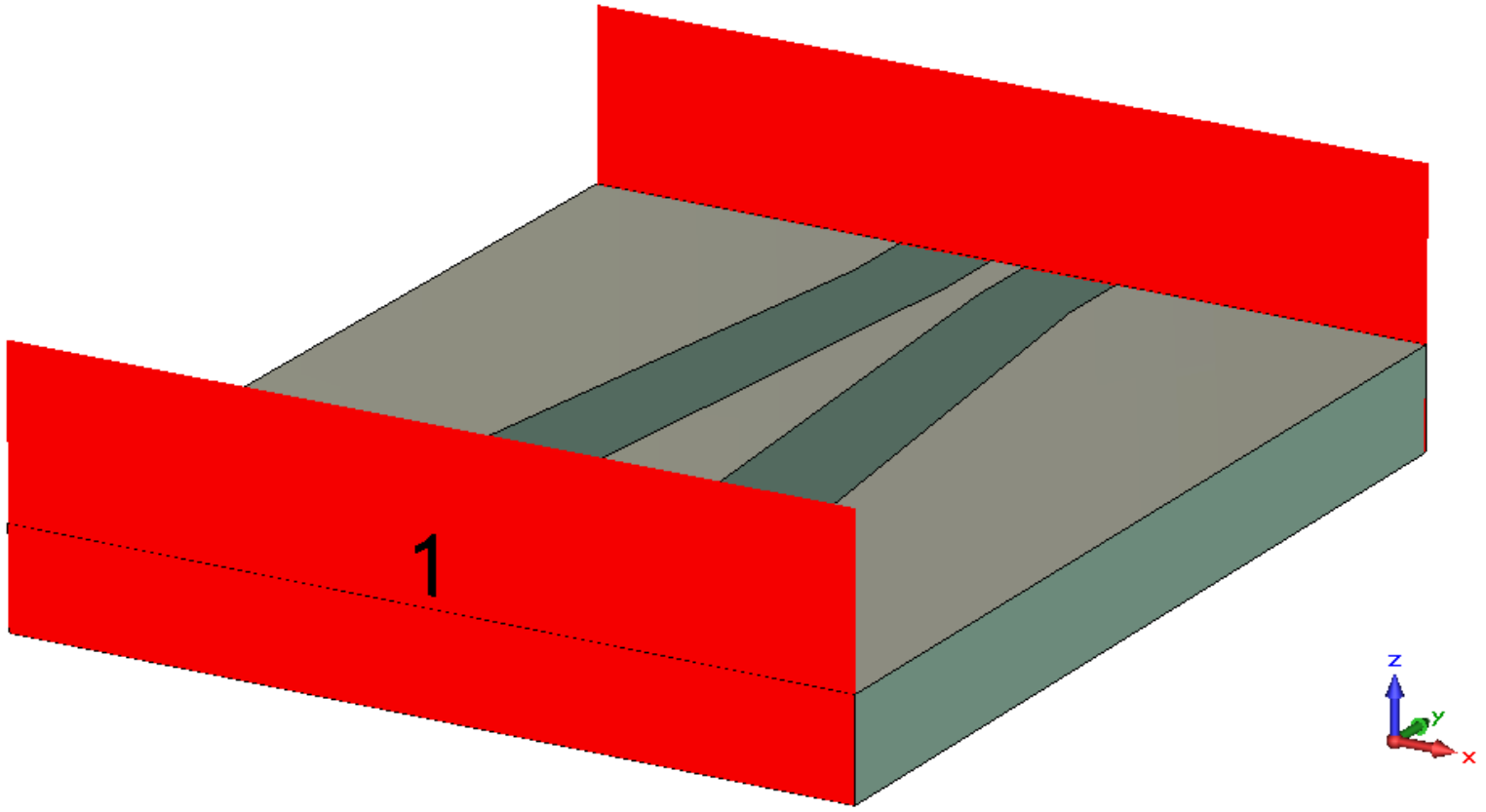
选好模板之后，默认单位为GHz，mm。

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模型

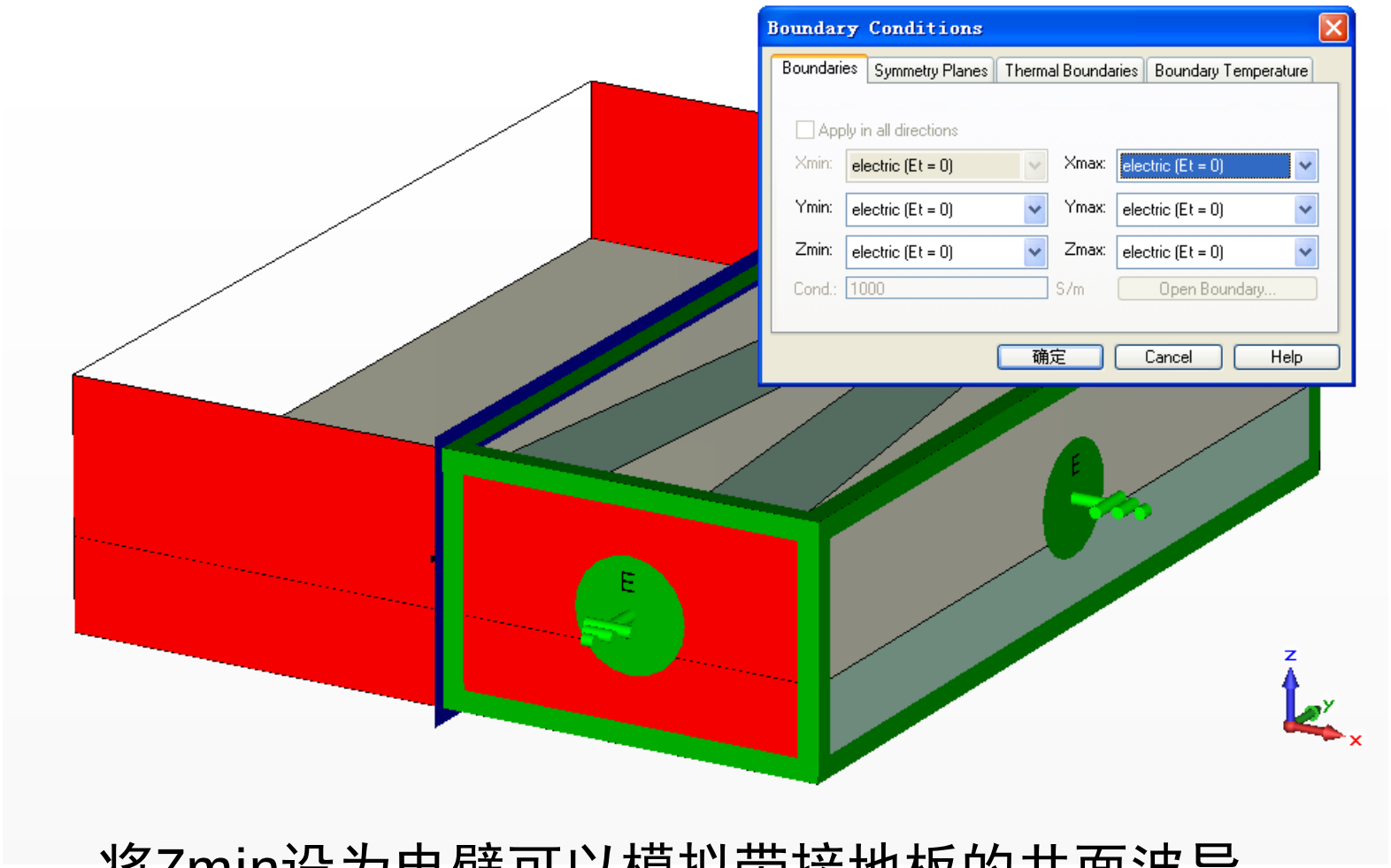


端口设置



若中心带线宽度为W的话，则端口长度至少要保证2-3个W，高度至少为1个W。详情可见CST丛书3 P88-90。

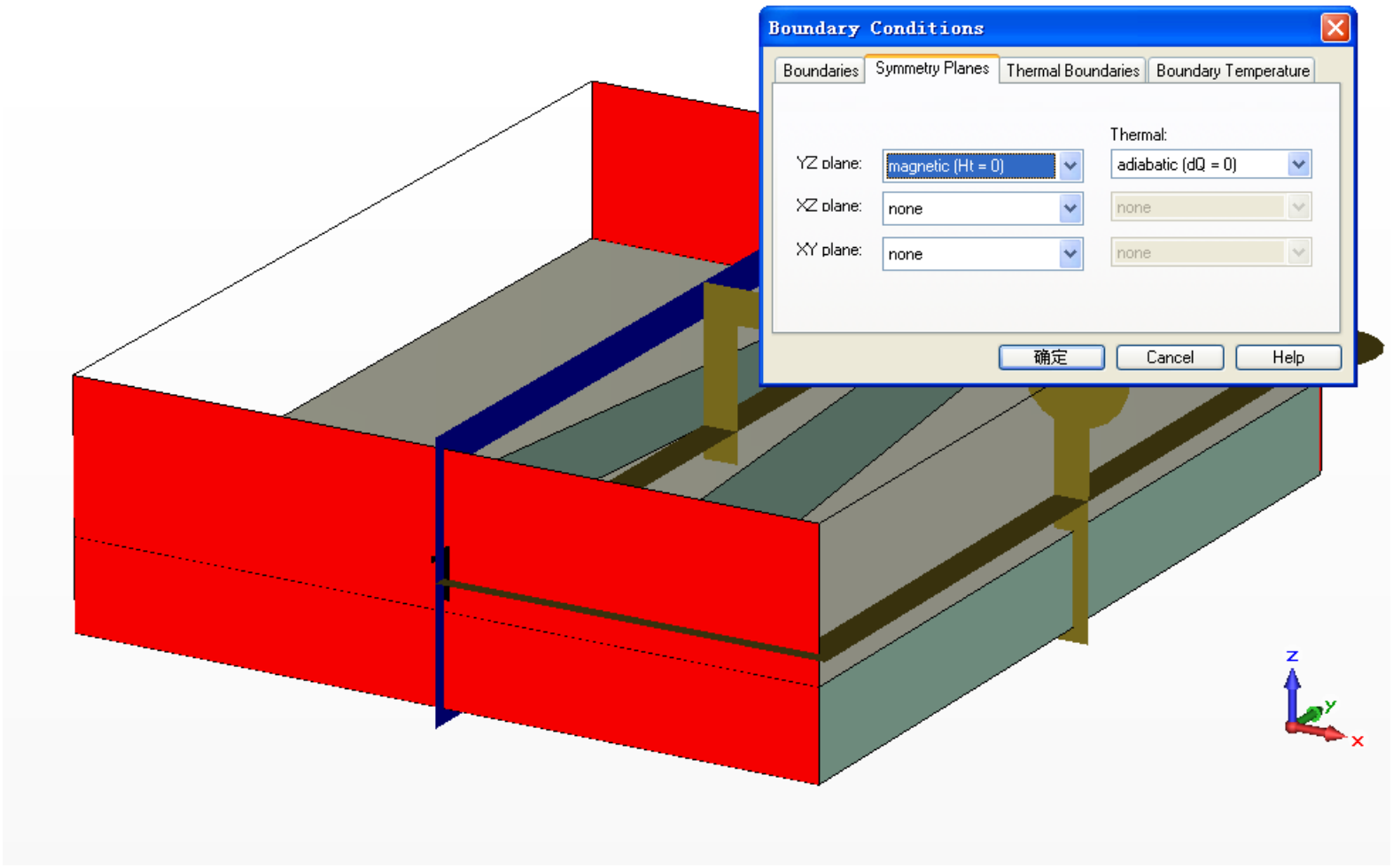
边界条件设置



将Zmin设为电壁可以模拟带接地板的共面波导。

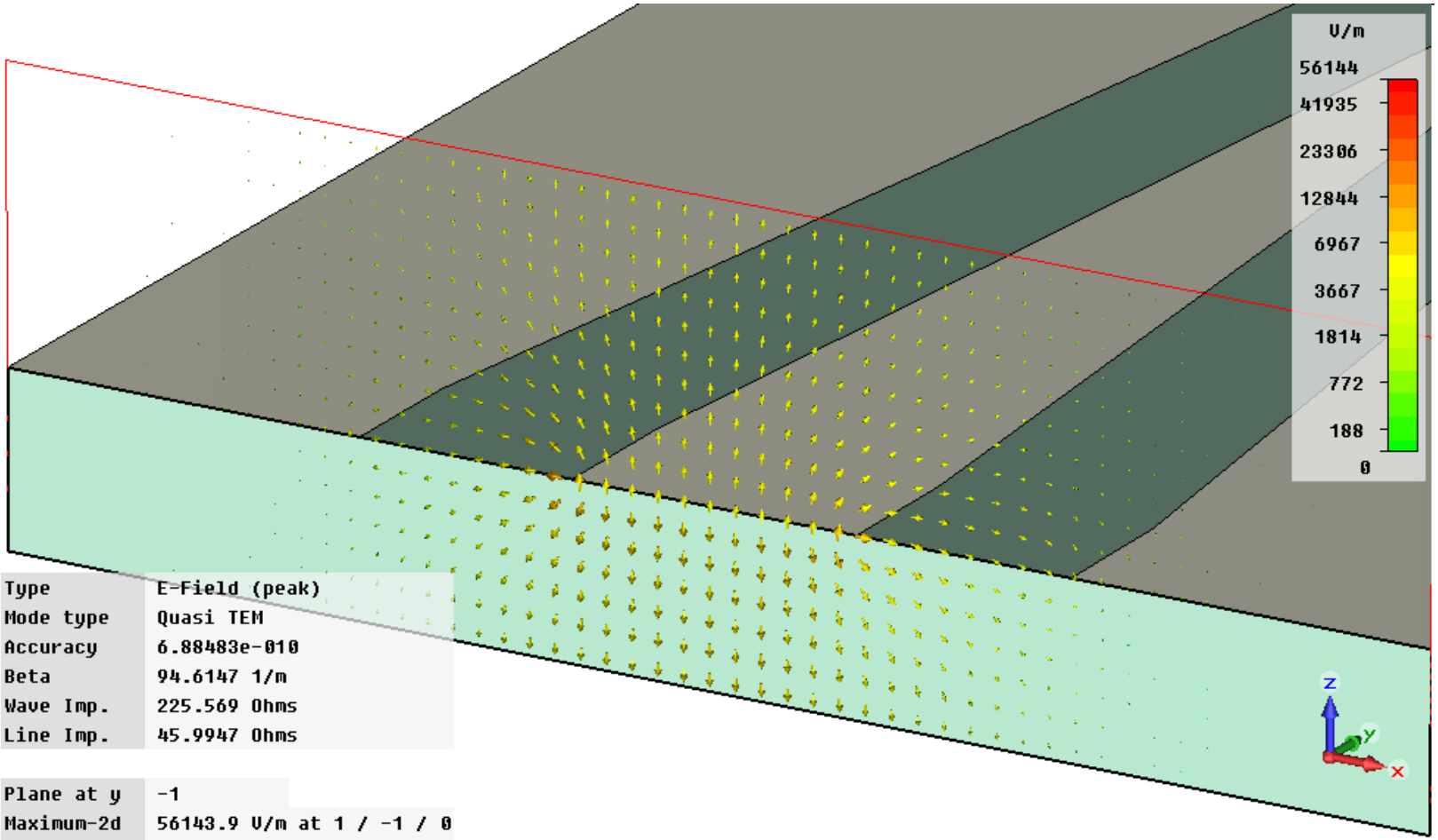
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采用对称面减少仿真时间



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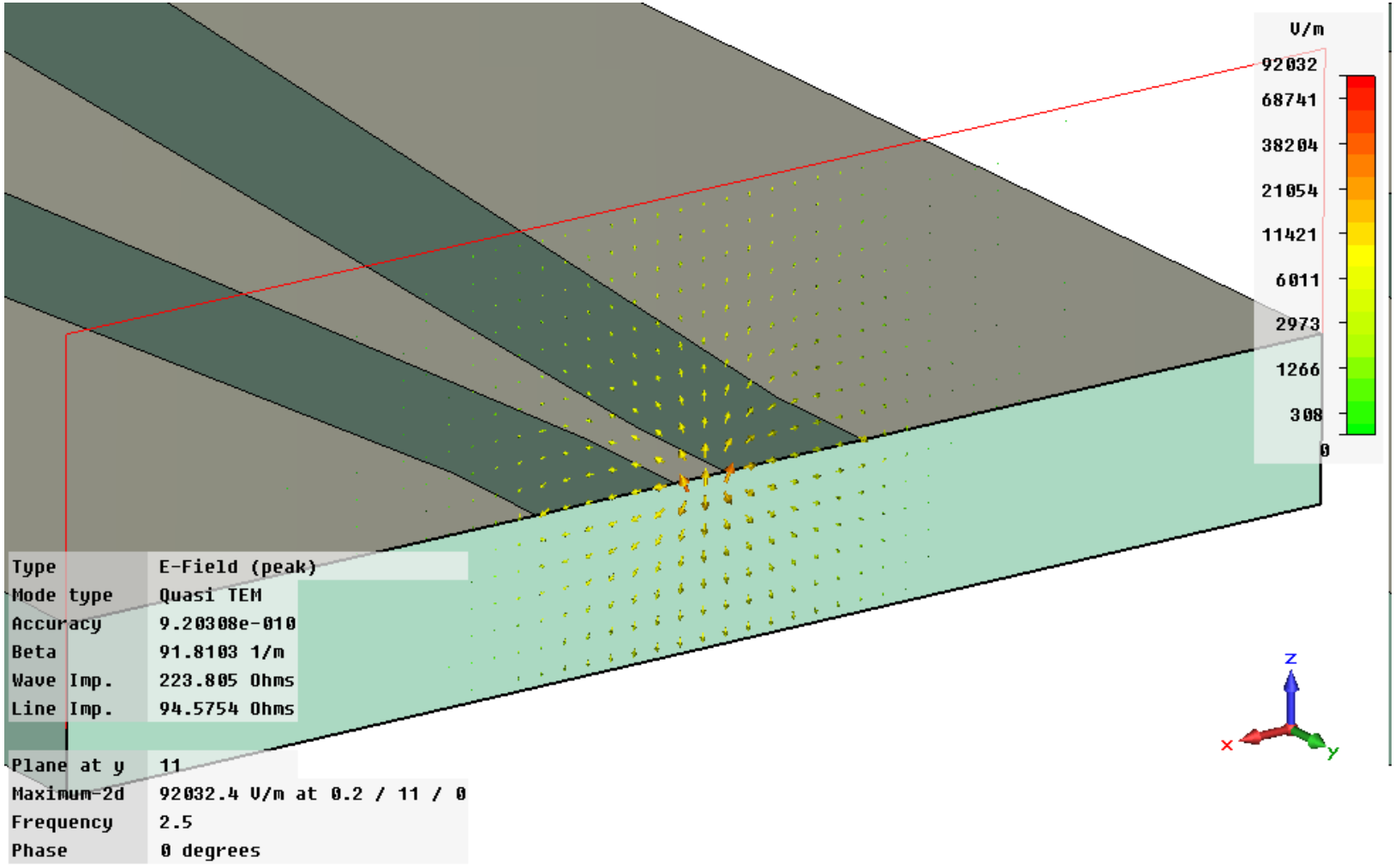
端口1模式场



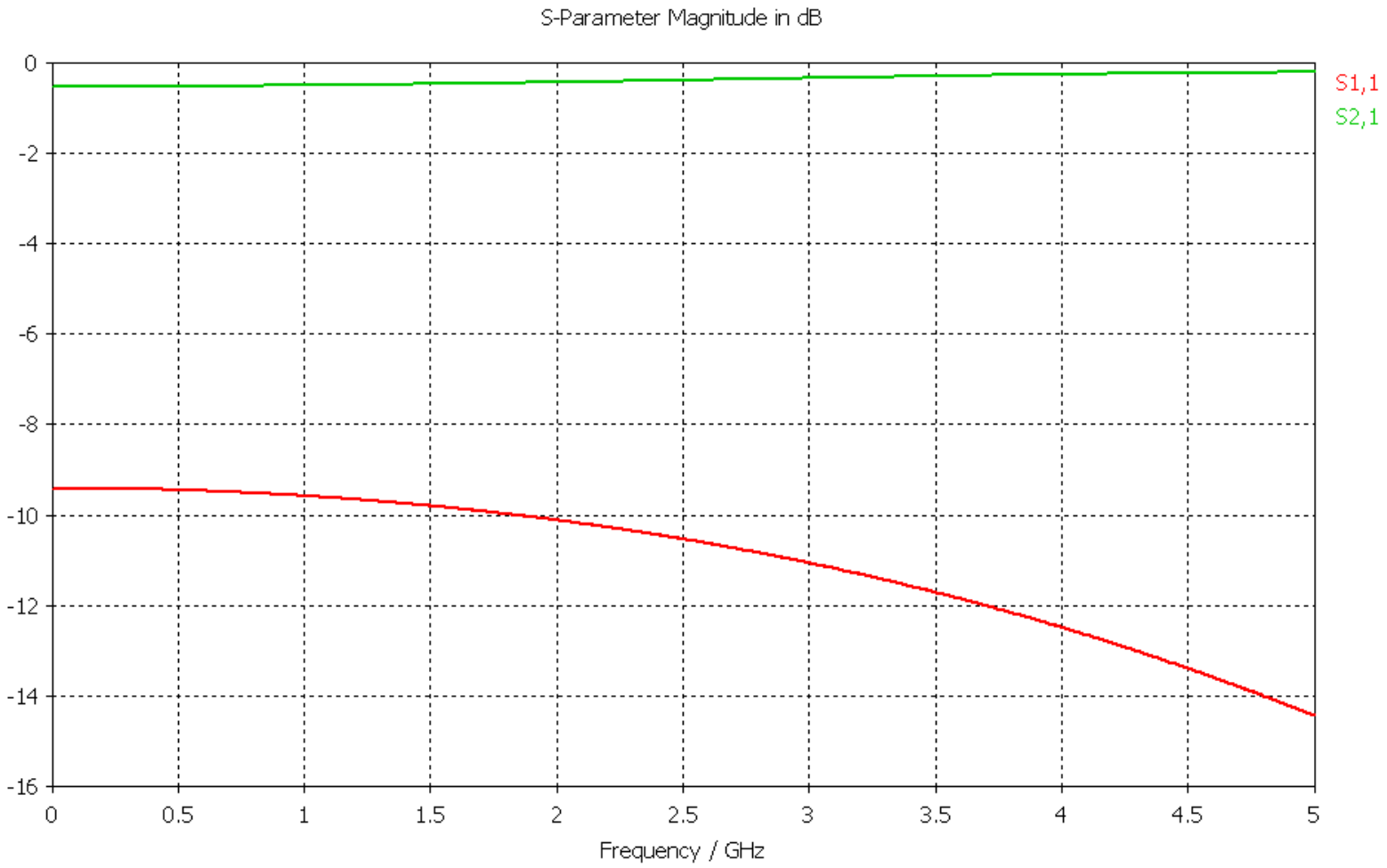
Type	E-Field (peak)
Mode type	Quasi TEM
Accuracy	6.88483e-010
Beta	94.6147 1/m
Wave Imp.	225.569 Ohms
Line Imp.	45.9947 Ohms

Plane at y	-1
Maximum-2d	56143.9 V/m at 1 / -1 / 0
Frequency	2.5
Phase	0 degrees

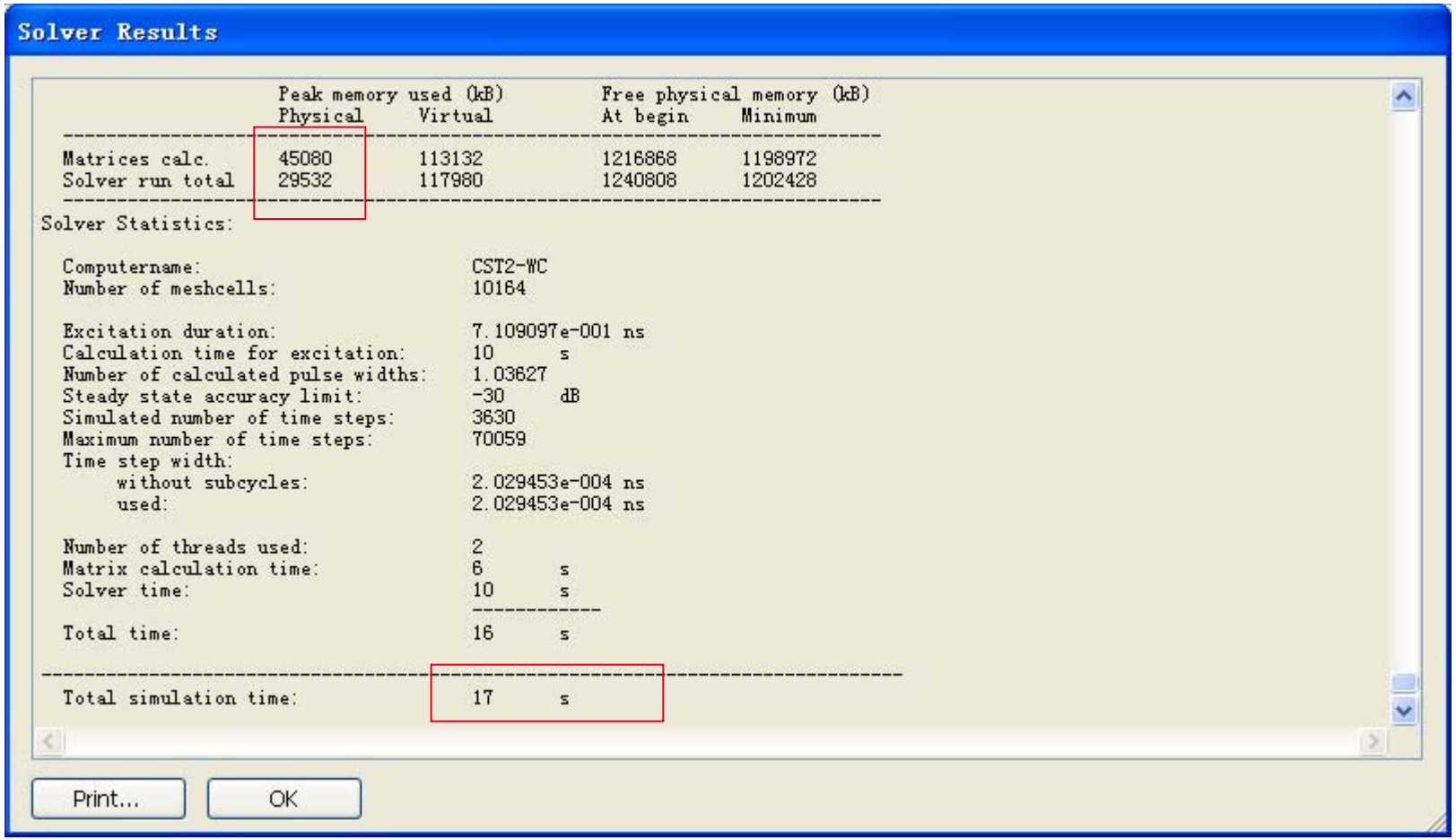
端口2模式场


V/m
92032
68741
38204
21054
11421
6011
2973
1266
308
0
Type E-Field (peak)
Mode type Quasi TEM
Accuracy 9.20308e-010
Beta 91.8103 1/m
Wave Imp. 223.805 Ohms
Line Imp. 94.5754 Ohms
Plane at y 11
Maximum-2d 92032.4 V/m at 0.2 / 11 / 0
Frequency 2.5
Phase 0 degrees
z
x
y

S11、S21


S-Parameter Magnitude in dB
0
-2
-4
-6
-8
-10
-12
-14
-16
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Frequency / GHz
S1,1
S2,1

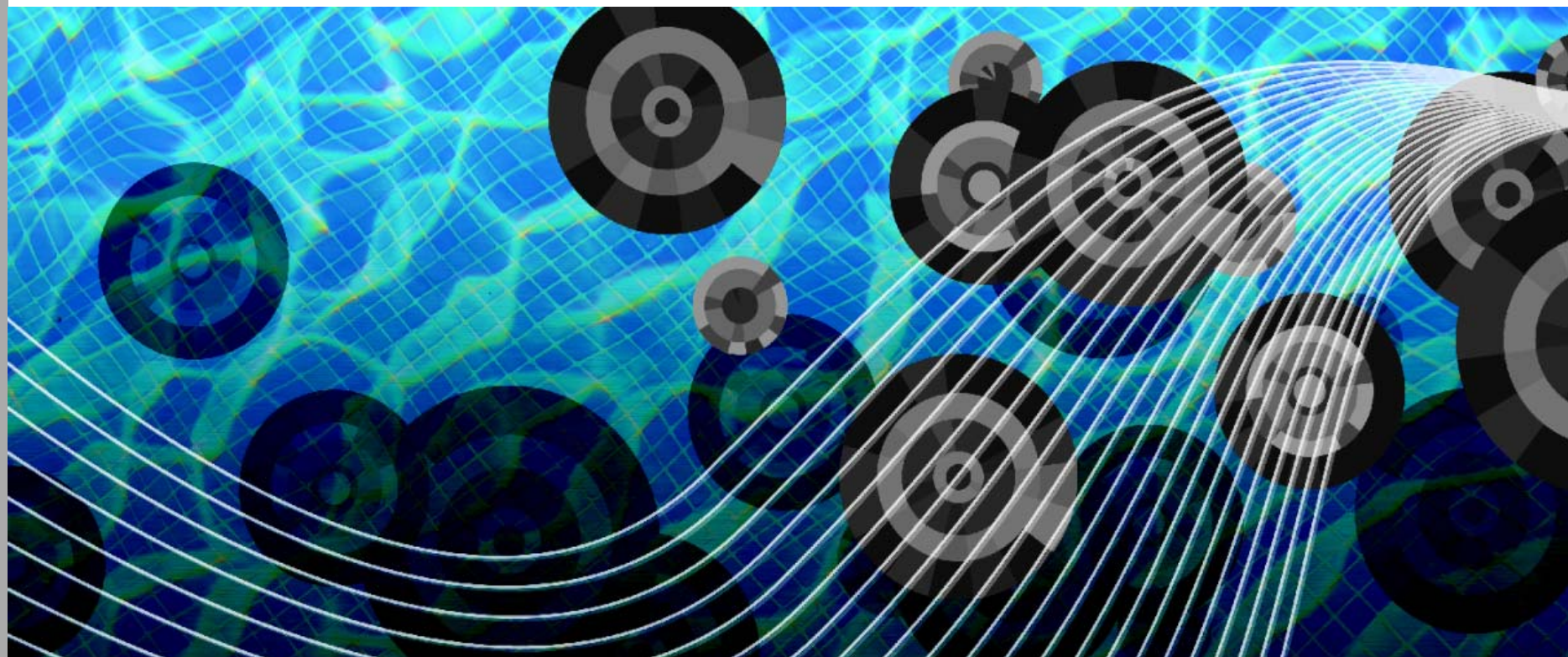
占用系统资源



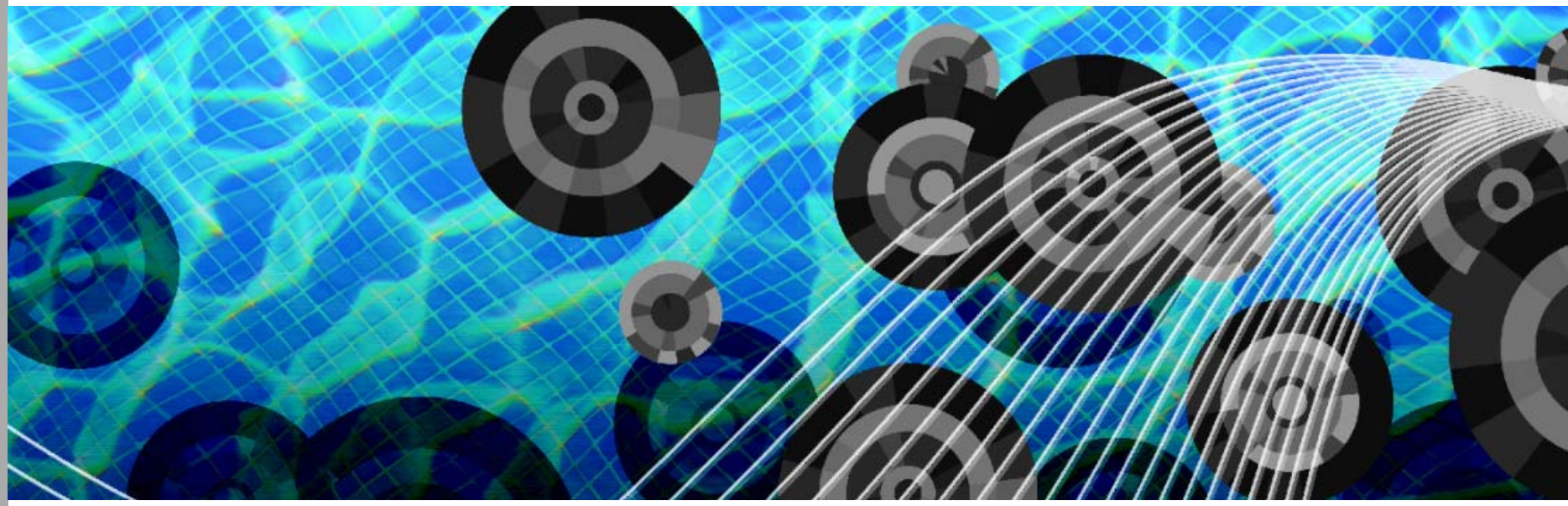
建立矩阵占用内存45M，求解占用内存29M，仿真求解计算时间17s。

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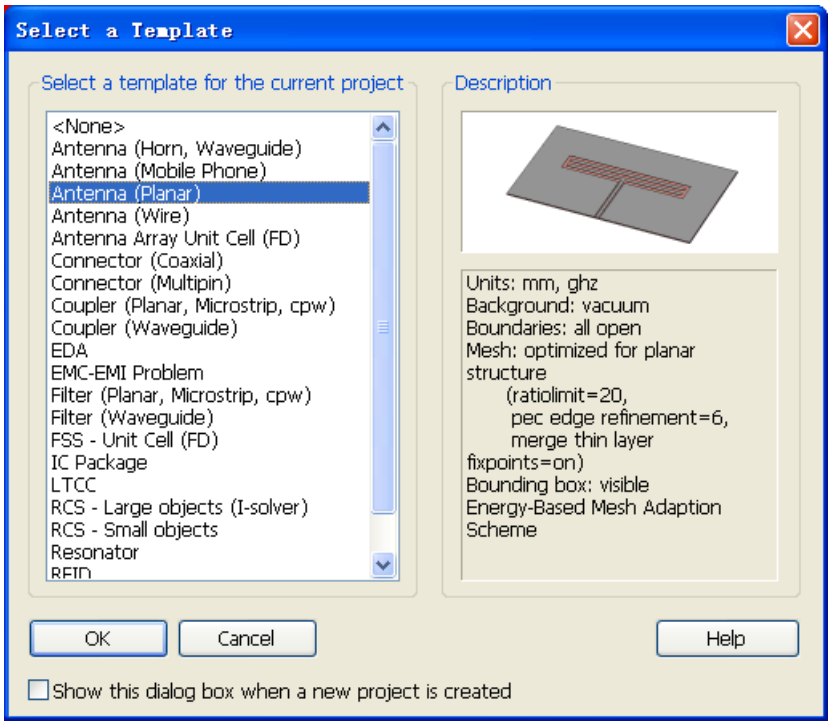
鳍线

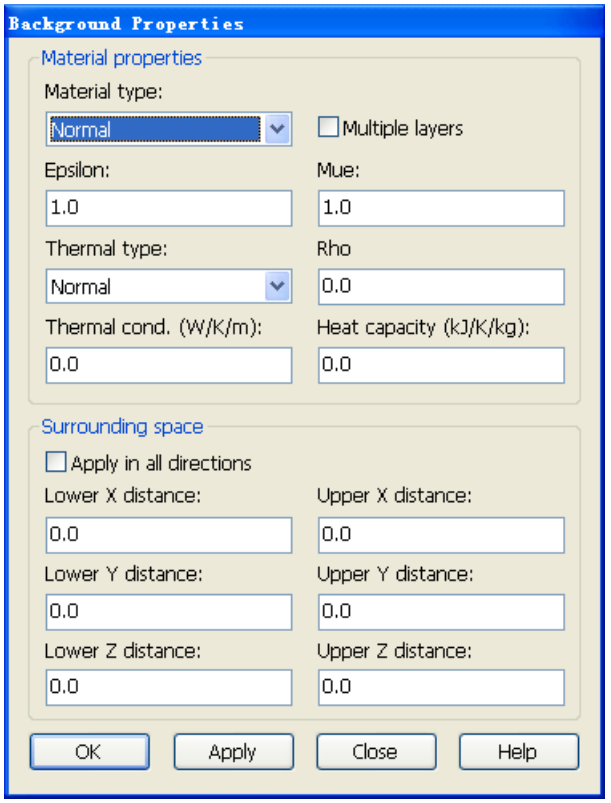


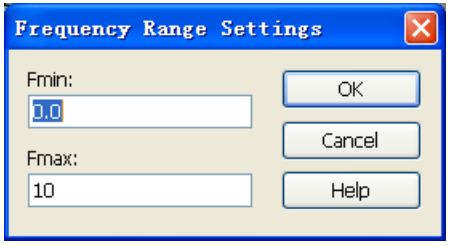
单侧鳍线



选择模板、背景条件、频带范围

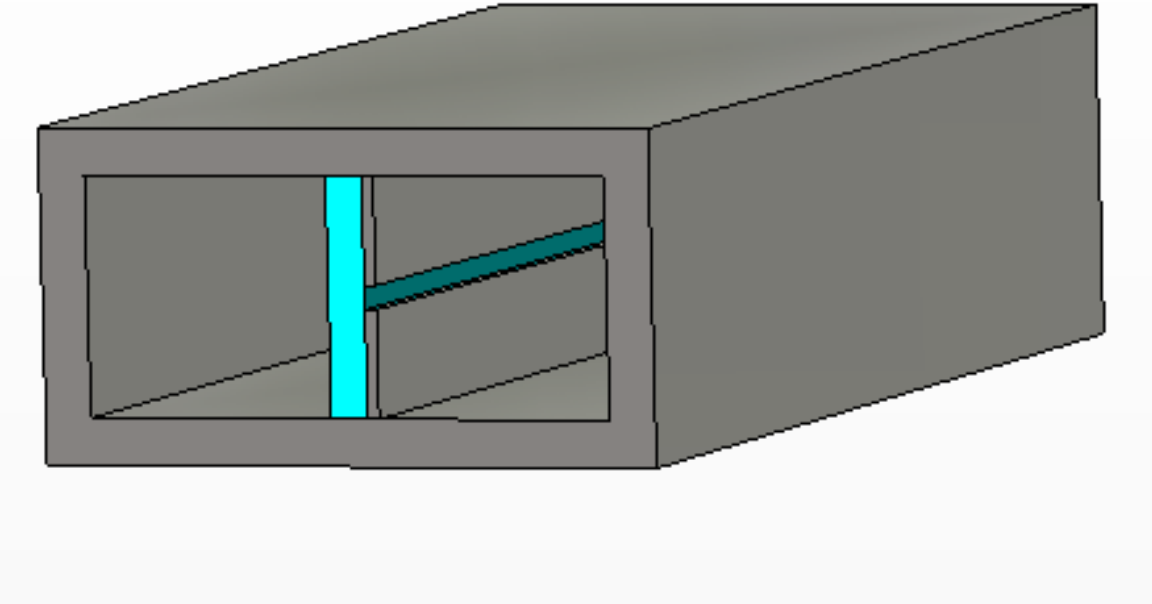




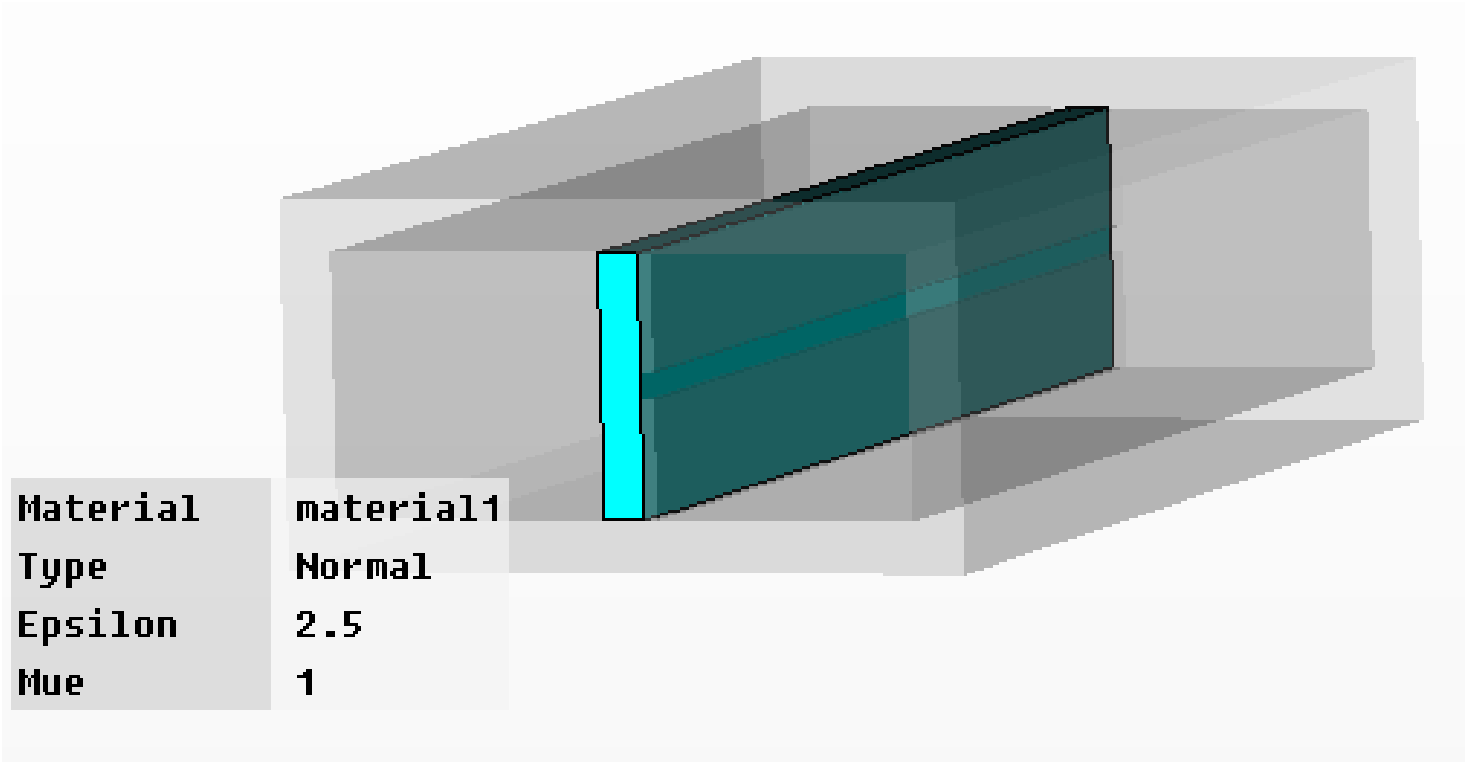


选好模板之后，默认单位为GHz，mm。

单侧鳍线模型



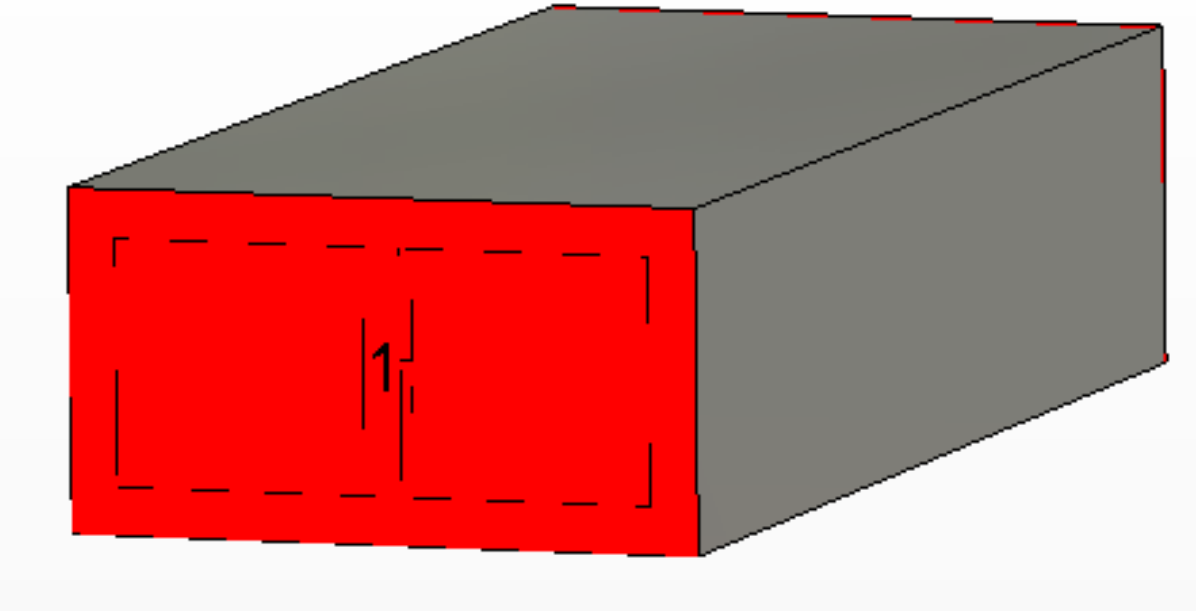
介质板材质



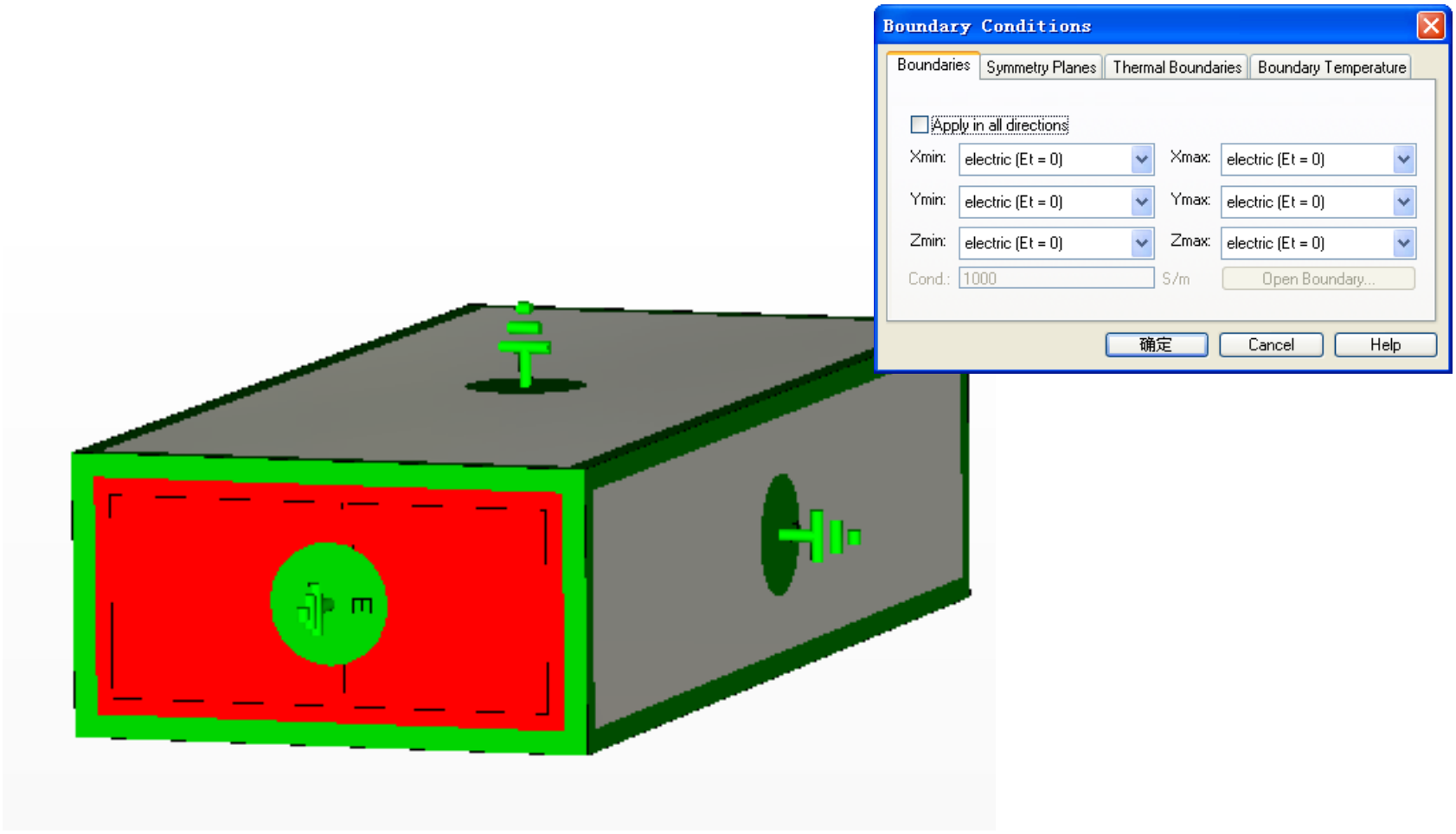
Material	material1
Type	Normal
Epsilon	2.5
Mue	1

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端口设置

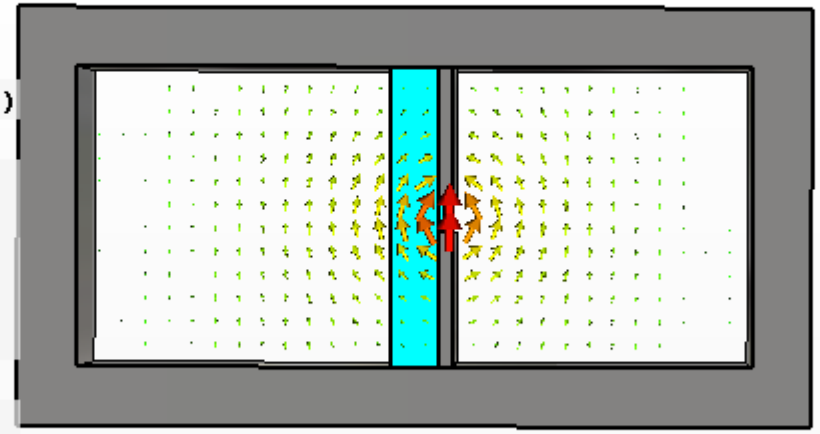


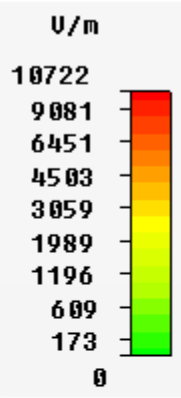
边界条件


Boundary Conditions
Boundaries
Symmetry Planes
Thermal Boundaries
Boundary Temperature
Apply in all directions
Xmin:
electric (Et = 0)
Xmax:
electric (Et = 0)
Ymin:
electric (Et = 0)
Ymax:
electric (Et = 0)
Zmin:
electric (Et = 0)
Zmax:
electric (Et = 0)
Cond.:
1000
S/m
Open Boundary...
确定
Cancel
Help

端口模式场

Type	E-Field (peak)
Mode type	TE
Accuracy	5.17472e-014
Fcutoff	1.81307
Beta	97.66 1/m
Wave Imp.	404.243 Ohms
Plane at z	0
Maximum-2d	10722.5 V/m at -0.8001 / -0.508 / 0
Frequency	5
Phase	0 degrees

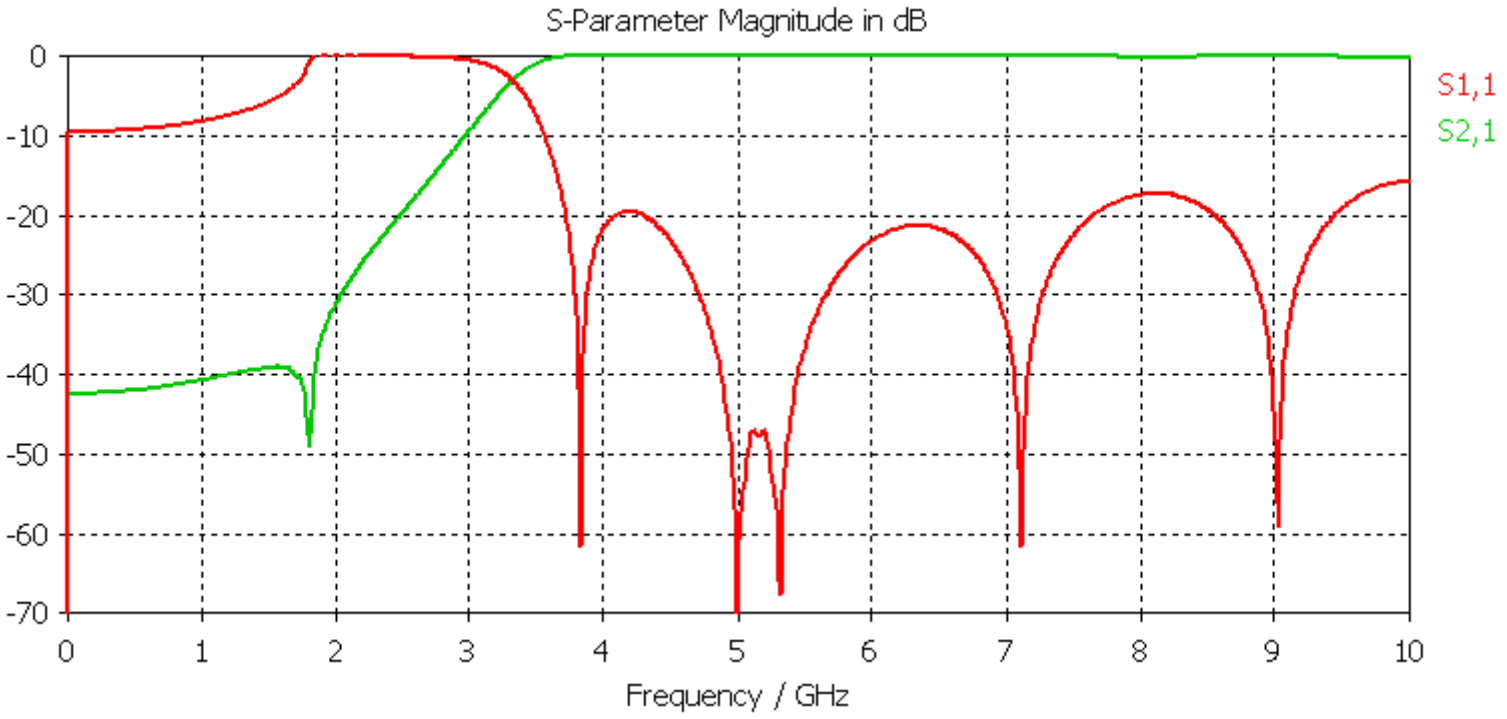




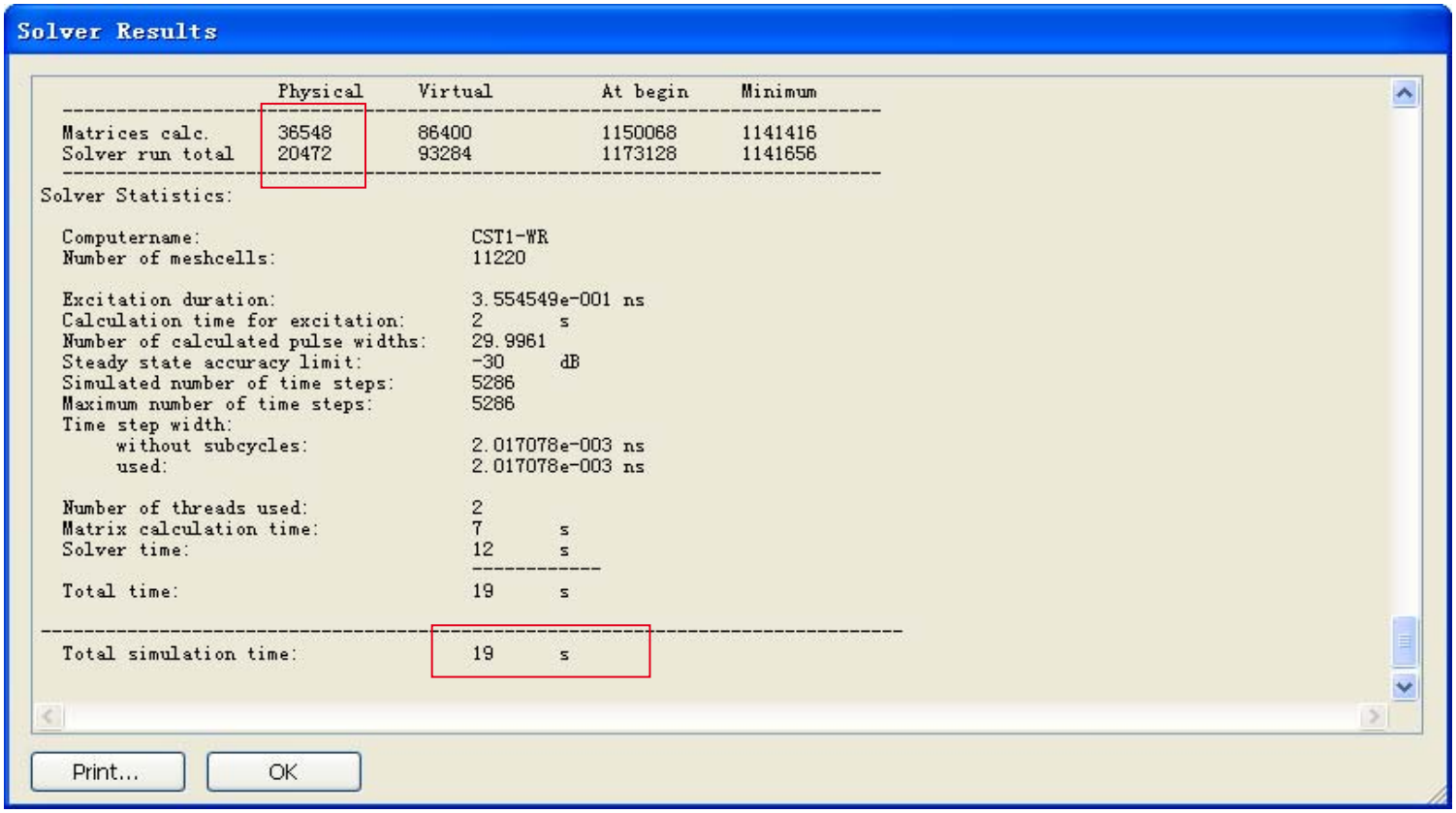


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S11 S21 dB

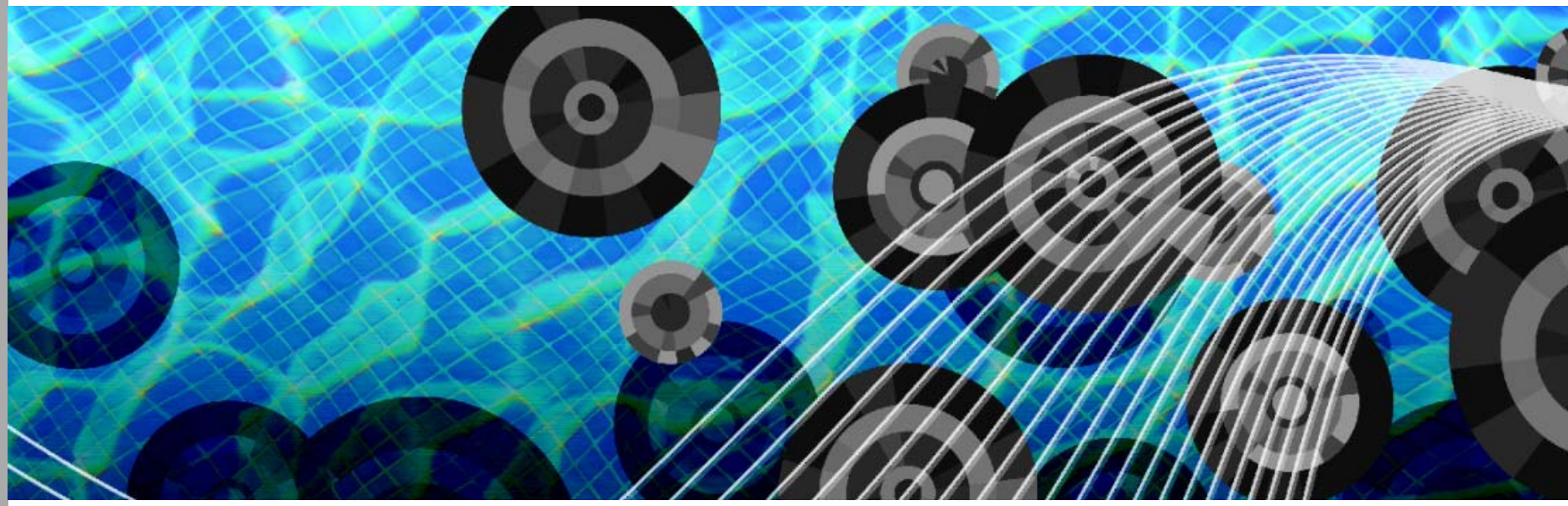

S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
-70
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1
S2,1

占用系统资源

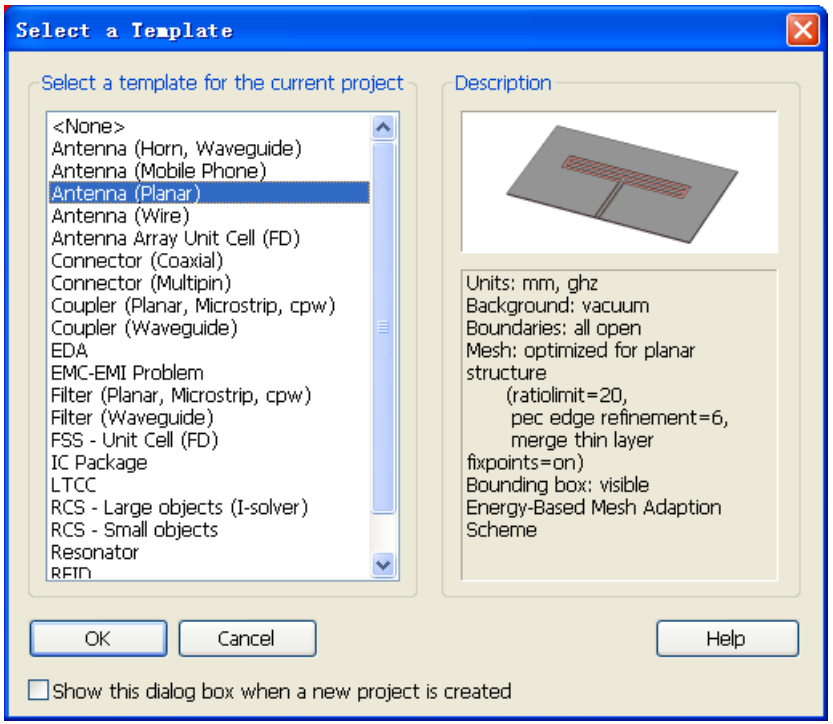


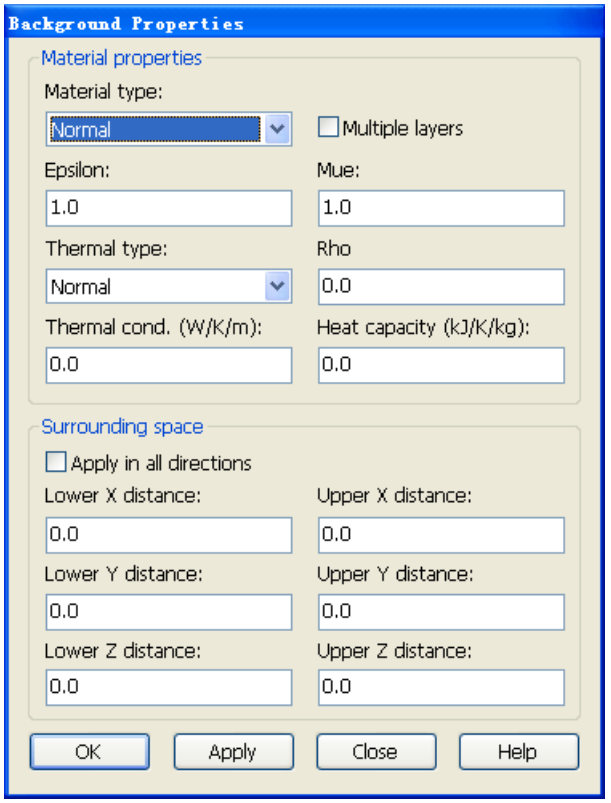
建立矩阵占用内存35M，求解占用内存20M，求解器计算时间19s。

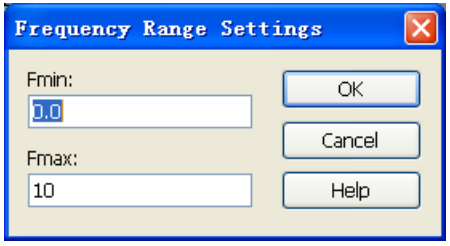
双侧鳍线



选择模板、背景条件、频带范围

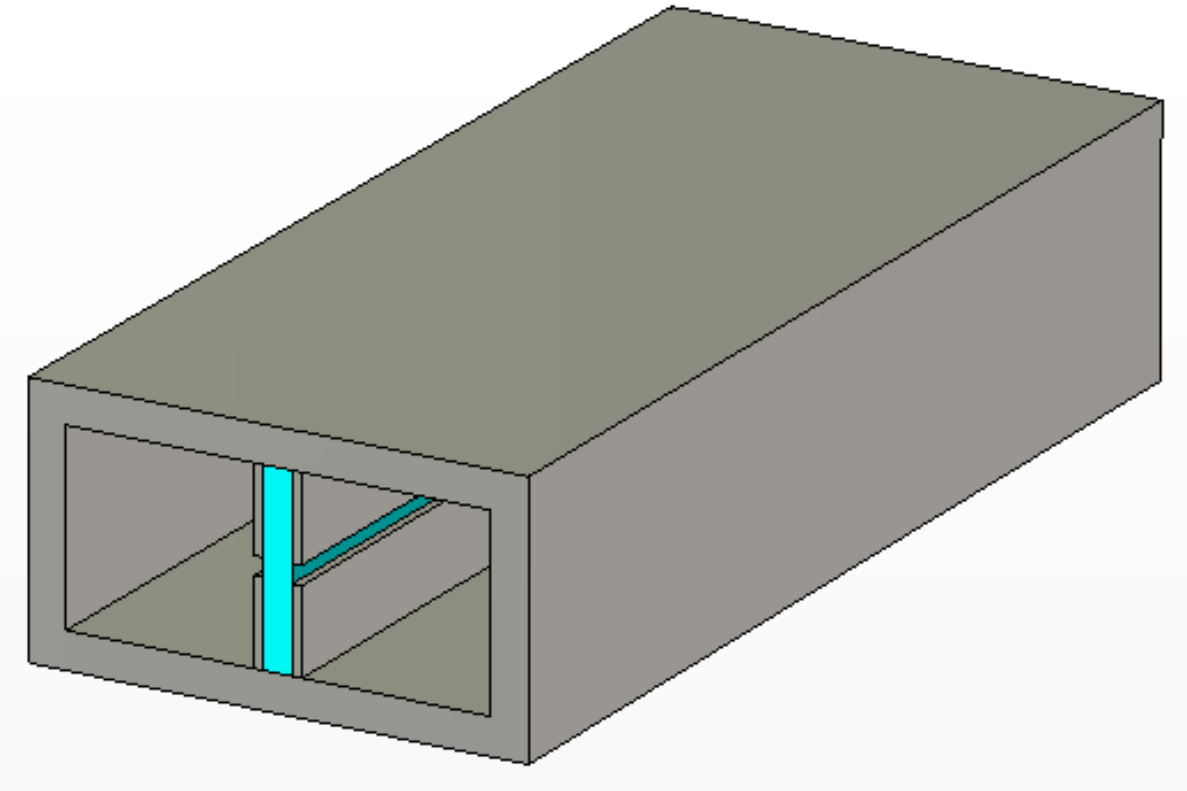




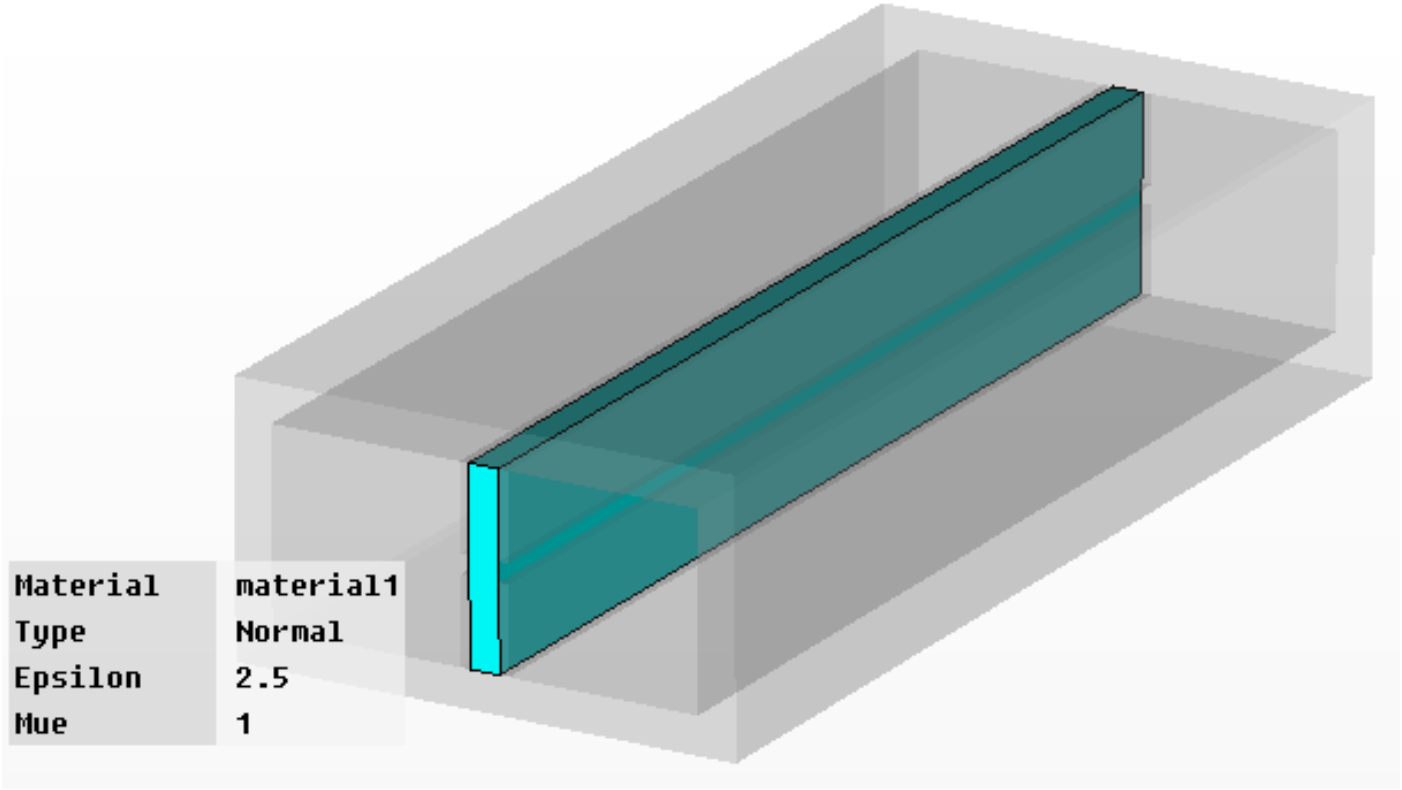


选好模板之后，默认单位为GHz，mm。

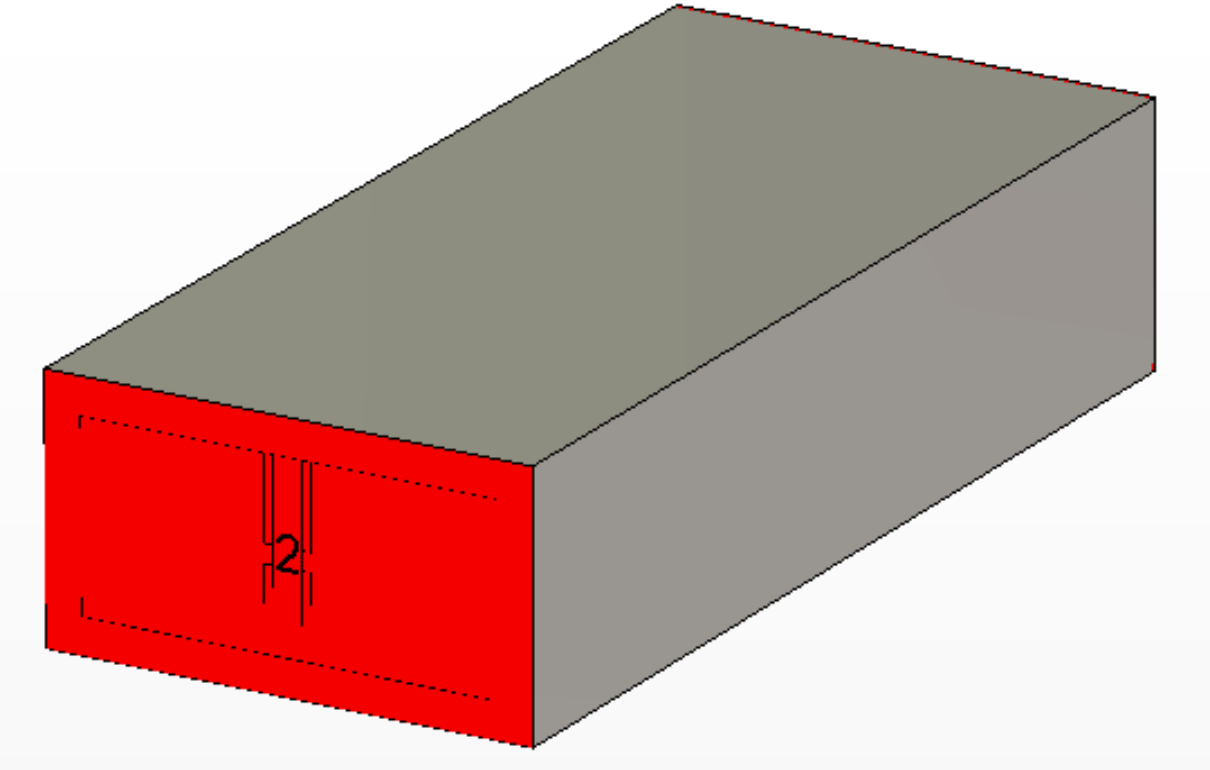
双侧鳍线模型



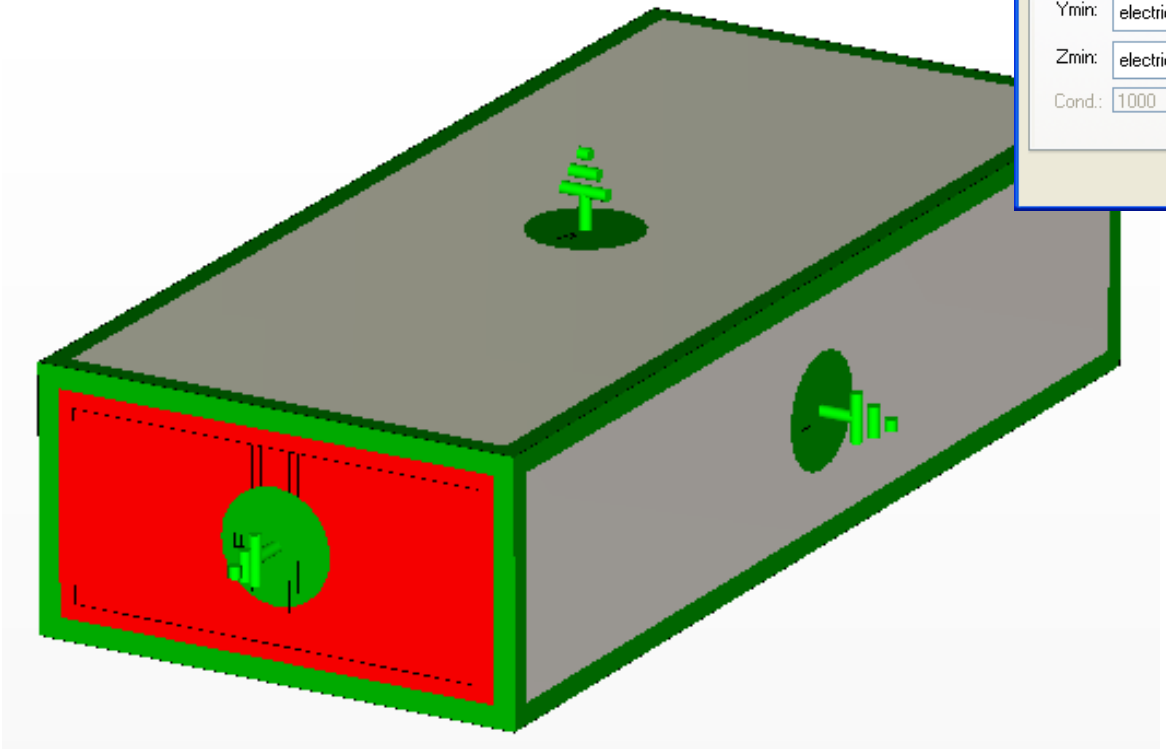
介质板材质

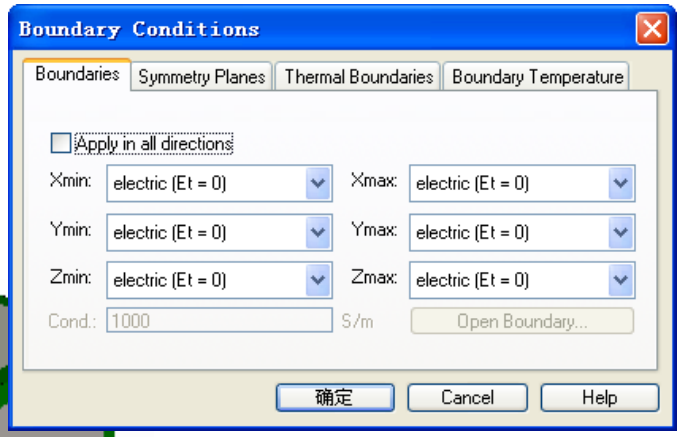


端口设置



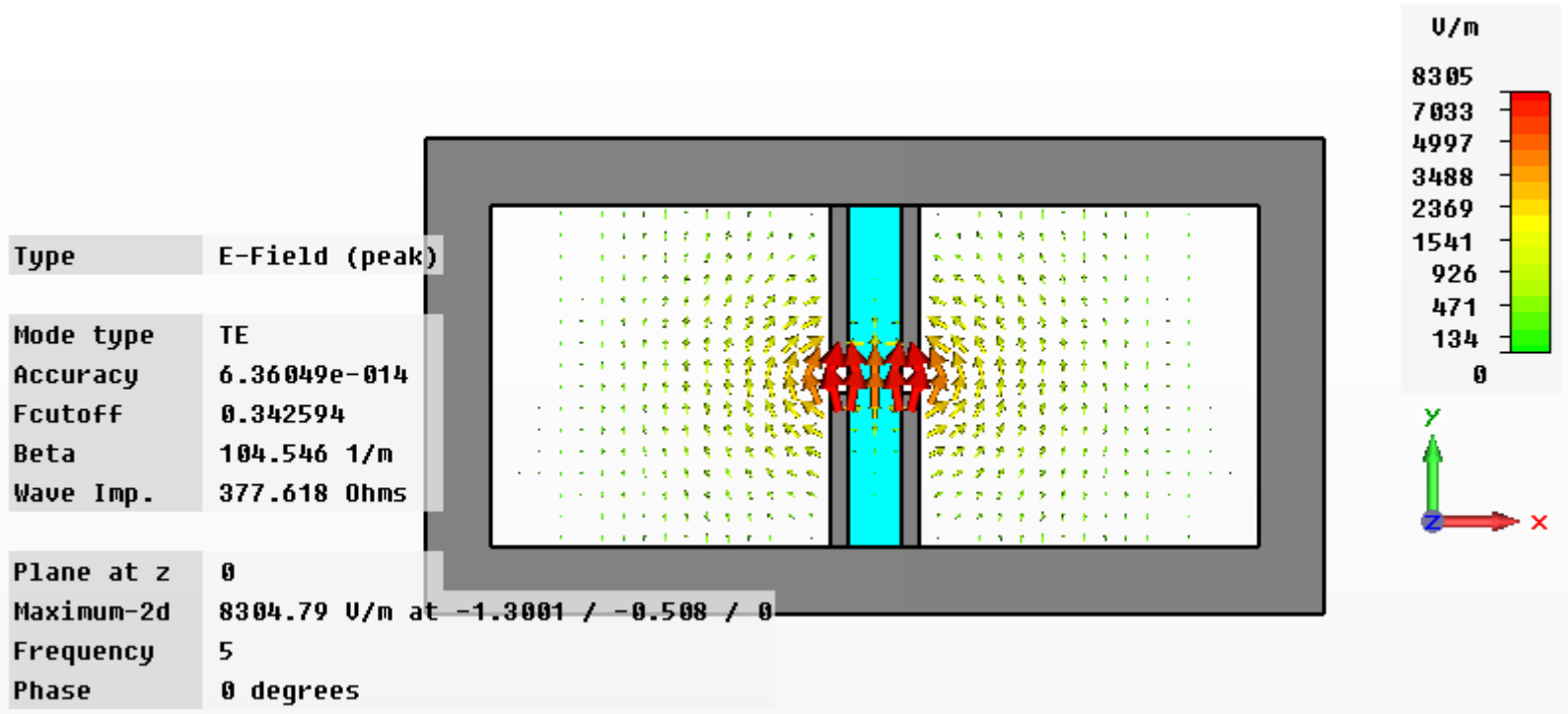
边界条件





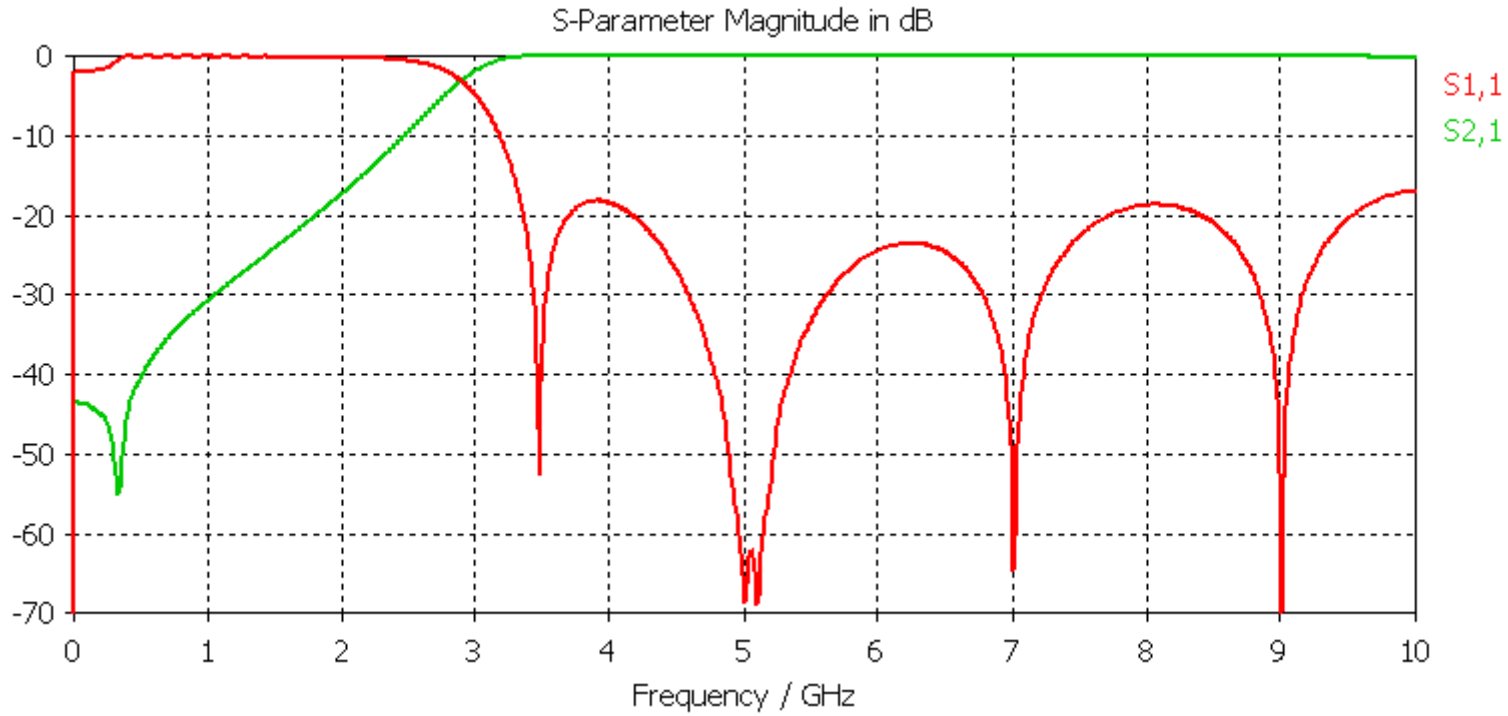
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端口模式场

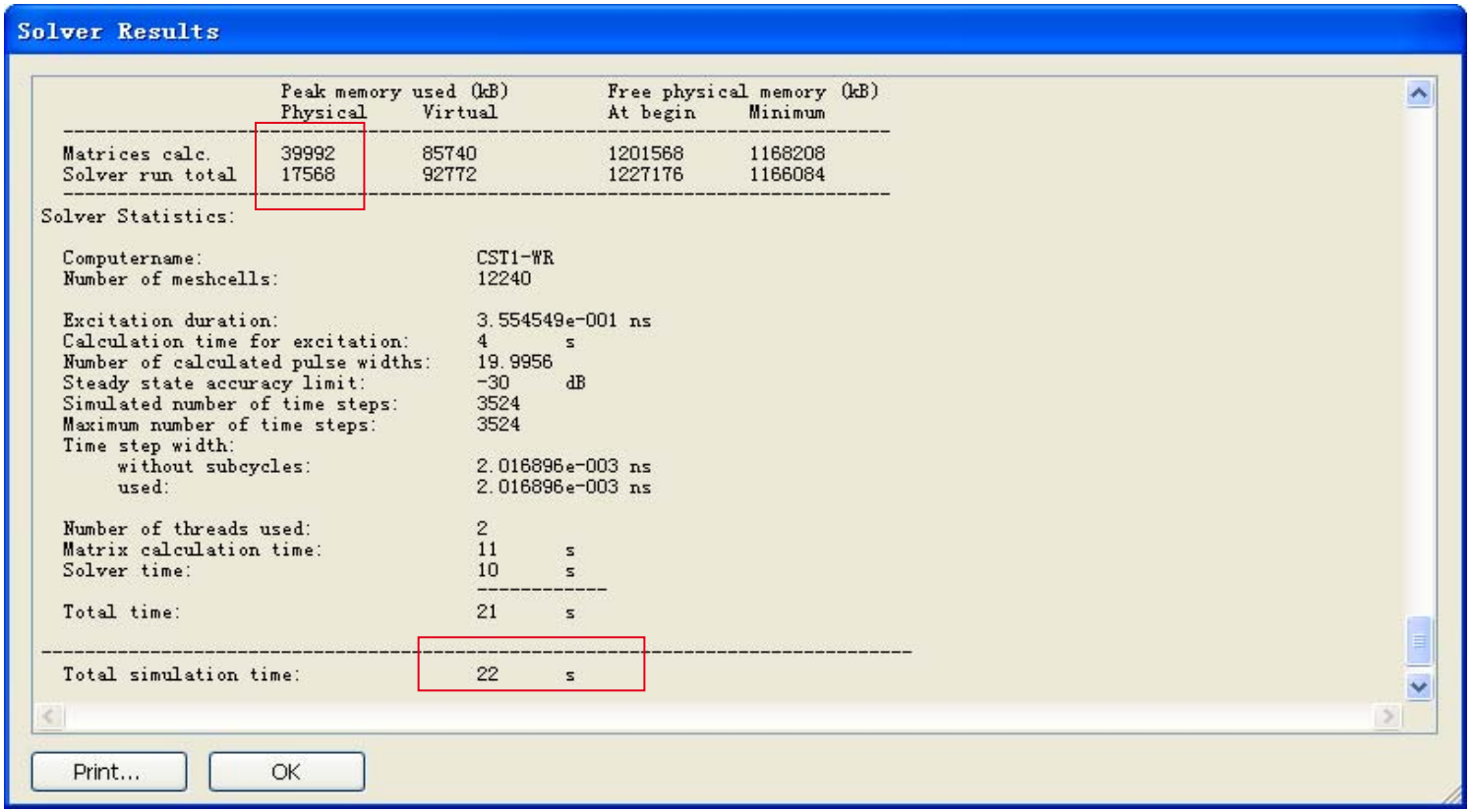


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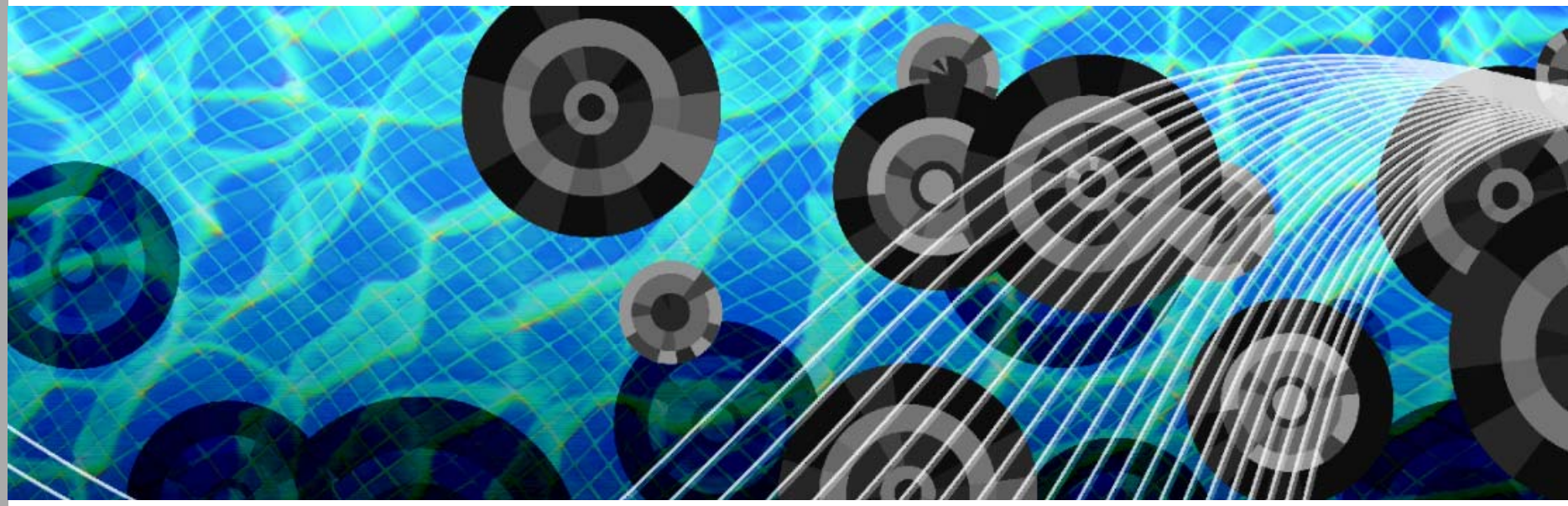
S11 S21 dB


S-Parameter Magnitude in dB
0
-10
-20
-30
-40
-50
-60
-70
0
1
2
3
4
5
6
7
8
9
10
Frequency / GHz
S1,1
S2,1

占用系统资源



建立矩阵占用内存39M，求解占用内存17M，求解器计算时间22s。



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